



US 20250281288A1

(19) **United States**(12) **Patent Application Publication**
ALBITOV et al.(10) **Pub. No.: US 2025/0281288 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **TRANSLUMINAL DELIVERY SYSTEM**

(60) Provisional application No. 63/120,808, filed on Dec. 3, 2020.

(71) Applicant: **CARDIOVALVE LTD.**, Or Yehuda (IL)(72) Inventors: **Michael ALBITOV**, Kiryat Ono (IL);
Maxim KARALNIK, Karmiel (IL);
Meni IAMBERGER, Kfar Saba (IL);
Ilia Hariton, Zichron Yaakov (IL); **Tal HAMMER**, Ramat Gan (IL)(73) Assignee: **CARDIOVALVE LTD.**, Or Yehuda (IL)(21) Appl. No.: **18/265,025**(22) PCT Filed: **Dec. 2, 2021**(86) PCT No.: **PCT/IL2021/051433**

§ 371 (c)(1),

(2) Date: **Jun. 2, 2023****Related U.S. Application Data**

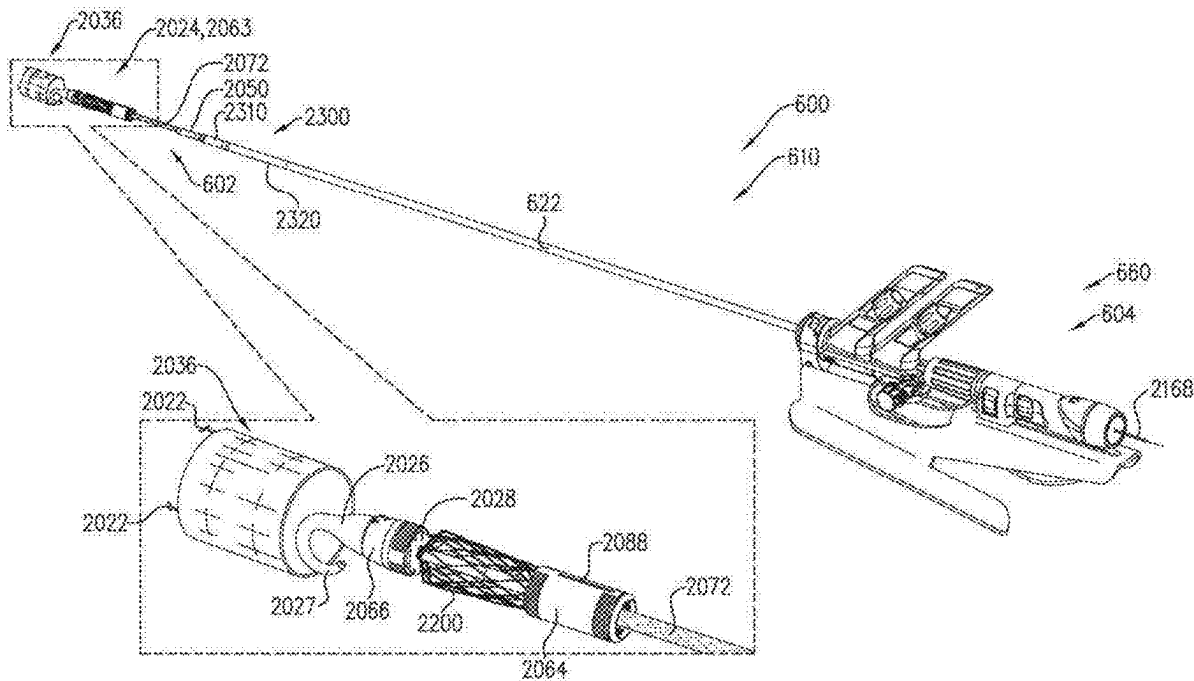
(63) Continuation-in-part of application No. 17/399,594, filed on Aug. 11, 2021, now Pat. No. 12,357,459.

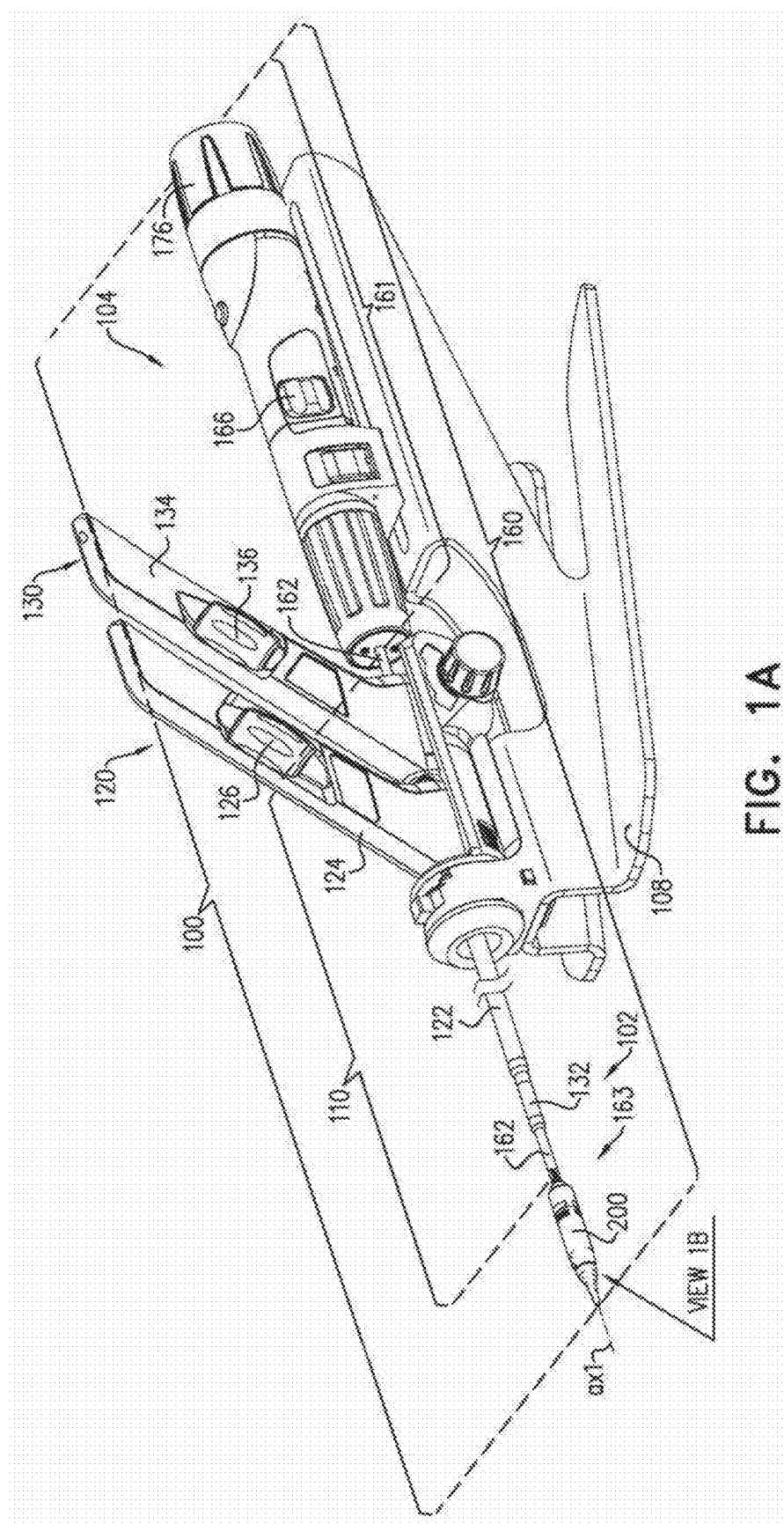
Publication Classification(51) **Int. Cl.****A61F 2/24** (2006.01)**A61F 2/95** (2013.01)(52) **U.S. Cl.**CPC **A61F 2/2436** (2013.01); **A61F 2/2439**
(2013.01); **A61F 2/9524** (2020.05); **A61F**
2220/0033 (2013.01)

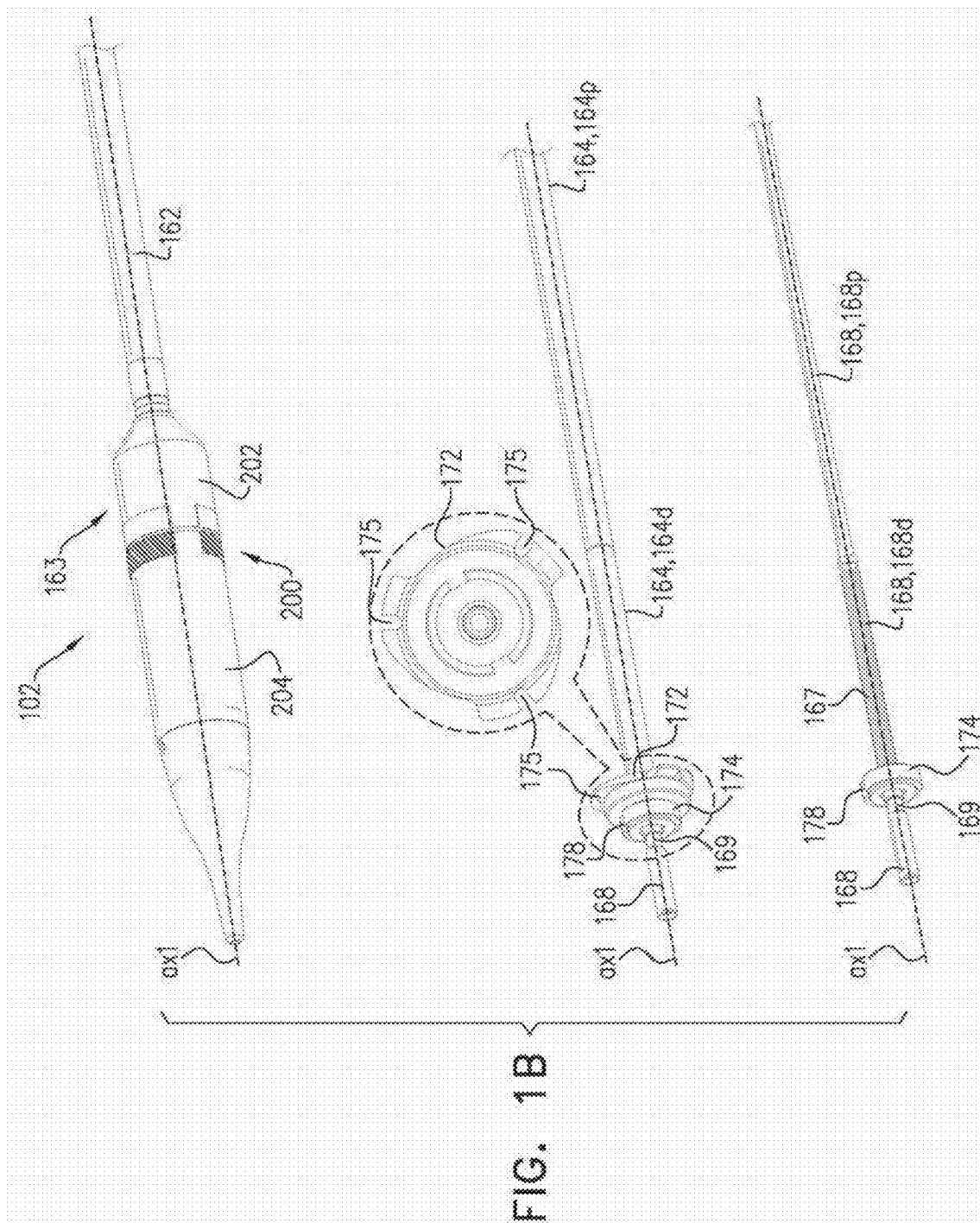
(57)

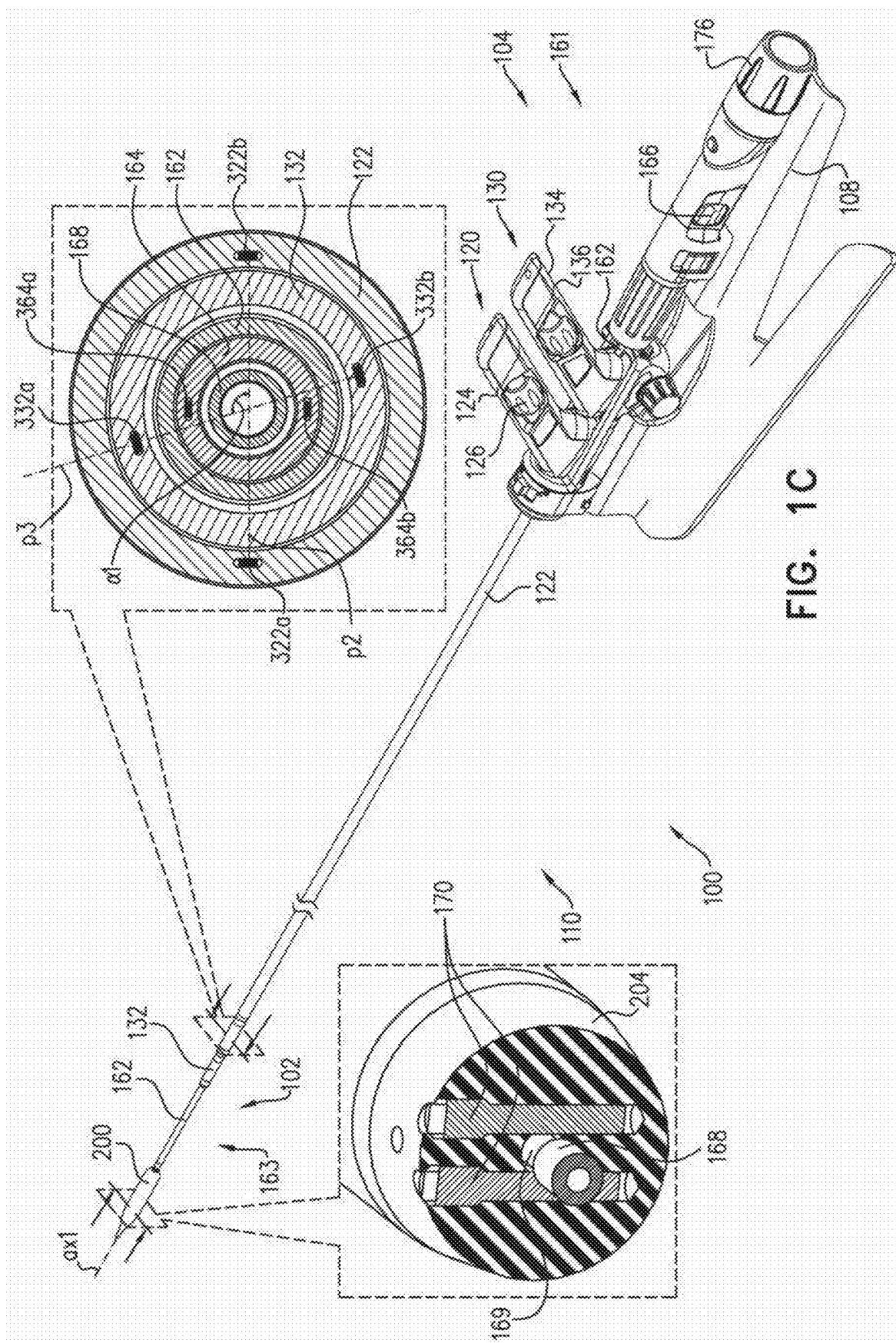
ABSTRACT

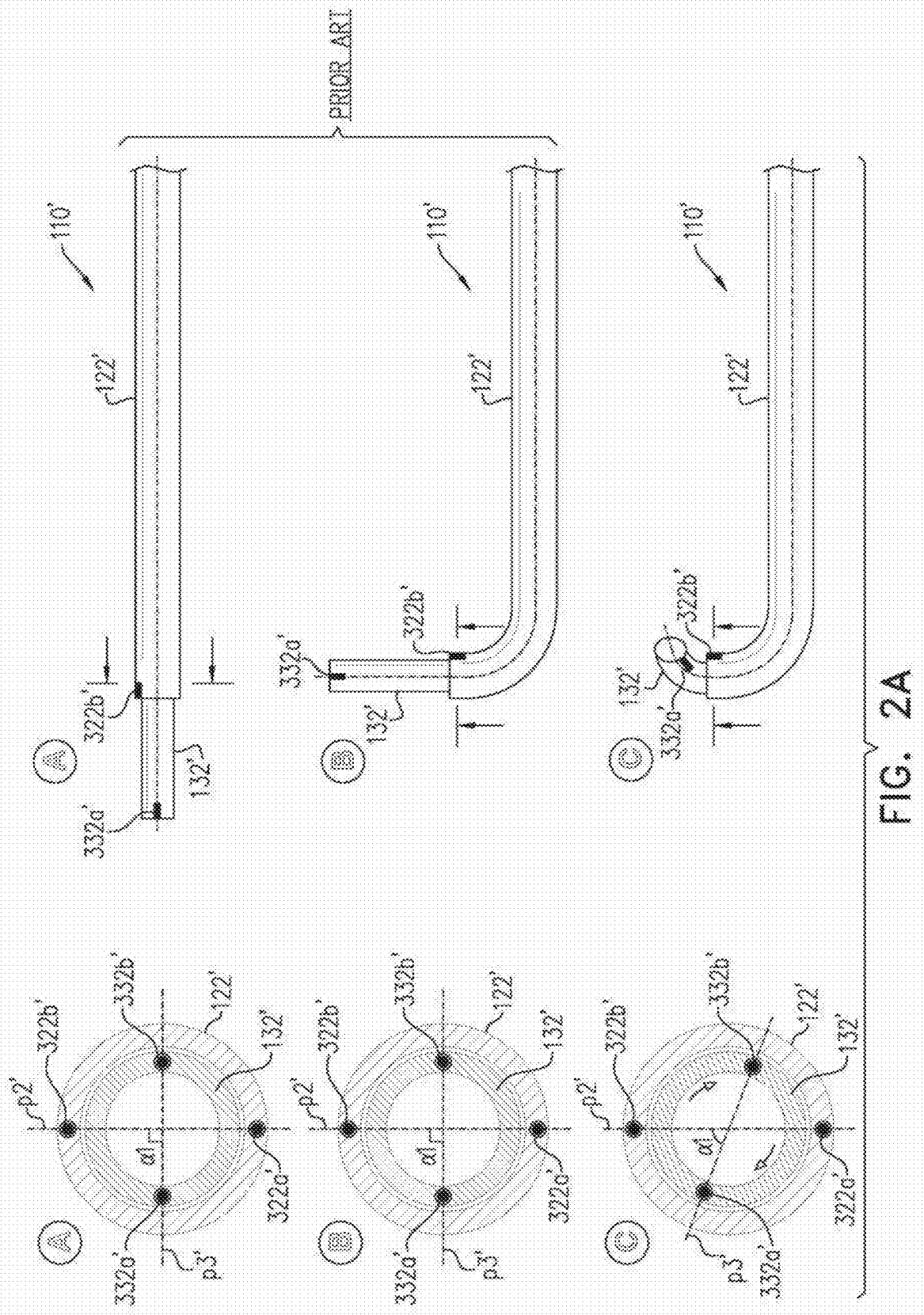
A percutaneous delivery tool includes a shaft, and proximal and distal capsules which move axially with respect to the shaft. A prosthetic valve includes a tubular portion, an upstream support portion extending therefrom and a plurality of flanges each having a respective flange end-portion. The prosthetic valve is restrainable by the delivery tool, such that the shaft and a downstream end of the tubular portion are disposed within the distal capsule, and the upstream support portion and the flange end-portions are disposed within the proximal capsule. Other embodiments are also described.

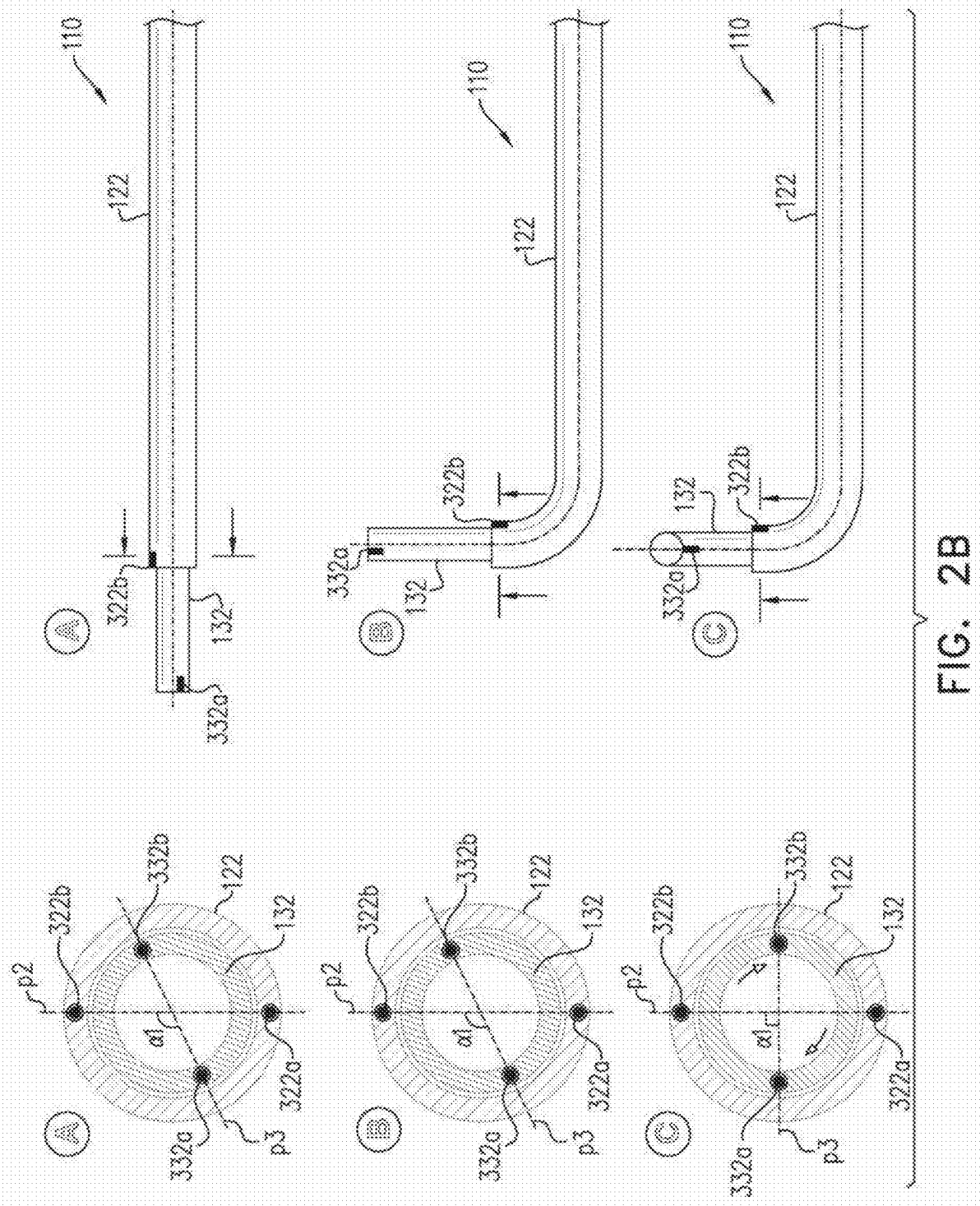


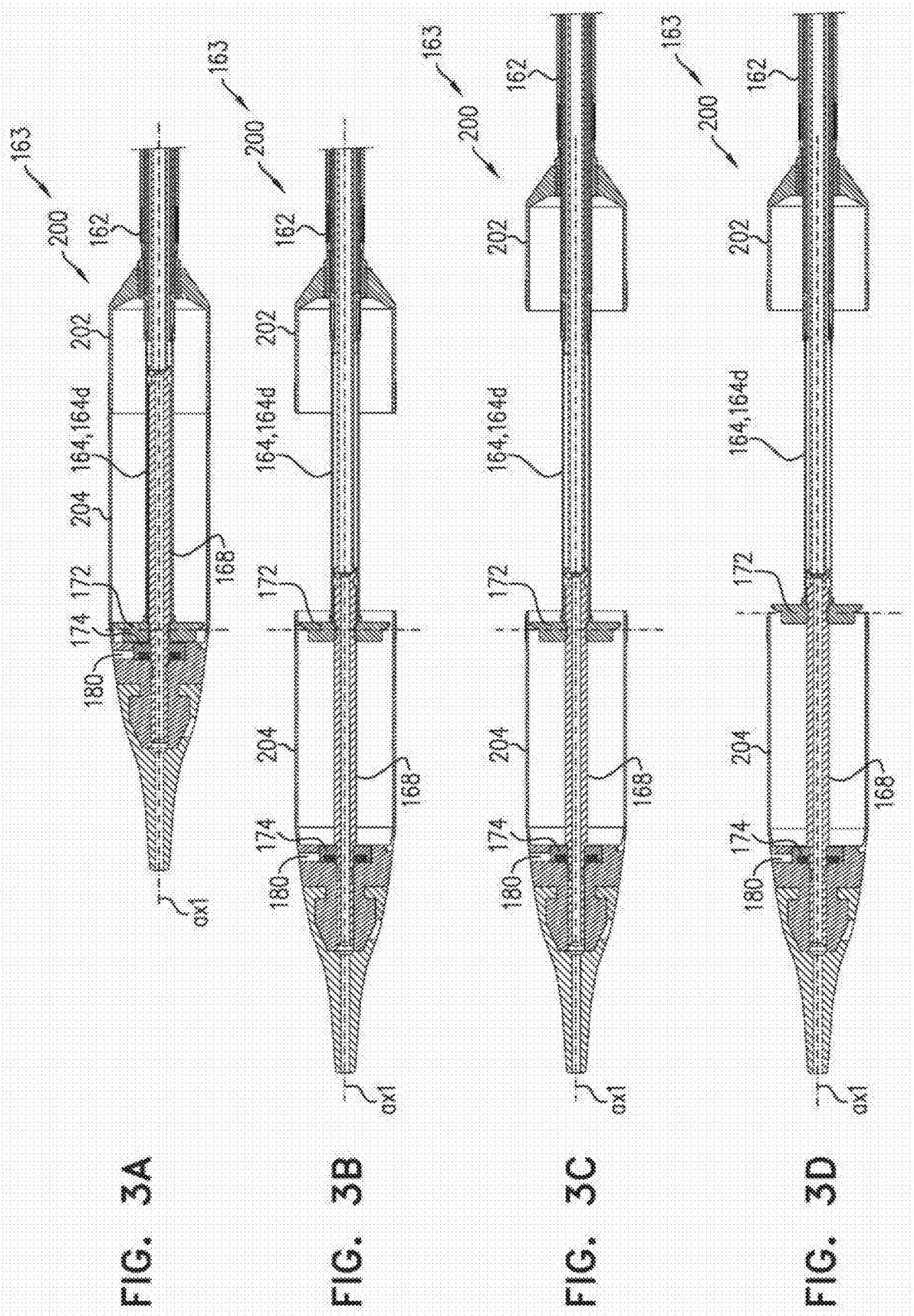


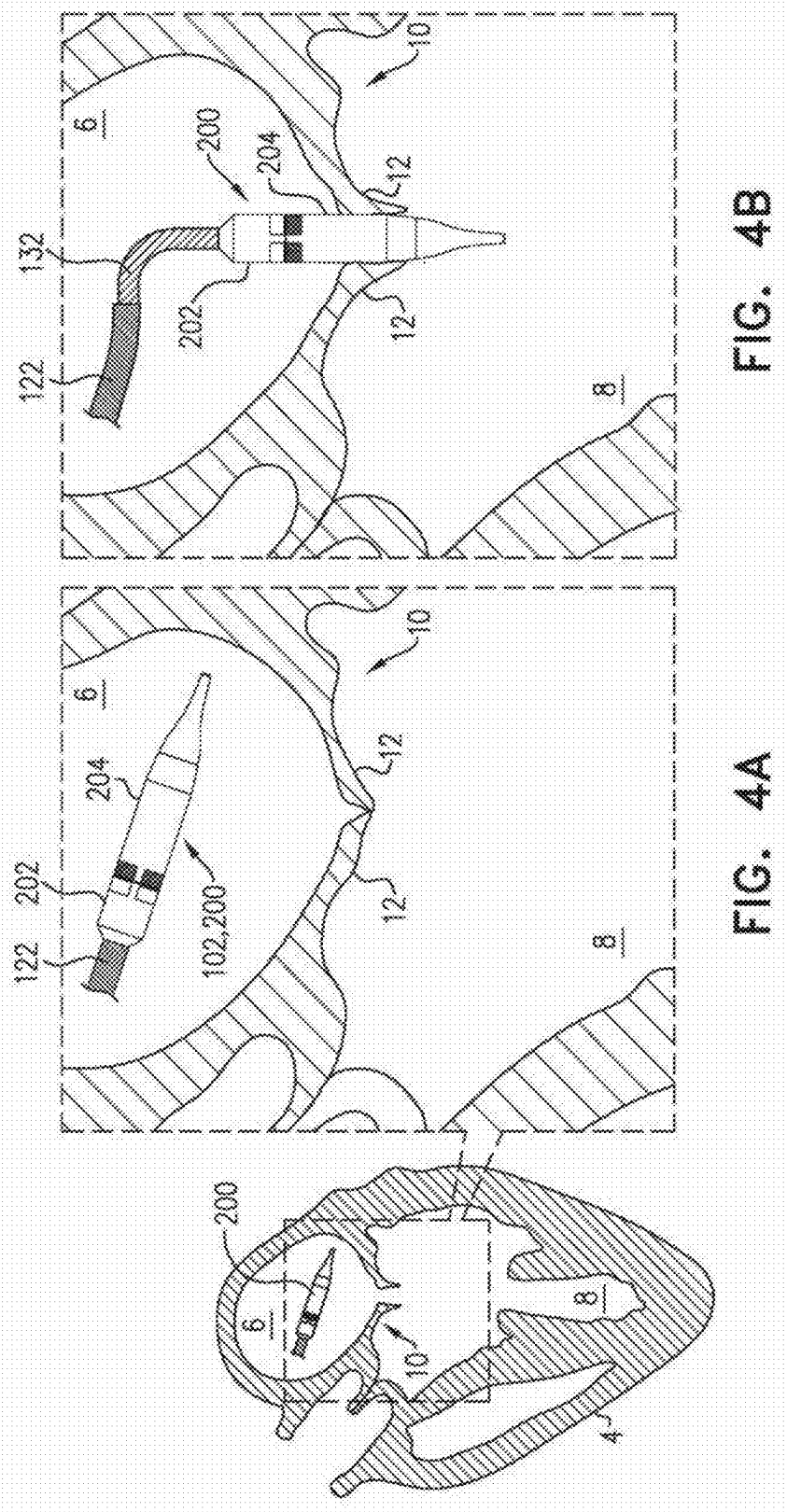


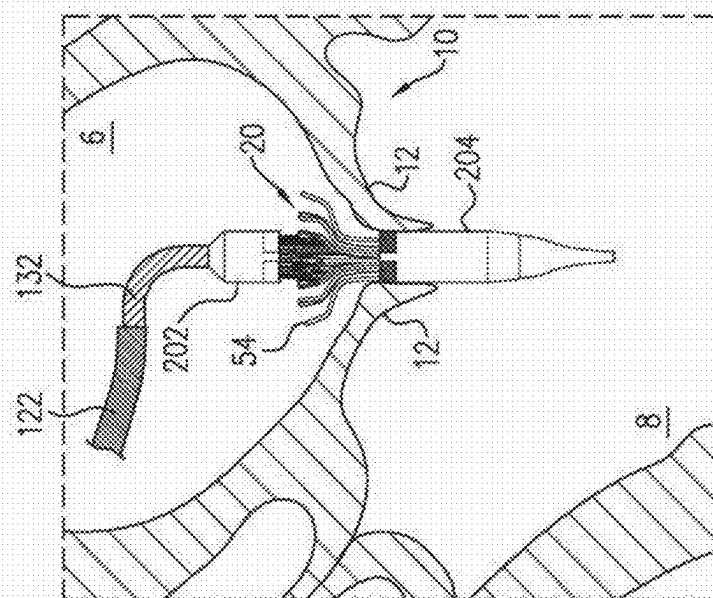
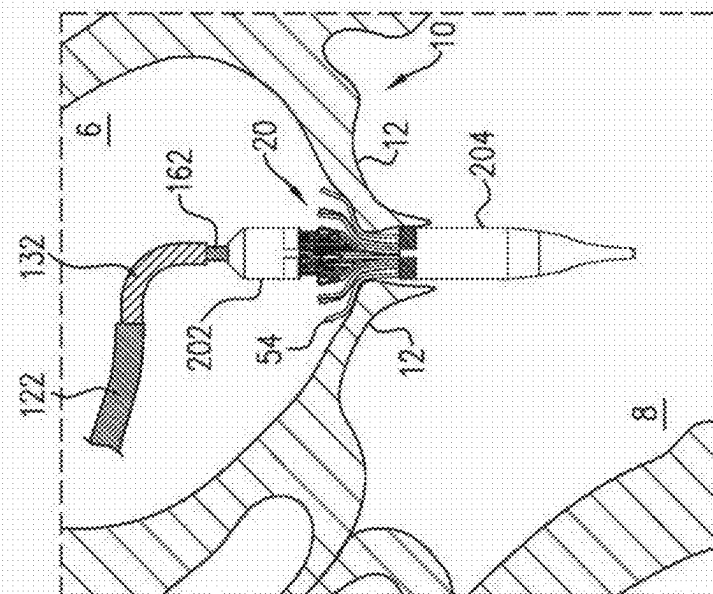












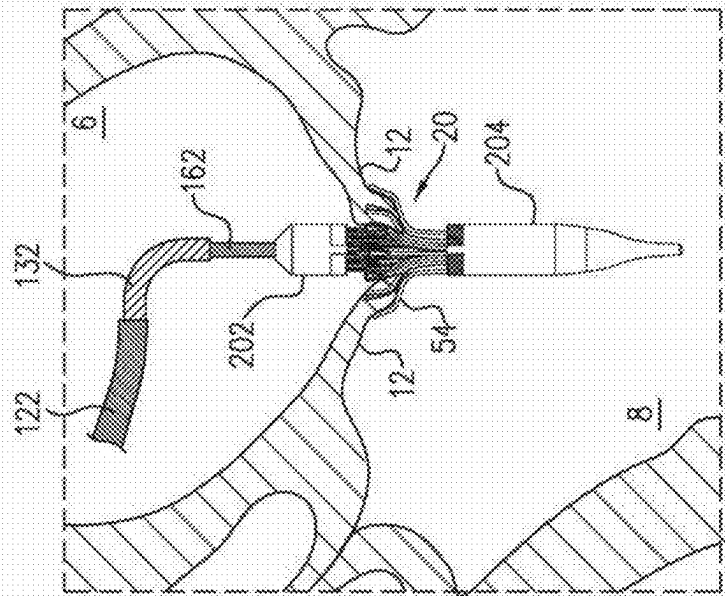


FIG. 4F

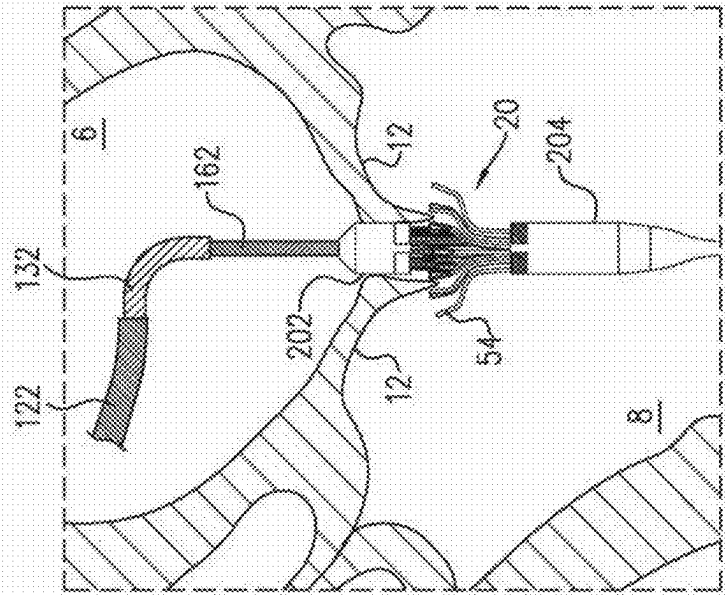
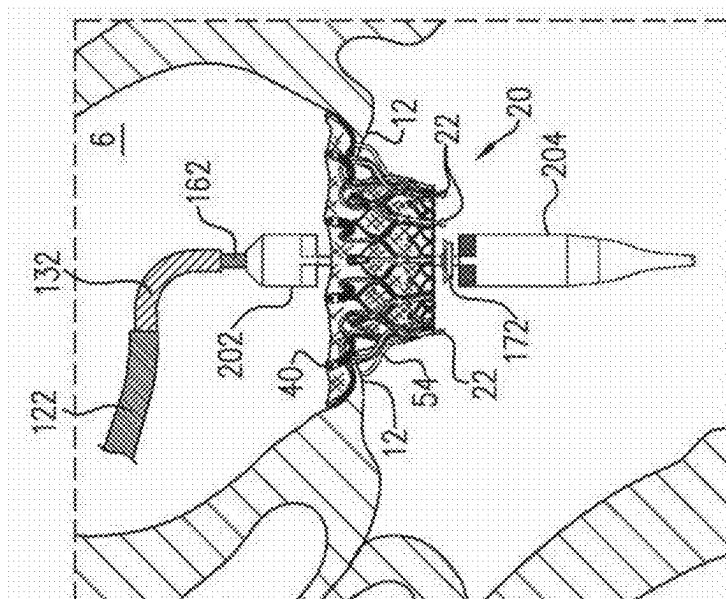
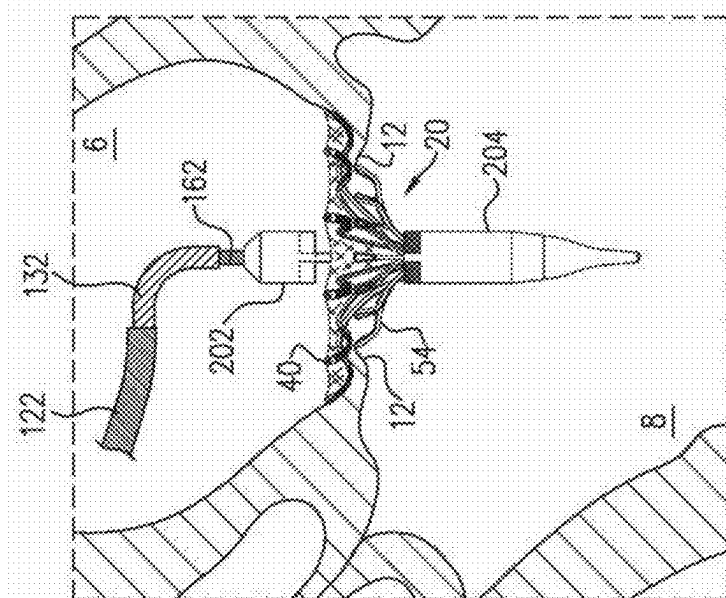


FIG. 4E



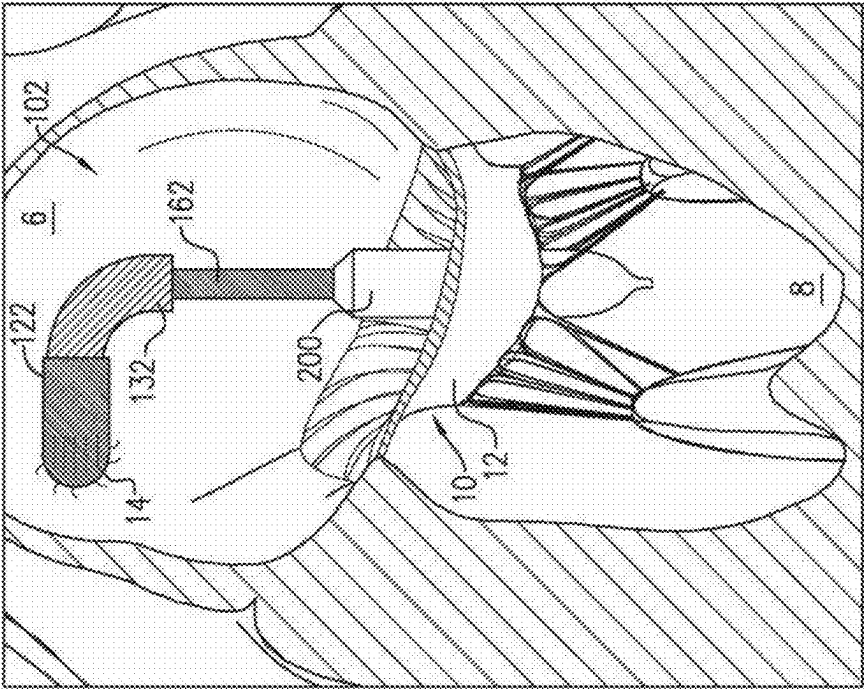


FIG. 5B

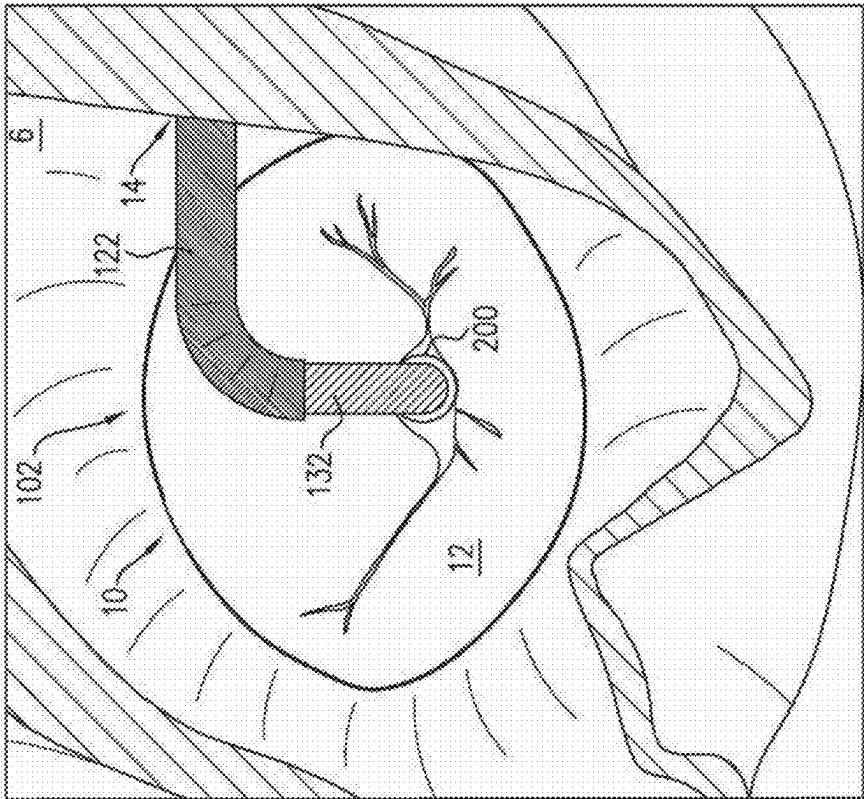


FIG. 5A

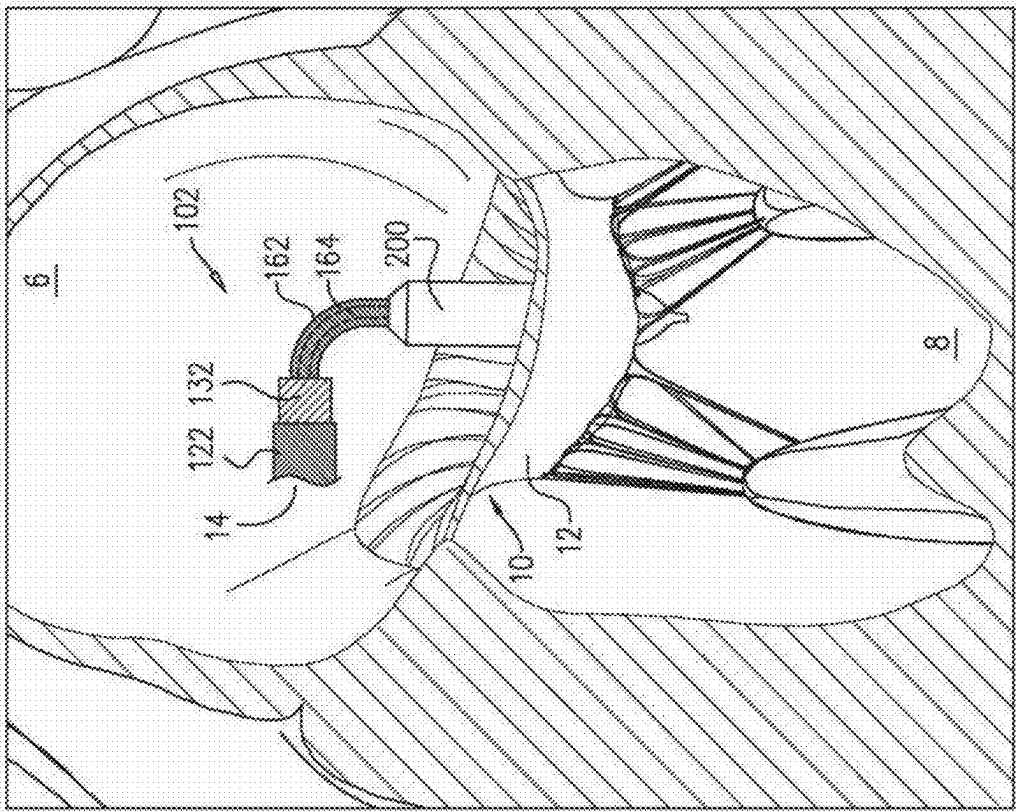


FIG. 7

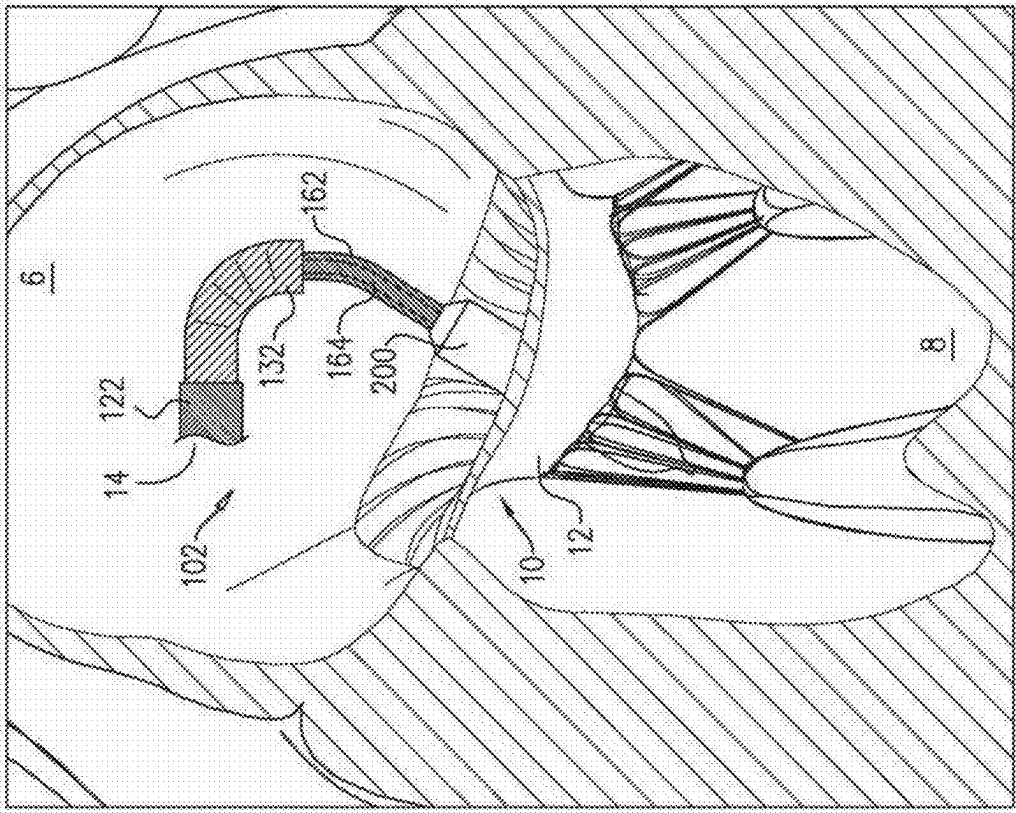
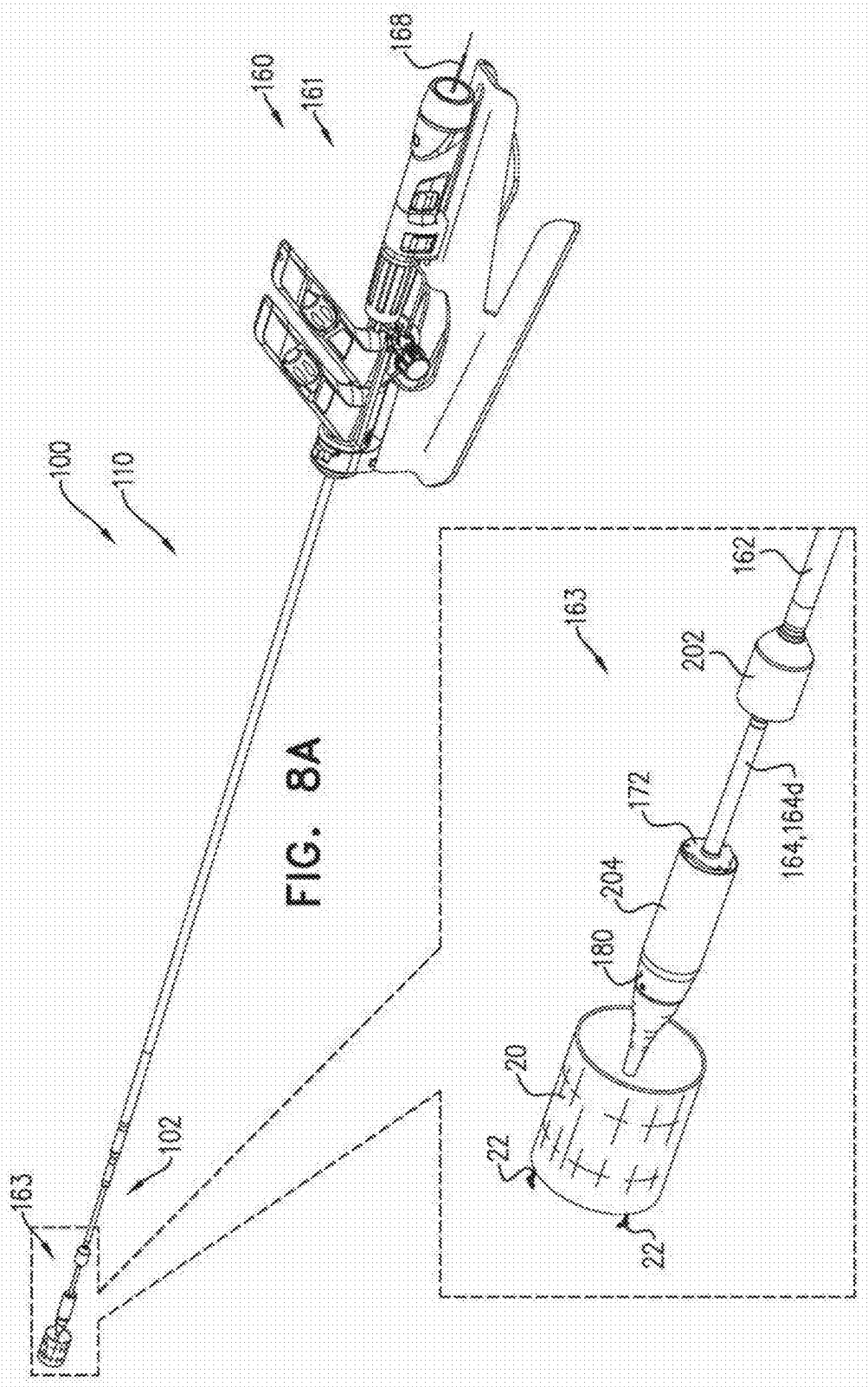


FIG. 6



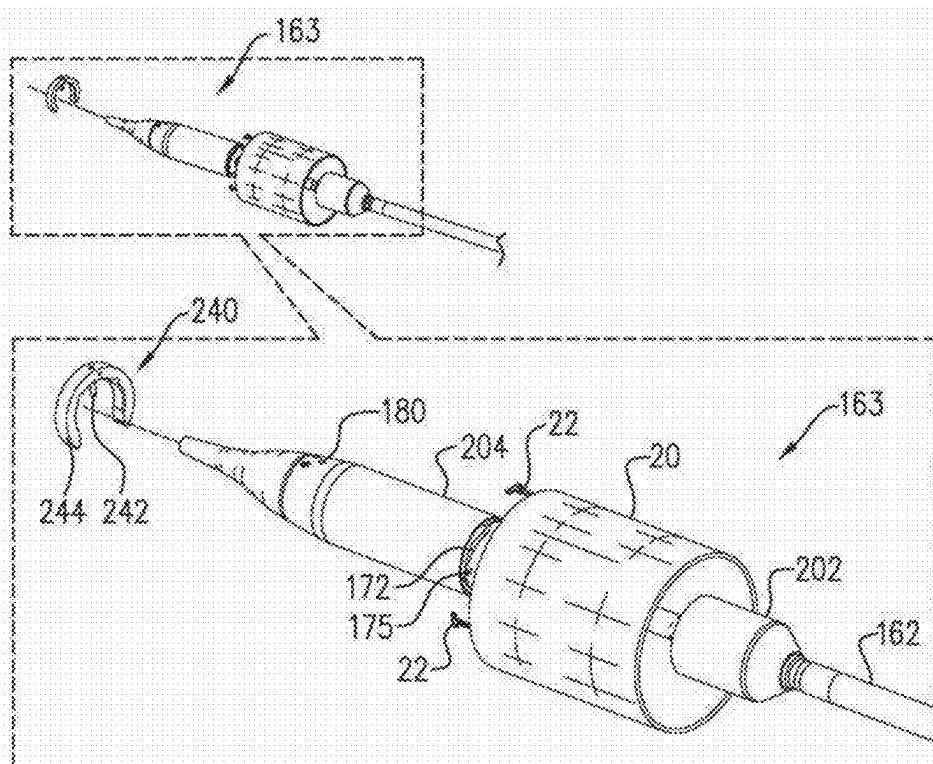


FIG. 8B

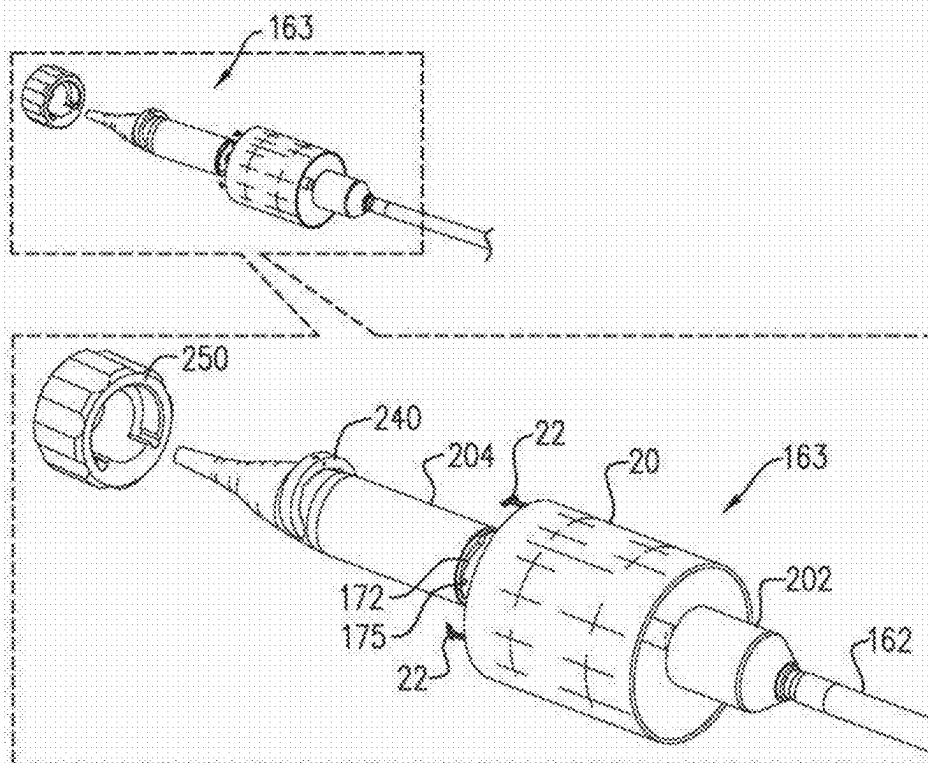


FIG. 8C

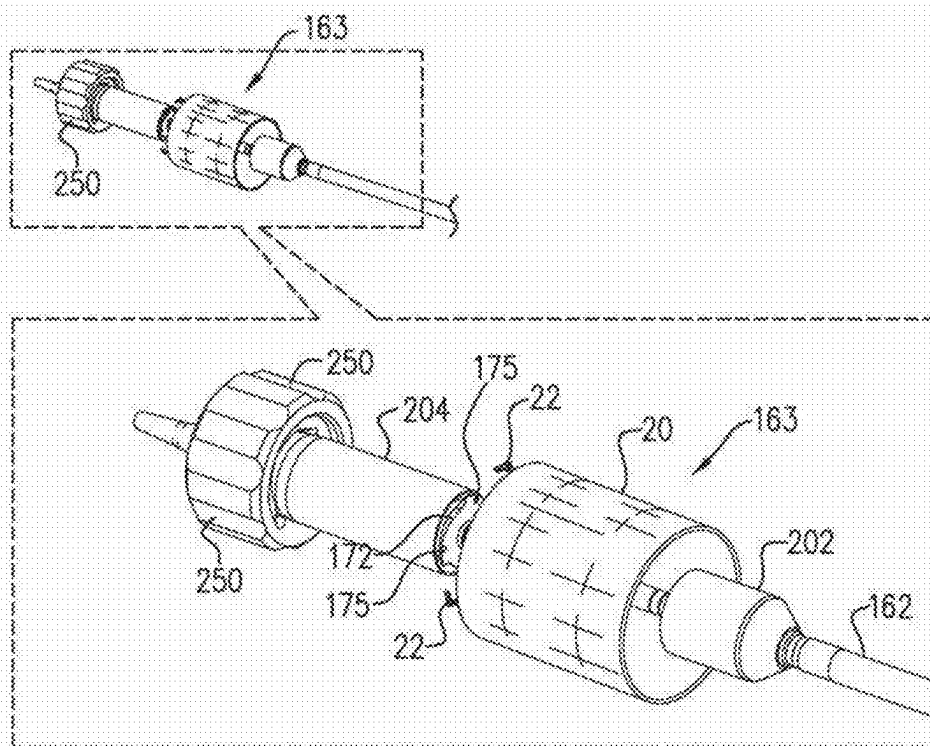


FIG. 8D

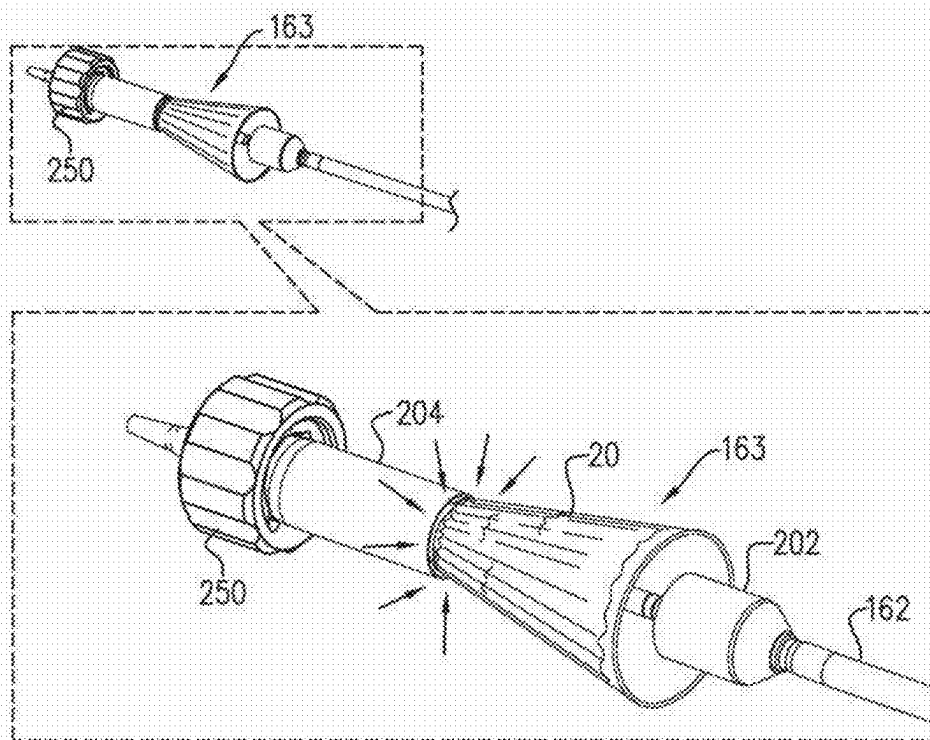


FIG. 8E

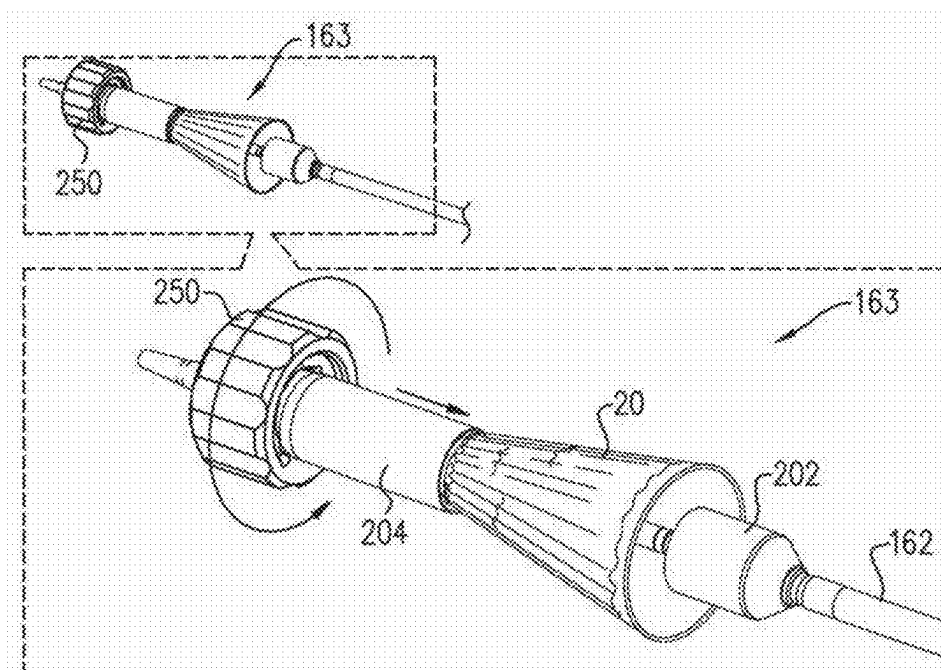


FIG. 8F

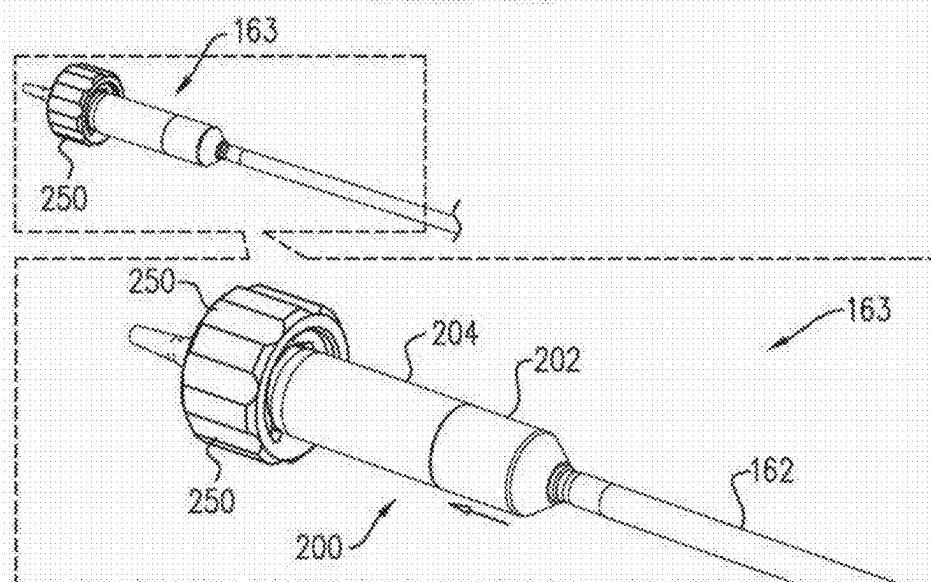


FIG. 8G

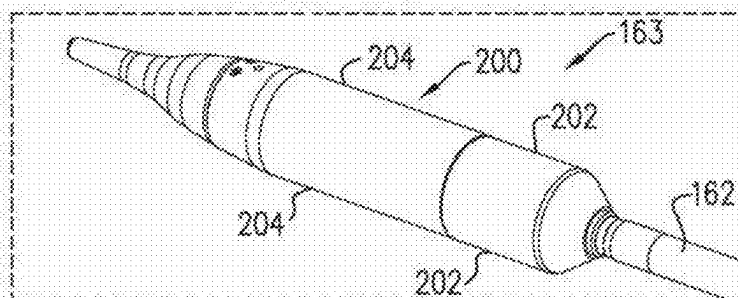
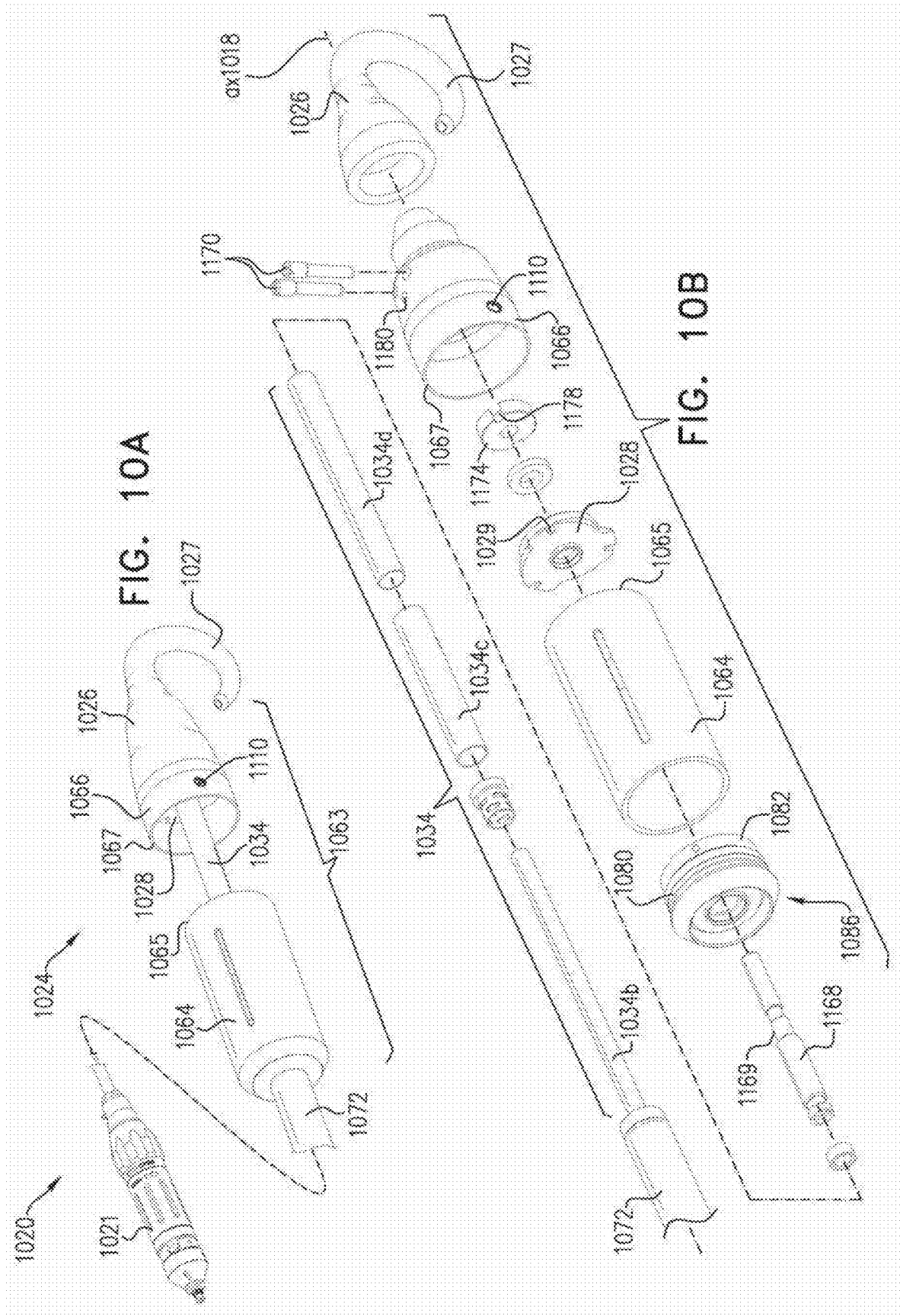
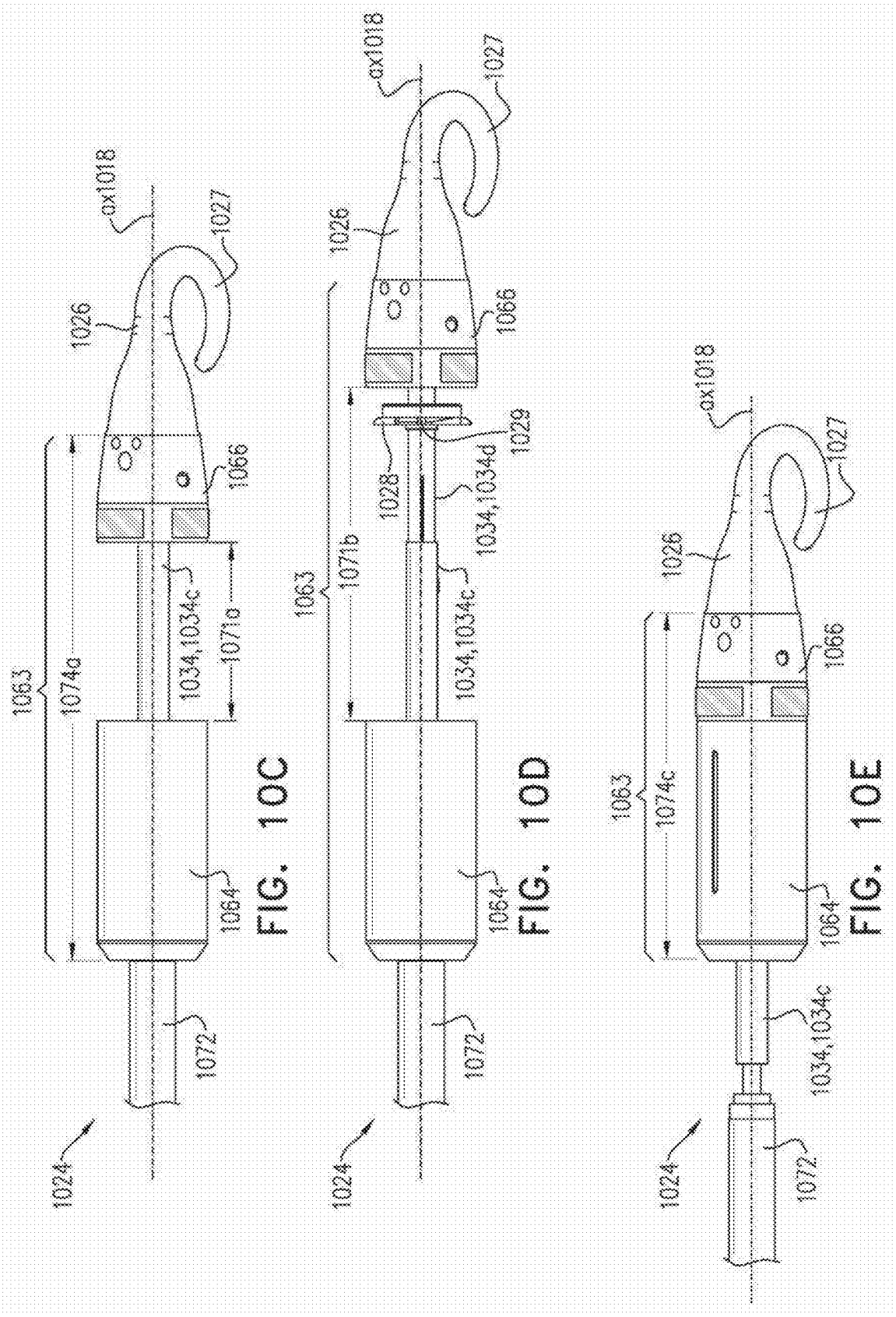
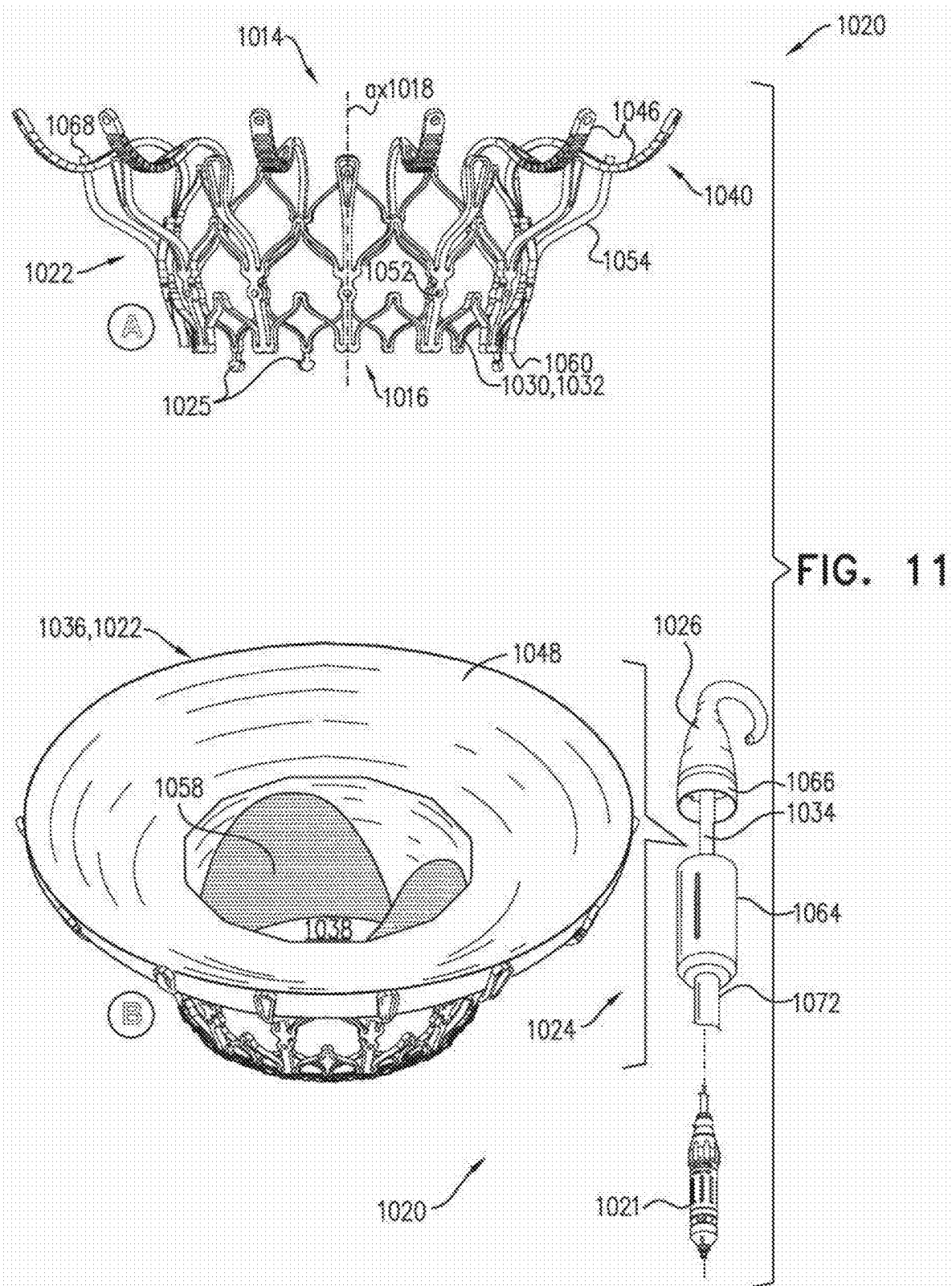
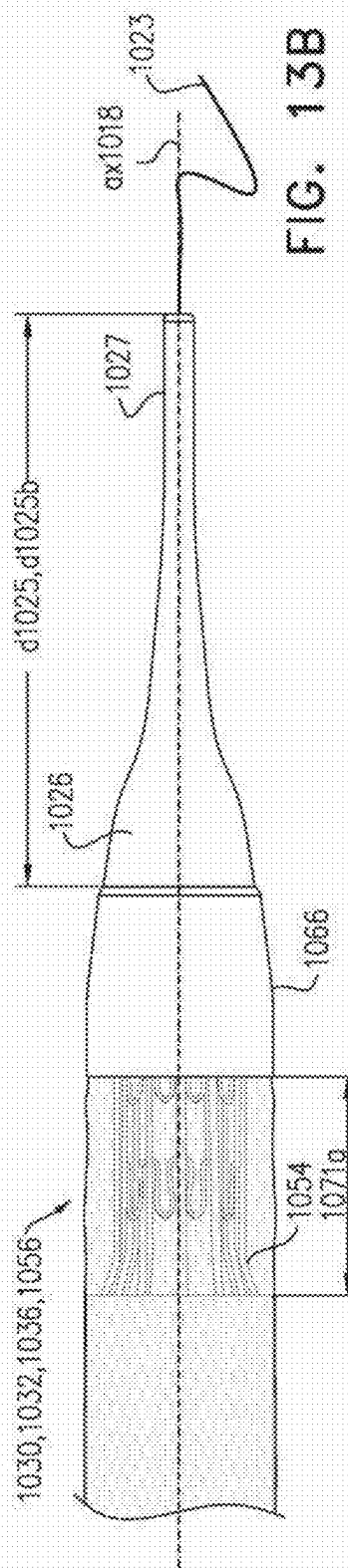
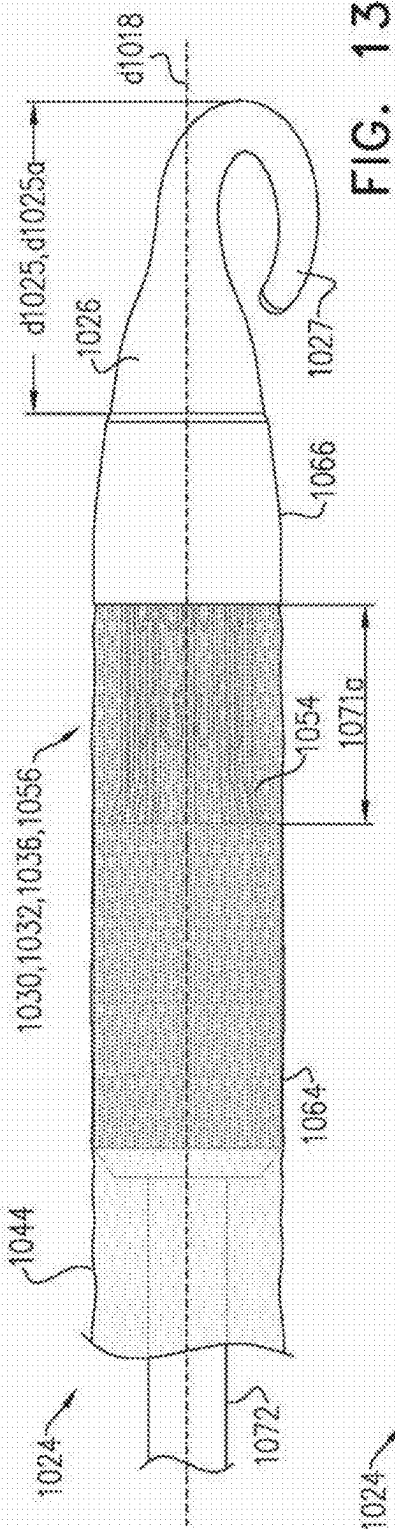
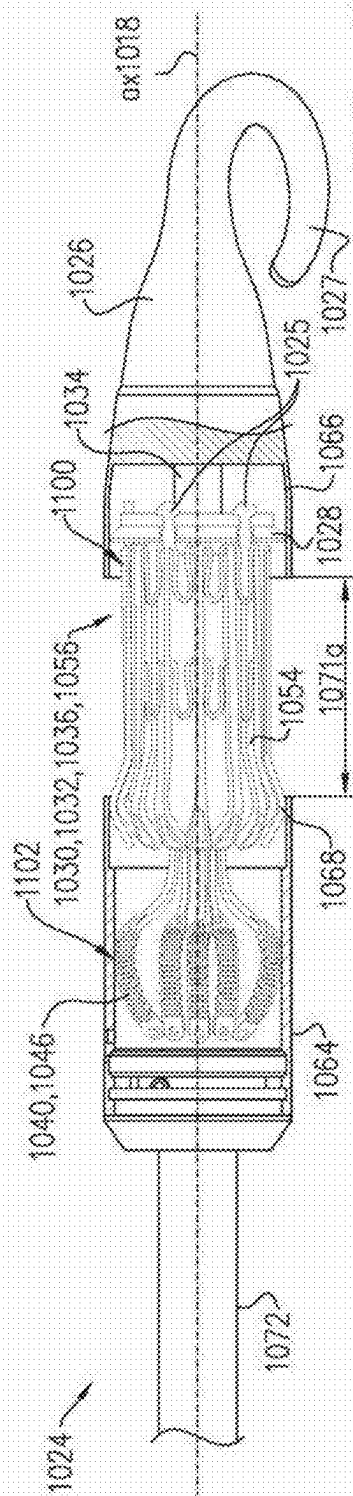


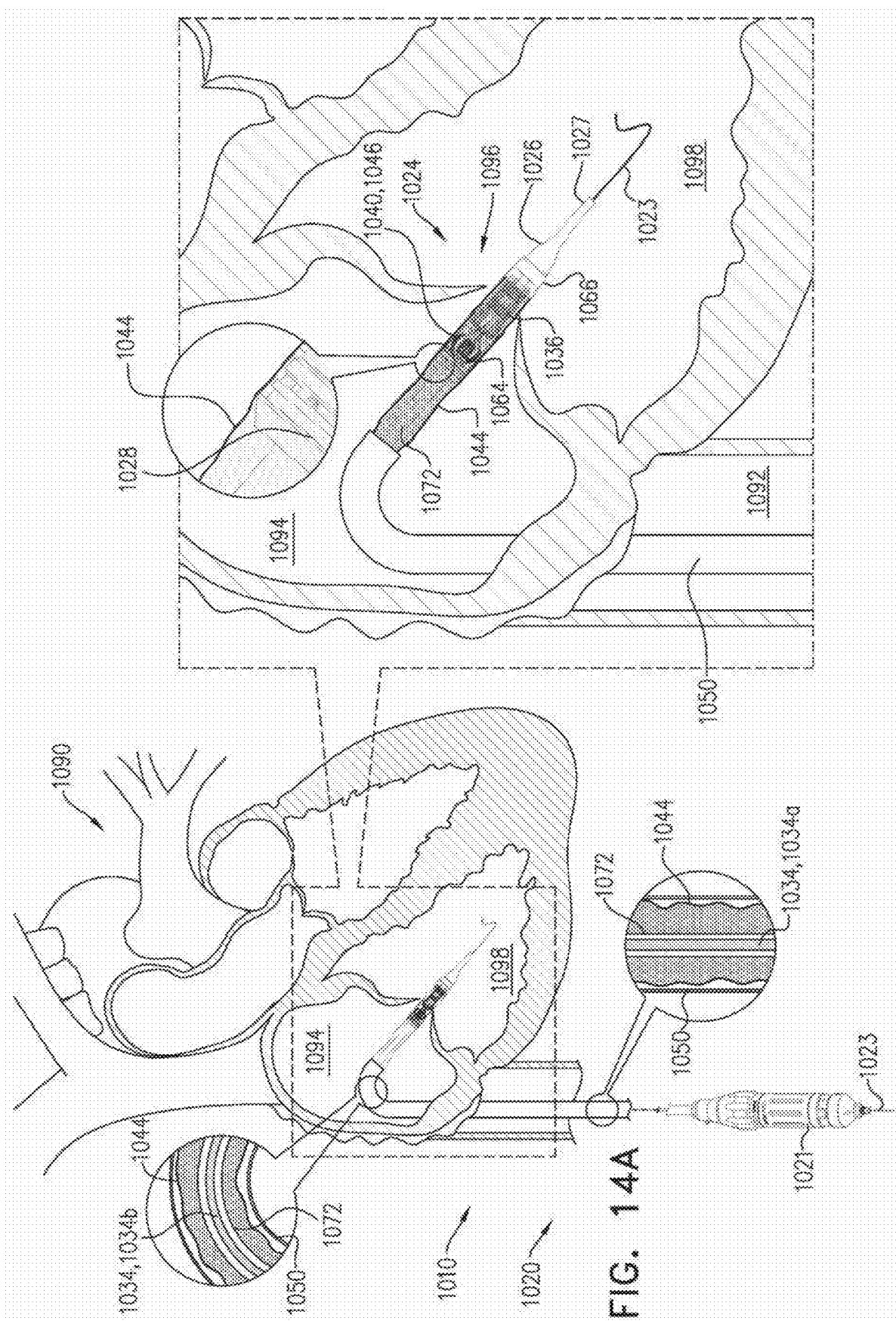
FIG. 9

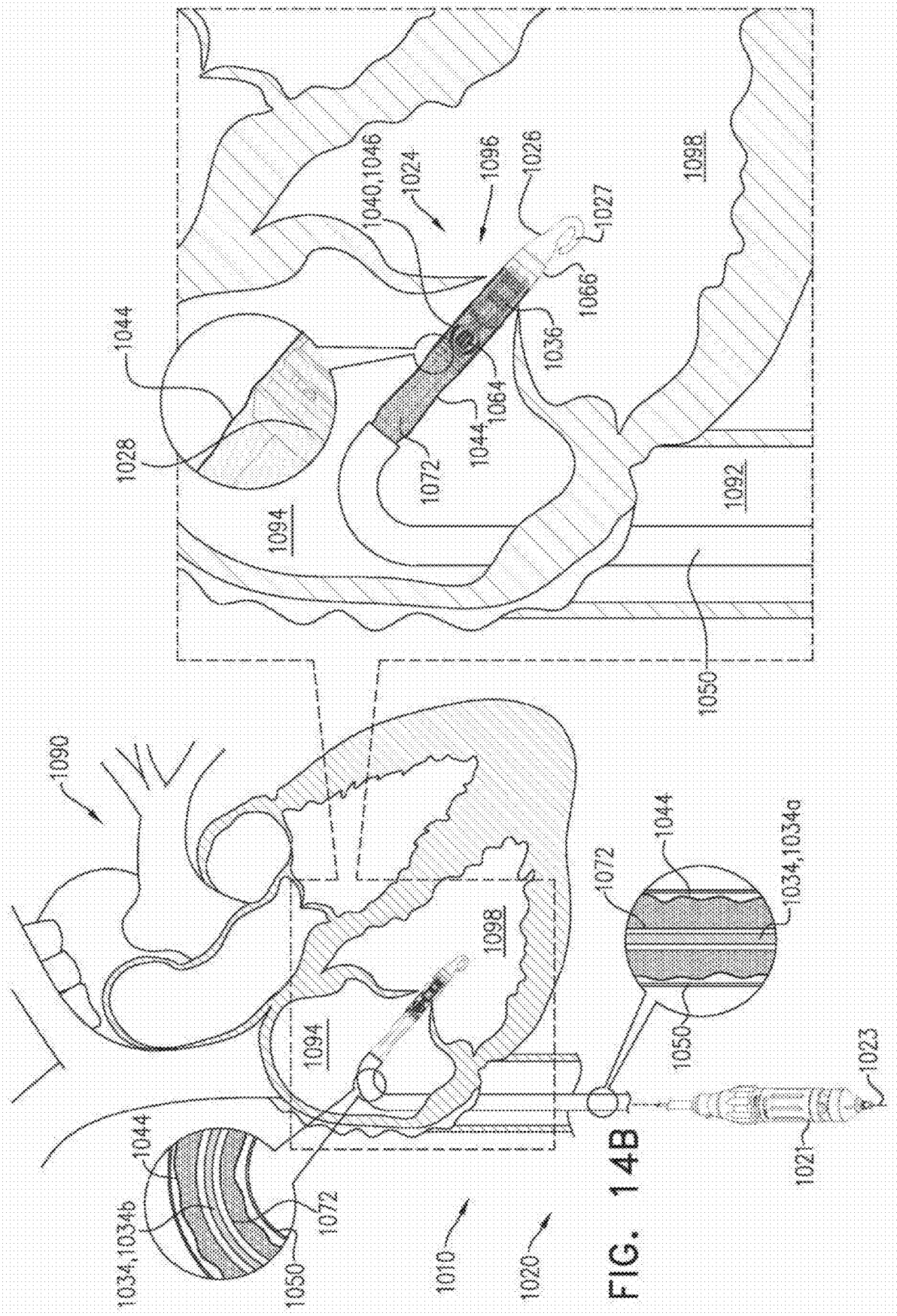


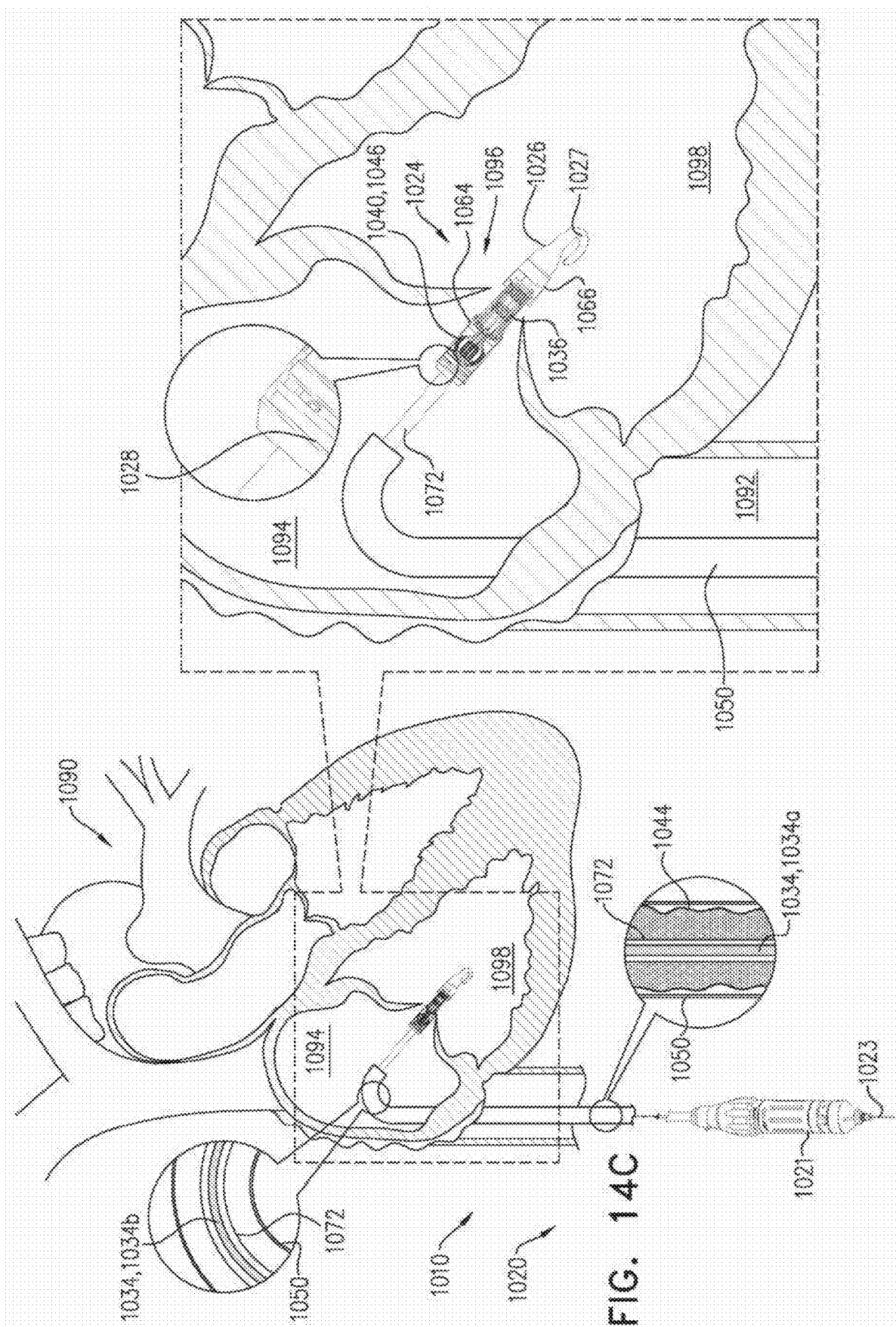












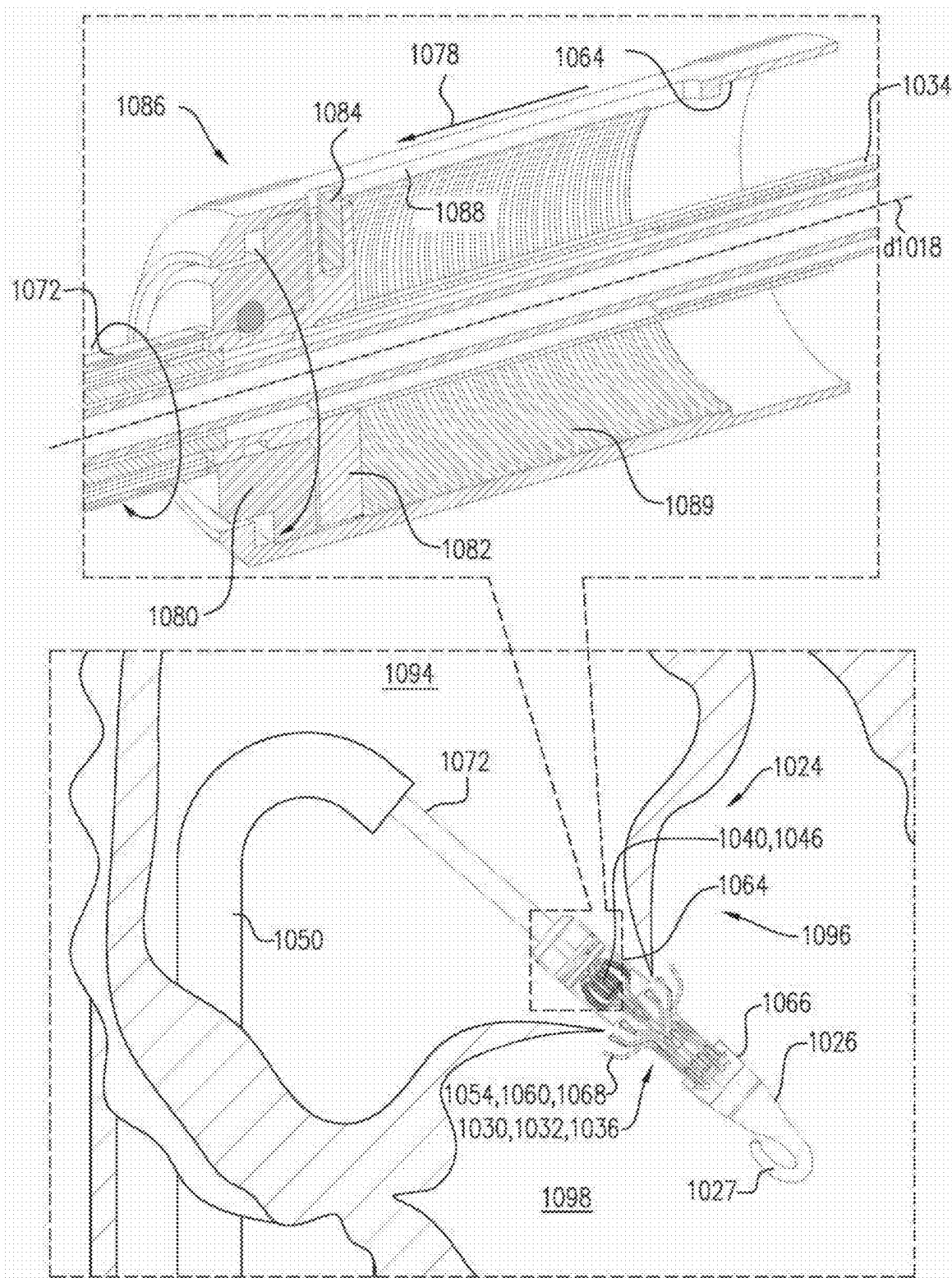


FIG. 14D

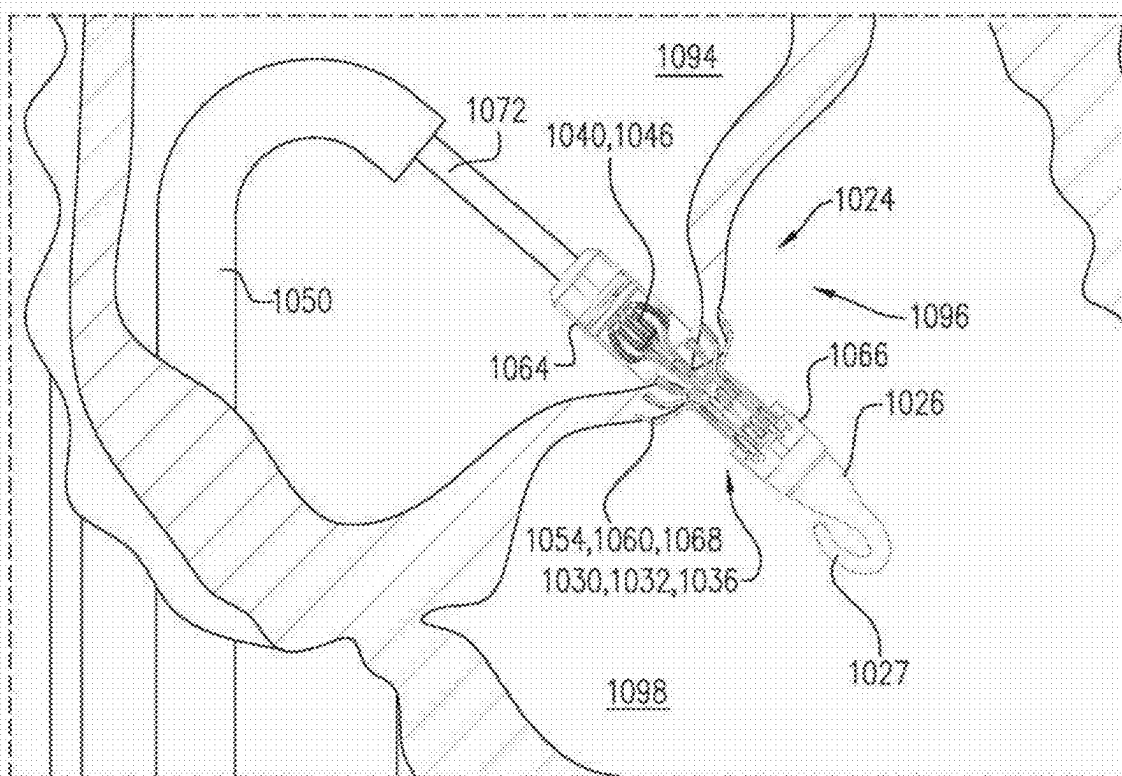


FIG. 14E

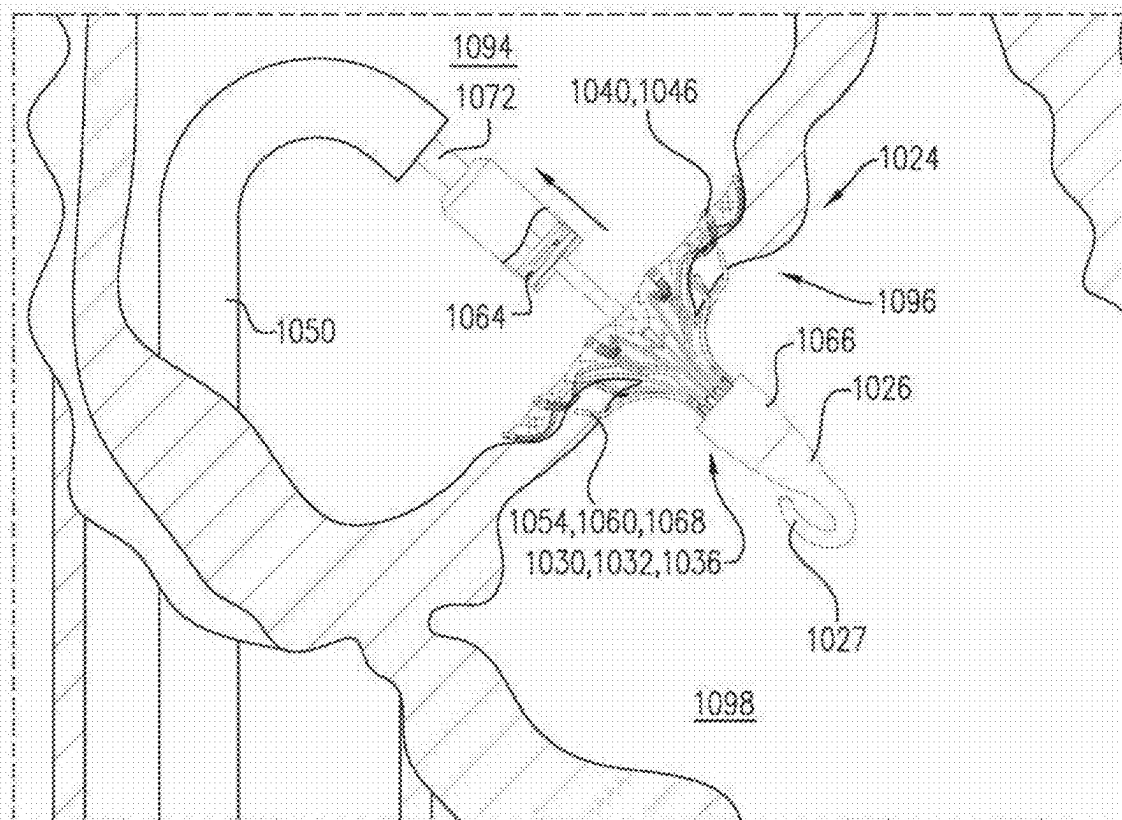


FIG. 14F

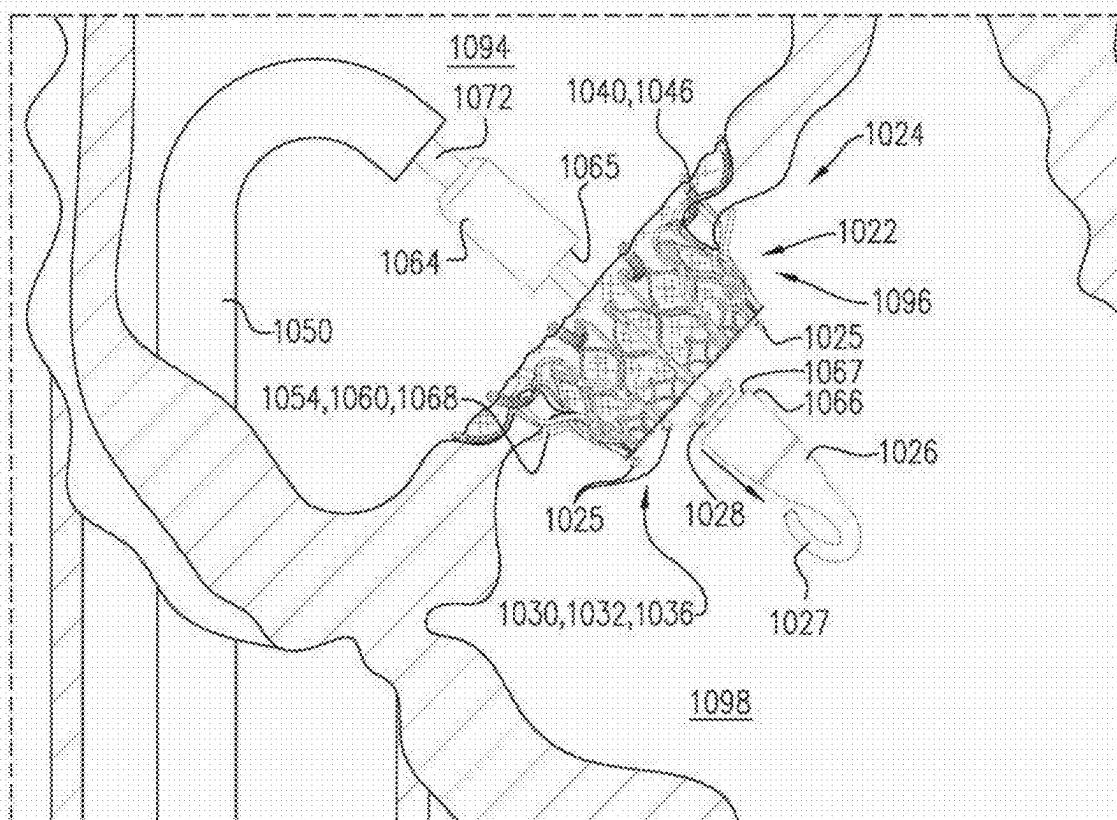


FIG. 14G

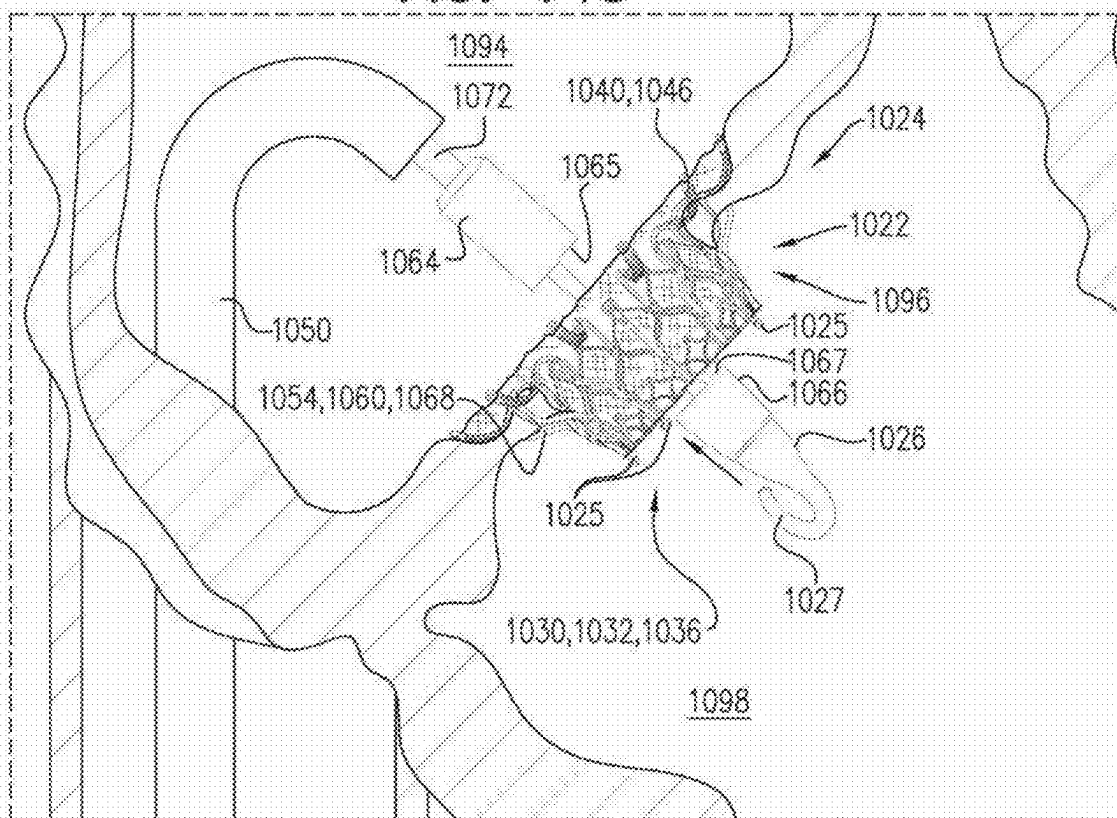


FIG. 14H

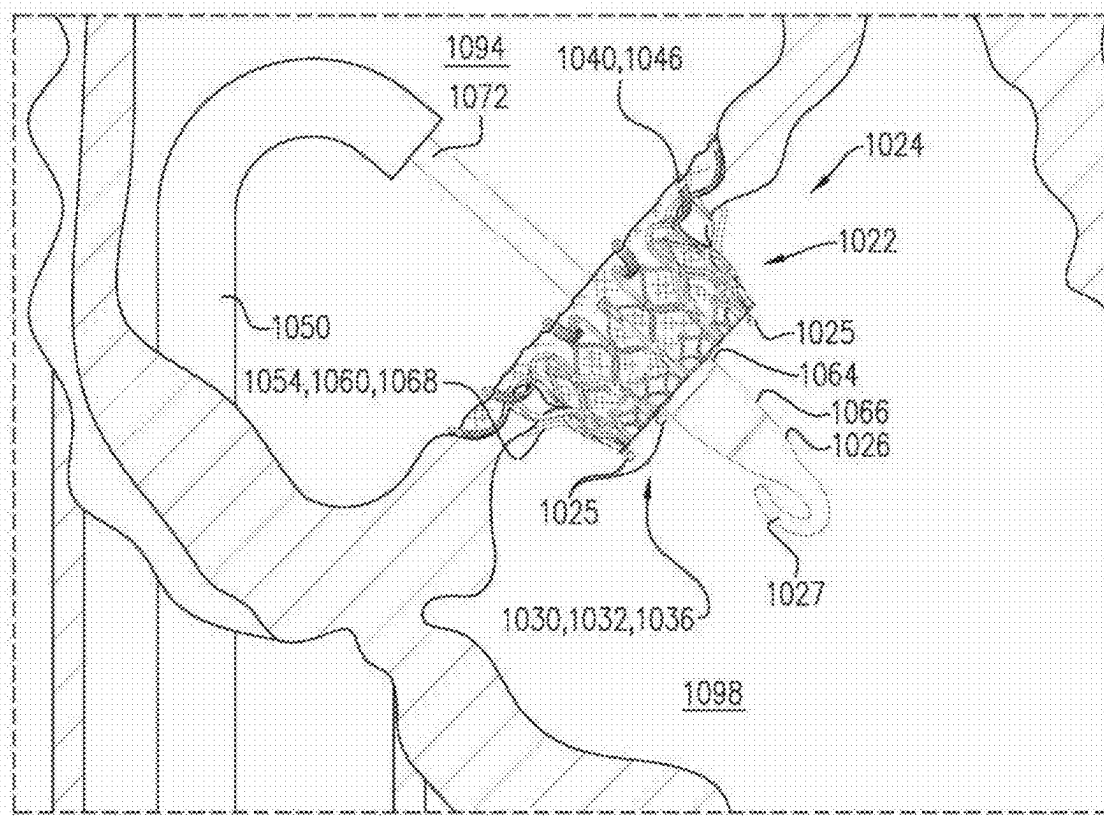


FIG. 14I

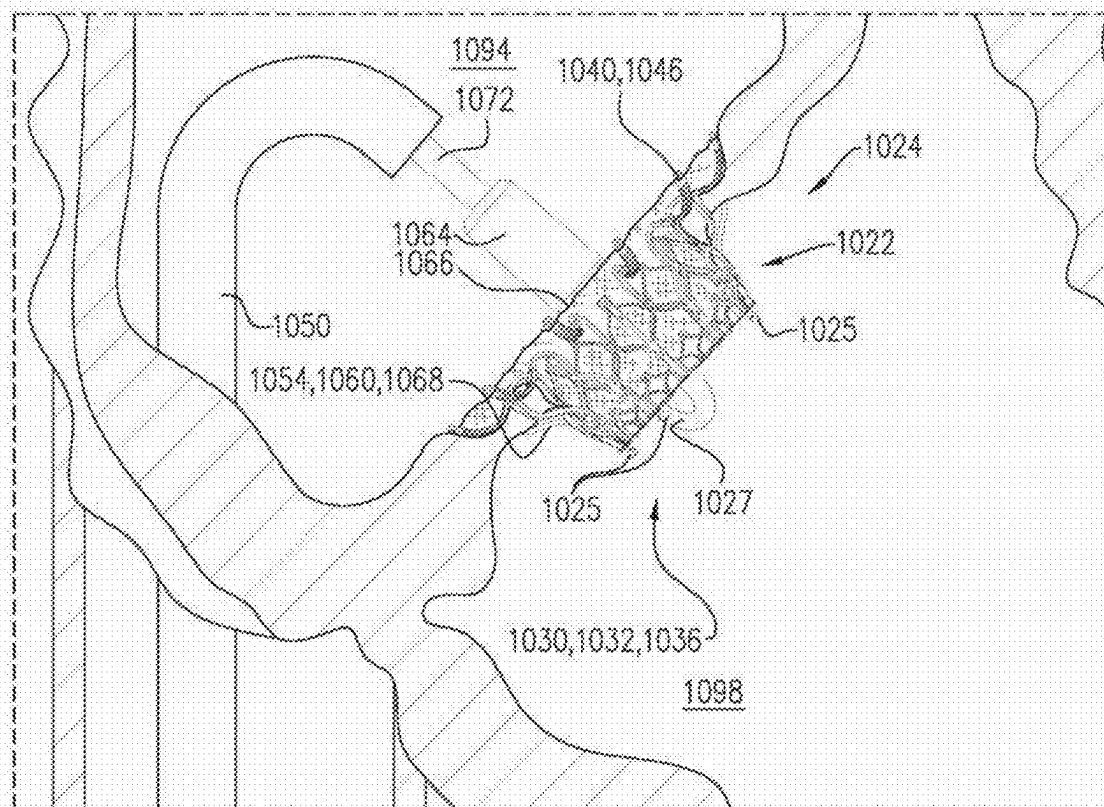
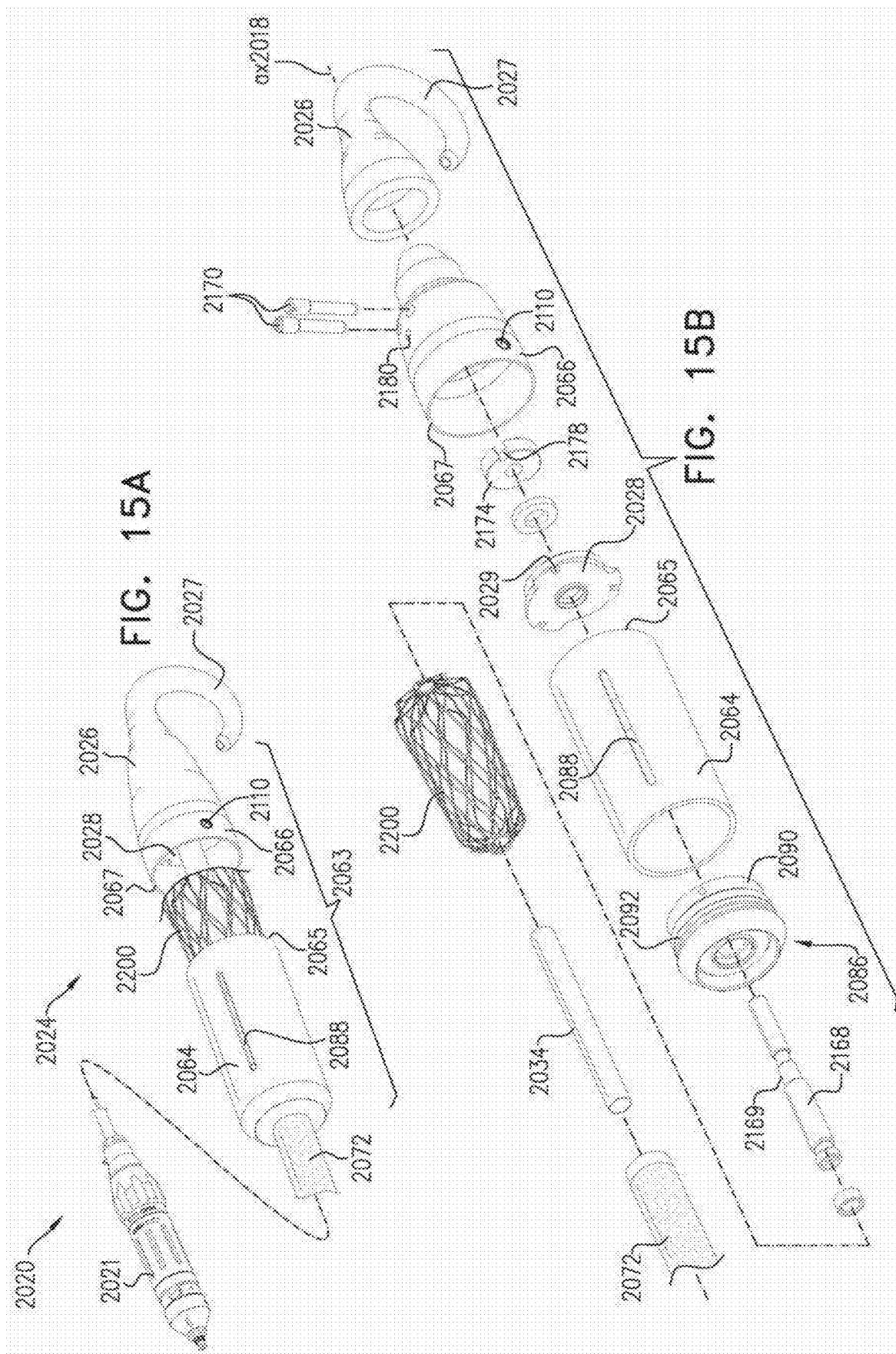
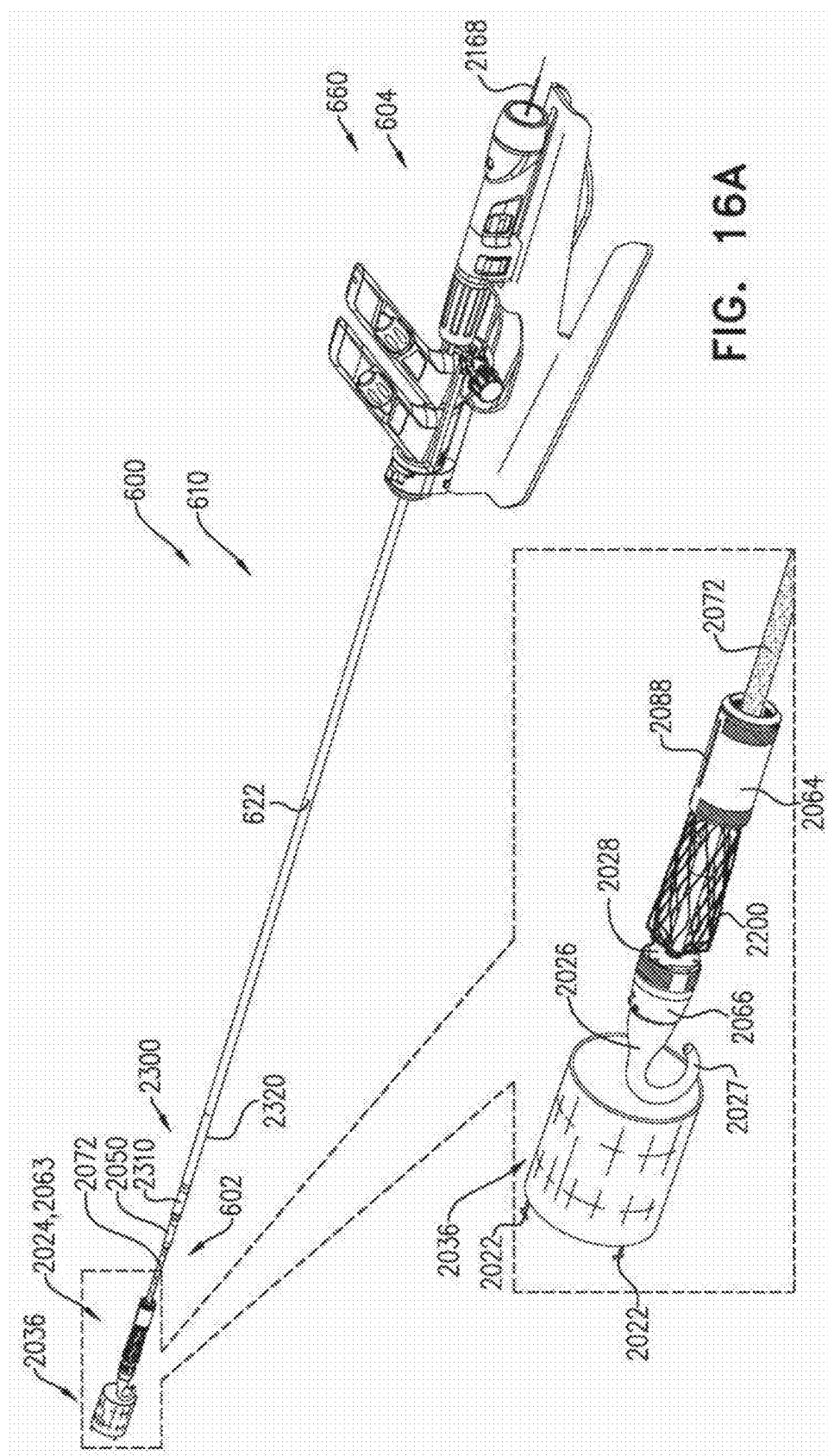


FIG. 14J





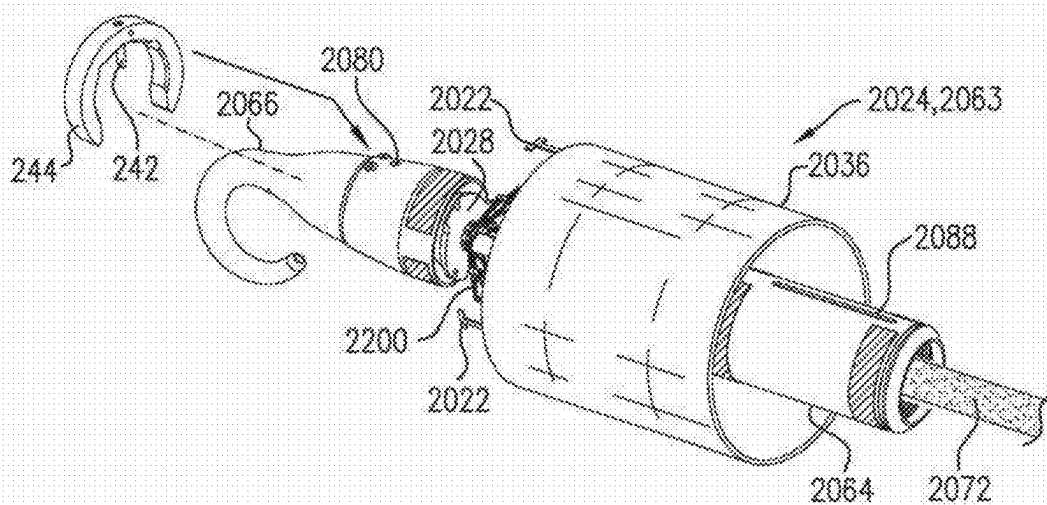


FIG. 16B

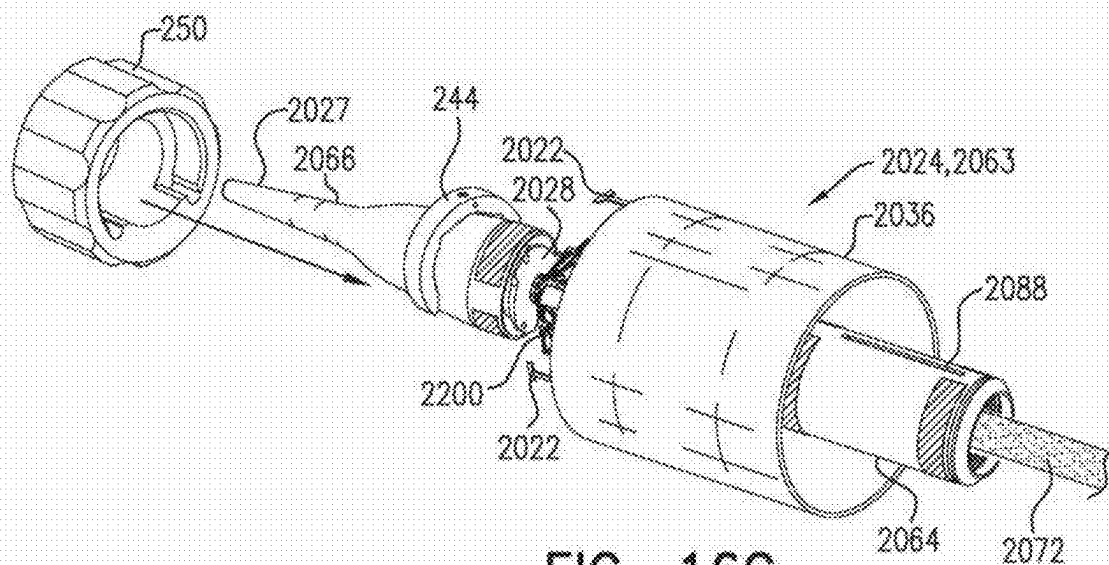
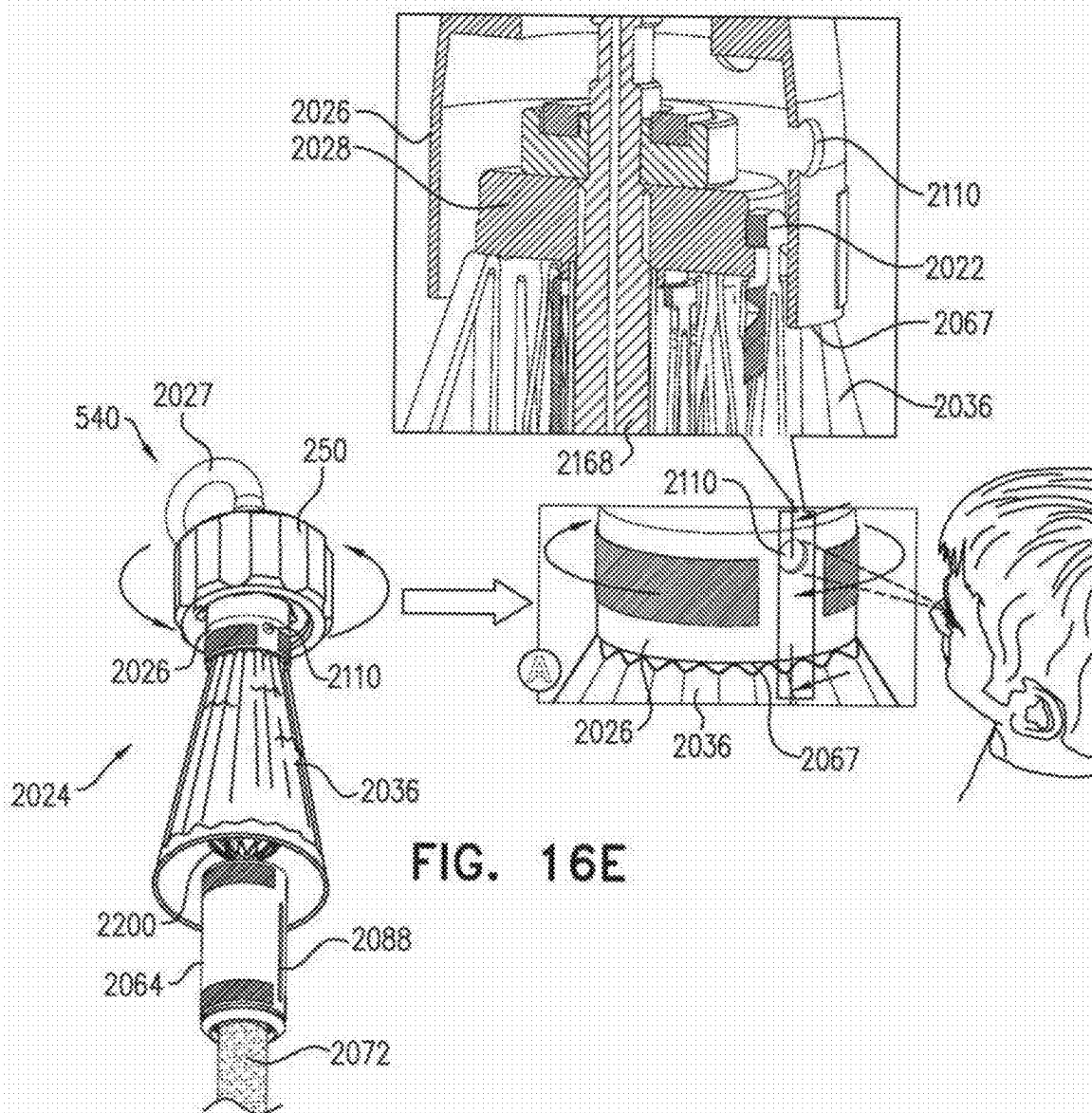
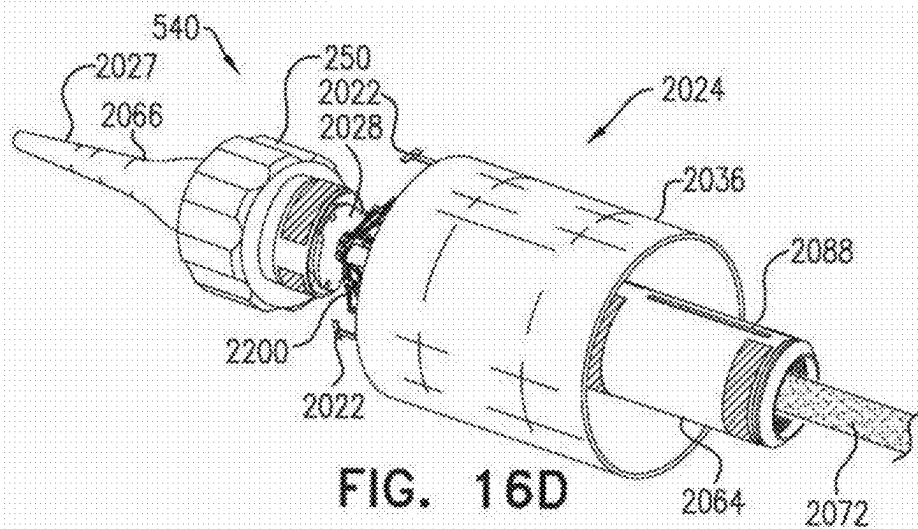
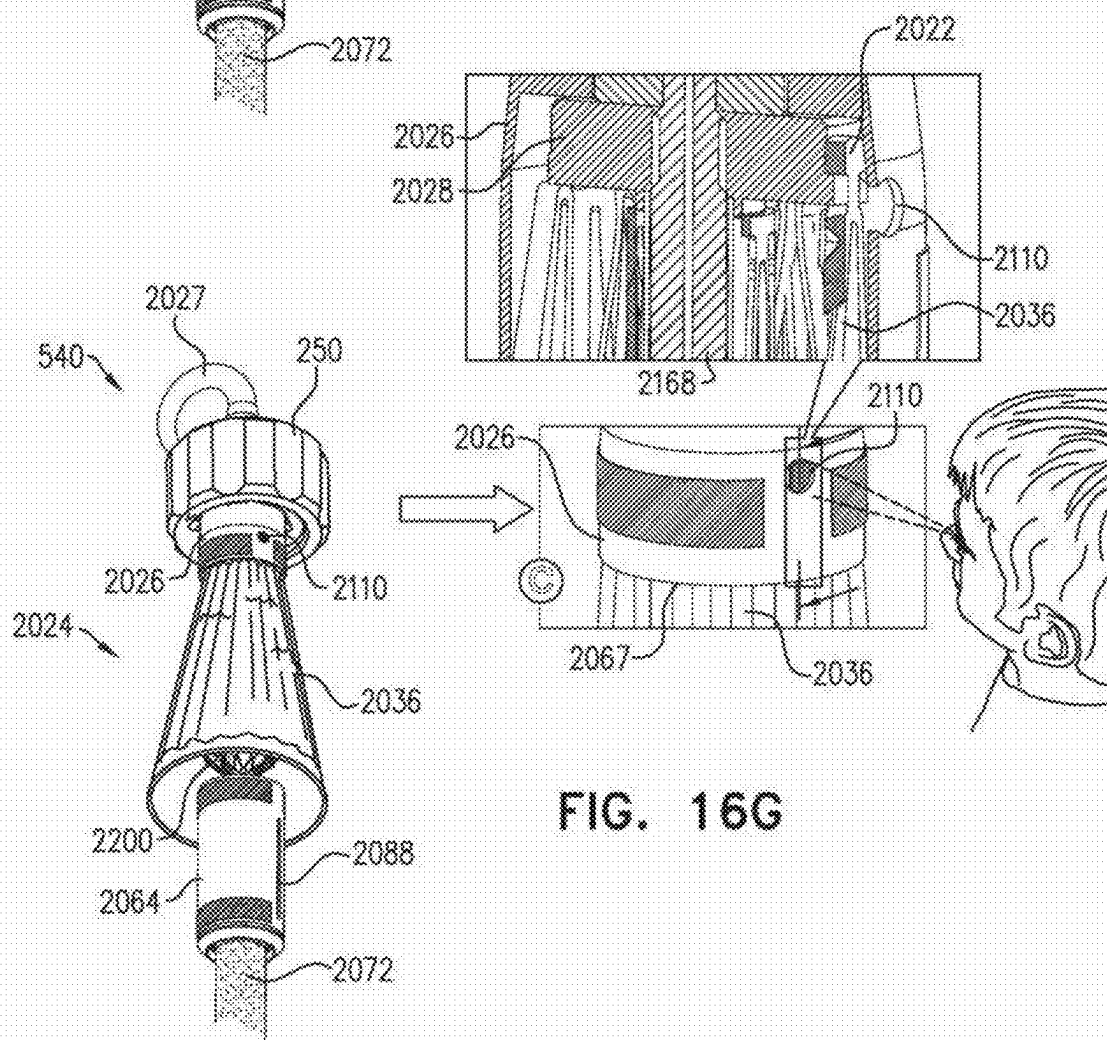
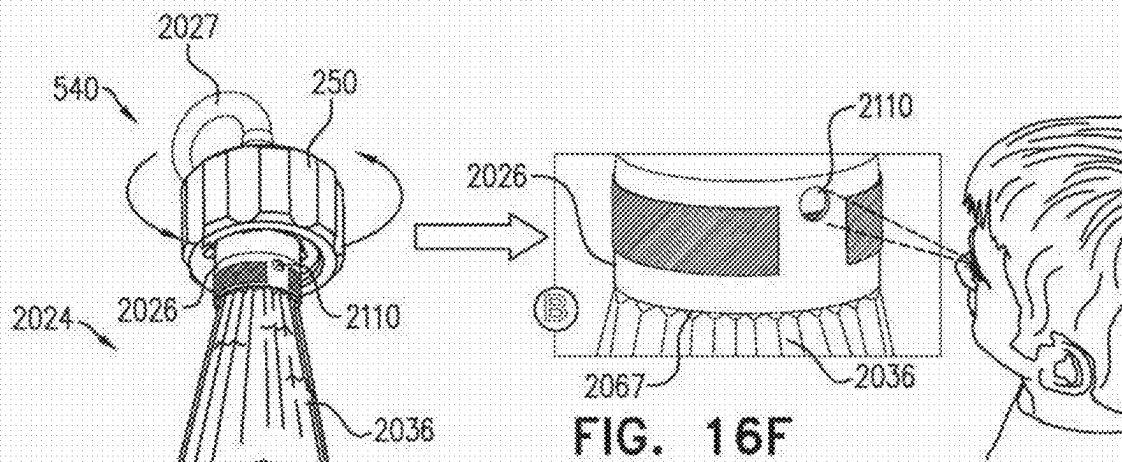
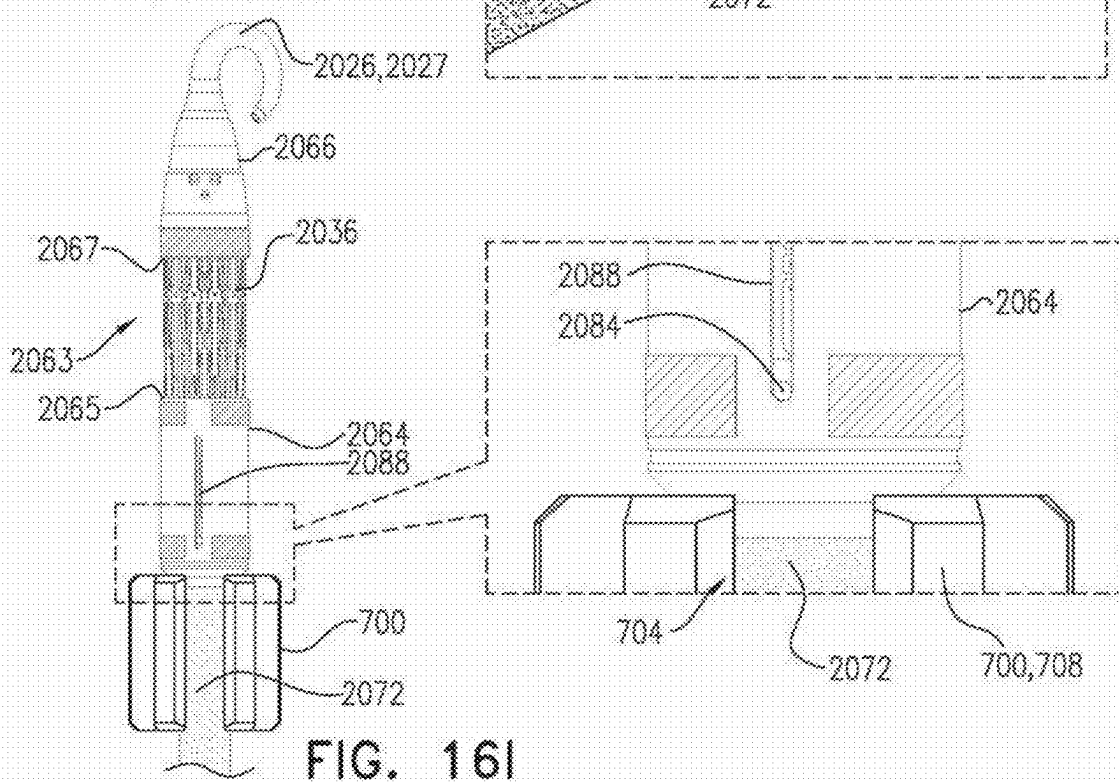
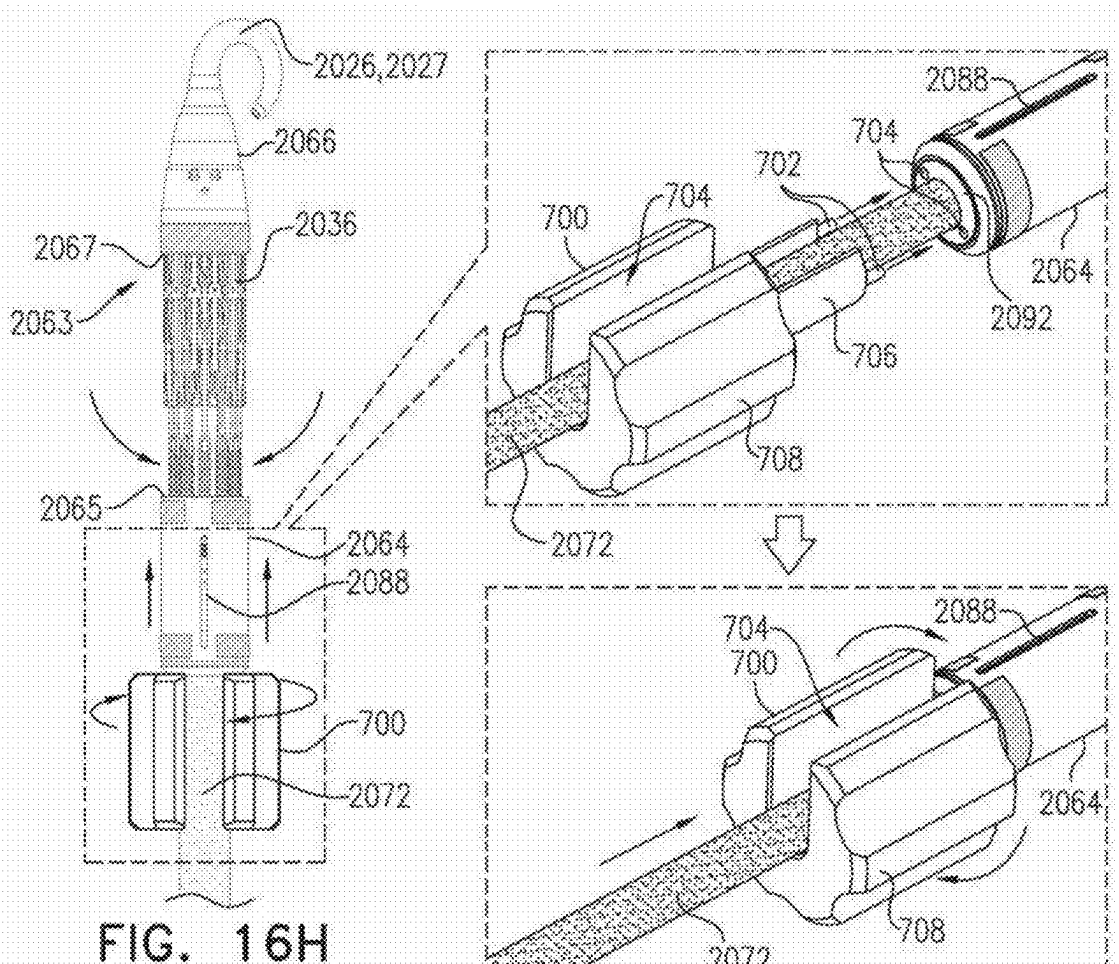
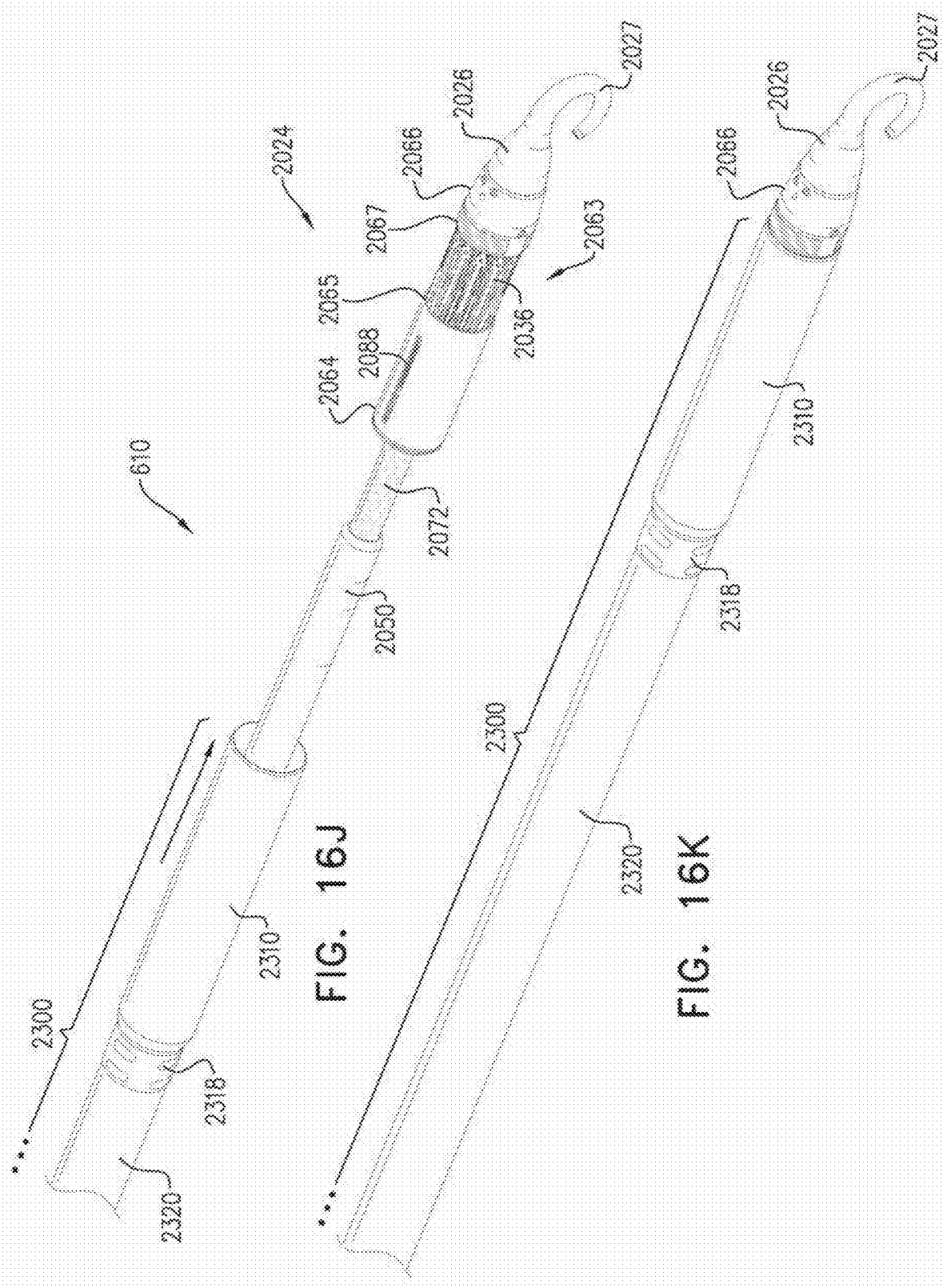


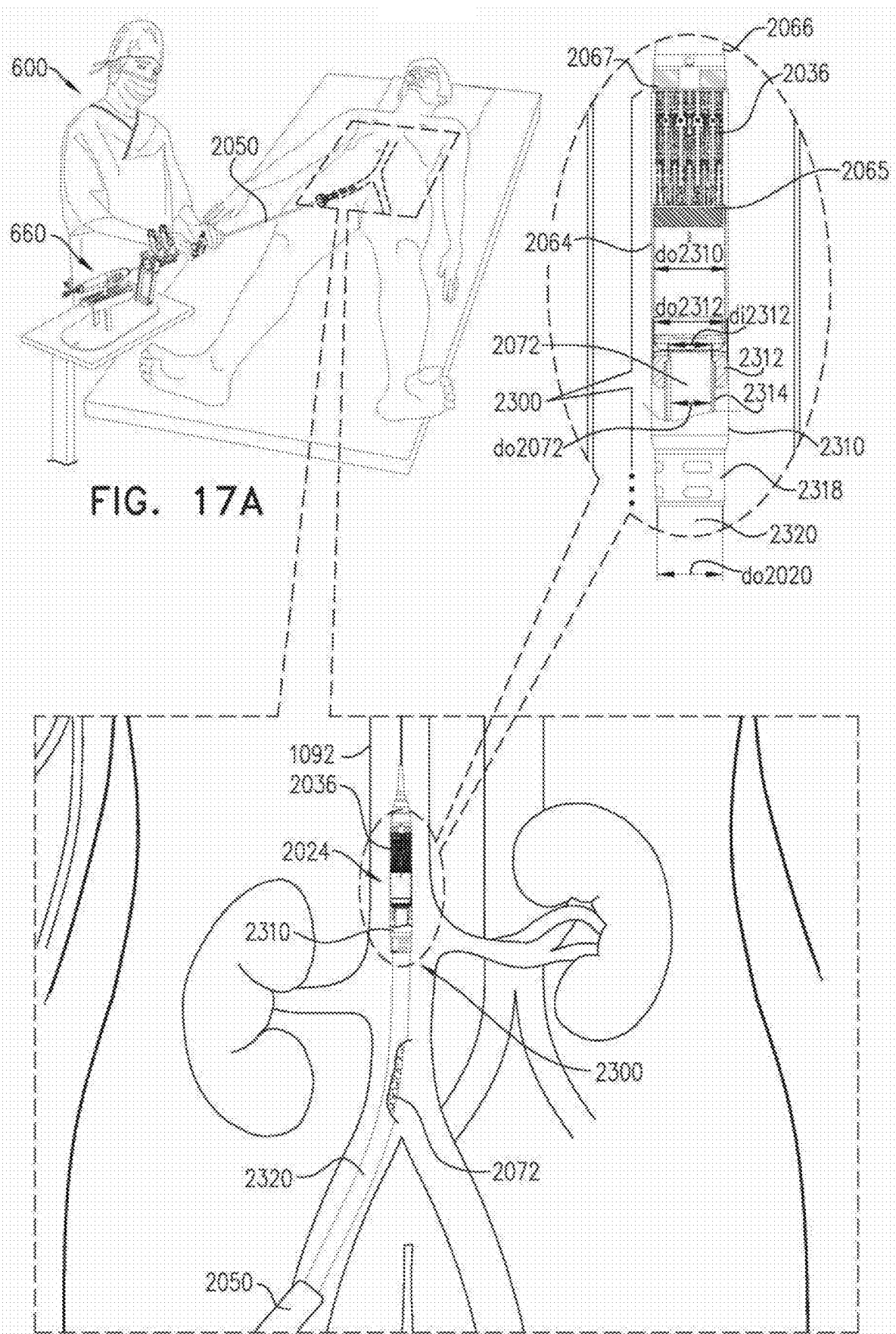
FIG. 16C

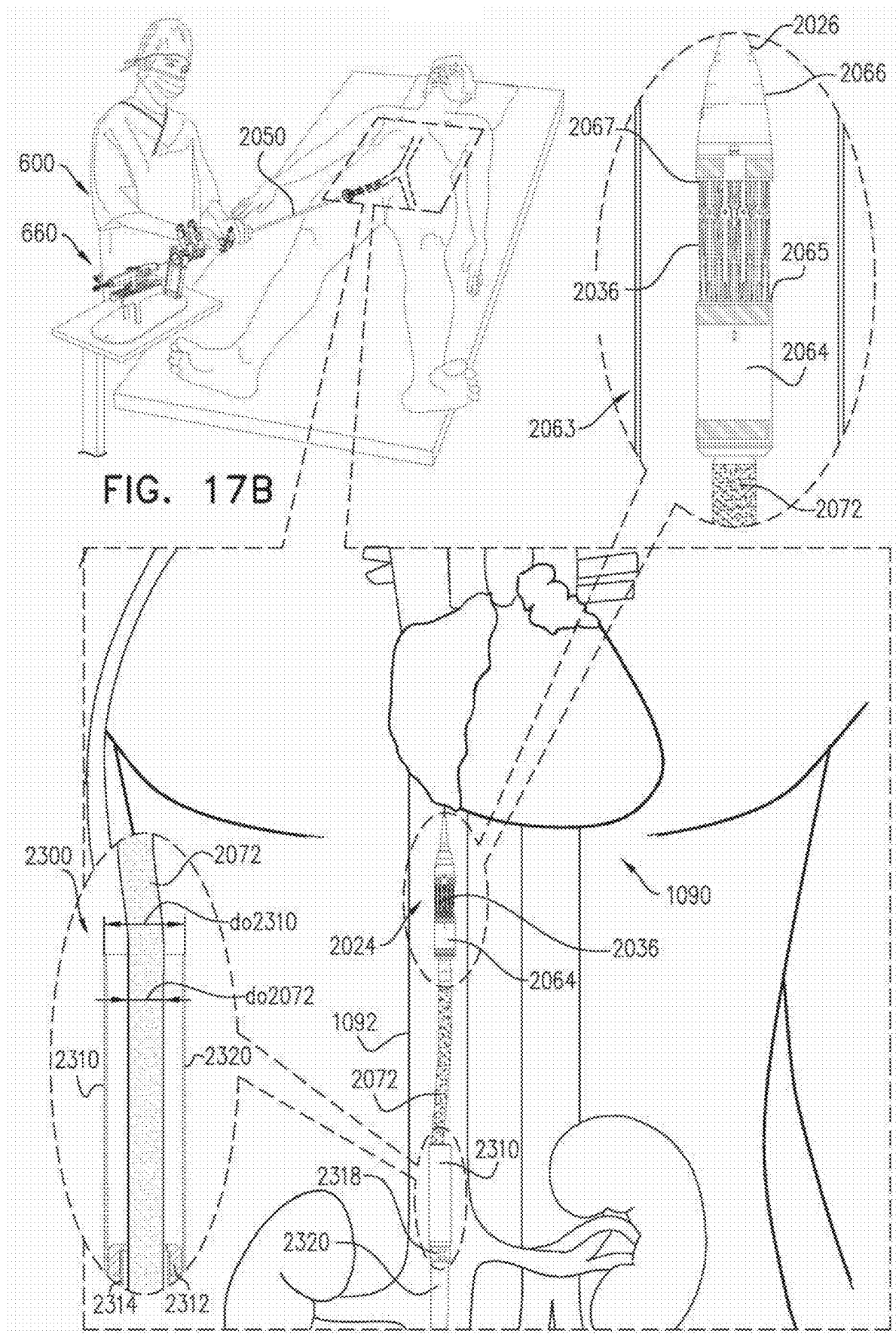


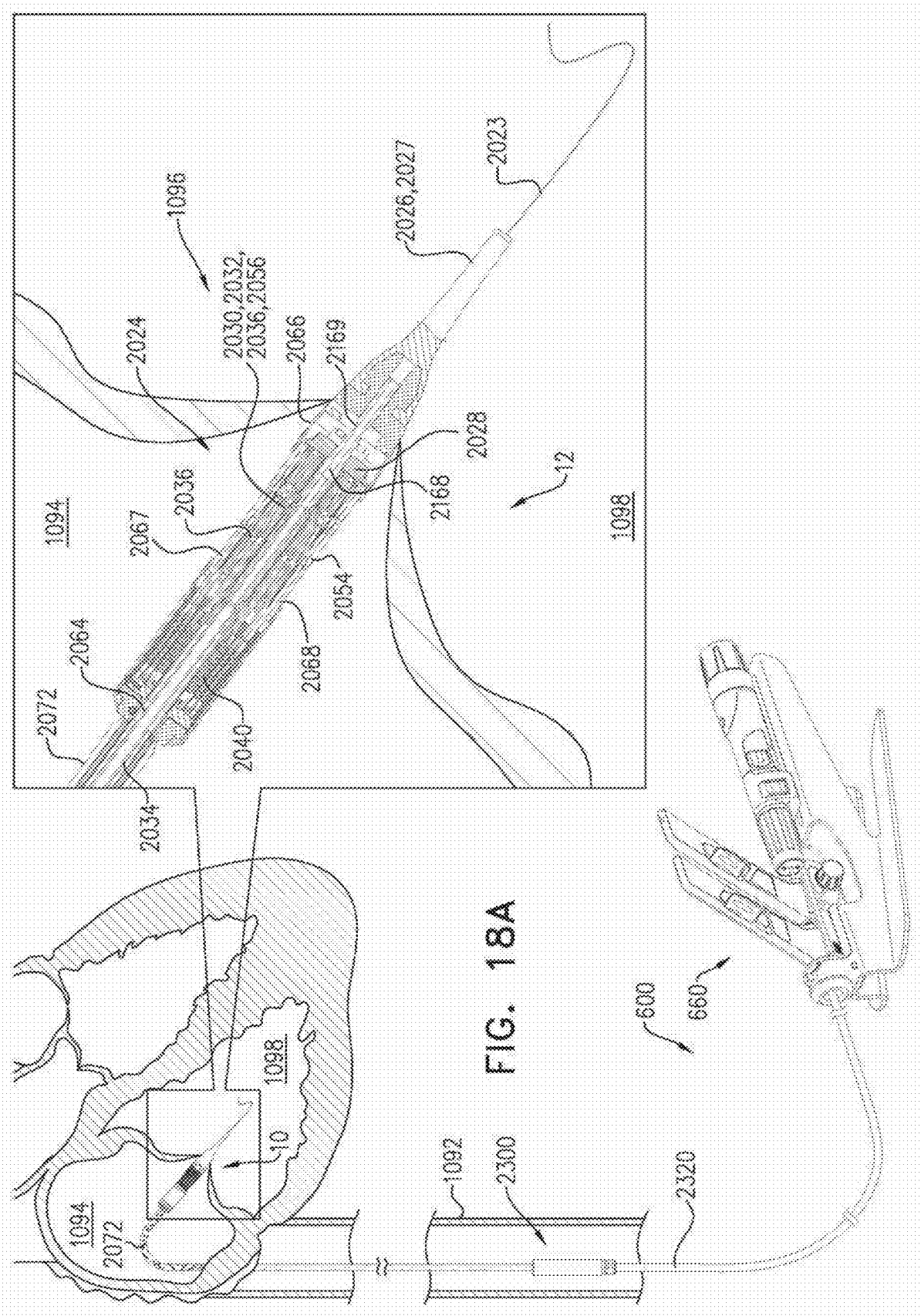


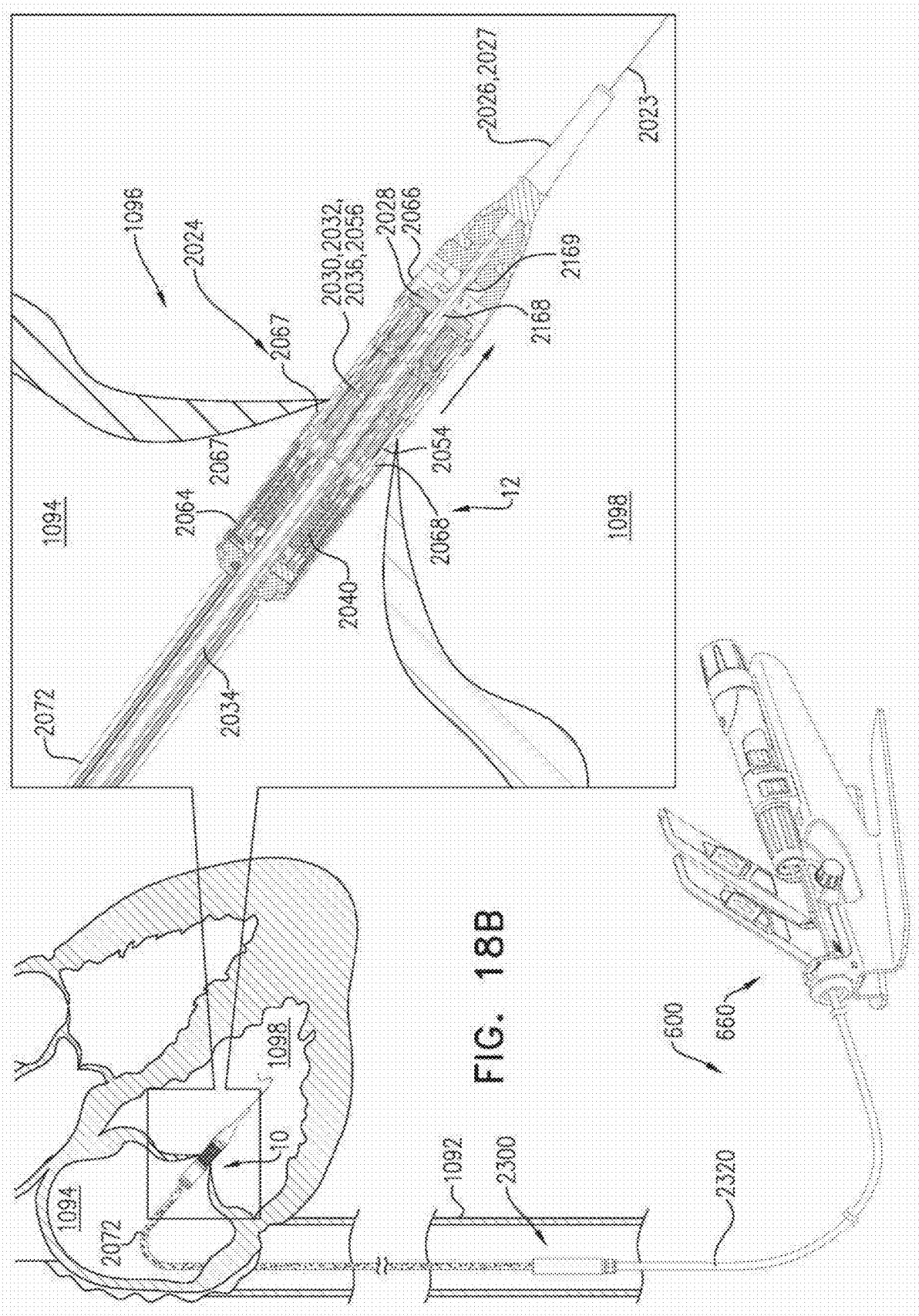


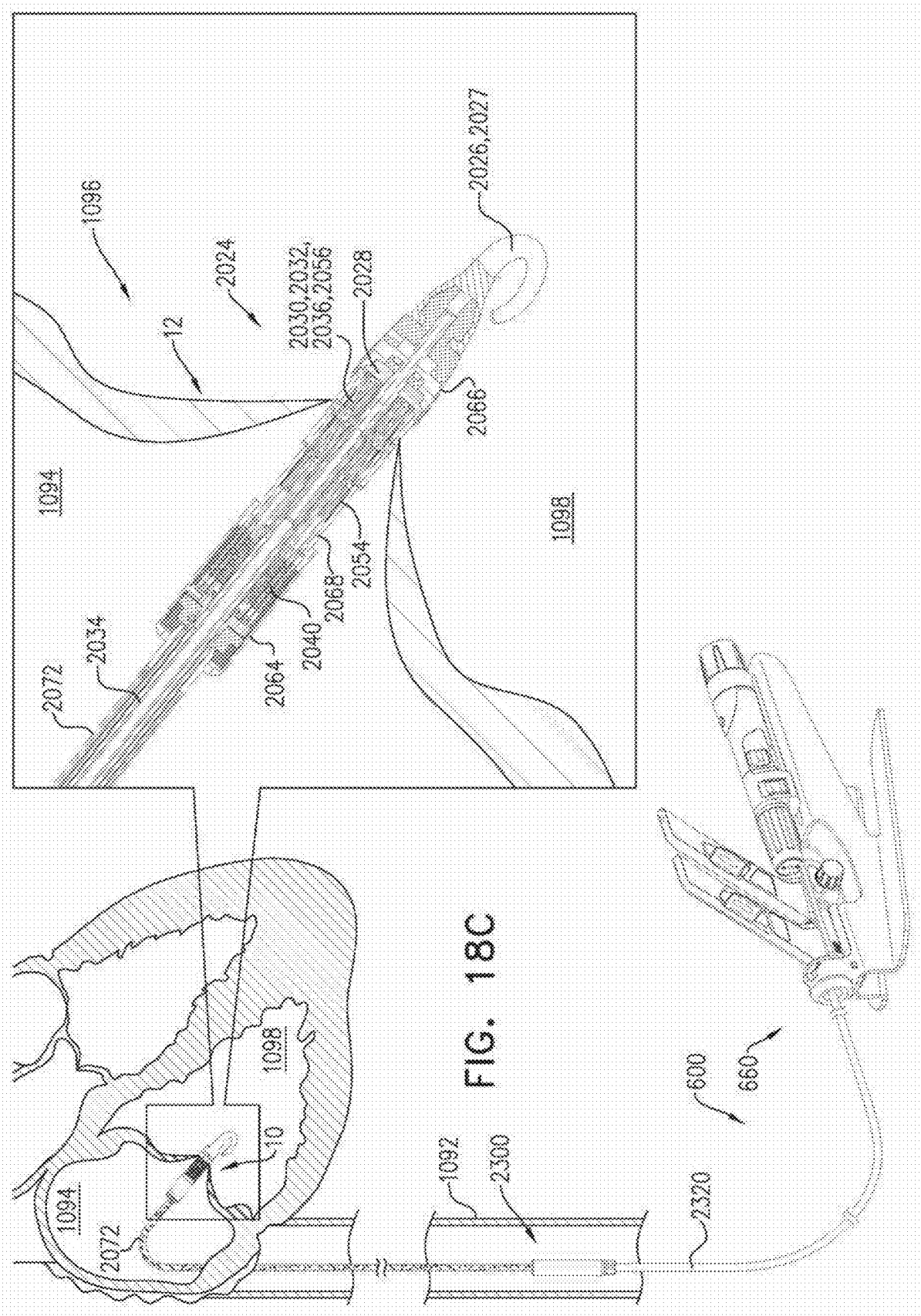


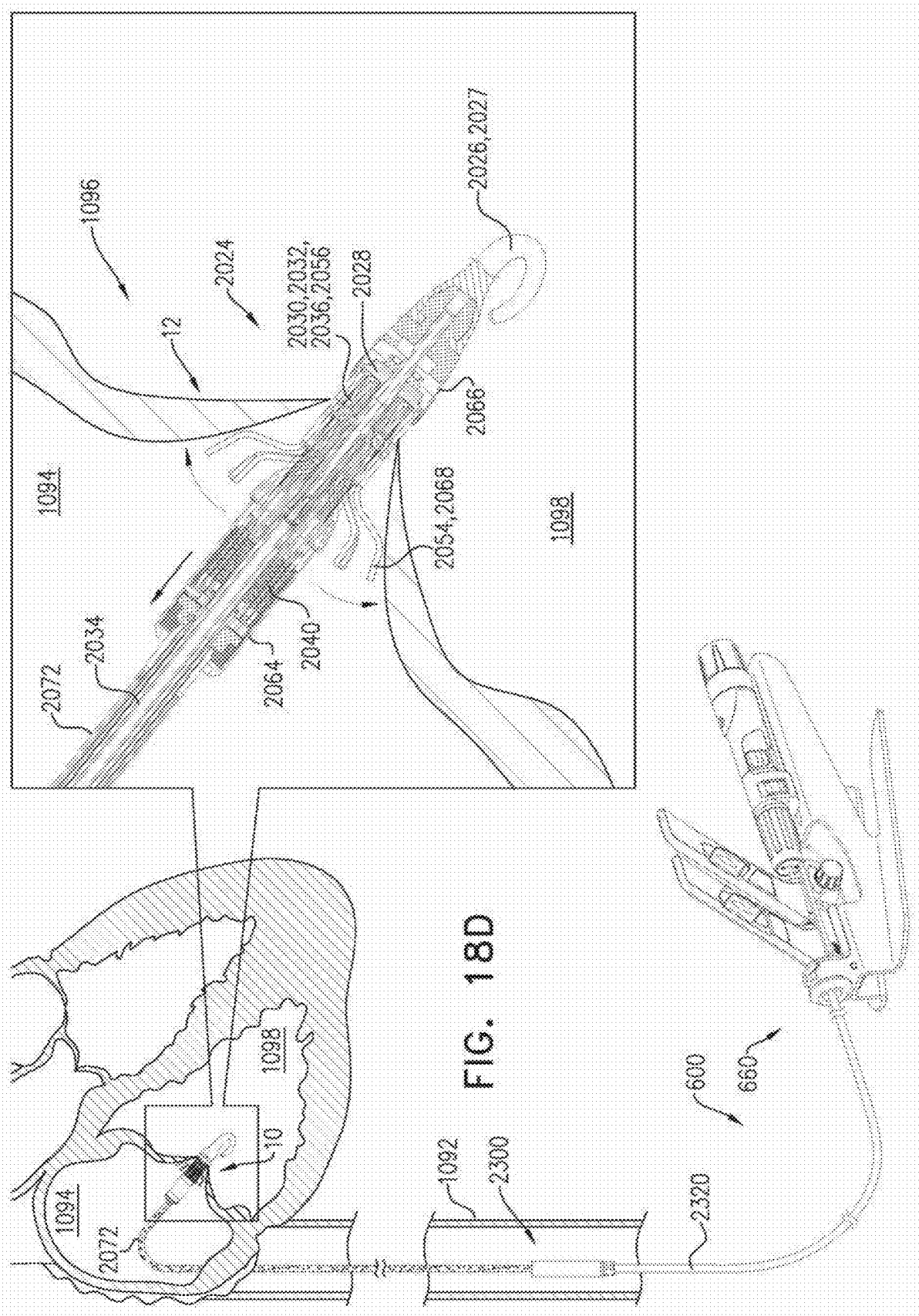


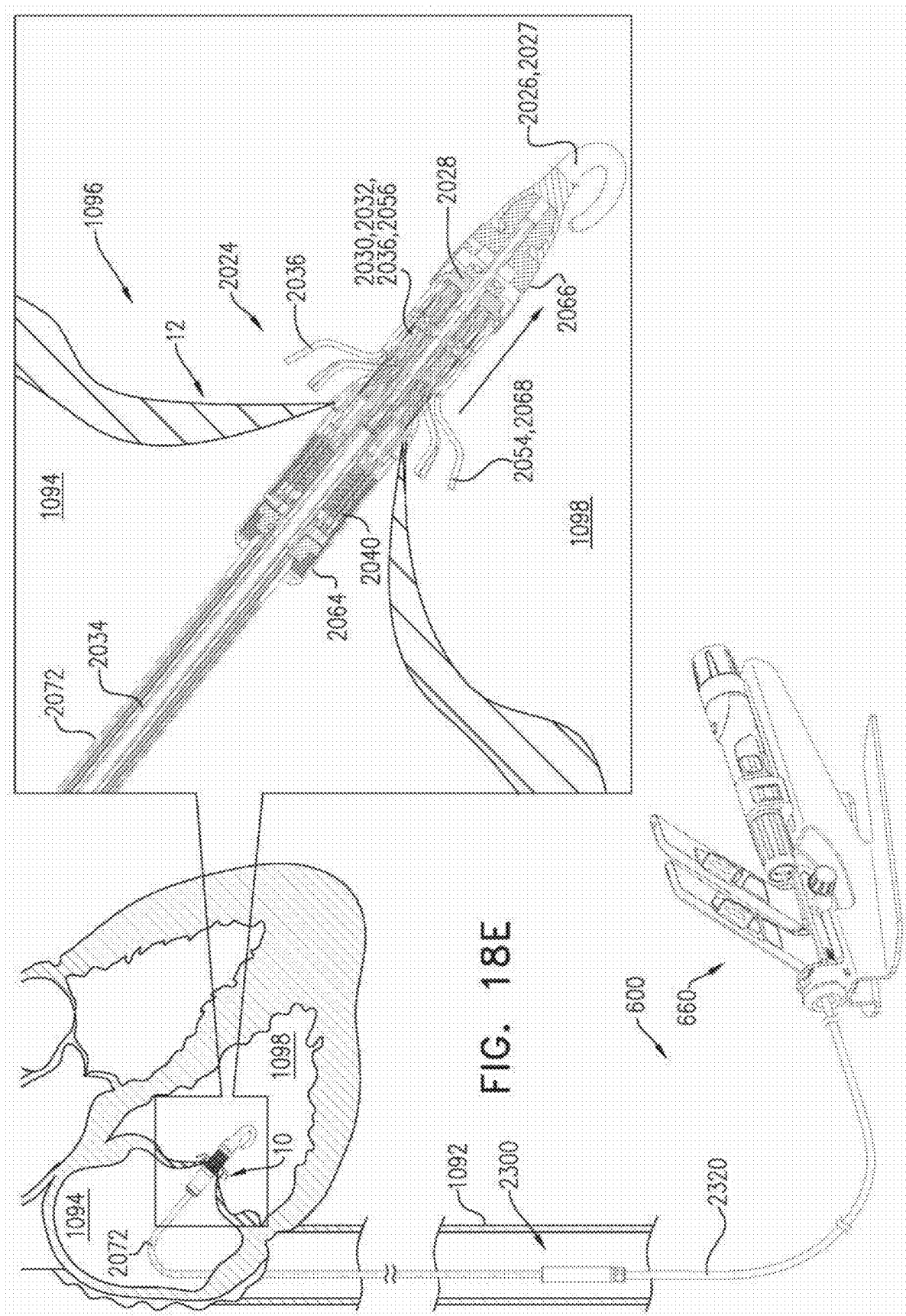


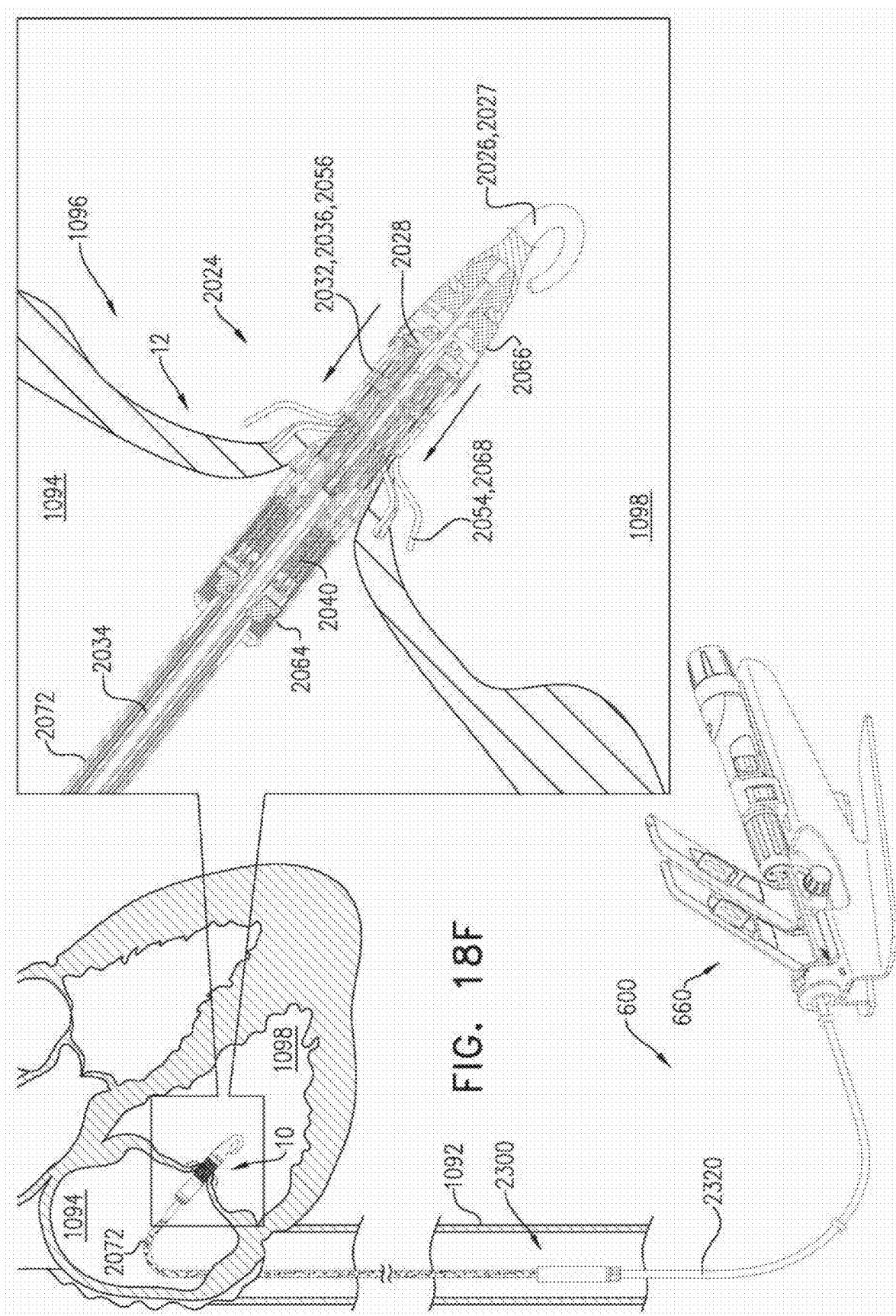












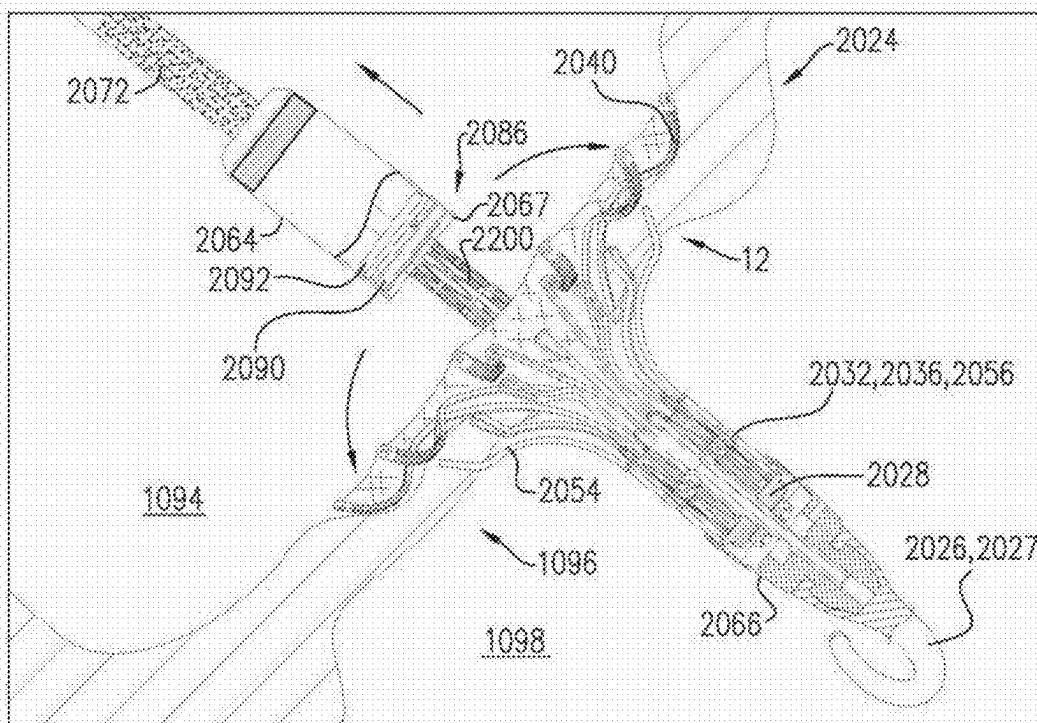


FIG. 18G

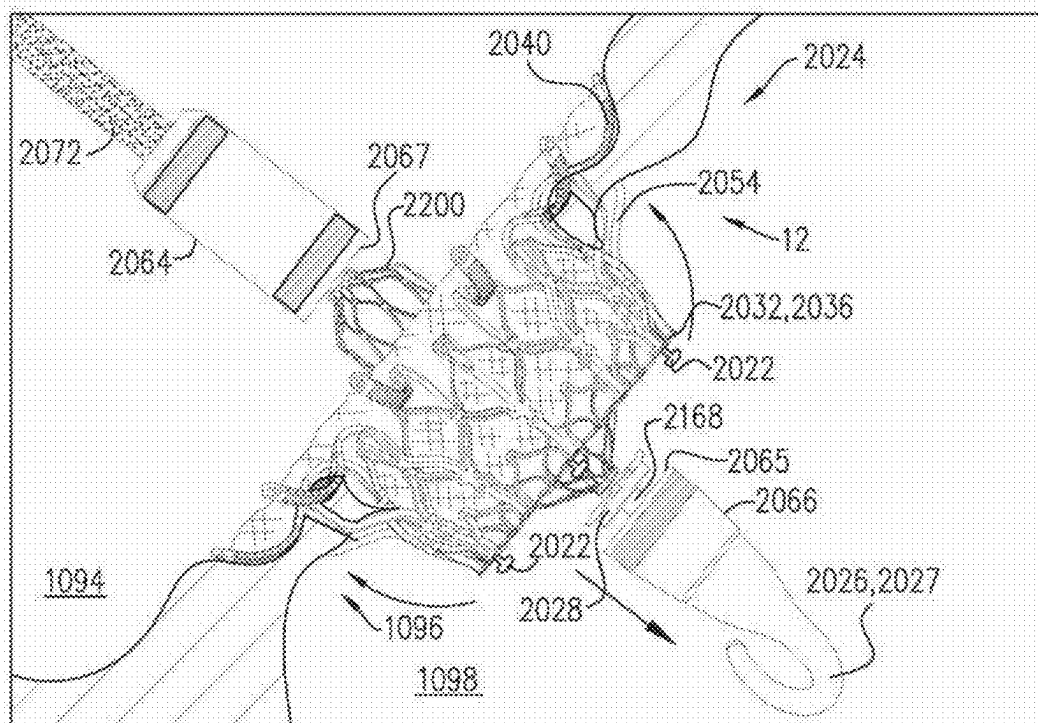


FIG. 18H

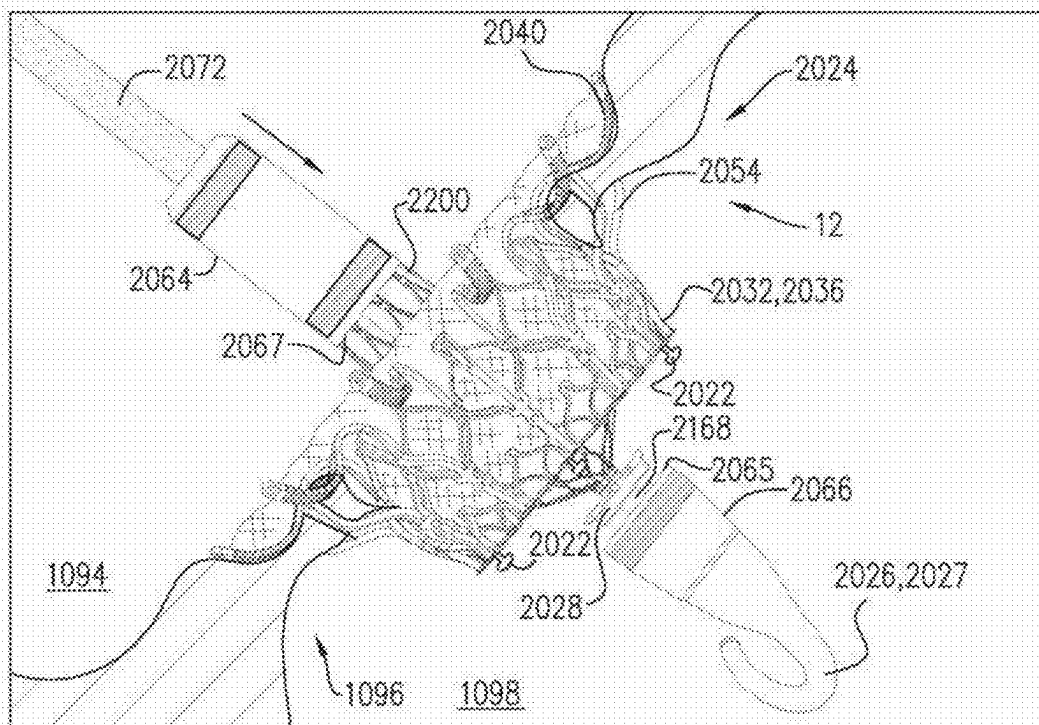


FIG. 18I

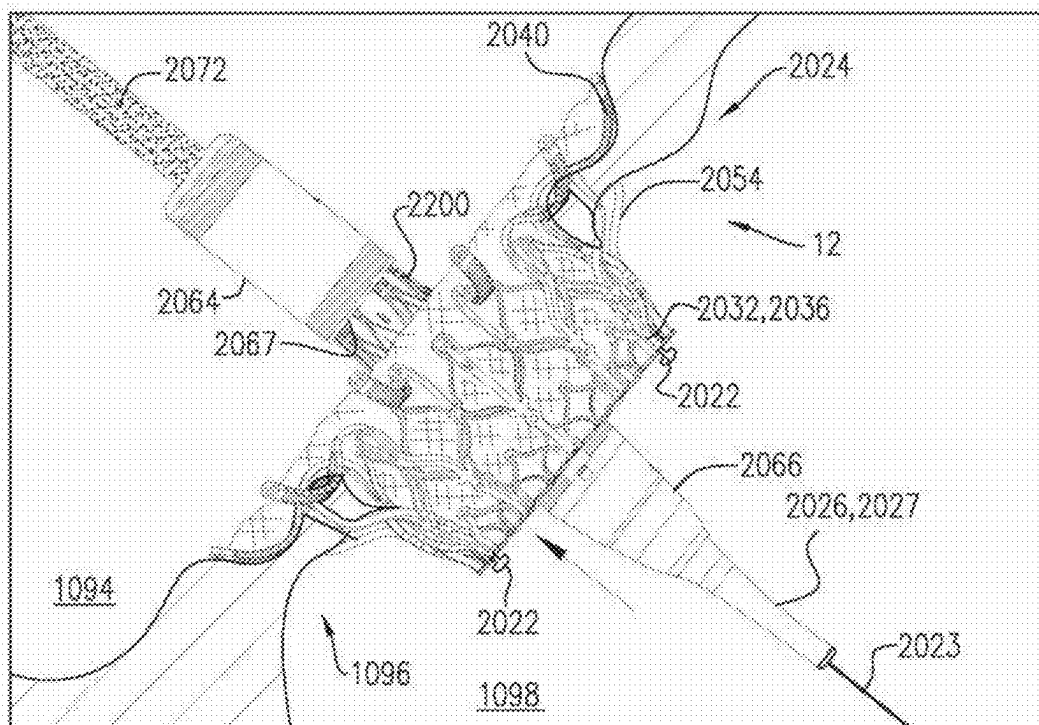


FIG. 18J

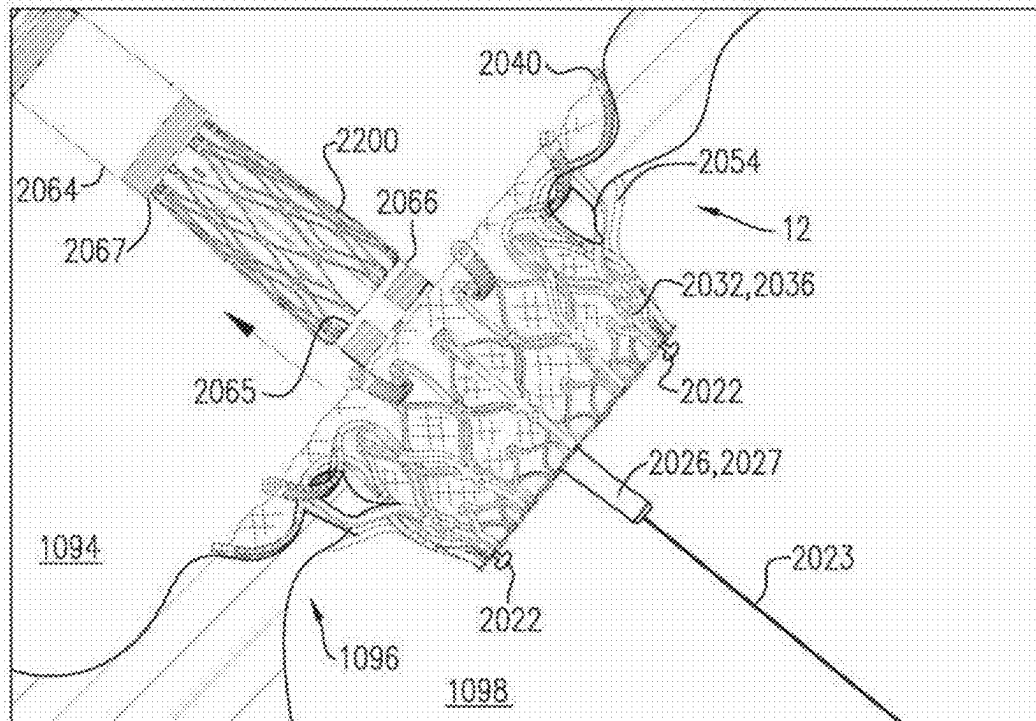


FIG. 18K

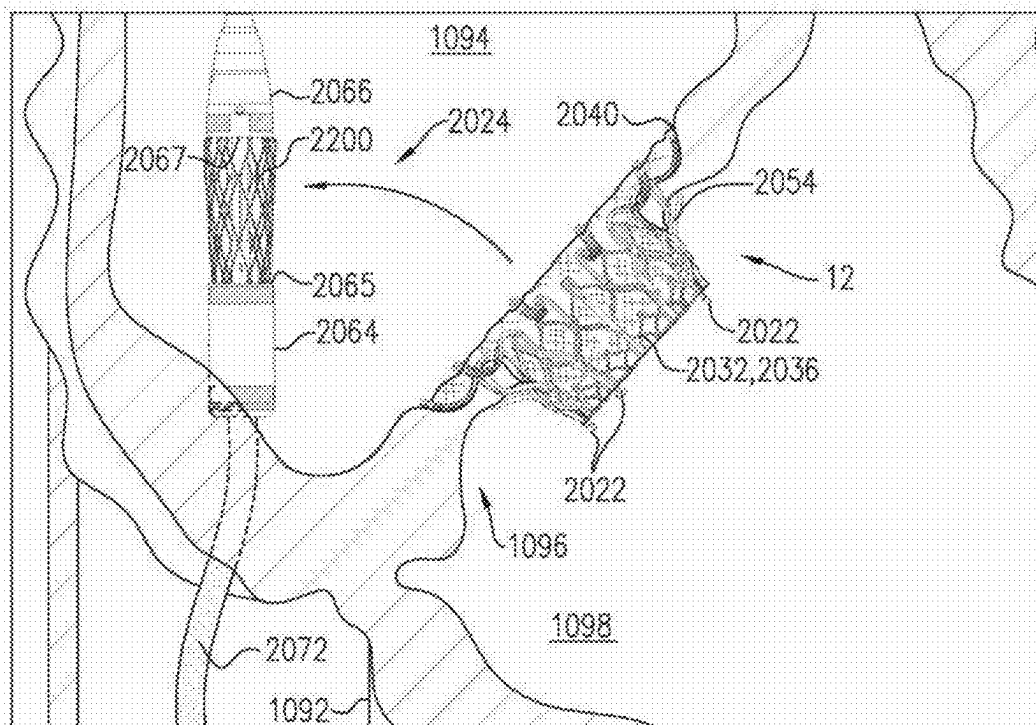


FIG. 18L

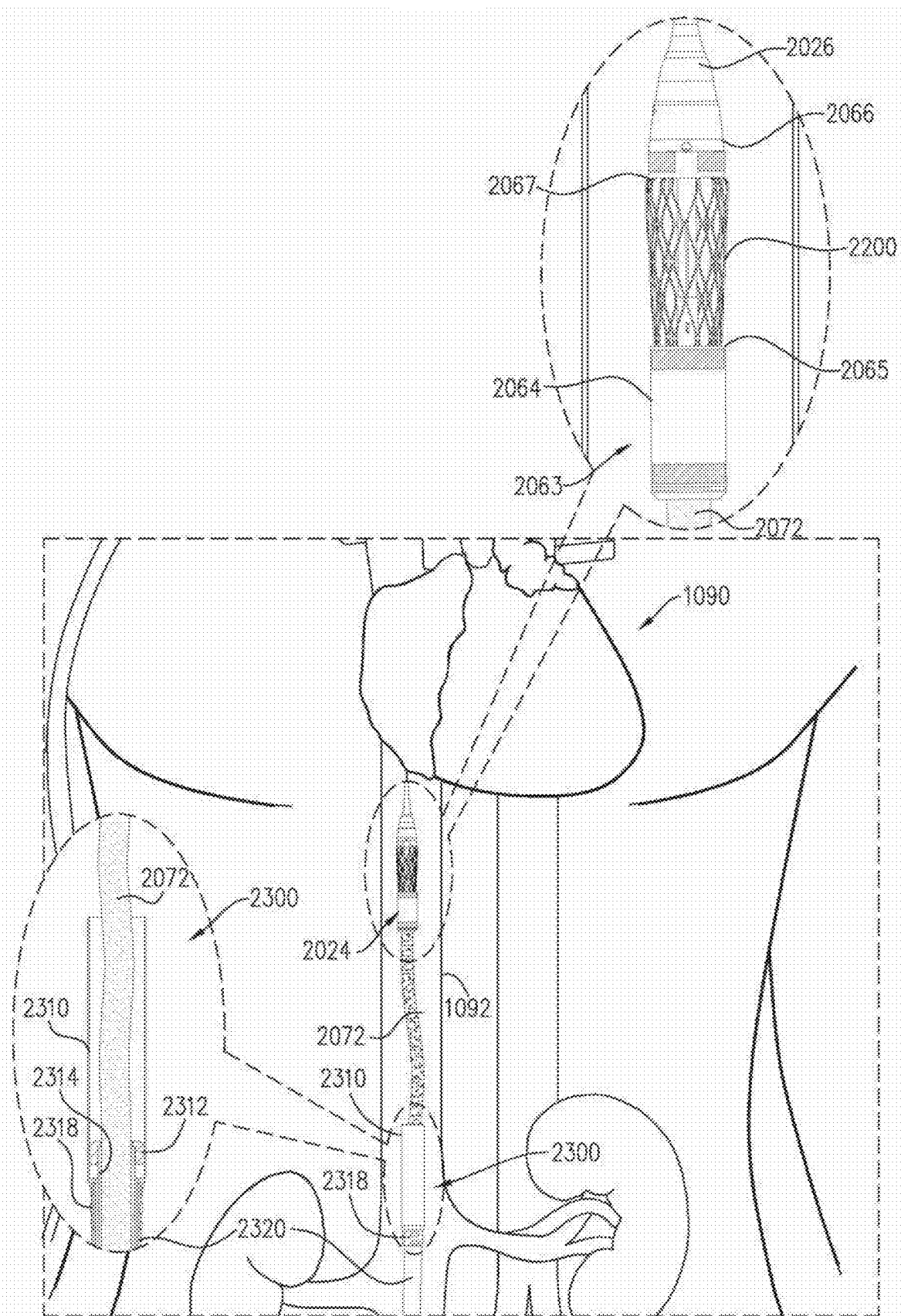


FIG. 18M

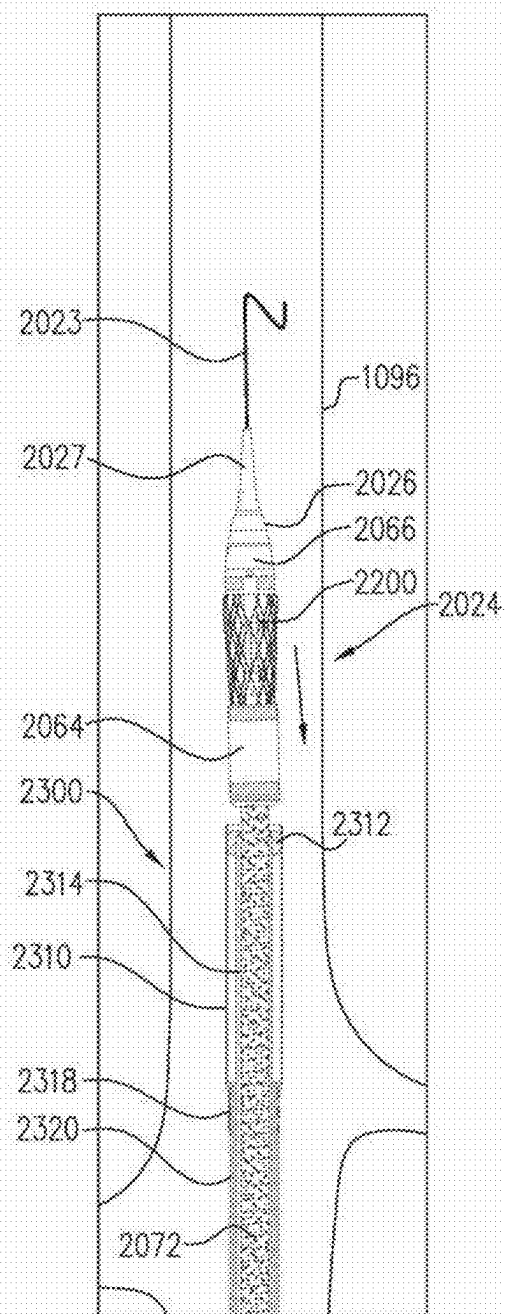


FIG. 18N

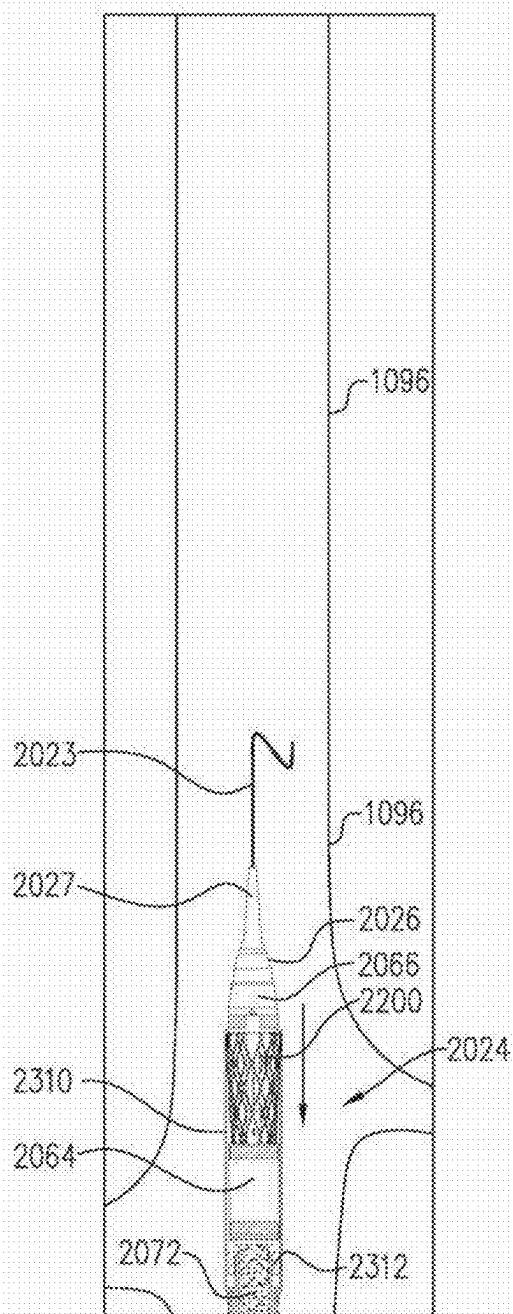
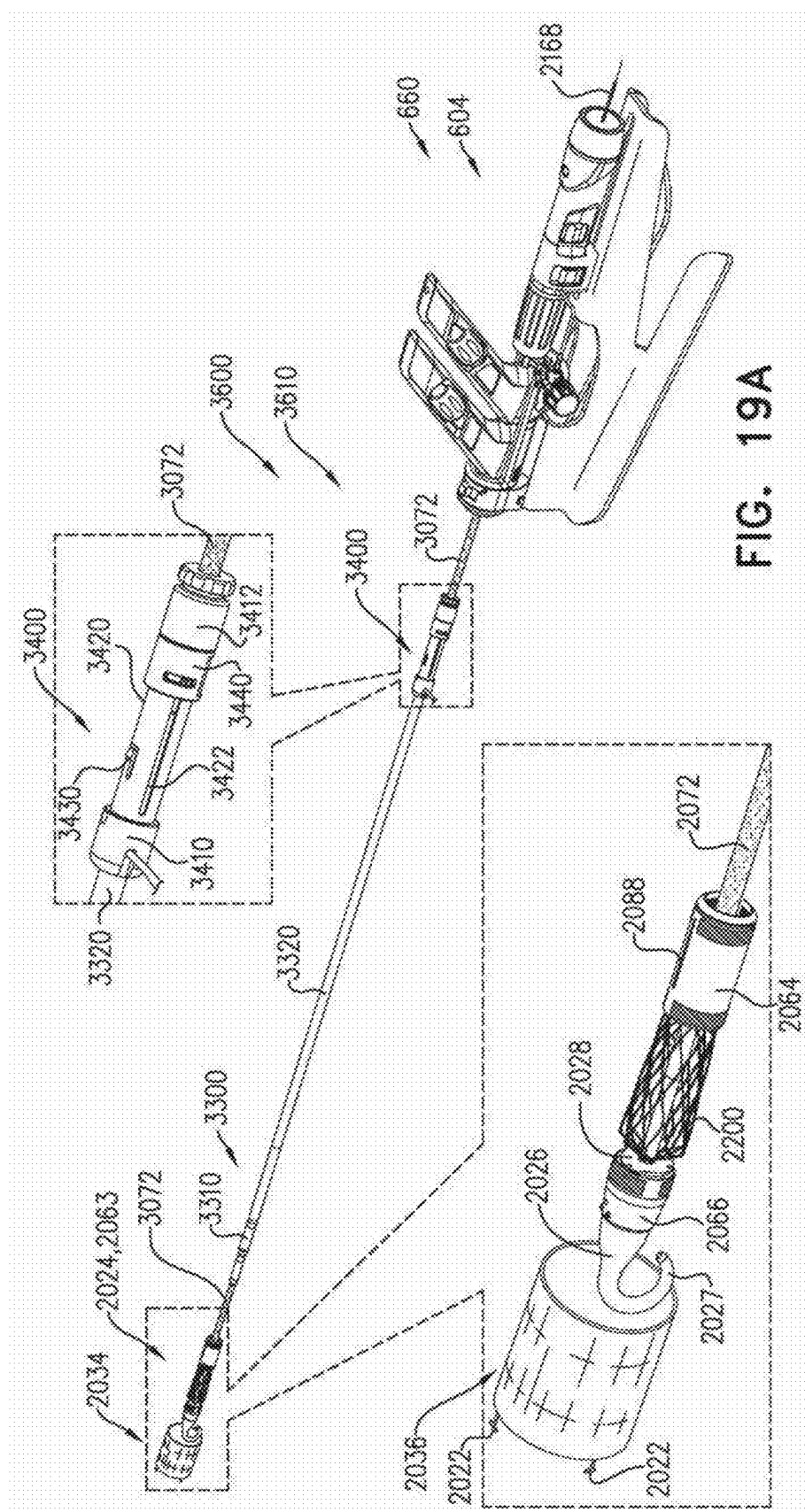
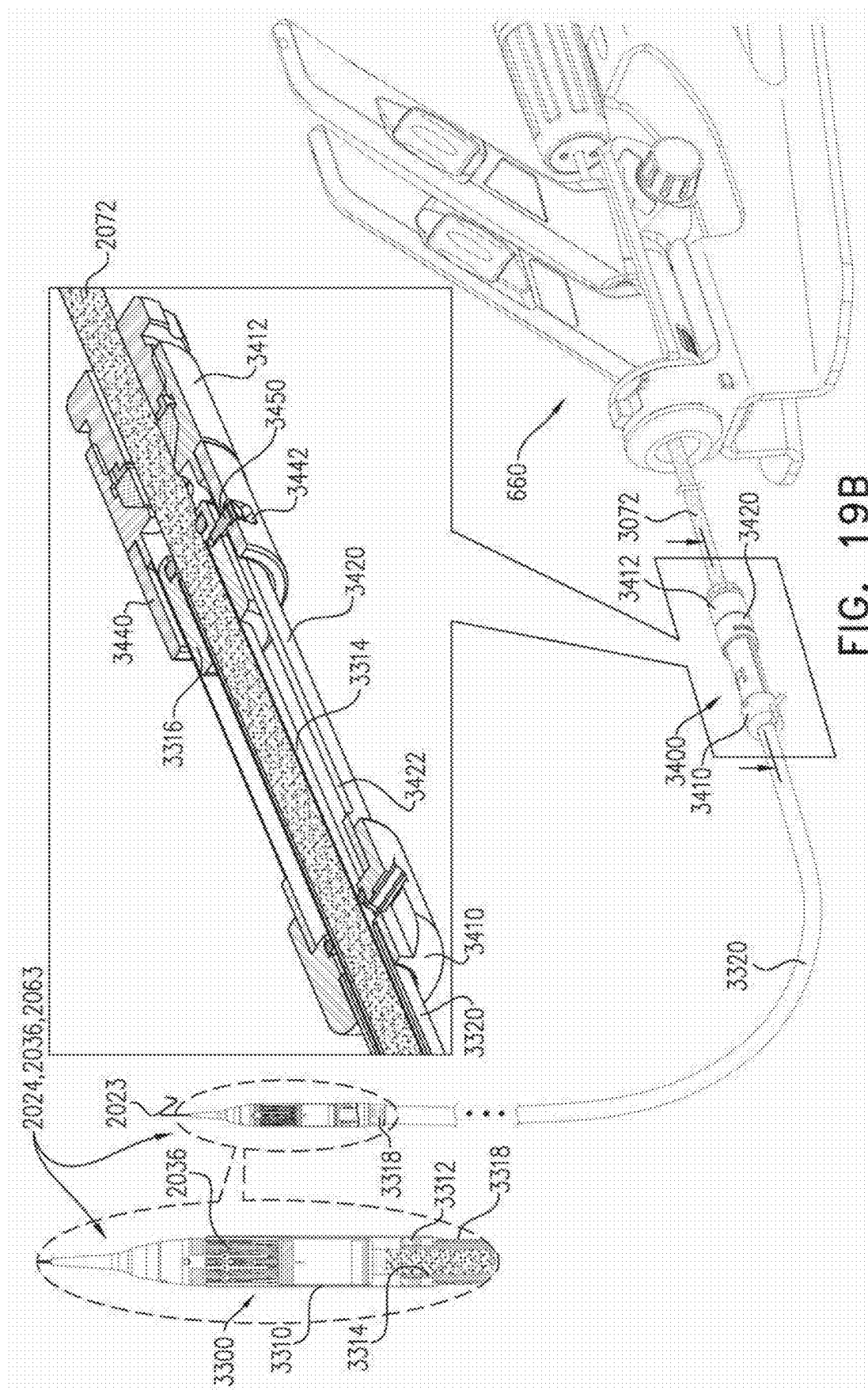
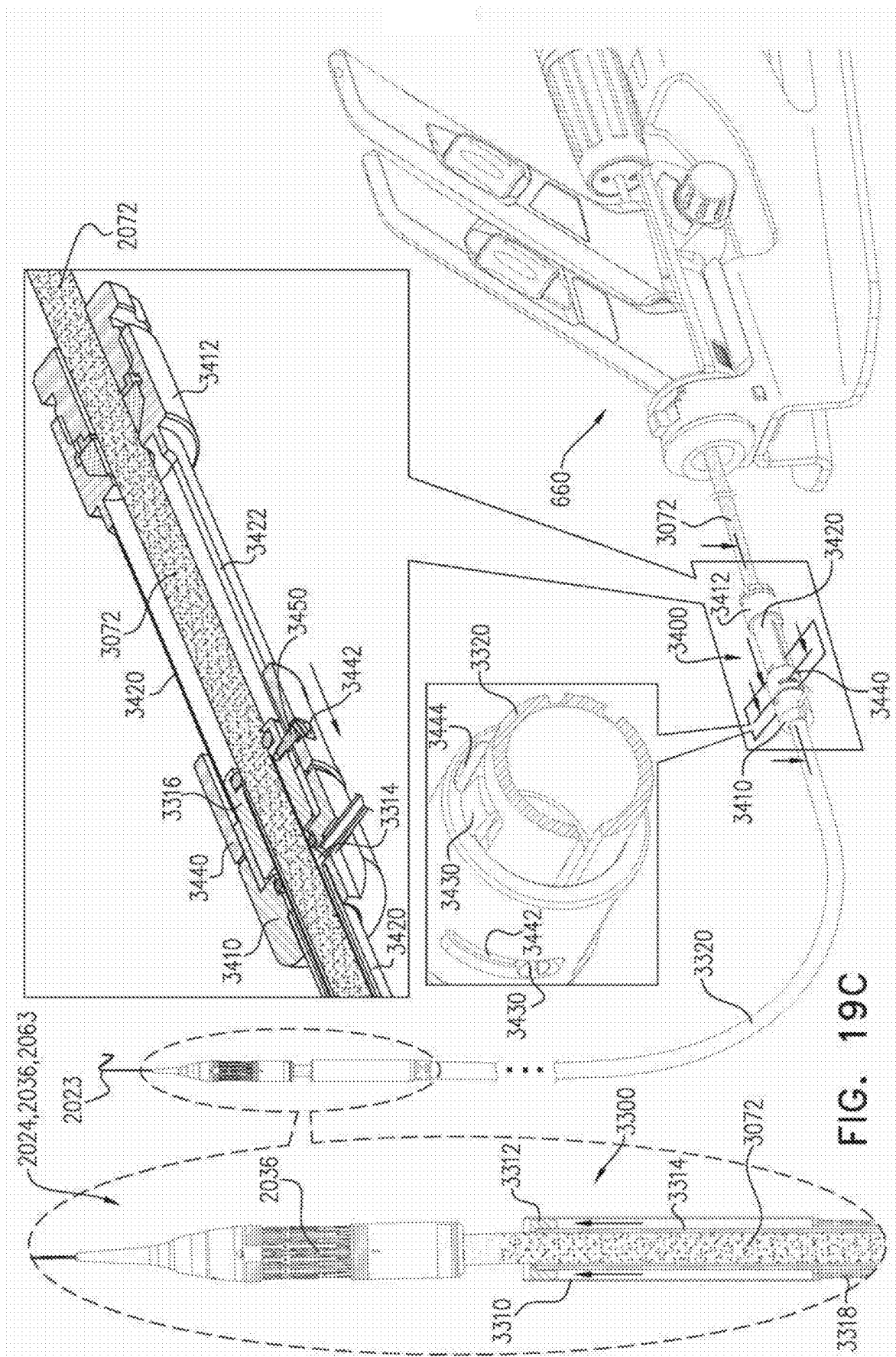
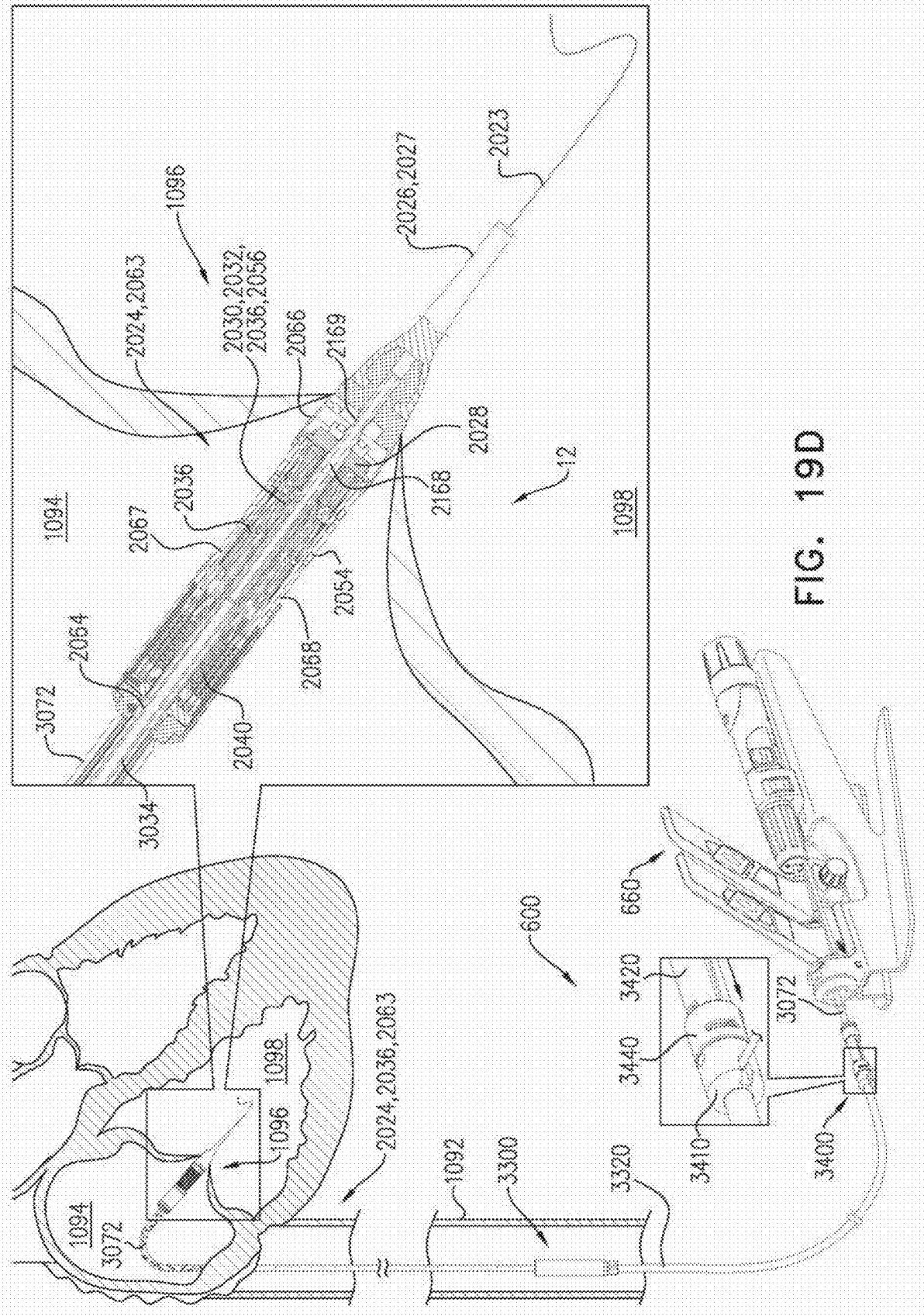


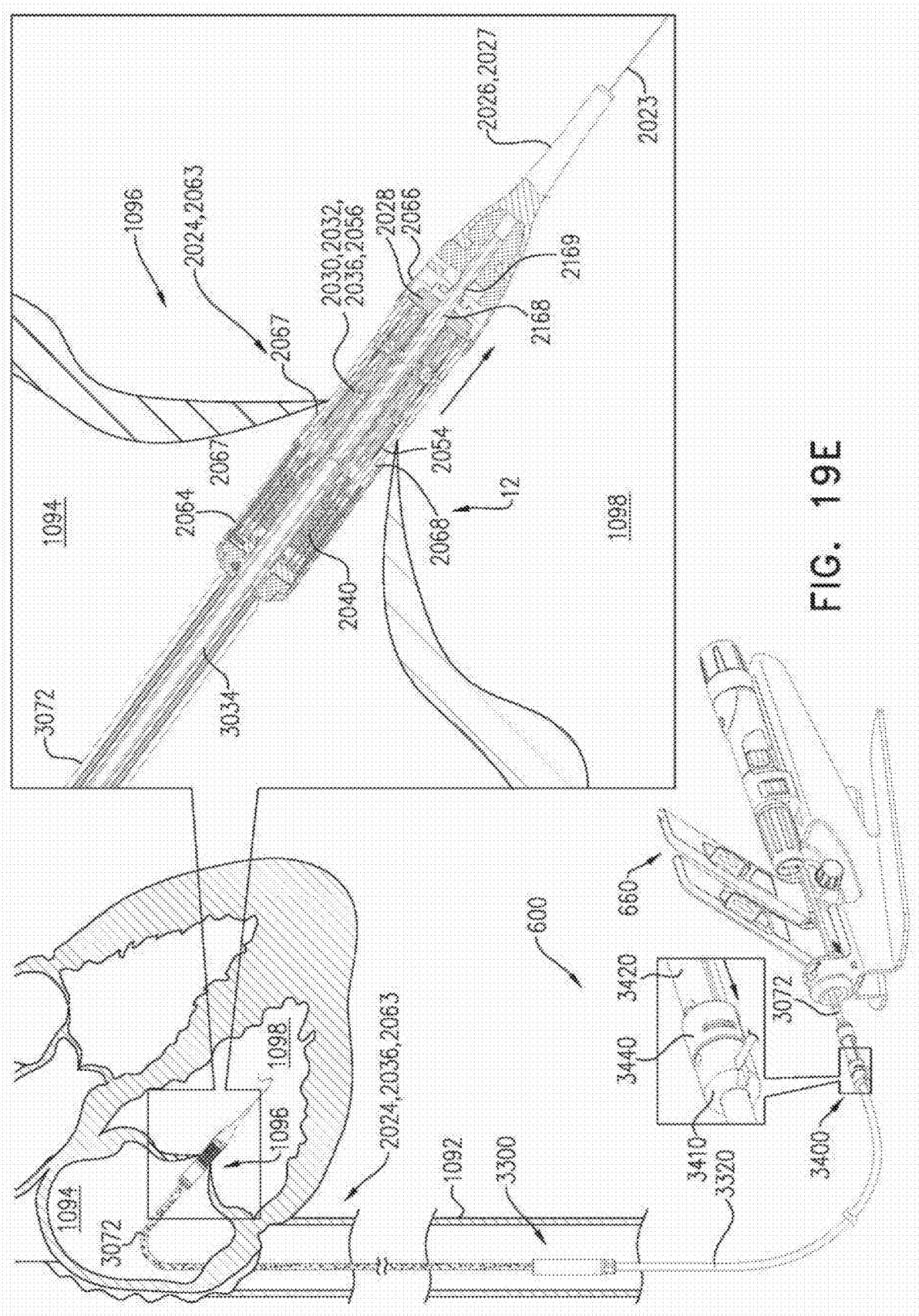
FIG. 18O











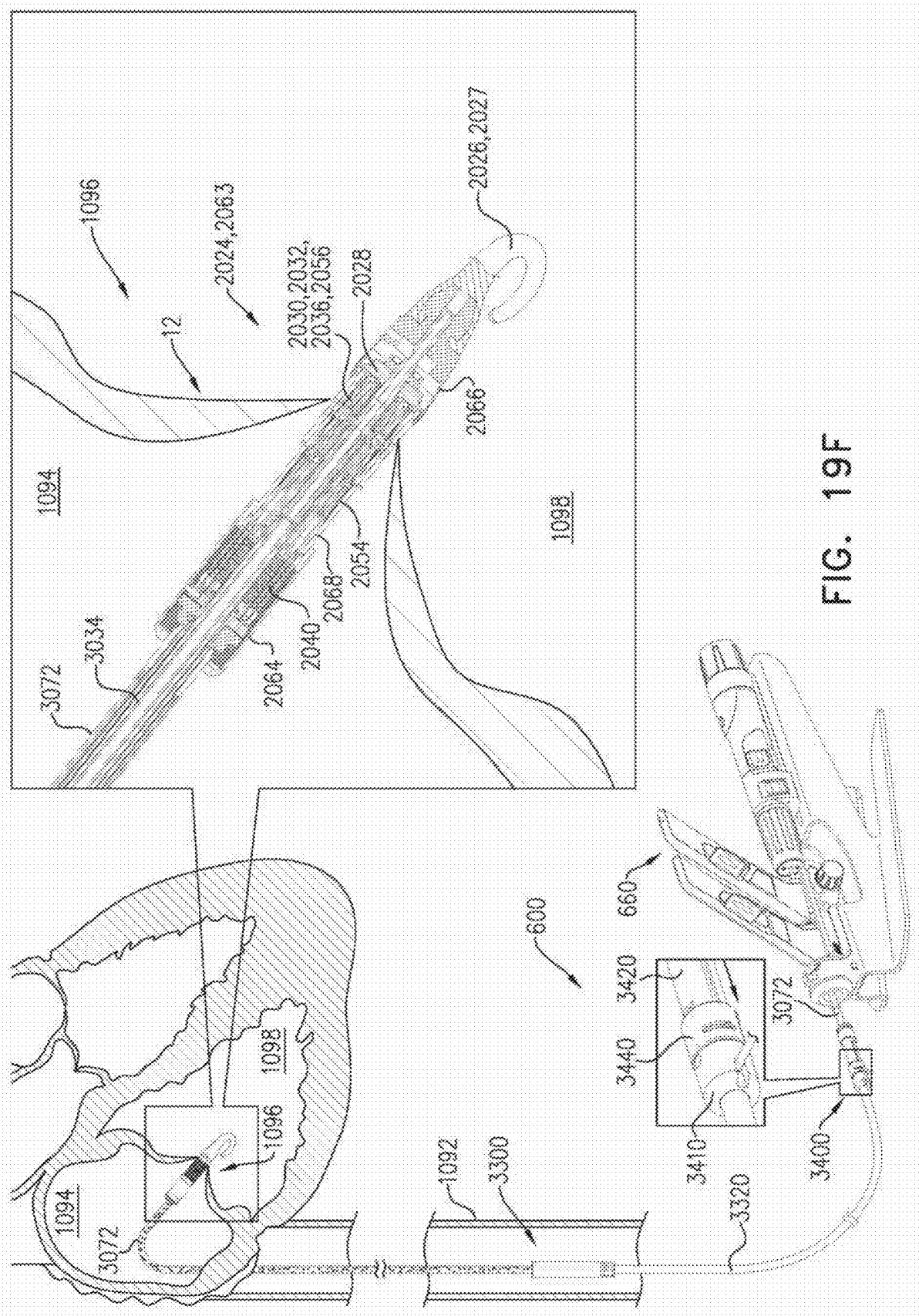


FIG. 19F

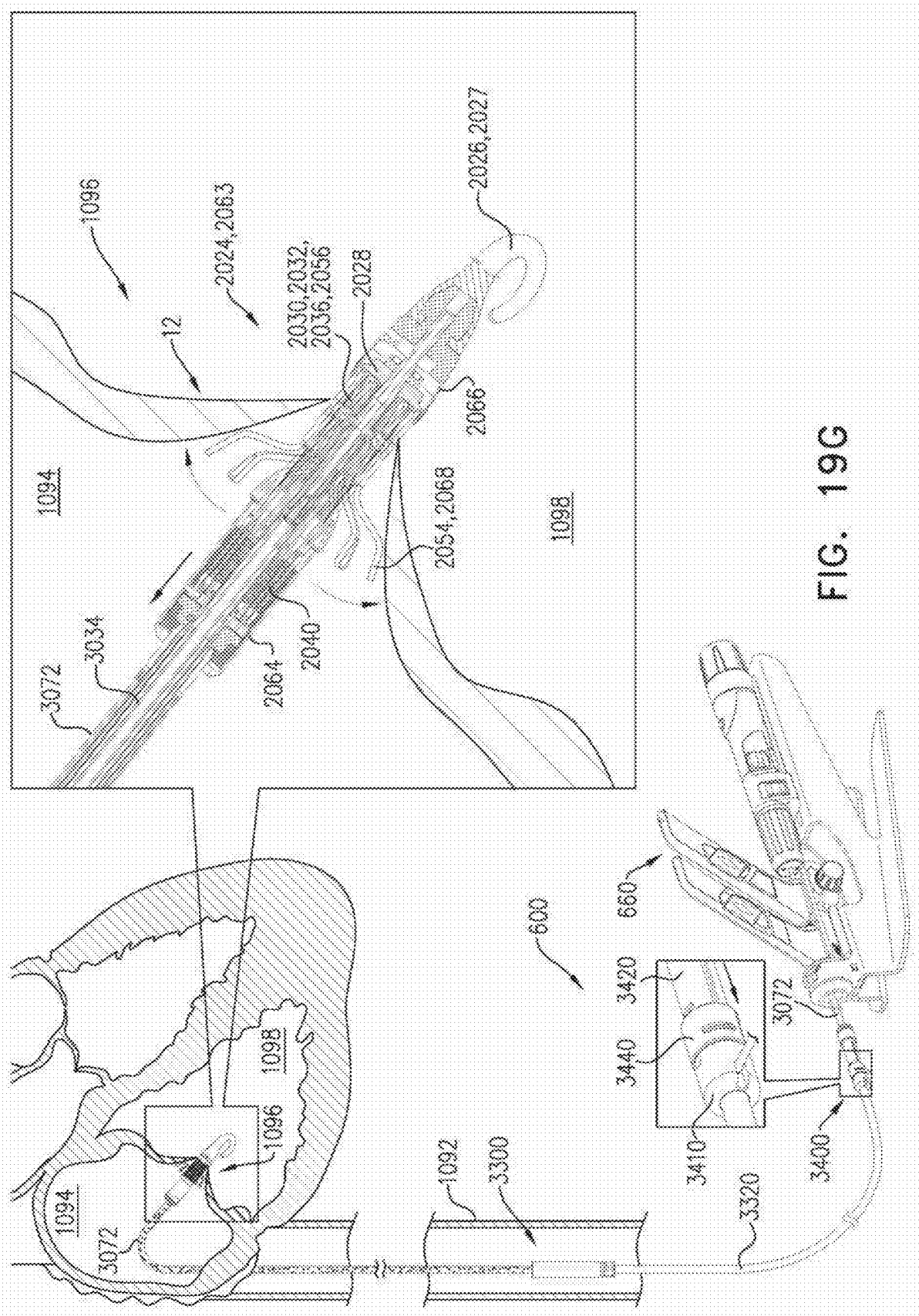
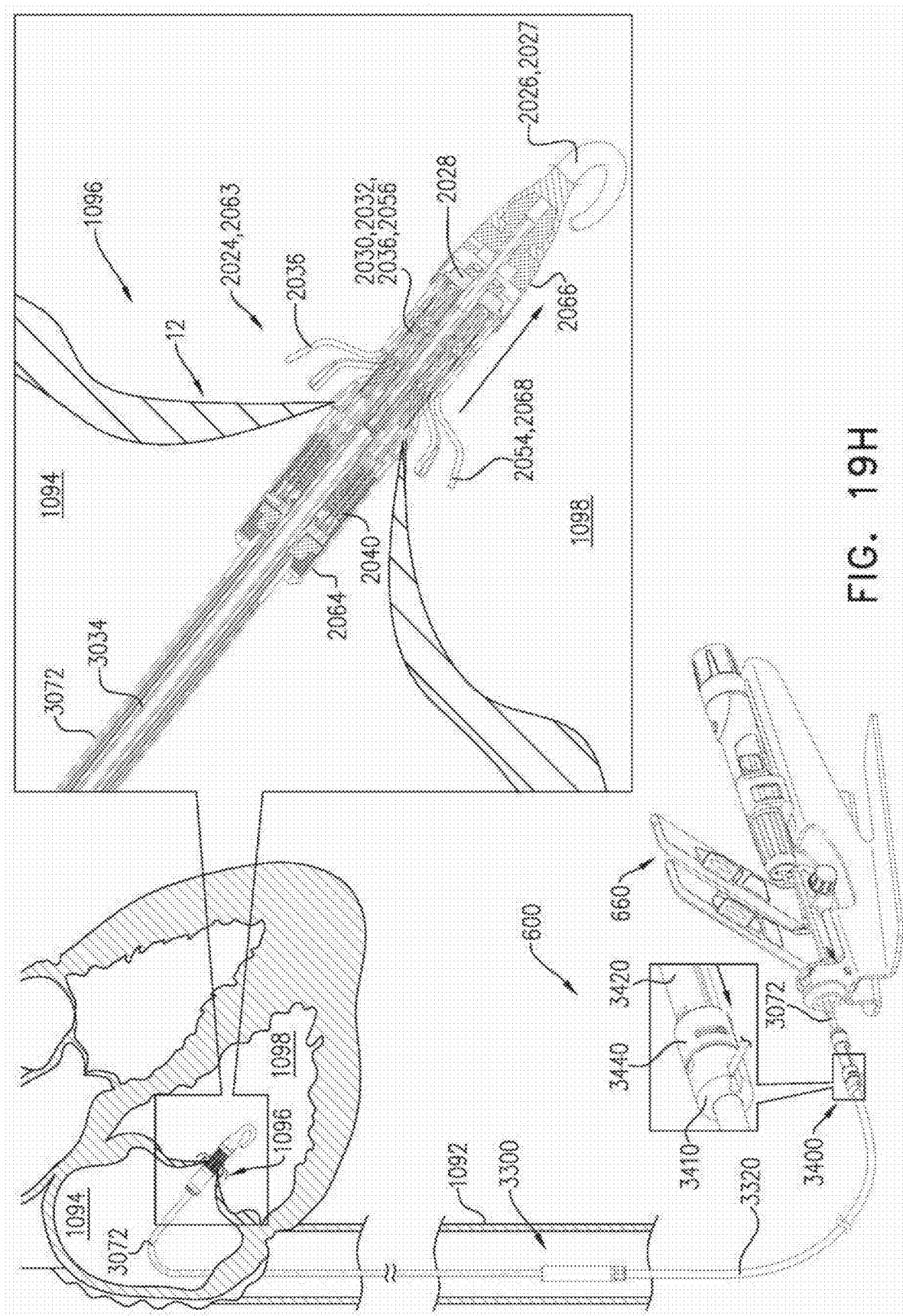
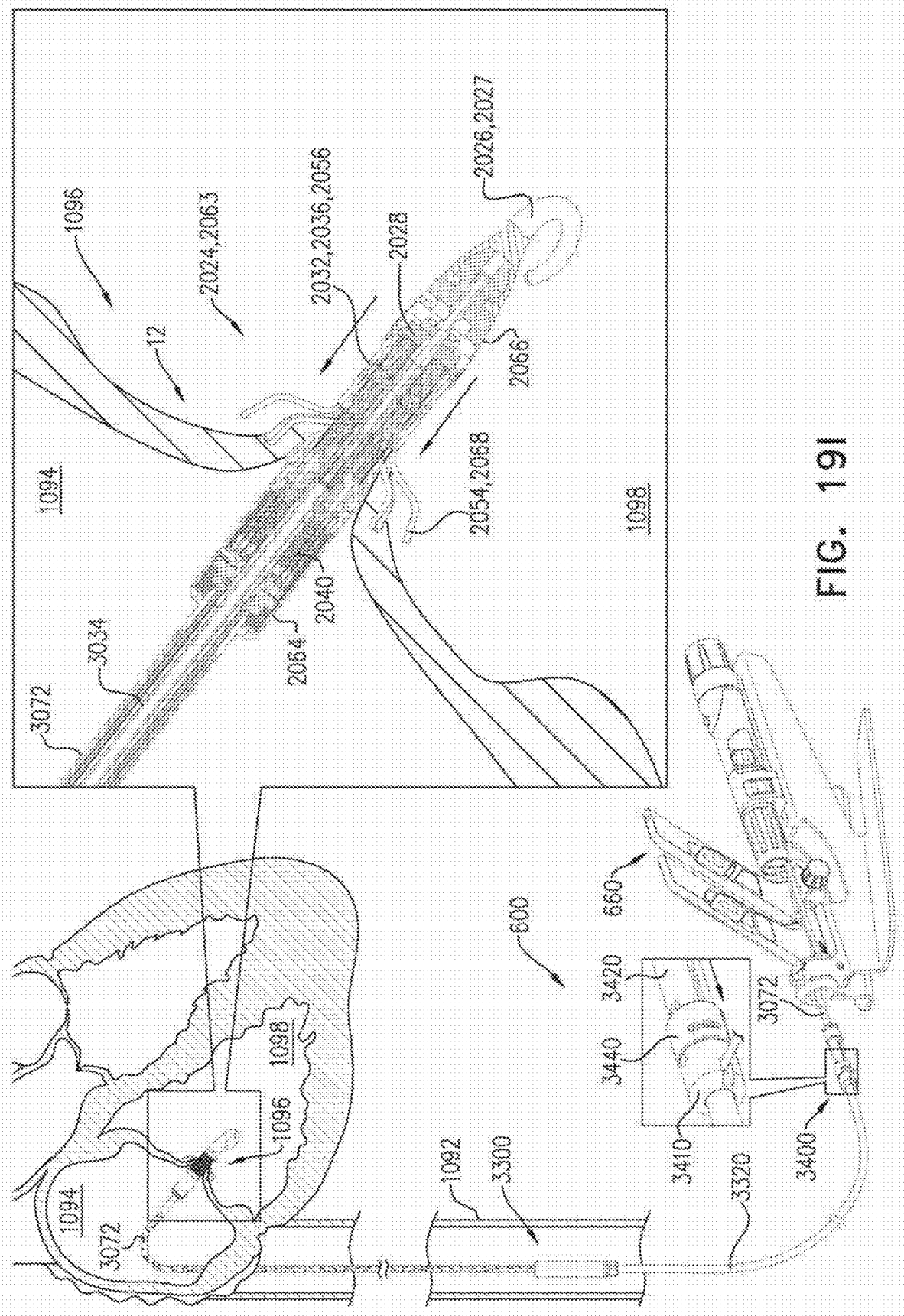


FIG. 19G





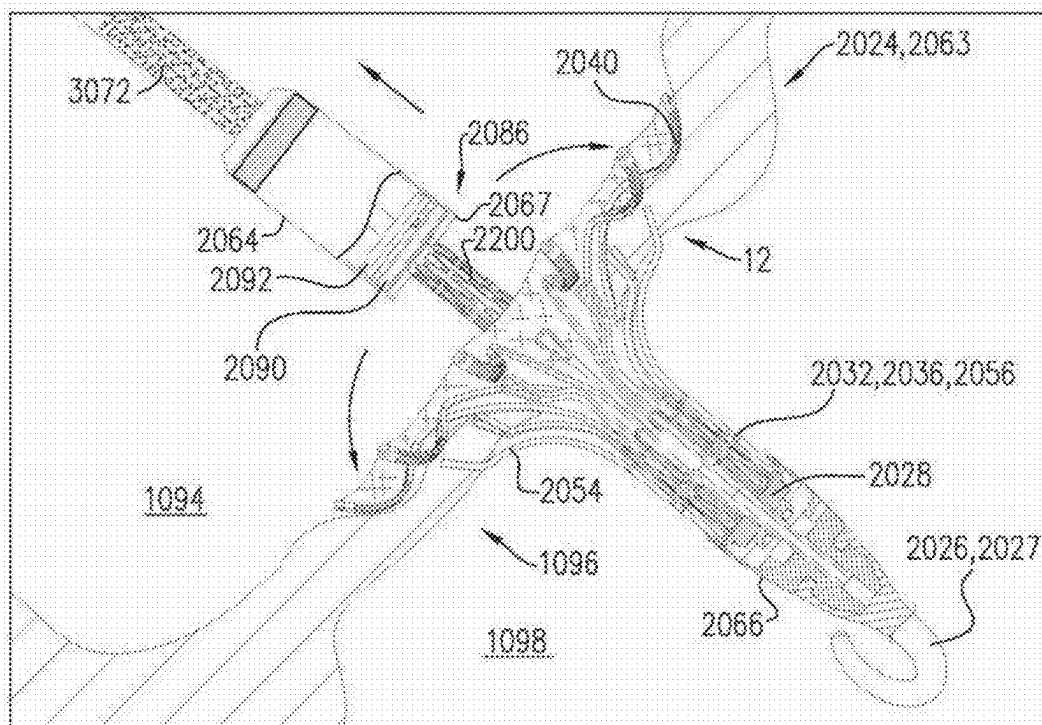


FIG. 19J

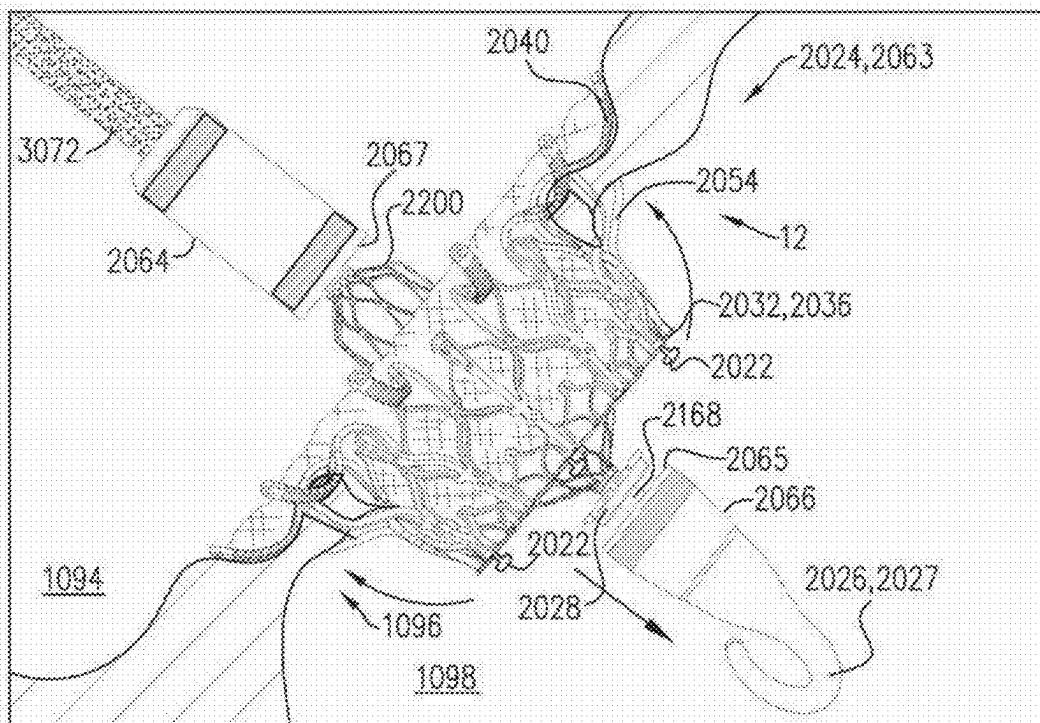


FIG. 19K

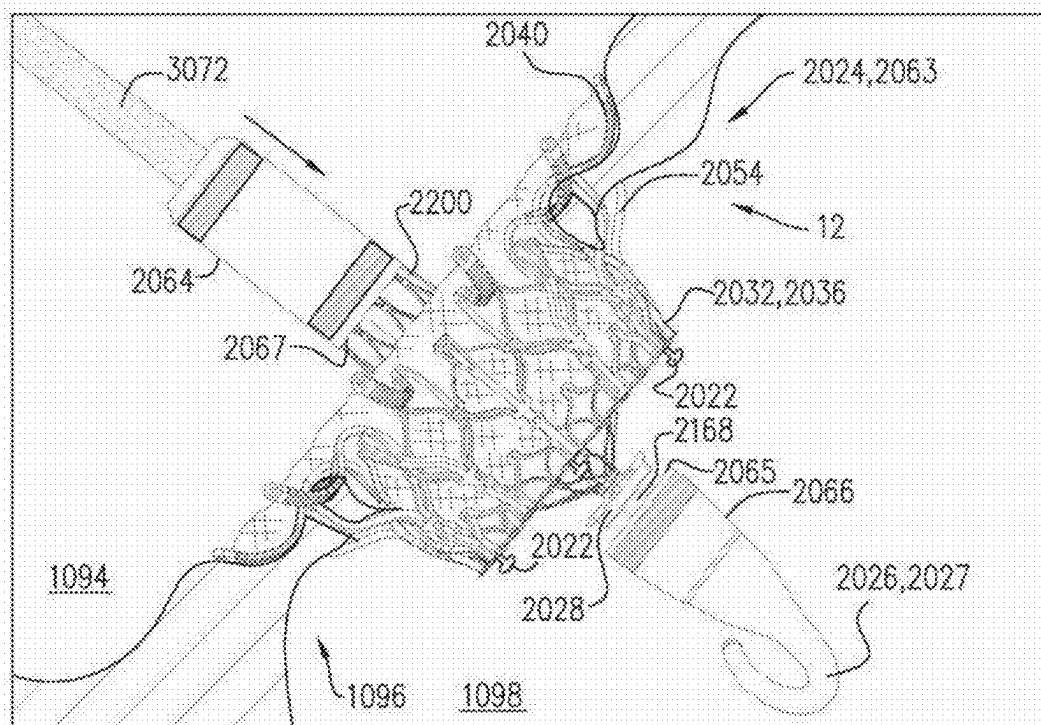


FIG. 19L

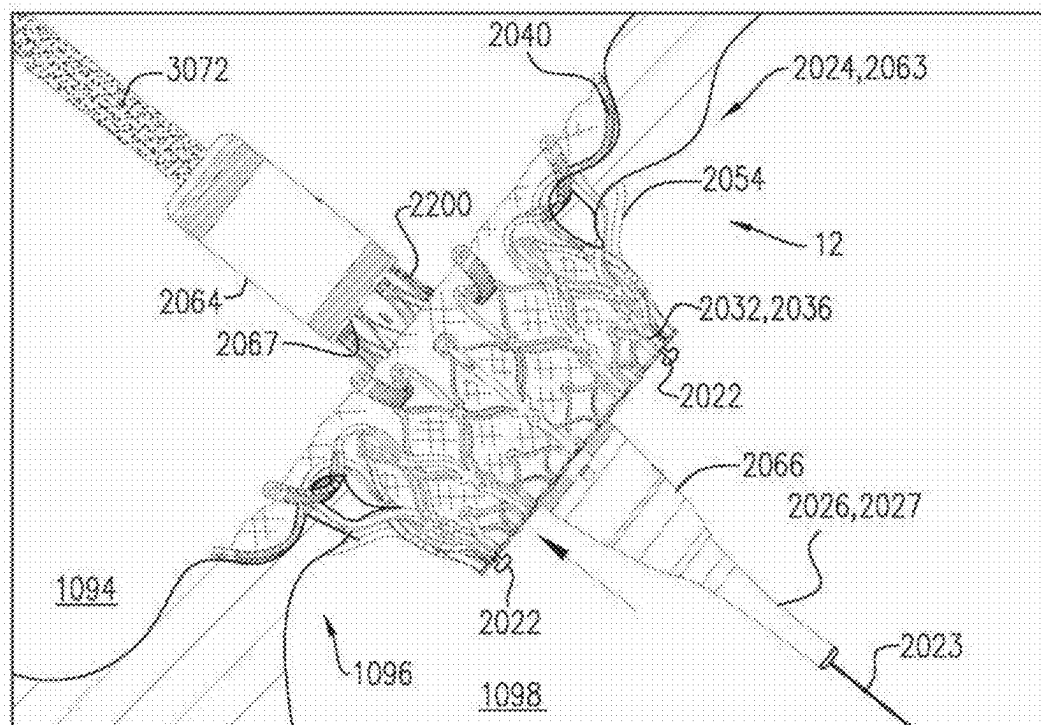


FIG. 19M

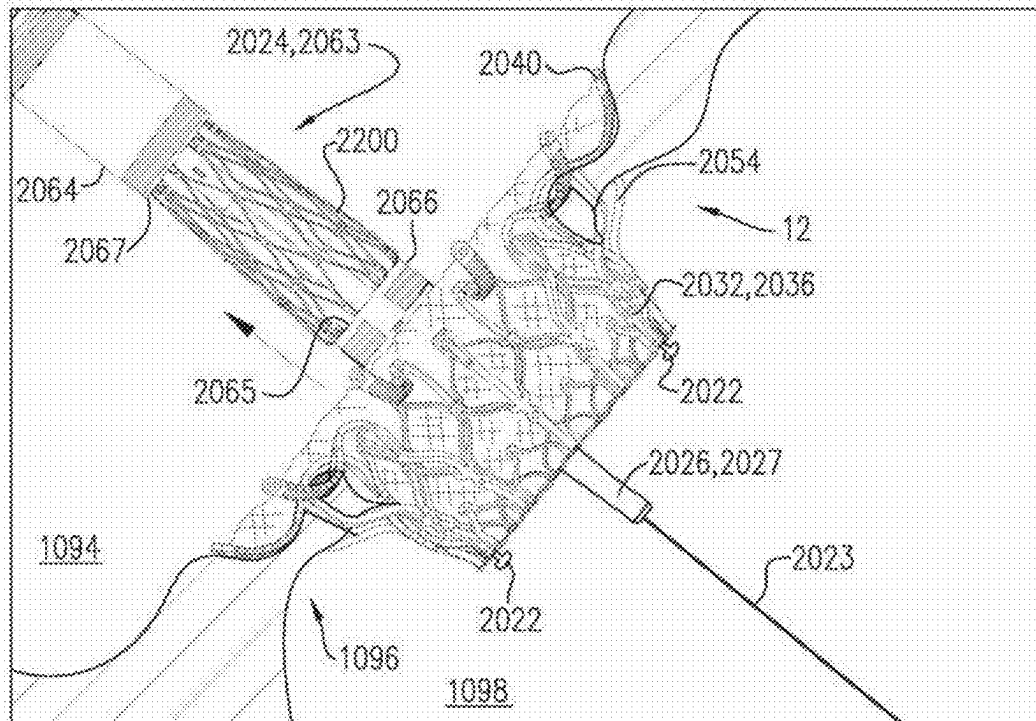


FIG. 19N

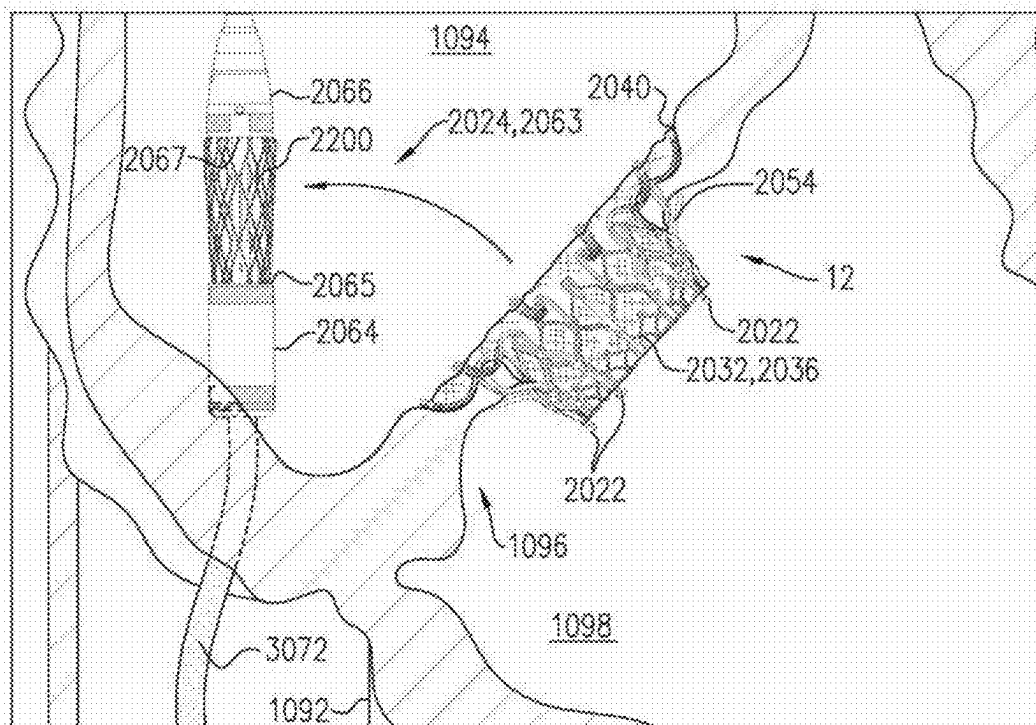
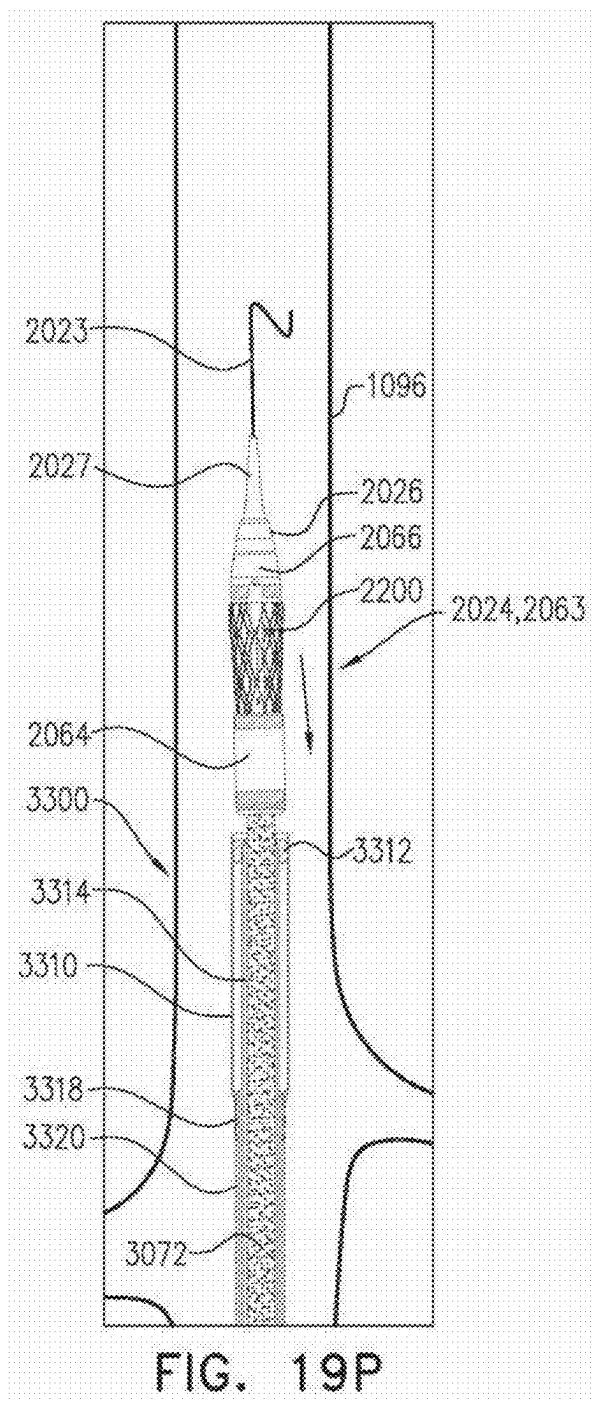
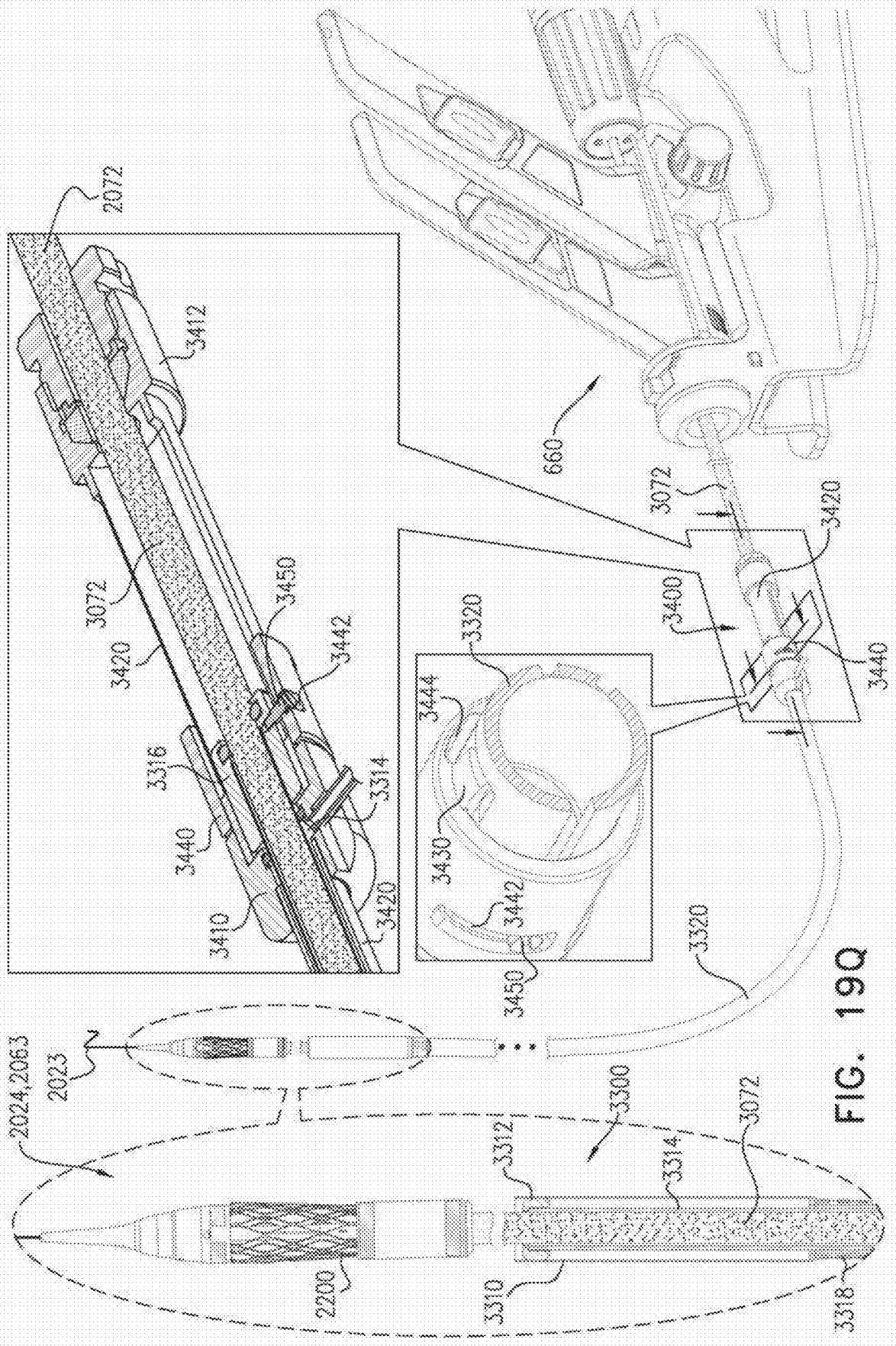


FIG. 190





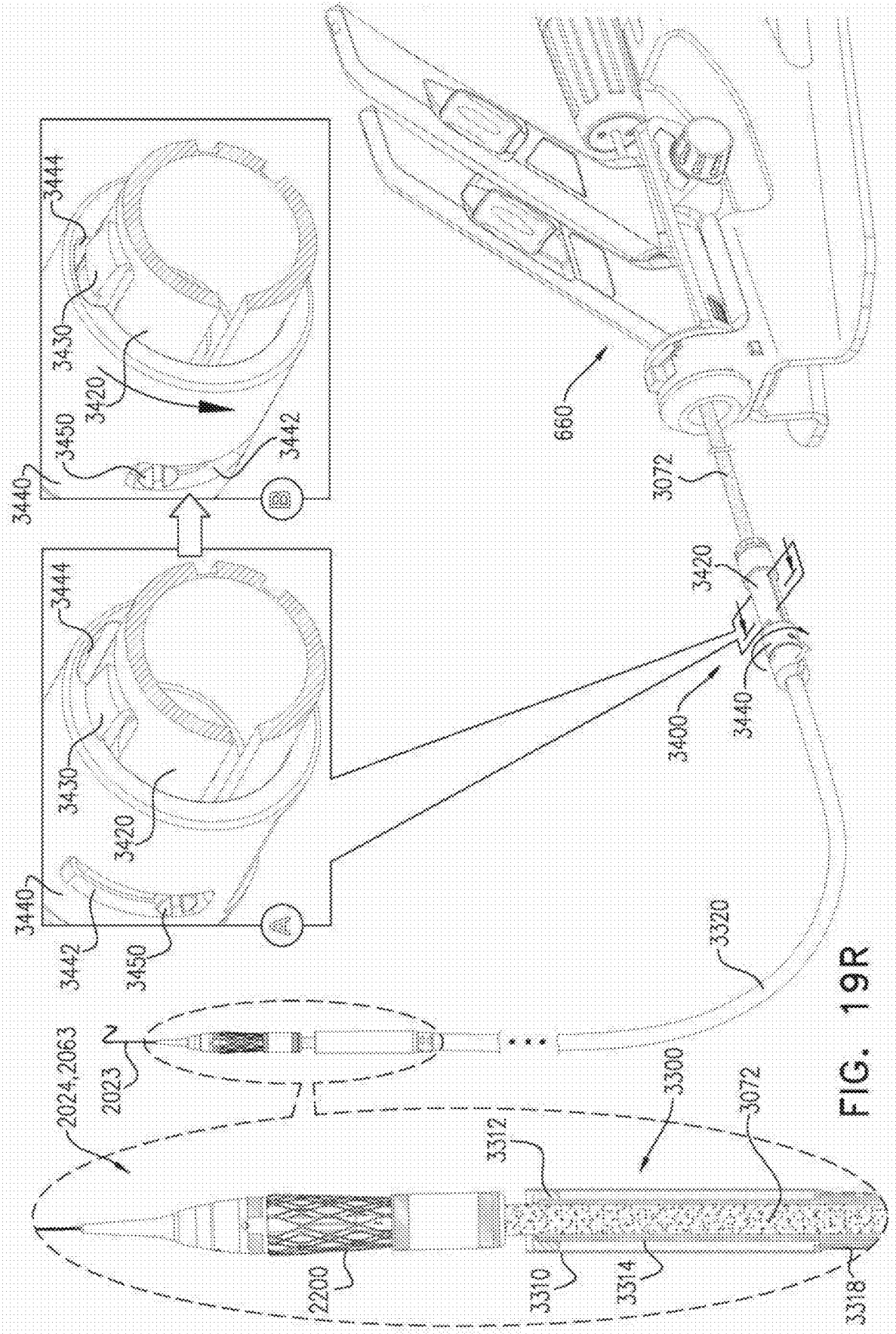
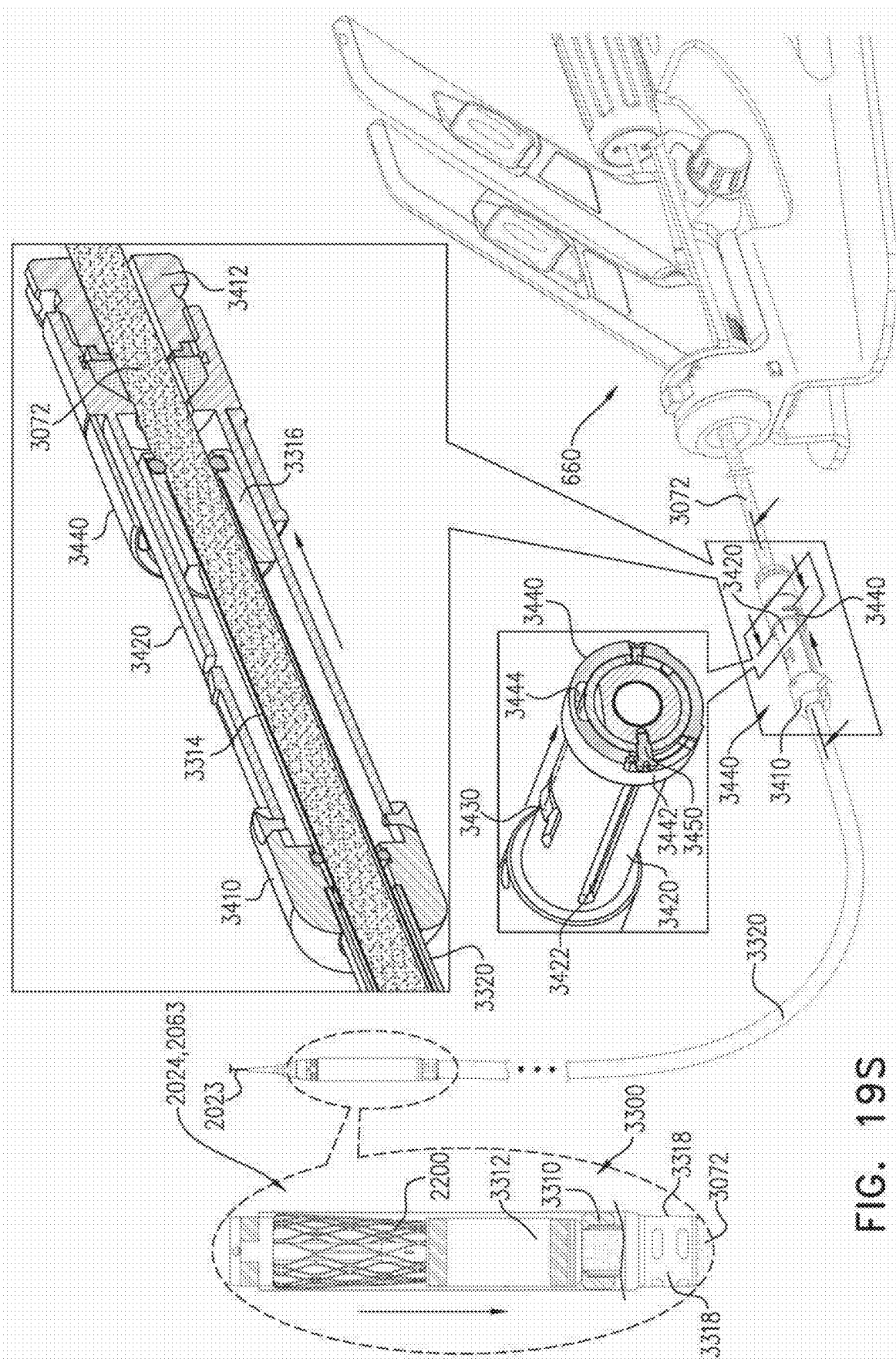
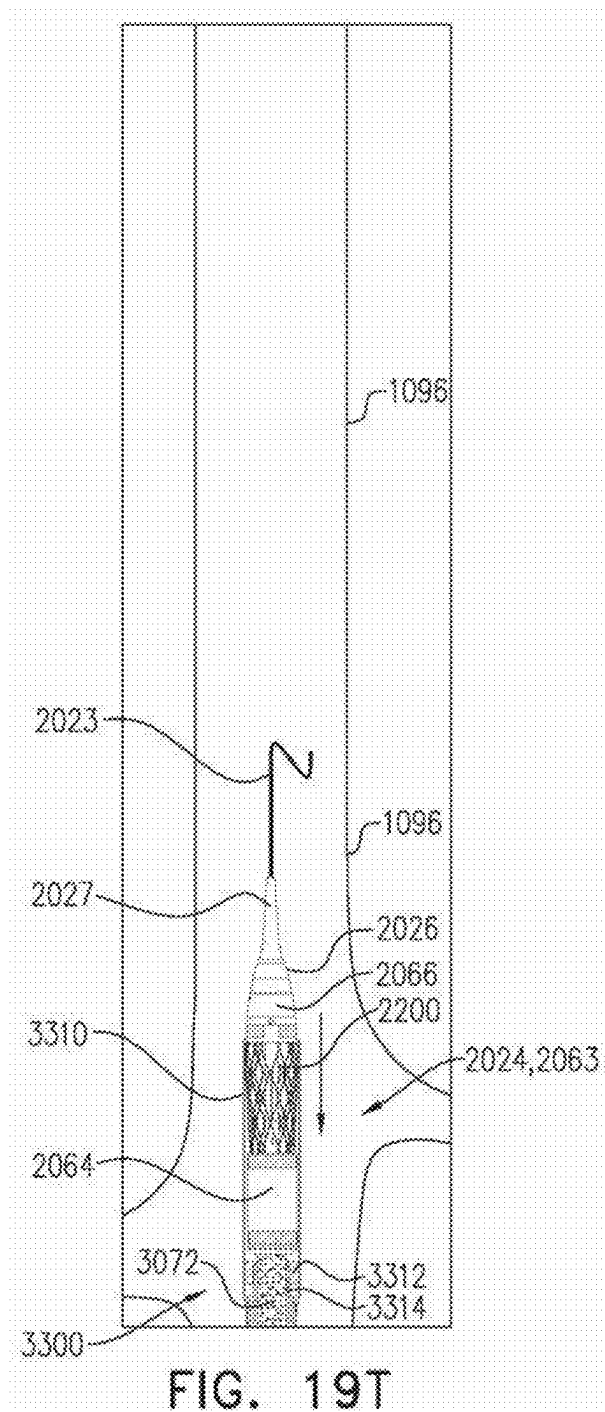
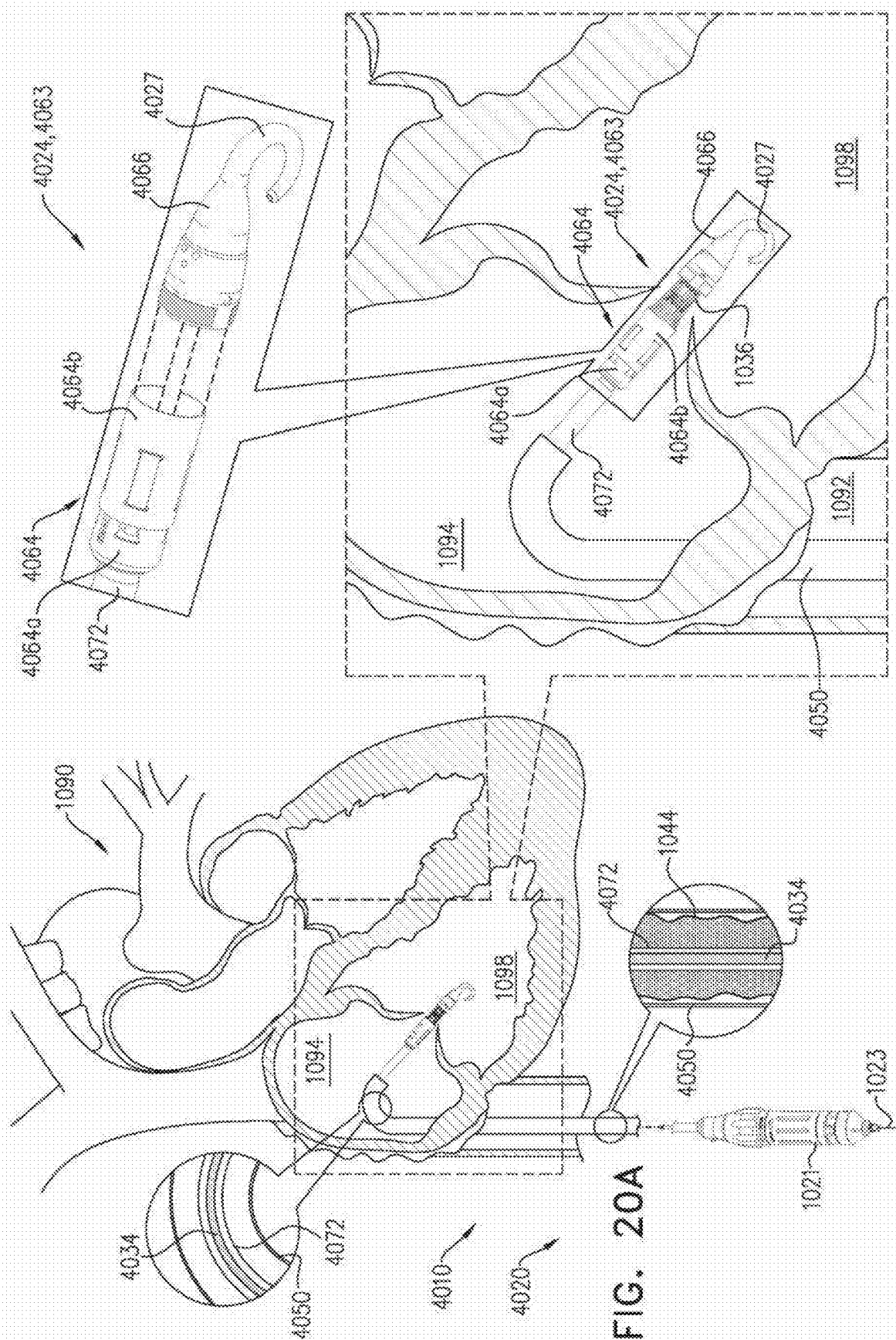


FIG. 19R



SGE





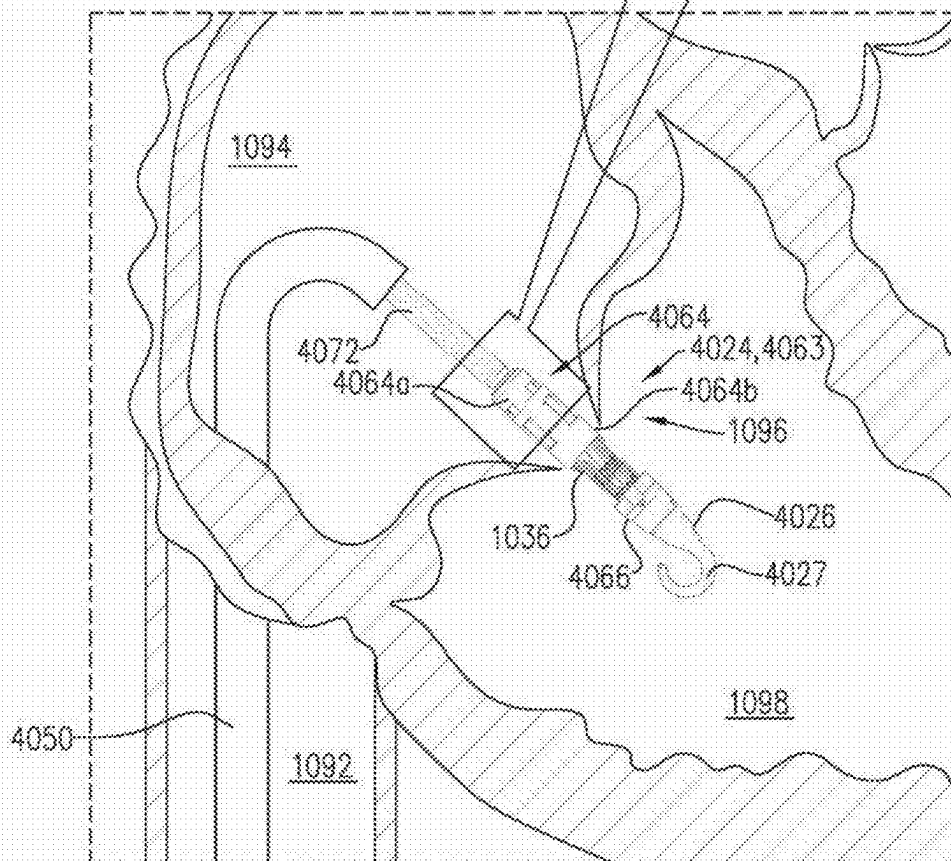
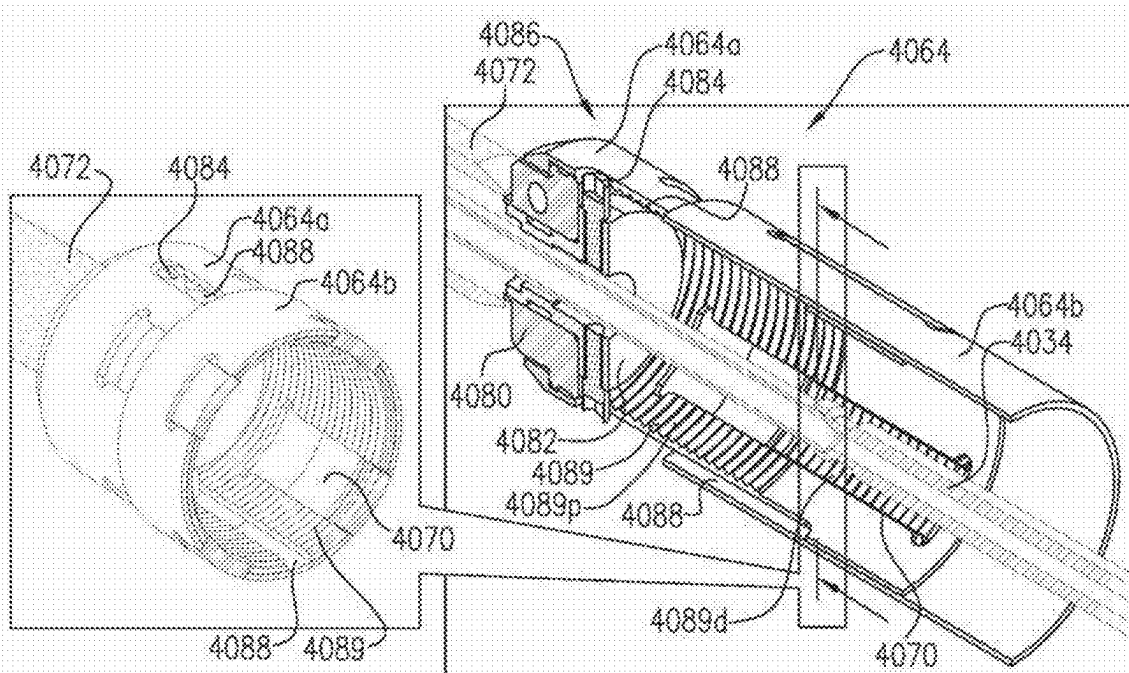


FIG. 20B

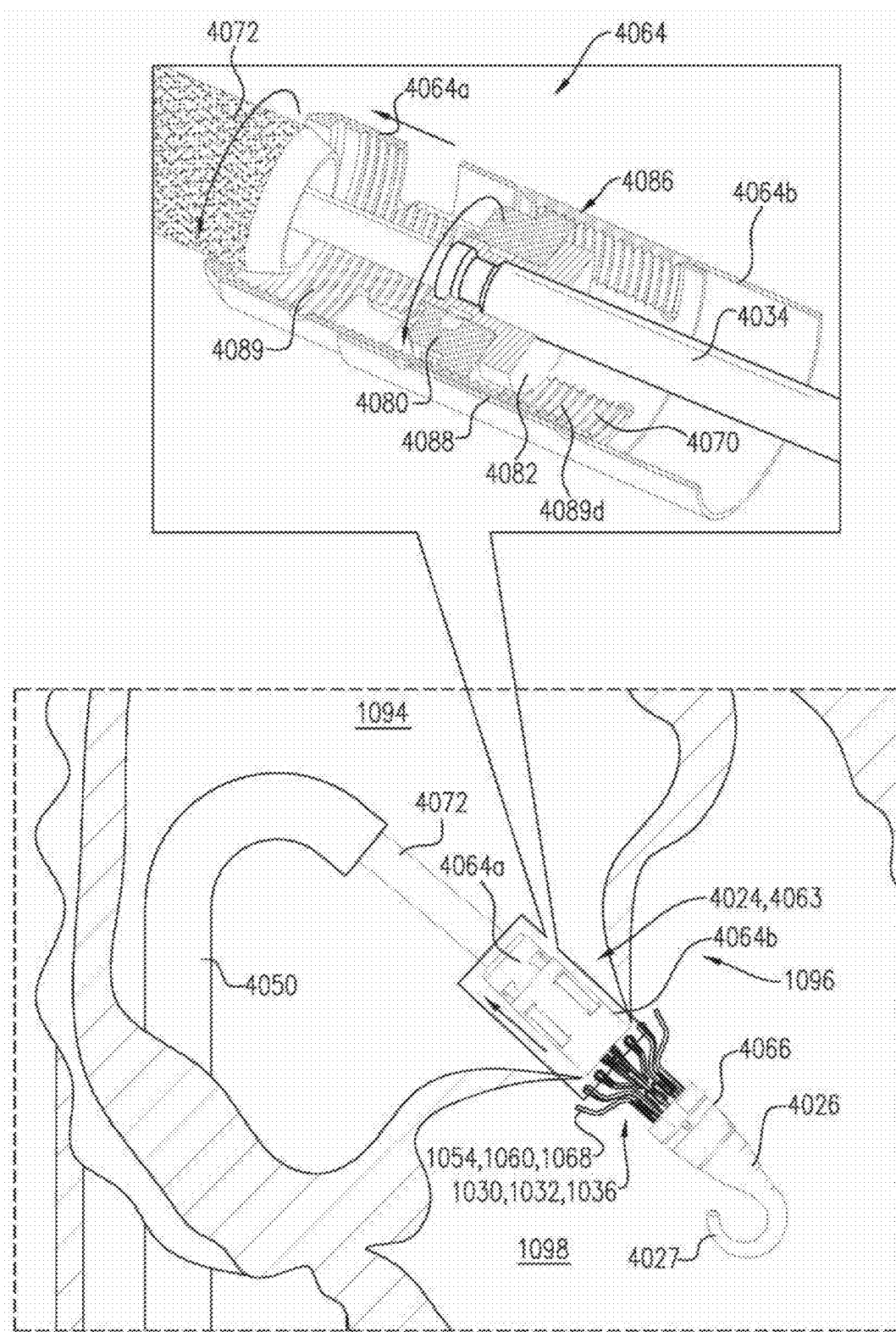


FIG. 20C

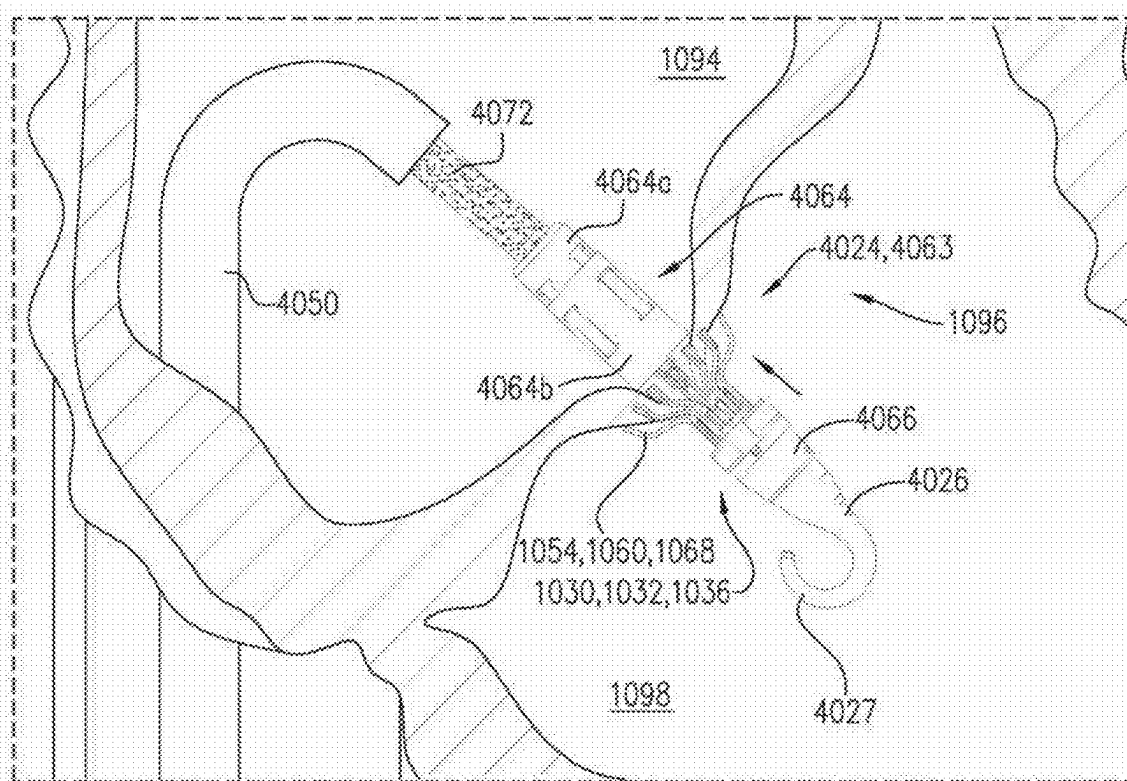


FIG. 20D

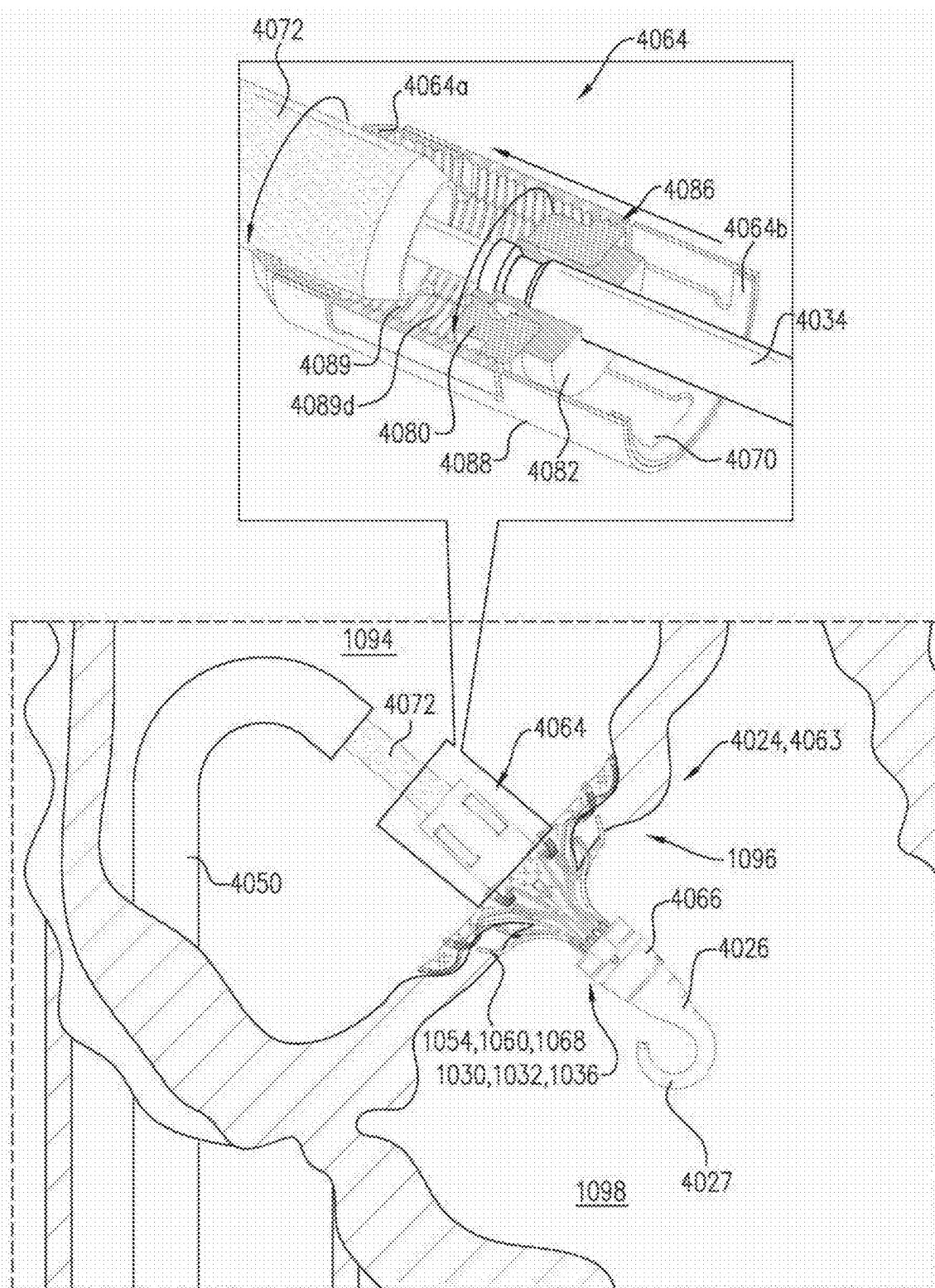


FIG. 20E

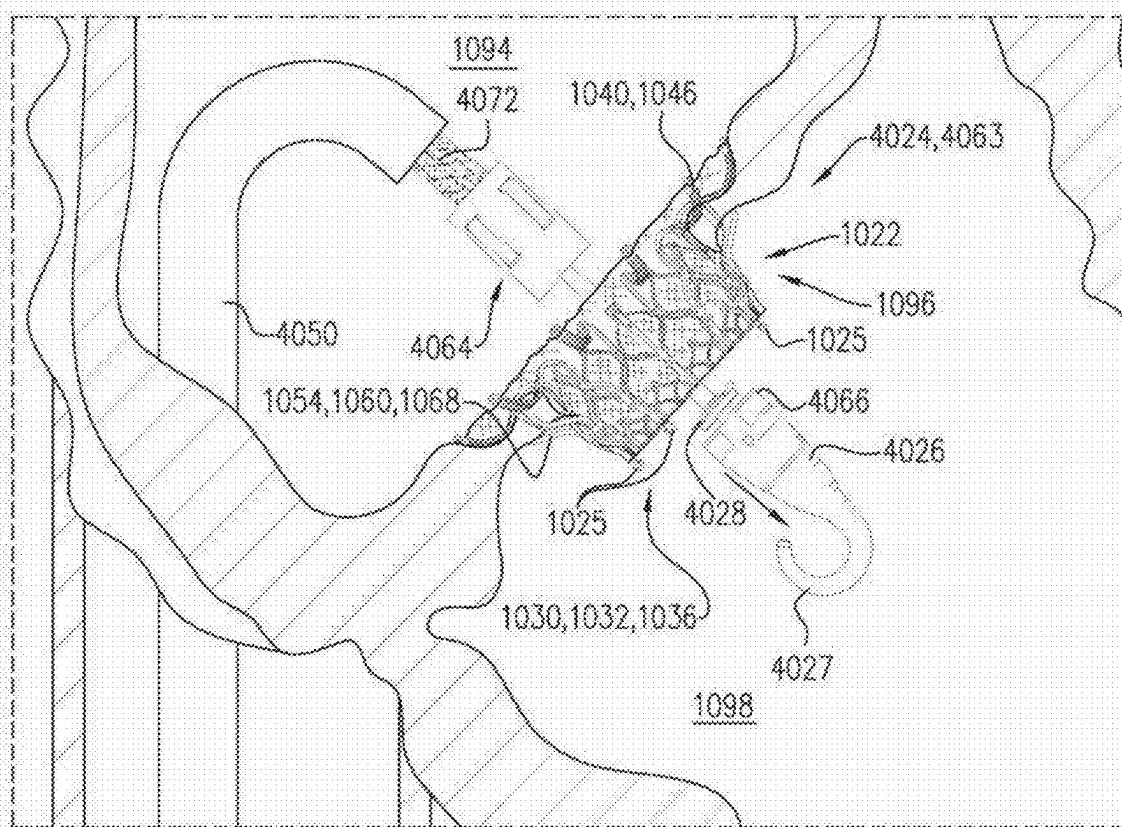


FIG. 20F

TRANSLUMINAL DELIVERY SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority from and is a Continuation in Part of U.S. application Ser. No. 17/399,594, filed Aug. 11, 2021, entitled “TRANSLUMINAL DELIVERY SYSTEM,” which claims priority from US Provisional Application 63/120,808. Filed Dec. 3, 2020, entitled. “TRANSLUMINAL DELIVERY SYSTEM.” Each of the applications mentioned above is assigned to the assignee of the present application and is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] Some applications of the present invention relate in general to transluminal implant-delivery systems. More specifically, some applications of the present invention relate to prosthetic heart valves, and transluminal delivery systems therefor.

BACKGROUND

[0003] Dilatation of the annulus of a heart valve, such as that caused by ischemic heart disease, prevents the valve leaflets from fully coapting when the valve is closed. Regurgitation of blood from the ventricle into the atrium results in increased total stroke volume and decreased cardiac output, and ultimate weakening of the ventricle secondary to a volume overload and a pressure overload of the atrium.

SUMMARY OF THE INVENTION

[0004] Applications of the present invention are directed to apparatus and methods for delivering an implant to a subject.

[0005] For some applications, aspects of the present invention include a transluminal delivery tool that includes a multi-catheter system and an implantation instrument. The implantation instrument has a distal part that is configured to be advanced into the subject, as well as a proximal part that includes an extracorporeal control system.

[0006] The catheter system typically includes a first catheter unit and a second catheter unit, each catheter unit including a respective catheter that is mounted at a proximal end thereof to a respective handle. Selective adjustment of the axial and/or rotational position of each handle facilitates adjustment of the axial and/or rotational position of the corresponding catheter. Typically, the respective handles, and the proximal portion of the implantation instrument, are mounted on a mount for stabilization during use.

[0007] For some applications, a first catheter extends distally from within a second catheter. For some such applications, sliding a first-catheter distal portion distally over a second-catheter distal portion, ensheathes the second-catheter distal portion within the first-catheter distal portion, and sliding the first-catheter distal portion proximally over the second-catheter distal portion exposes the second-catheter distal portion from the first catheter.

[0008] Typically for such applications, actuation of a first-catheter controller actively bends the first-catheter distal portion via a first-catheter control element, and actuation of a second-controller controller actively bends a second-catheter distal portion via a second-catheter control element.

[0009] For some applications, each control element includes a pull wire that extends from a respective controller, through a secondary lumen of the respective catheter, to a distal portion of the catheter, to which the pull wire is fixed.

[0010] For some applications, each catheter (e.g., a distal end thereof) is bendable, by actuation of the respective controller, along a respective steering plane. For some such applications, the second catheter is rotationally oriented with respect to the first catheter such that, while the first-catheter distal end is bent in a first-catheter steering plane, bending of the second-catheter distal end causes the second-catheter distal end to rotate with respect to the first-catheter distal end such that the second-catheter steering plane moves toward being perpendicular to the first-catheter steering plane.

[0011] For some applications, a distal pail of the second catheter is coupled to a capsule assembly. For some such applications, the capsule assembly includes a proximal capsule and a distal capsule. For example, each capsule may have a respective open end that faces the open end of the other capsule.

[0012] For some applications, the distal part of the delivery tool includes a plurality of coaxial tubular members that extend distally from the proximal portion of the instrument (e.g., through the second catheter of the catheter system). Typically for such applications, a capsule catheter extends distally through the second catheter and out an open distal end of the second catheter. Further typically for such applications, a shaft extends distally from the proximal portion of the delivery tool, through the capsule catheter and out of the open end of the proximal capsule. For example, a mount (e.g., to which the implant may be engaged may be fixedly coupled to a distal end of the shaft.

[0013] For some such applications, a rod extends out of a distal end of the shaft, such that a distal portion of the rod is disposed outside of the distal end of the shaft. Typically for such applications, the rod is operatively coupled to the shaft such that the rod may be screwed through the shaft.

[0014] Typically, the implant may be ensheathed (e.g., restrained from expanding) within the capsule assembly. For some applications, the implant comprises a proximal-implant portion, a distal-implant portion and a flange. Typically for such applications, the implant is ensheathed during delivery of the delivery tool such that the proximal-implant portion and at least a flange end-portion of the flange are restrained within the proximal capsule. Further typically for such applications, the distal-implant portion of the implant is restrained within the distal capsule.

[0015] For some applications, the implant may be unsheathed from the distal capsule by moving the distal capsule linearly off of the implant (e.g., without screwing the distal capsule with respect to the rod). For some such applications, the distal capsule is rotationally coupled to the distal portion of the rod, such that rotation of the rod does not rotate the distal capsule. For example, the distal capsule may be advanced distally off of the implant by screwing the rod through the shaft while the distal capsule is axially locked with respect to the rod, yet rotationally coupled to the rod.

[0016] For some such applications, the capsule assembly includes a plurality of pins that are axially aligned with a circumferential recess defined by the rod. Typically for such applications, the pins inhibit rotation of the distal capsule by traversing the distal capsule sufficiently closely to the rod to inhibit axial movement of the rod with respect to the pins,

while providing sufficient clearance between the pins and the rod (e.g., the recess defined thereby) to allow the rod to rotate with respect to the pins.

[0017] For some applications, the distal capsule is reversibly rotationally lockable or unlockable with respect to the rod, such that the implant may be ensheathed in the distal capsule by moving the distal capsule helically over the implant, and unsheathed by moving the distal capsule linearly off of the implant. For example, an accessory may be introduced (e.g., defining a detent shaped to fit into the recess defined by the rod and the distal capsule) configured to rotationally lock the rod with respect to the distal capsule. In this way, the accessory may be attached for ensheathing of the implant within the distal capsule, and the accessory subsequently removed before unsheathing (e.g., before transluminal delivery) of the implant.

[0018] For some applications, aspects of the present invention include a delivery system that comprises a delivery tool and the prosthetic valve. For some such applications, the delivery tool is used to deliver the prosthetic valve to a native valve of a heart of the subject. For example, the native valve may be a tricuspid valve.

[0019] For some applications, the delivery tool has a proximal portion and a distal portion. For some such applications, the distal portion comprises a proximal capsule and a distal capsule, each of the capsules defining a respective open end.

[0020] Typically for such applications, the open end of the proximal capsule faces the proximal end of the distal capsule. For some such applications, the open end of the proximal capsule may face the open end of the distal capsule. For example, while the delivery tool is in a delivery state for transluminally delivering the delivery tool to the heart, an inter-capsule gap may separate between the open end of the proximal capsule and the open end of the distal capsule.

[0021] For some such applications, the prosthetic valve comprises a tubular portion that defines a lumen, within which a plurality of prosthetic leaflets are disposed. Typically for such applications, the prosthetic valve further comprises an upstream support portion that extends from the tubular portion, and defines a plurality of flanges. Each flange is coupled to the tubular portion at a coupling point, from which the flange extends to flange end-portion.

[0022] For some applications, while the delivery tool is in the delivery state, the prosthetic valve is restrained in a compressed state by the delivery tool. For some such applications, while the delivery tool is in the delivery state, the tubular portion of the prosthetic valve is engaged with a mount of the delivery tool. Alternatively or in addition, the prosthetic valve may be engaged with a portion of the shaft, mutatis mutandis.

[0023] For some applications, while the delivery tool is in the delivery state, the mount and a downstream end of the tubular portion are disposed within the distal capsule, and the upstream support portion and the flange end-portions are disposed within the proximal capsule. Further typically, while the delivery tool is in the delivery state, a segment of the tubular portion is disposed at the inter-capsule gap.

[0024] For some applications, the delivery tool is transluminally advanced to a ventricle of the heart, such that the distal capsule is disposed within the ventricle. For some such

applications, the delivery tool comprises a flexible sheath, and the sheath is retracted, exposing the proximal capsule from the sheath.

[0025] For some applications, the delivery system comprises a guidewire along which the delivery tool is transluminally advanced to the heart. For some such applications, the delivery tool comprises a nosecone having a flexible distal end-portion. For example, the distal end-portion may have a relaxed curled shape, and the distal end-portion may be straightened when the guidewire occupies the distal end-portion.

[0026] For some applications, the delivery tool comprises a shaft that comprises a rigid proximal shaft segment that extends from the proximal portion of the delivery tool to the distal portion of the delivery tool. For some such applications, the shaft comprises a flexible shaft segment that extends from the rigid proximal shaft segment to a rigid distal shaft segment. Typically for such applications, the rigid distal shaft segment extends through at least part of the proximal capsule and/or of the distal capsule.

[0027] Typically, the proximal capsule is then proximally retracted, such that the flange end-portions are released from the proximal capsule and expand radially outward. For some applications, the delivery tool comprises a disc-assembly comprising a proximal disc that is rotatably coupled to a distal disc. The proximal disc defines outer threading that is complementary to inner threading defined by the proximal capsule. Typically for such applications, rotation of a capsule catheter that is fixedly coupled to the proximal disc, screws the proximal capsule over the disc-assembly, with respect to the mount.

[0028] Further typically, the distal portion is then retracted, such that the flange end-portions contact tissue of the native valve. The proximal capsule is then retracted, such that the upstream support portion is released from the proximal capsule, and expands radially outward. In this way, tissue of the native valve is squeezed between the upstream support portion and the flange end-portions.

[0029] Subsequently, the distal capsule is advanced with respect to the mount, thereby releasing the mount and the downstream end of the tubular portion from the distal capsule. Thus, the tubular portion expands radially outward at the native valve, such that the prosthetic valve assumes an expanded state.

[0030] Subsequently, the proximal capsule is advanced toward the mount, such that the open end of the proximal capsule abuts the distal capsule. For some applications, the open end of the proximal capsule abuts the open end of the distal capsule. For some applications, the distal capsule is retracted toward the mount, prior to advancing the proximal capsule. While the proximal capsule abuts the distal capsule, the distal portion of the delivery tool is retracted through the lumen of the tubular portion.

[0031] There is therefore provided, in accordance with an application of the present invention. Apparatus for use at a heart of a subject, the apparatus including:

[0032] a delivery tool dimensioned for percutaneous delivery to the heart, the delivery tool having a distal portion that defines a central longitudinal axis at the distal portion and includes:

[0033] a shaft; and

[0034] a proximal capsule and a distal capsule, each of the capsules:

- [0035] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and
- [0036] coupled to the shaft in a manner that allows axial movement of the capsule with respect to the shaft, along the central longitudinal axis at the distal portion; and
- [0037] a prosthetic heart valve including:
 - [0038] a tubular portion that defines a lumen; and
 - [0039] a plurality of prosthetic leaflets disposed within the lumen, and;
- [0040] the prosthetic heart valve is restrainable in a compressed state by the delivery tool, such that a downstream end of the tubular portion is disposed within the distal capsule, and
- [0041] the distal capsule is shaped so as to define an opening for visualizing ensheathing of at least a portion of the downstream end of the tubular portion within the distal capsule.
- [0042] In an application, the opening defines a window.
- [0043] In an application, the delivery tool includes a mount surrounding the shaft and configured to engage the downstream end of the tubular portion, and the opening is configured to allow visualizing of the mount and the downstream end of the tubular portion.
- [0044] In an application, the mount is shaped so as to define one or more slots, and the downstream end of the tubular portion is shaped so as to define one or more adaptors, each one of the adaptors being configured to be received within a respective one of the one or more slots so as to facilitate engaging between the mount and the downstream end of the tubular portion.
- [0045] In an application, in the compressed state of the prosthetic heart valve, the distal capsule maintains coupling between the downstream end of the tubular portion and the mount by surrounding the one or more adaptors and maintaining each one of the one or more adaptors within the respective slot of the mount.
- [0046] In an application, the prosthetic heart valve includes:
 - [0047] an upstream support portion that extends from the tubular portion; and
 - [0048] a plurality of flanges, each of the flanges coupled to the tubular portion at a respective coupling point that is downstream of the upstream support portion, and extends from the coupling point to a respective flange end-portion of the flange.
- [0049] In an application, the prosthetic heart valve is restrainable in the compressed state by the delivery tool such that the upstream support portion and the flange end-portions are disposed within the proximal capsule.
- [0050] There is further provided, in accordance with an application of the present invention, a method for preparing a prosthetic heart valve for implantation, the method including:
 - [0051] using a crimping tool, crimping the prosthetic heart valve around a distal portion of a shaft of a delivery tool; and
 - [0052] subsequently to the crimping, ensheathing the prosthetic heart valve in a capsule by extracorporeally (i) coupling an ensheathing tool directly to the distal portion and (ii) applying a rotational force to the ensheathing tool to effect linear movement of the capsule with respect to the prosthetic heart valve.
- [0053] In an application:
 - [0054] subsequently to the ensheathing, advancing the ensheathed prosthetic heart valve and the distal portion of the delivery tool into a subject, while retaining a proximal portion of the delivery tool outside of the subject; and
 - [0055] subsequently, deploying the prosthetic heart valve within a heart of the subject from the capsule by extracorporeally applying an unsheathing force to a controller at the proximal portion of the delivery tool.
- [0056] In an application:
 - [0057] the capsule is a distal capsule,
 - [0058] the delivery tool includes a proximal capsule, each of the proximal and distal capsules:
 - [0059] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and
 - [0060] being coupled to the shaft in a manner that allows axial movement of the capsule with respect to the shaft, along a central longitudinal axis at the distal portion; and
 - [0061] ensheathing the prosthetic heart valve in the capsule includes:
 - [0062] ensheathing a downstream end of the prosthetic heart valve in the distal capsule; and
 - [0063] subsequently, ensheathing an upstream end of the prosthetic heart valve in the proximal capsule.
- [0064] In an application, the ensheathing tool is a distal-capsule ensheathing tool and ensheathing the downstream end of the prosthetic heart valve includes applying a first ensheathing force to the distal portion of the delivery tool using the distal-capsule ensheathing tool directly coupled to the distal capsule.
- [0065] In an application, the method includes coupling the distal-capsule ensheathing tool directly to the distal capsule.
- [0066] In an application, ensheathing the upstream end of the prosthetic heart valve in the proximal capsule includes applying a second ensheathing force to the distal portion of the delivery tool using a proximal-capsule ensheathing tool directly coupled to the distal portion.
- [0067] In an application, the method includes:
 - [0068] subsequently to the ensheathing the of the upstream end of the prosthetic heart valve, advancing the ensheathed prosthetic heart valve and the distal portion of the delivery tool into a heart of a subject, while retaining a proximal portion of the delivery tool outside of the subject; and
 - [0069] subsequently, deploying the prosthetic heart valve within the heart of the subject from the proximal and distal capsules by extracorporeally applying an unsheathing force to a controller at the proximal portion of the delivery tool.
- [0070] In an application, the method includes, during the ensheathing of the downstream end of the prosthetic heart valve in the distal capsule, visualizing the ensheathing of at least a portion of the downstream end of the prosthetic heart valve within the distal capsule through an opening defined in the distal capsule for visualizing the ensheathing.
- [0071] In an application, the delivery tool includes a mount surrounding the shaft and configured to engage the downstream end of the prosthetic heart valve, and visualizing the ensheathing of the at least the portion of the downstream end of the prosthetic heart valve within the

distal capsule includes visualizing the mount and the downstream end of the prosthetic heart valve.

[0072] In an application:

[0073] the mount is shaped so as to define one or more slots,

[0074] the downstream end of the prosthetic heart valve is shaped so as to define one or more adaptors, each one of the adaptors being configured to be received within a respective one of the one or more slots so as to facilitate engaging between the mount and the downstream end of the prosthetic heart valve, and

[0075] ensheathing the downstream end of the prosthetic heart valve in the distal capsule includes ensheathing the downstream end of the prosthetic heart valve such that the one or more adapters fit within the one or more slots.

[0076] In an application, ensheathing the downstream end of the prosthetic heart valve in the distal capsule includes maintaining coupling between the downstream end of the prosthetic heart valve and the mount by the ensheathing of the downstream end of the prosthetic heart valve in the distal capsule.

[0077] There is further provided, in accordance with an application of the present invention, a method for preparing a prosthetic heart valve for implantation, the method including:

[0078] using a crimping tool, crimping the prosthetic heart valve around a distal portion of a shaft of a delivery tool; and

[0079] subsequently to the crimping, ensheathing a downstream end of the prosthetic heart valve in a capsule, and during the ensheathing, visualizing the ensheathing of at least a portion of the downstream end of the prosthetic heart valve within the capsule through an opening defined in the capsule for visualizing the ensheathing.

[0080] In an application:

[0081] the capsule is a distal capsule, and the opening is defined in the distal capsule,

[0082] the delivery tool includes a proximal capsule, each of the capsules:

[0083] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and

[0084] being coupled to the shaft in a manner that allows axial movement of the capsule with respect to the shaft, along a central longitudinal axis at the distal portion; and ensheathing the prosthetic heart valve in the capsule includes:

[0085] ensheathing a downstream end of the prosthetic heart valve in the distal capsule; and

[0086] subsequently, ensheathing an upstream end of the prosthetic heart valve in the proximal capsule subsequently to the ensheathing of the downstream end of the prosthetic heart valve in the distal capsule.

[0087] In an application, ensheathing the downstream end of the prosthetic heart valve includes applying a first ensheathing force to the distal portion of the delivery tool using a distal-capsule ensheathing tool directly coupled to the distal capsule.

[0088] In an application, the method includes coupling the distal-capsule ensheathing tool directly to the distal capsule.

[0089] In an application, the method includes ensheathing an upstream end of the prosthetic heart valve in the proximal

capsule subsequently to the ensheathing of the downstream end of the prosthetic heart valve in the distal capsule by applying a second ensheathing force to the distal portion of the delivery tool using a proximal-capsule ensheathing tool directly coupled to the distal portion.

[0090] In an application, the method includes:

[0091] subsequently to the ensheathing the of the upstream end of the prosthetic heart valve, advancing the ensheathed prosthetic heart valve and the distal portion of the delivery tool into a heart of a subject, while retaining a proximal portion of the delivery tool outside of the subject; and

[0092] subsequently, deploying the prosthetic heart valve within the heart of the subject from the capsule by extracorporeally applying an unsheathing force to a controller at the proximal portion of the delivery tool.

[0093] In an application, the delivery tool includes a mount surrounding the shaft and configured to engage the downstream end of the prosthetic heart valve, and visualizing the ensheathing of the at least the portion of the downstream end of the prosthetic heart valve within the distal capsule includes visualizing the mount and the downstream end of the prosthetic heart valve.

[0094] In an application:

[0095] the mount is shaped so as to define one or more slots.

[0096] the downstream end of the prosthetic heart valve is shaped so as to define one or more adaptors, each one of the adaptors being configured to be received within a respective one of the one or more slots so as to facilitate engaging between the mount and the downstream end of the prosthetic heart valve, and

[0097] ensheathing the downstream end of the prosthetic heart valve in the distal capsule includes crimping the downstream end of the prosthetic heart valve such that the one or more adapters fit within the one or more slots:

[0098] In an application, ensheathing the downstream end of the prosthetic heart valve in the distal capsule includes maintaining coupling between the downstream end of the prosthetic heart valve and the mount by the ensheathing of the downstream end of the prosthetic heart valve in the distal capsule.

[0099] There is further provided, in accordance with an application of the present invention, apparatus, including:

[0100] a delivery tool for use with a prosthetic heart valve, the delivery tool including:

[0101] a tubular shaft;

[0102] a rod:

[0103] extending from within the shaft out of a distal end of the shaft such that a distal portion of the rod is disposed outside of the distal end of the shaft, and

[0104] operatively coupled to the shaft such that rotational movement of the rod with respect to the shaft is converted into axial movement of the rod with respect to the shaft; and

[0105] a proximal capsule and a distal capsule, each of the capsules:

[0106] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and

- [0107] coupled to the shaft in a manner that allows axial movement of the capsule with respect to the shaft, along a central longitudinal axis at the distal portion; and
- [0108] a first accessory, including a detent, the first accessory being couplable to the distal capsule such that the detent rotationally locks the distal capsule to the rod; and
- [0109] a second accessory being operatively couplable to the proximal capsule such that rotational movement of the second accessory with respect to the shaft is converted into axial movement of the distal capsule with respect to the shaft.
- [0110] In an application, the prosthetic heart valve includes:
- [0111] a tubular portion that defines a lumen; and
- [0112] a plurality of prosthetic leaflets disposed within the lumen, and:
- [0113] the prosthetic heart valve is restrainable in a compressed state by the delivery tool, such that a downstream end of the tubular portion is disposed within the distal capsule, and an upstream end of the tubular portion is disposed within the proximal capsule.
- [0114] In an application, the second accessory includes a cuff shaped to surround the shaft, and the cuff includes:
- [0115] a user grip for facilitating rotating of the second accessory with respect to the shaft, and
- [0116] a distal coupling portion configured to reversibly couple the second accessory to the distal portion.
- [0117] In an application, the distal coupling portion is sized to fit within an opening in the distal portion so as to couple the second accessory to the distal portion.
- [0118] In an application, the distal coupling portion includes one or more pins shaped so as to fit within respective holes defined by the distal portion, to couple the second accessory to the distal portion.
- [0119] There is further provided, in accordance with an application of the present invention, apparatus for use with a prosthetic heart valve, the apparatus including:
- [0120] a delivery tool for delivering the prosthetic heart valve to a heart of a subject, the delivery tool including:
- [0121] a catheter system including one or more catheters, the catheter system having a catheter-system outer diameter;
- [0122] a housing for housing the prosthetic heart valve, the housing being disposed at a distal portion of the catheter system and having a housing inner diameter that is greater than a diameter of at least one of the catheters of the catheter system;
- [0123] a catheter alignment mechanism including:
- [0124] an elongate oversheath shaped so as to define an elongate-oversheath lumen for slidable passage therethrough of the catheter system;
- [0125] a supplemental tube coupled to a distal end of the elongate oversheath, the supplemental tube shaped so as to define a supplemental-tube lumen sized for encasing at least a portion of the housing during (1) at least a portion of transluminally delivering the distal portion of the delivery tool to the heart, and (2) retracting of the housing out of a body of the subject;
- [0126] an alignment tube disposed between the oversheath and the one or more catheters of the catheter system during the delivery state; and
- [0127] an aligner disposed between the alignment tube and the supplemental tube, and configured to align the one or more catheters of the catheter system with respect to the supplemental tube.
- [0128] In an application, the aligner includes a ring.
- [0129] In an application, the aligner and a distal portion of the alignment tube are axially slidable within the supplemental-tube lumen from a proximal-to-distal direction by distally advancing the alignment tube along the one or more catheters to distally advance the aligner in order to align the one or more catheters of the catheter system with respect to the supplemental tube prior to the encasing of the at least the portion of the housing within the supplemental tube.
- [0130] In an application, the catheter alignment mechanism:
- [0131] has an unlocked state in which the aligner is axially slidable within the supplemental-tube lumen along a proximal-to-distal axis, and
- [0132] has a locked state in which the aligner is fixedly disposed within the supplemental tube.
- [0133] In an application:
- [0134] in the unlocked state, the aligner and the distal portion of the alignment tube are axially slidable within the supplemental-tube lumen along the proximal-to-distal axis, and
- [0135] in the locked state, the aligner and the distal portion of the alignment tube are fixedly disposed within the supplemental tube.
- [0136] In an application, the delivery tool includes an aligner locking mechanism that is operatively connected to the catheter alignment mechanism to transition the catheter alignment mechanism from the locked state to the unlocked state.
- [0137] In an application, the housing has a housing outer diameter that is larger than an outer diameter of at least one of the catheters of the catheter system.
- [0138] In an application, the supplemental tube has a supplemental-tube outer diameter that is larger than an outer diameter of the elongate oversheath.
- [0139] In an application, the elongate oversheath has an elongate-oversheath outer diameter, and the supplemental tube has a supplemental-tube outer diameter that is larger than the elongate-oversheath outer diameter.
- [0140] In an application, the alignment tube is slidable within the elongate-oversheath lumen and within the supplemental-tube lumen such that the aligner is slidable between the supplemental tube and the one or more catheters.
- [0141] In an application, the aligner includes a ring and has an inner diameter that is 0.05-3.0 mm larger than an outer diameter of a largest catheter of the one or more catheters that passes through the aligner.
- [0142] In an application, the supplemental tube has a supplemental-tube inner diameter, and the alignment tube has an alignment-tube outer diameter that is 1.9-5.5 mm smaller than the supplemental-tube inner diameter.
- [0143] In an application:
- [0144] the housing includes a proximal capsule and a distal capsule, each of the capsules having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule,
- [0145] the prosthetic heart valve is restrainable in a compressed state by the delivery tool within the housing in a manner in which an upstream portion of the prosthetic heart valve is ensheathed by the proximal

capsule and a downstream portion of the prosthetic heart valve is ensheathed by the distal capsule, and

[0146] the supplemental-tube lumen is sized for encasing the proximal capsule and at least a proximal portion of the distal capsule.

[0147] In an application, the delivery tool is configured such that during the delivery state, an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule, and a segment of the prosthetic heart valve is disposed at the inter-capsule gap.

[0148] In an application:

[0149] during entry of the delivery tool within the body of the subject, the supplemental tube surrounds the proximal capsule and the at least the proximal portion of the distal capsule, and the segment of the prosthetic heart valve disposed at the inter-capsule gap, and

[0150] subsequently to the entry, the proximal capsule and the at least the proximal portion of the distal capsule and the prosthetic heart valve are exposed from within the supplemental tube and are advanceable toward the heart by advancement of at least one catheter of the catheter system.

[0151] In an application, the prosthetic heart valve includes:

[0152] a tubular portion that defines a lumen;

[0153] a plurality of prosthetic leaflets disposed within the lumen;

[0154] an upstream support portion that extends from the tubular portion; and

[0155] a plurality of flanges, each of the flanges coupled to the tubular portion at a respective coupling point that is downstream of the upstream support portion, and extending from the coupling point to a respective flange end-portion of the flange.

[0156] In an application:

[0157] the housing includes a proximal capsule and a distal capsule, each of the capsules having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule; and

[0158] the distal portion is configured such that while the delivery tool is in the delivery state, the prosthetic heart valve is engaged with the delivery tool such that:

[0159] a downstream end of the tubular portion is disposed within the distal capsule, and

[0160] the upstream support portion and the flange end-portions are disposed within the proximal capsule.

[0161] In an application, during entry of the delivery tool within the body of the subject, the supplemental tube surrounds the proximal capsule and at least a proximal portion of the distal capsule.

[0162] In an application, during extracting of the delivery tool from within the body of the subject, the supplemental tube surrounds the proximal capsule a distal end of the supplemental tube abuts the distal capsule.

[0163] There is further provided, in accordance with an application of the present invention, a method, including:

[0164] using a delivery tool, introducing into vasculature of a subject a prosthetic heart valve configured for implantation in a heart of the subject, the delivery tool including:

[0165] a catheter system including one or more catheters, the catheter system having a catheter-system outer diameter;

[0166] a housing for housing the prosthetic heart valve, the housing being disposed at a distal portion of the catheter system and having a housing inner diameter that is greater than a diameter of at least one of the catheters of the catheter system;

[0167] a catheter alignment mechanism including:

[0168] an elongate oversheath shaped so as to define an elongate-oversheath lumen for slidable passage therethrough of the catheter system;

[0169] a supplemental tube coupled to a distal end of the elongate oversheath, the supplemental tube shaped so as to define a supplemental-tube lumen sized for encasing at least a portion of the housing during (1) at least a portion of a delivery state for transluminally delivering the distal portion of the delivery tool to the heart, and (2) retracting of the housing out of a body of the subject;

[0170] an alignment tube disposed between the oversheath and the one or more catheters of the catheter system; and

[0171] an aligner disposed between the alignment tube and the supplemental tube, and positioned to align the one or more catheters of the catheter system with respect to the supplemental tube;

[0172] exposing the housing from within the supplemental tube;

[0173] advancing the housing to the heart by pushing the one or more catheter distally;

[0174] deploying the prosthetic heart valve from within the housing and during the exposing, implanting the prosthetic heart valve at the heart;

[0175] subsequently to the implanting, retracting proximally the housing toward the supplemental tube by proximally retracting one or more catheters;

[0176] aligning (i) one or more catheters of the catheter system with respect to the supplemental tube and thereby (ii) the housing with respect to the supplemental tube by moving the aligner, and by the moving, orienting the aligner to align the one or more catheters of the catheter system with respect to the supplemental tube;

[0177] subsequently to the aligning, retracting proximally the one or more catheters and by the retracting encasing the housing within the supplemental tube; and

[0178] subsequently to the encasing, extracting the delivery tool from within the body of the subject.

[0179] In an application:

[0180] the step of orienting the aligner includes transitioning the alignment mechanism from:

[0181] an unlocked state in which the aligner is axially slidable within the supplemental-tube lumen along a proximal-to-distal axis, to

[0182] a locked state in which the aligner is fixedly disposed within the supplemental tube.

[0183] In an application:

[0184] the step of orienting the aligner includes transitioning the alignment mechanism from:

[0185] an unlocked state in which the aligner and the distal portion of the alignment tube are axially slidable within the supplemental-tube lumen along a proximal-to-distal axis, to

[0186] a locked state in which the aligner and the distal portion of the alignment tube are fixedly disposed within the supplemental tube.

[0187] In an application:

[0188] the delivery tool includes an aligner locking mechanism that is operatively connected to the catheter alignment mechanism, and

[0189] the step of transitioning the alignment mechanism from the unlocked state to the locked state includes manipulating the aligner locking mechanism.

[0190] In an application, the method includes, prior to the step of encasing the housing within the supplemental tube, transitioning the alignment mechanism from:

[0191] an unlocked state in which the aligner is axially slidable within the supplemental-tube lumen along a proximal-to-distal axis, to

[0192] a locked state in which the aligner is fixedly disposed within the supplemental tube.

[0193] In an application:

[0194] the step of transitioning the alignment mechanism from the locked state to the unlocked state includes transitioning the alignment mechanism from:

[0195] a locked state in which the aligner and the distal portion of the alignment tube are fixedly disposed within the supplemental tube, to

[0196] an unlocked state in which the aligner and the distal portion of the alignment tube are axially slidable within the supplemental-tube lumen along a proximal-to-distal axis.

[0197] In an application:

[0198] the delivery tool includes an aligner locking mechanism that is operatively connected to the catheter alignment mechanism, and

[0199] the step of transitioning the alignment mechanism from the locked state to the unlocked state includes manipulating the aligner locking mechanism.

[0200] In an application, aligning includes straightening a portion of the one or more catheters such that the one or more catheters and the supplemental tube are concentrically disposed.

[0201] In an application, exposing the housing from within the supplemental tube includes pushing the one or more catheters distally while retaining the oversheath in place.

[0202] In an application, the aligning includes (i) aligning one or more catheters of the catheter system with respect to the supplemental tube and thereby (ii) aligning the housing with respect to the supplemental tube at a location within the vasculature that is outside the heart of the subject.

[0203] In an application, a distal portion of at least one catheter of the catheter system is configured to assume a curved orientation.

[0204] In an application, the distal portion is biased to assume the curved orientation in an absence of a force applied to the distal portion.

[0205] In an application, moving the aligner includes moving the aligner distally.

[0206] In an application, moving the aligner distally includes distally pushing the alignment tube.

[0207] In an application, the method includes retracting proximally the one or more catheters during the moving distally of aligner.

[0208] In an application, exposing the housing from within the supplemental tube includes retracting the oversheath proximally with respect to the one or more catheters.

[0209] In an application, exposing the housing from within the supplemental tube includes advancing the one or more catheters distally during the retracting of the oversheath proximally.

[0210] There is further provided, in accordance with an application of the present invention, apparatus for use at a heart of a subject, the apparatus including:

[0211] a delivery tool dimensioned for percutaneous delivery to the heart, the delivery tool having a distal portion that defines a distal portion axis and includes:

[0212] a shaft; and

[0213] a proximal capsule and a distal capsule, each of the capsules:

[0214] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and

[0215] being coupled to the shaft in a manner that facilitates axial movement of the capsule with respect to the shaft, along the distal portion axis; and

[0216] a prosthetic valve including:

[0217] a tubular portion that defines a lumen;

[0218] a plurality of prosthetic leaflets disposed within the lumen;

[0219] an upstream support portion that extends from the tubular portion; and

[0220] a plurality of flanges, each of the flanges:

[0221] coupled to the tubular portion at a respective coupling point that is downstream of the upstream support portion, and

[0222] extending from the coupling point to a respective flange end-portion of the flange; and

the prosthetic valve is restrainable in a compressed state by the delivery tool, such that:

[0223] the shaft and a downstream end of the tubular portion are disposed within the distal capsule, and

[0224] the upstream support portion and the flange end-portions are disposed within the proximal capsule.

[0225] In an application, the distal capsule is coupled to the shaft in a manner that facilitates proximal and distal movement of the distal capsule, with respect to the shaft.

[0226] In an application, the proximal and distal capsules are coupled to the shaft in a manner that facilitates proximal and distal movement of the proximal and distal capsules, with respect to the shaft.

[0227] In an application, the proximal capsule is coupled to the shaft in a manner that facilitates proximal and distal movement of the proximal capsule, with respect to the shaft.

[0228] In an application, the apparatus includes:

[0229] a capsule catheter extending proximally from the distal portion of the delivery tool; and

[0230] a disc-assembly coupled to the capsule catheter, the disc-assembly including:

[0231] a proximal disc fixedly coupled to the capsule catheter, the proximal disc shaped to define external threading, and

[0232] a distal disc, the distal disc rotatably coupled to the proximal disc.

[0233] In an application, the proximal capsule is shaped to define:

[0234] a longitudinal track configured to engage the distal disc, and

[0235] internal threading, the internal threading:

[0236] complementary to the external threading, and

[0237] traversing the longitudinal track; and the disc-assembly is disposed within the proximal capsule such that:

[0238] the external threading fits the internal threading, and

[0239] rotation of the capsule catheter in a first direction facilitates:

[0240] rotation of the proximal disc in the first direction, and

[0241] longitudinal movement of the proximal capsule, with respect to the disc-assembly.

[0242] In an application, the distal disc includes a locking pin that is disposed within the longitudinal track.

[0243] In an application, the proximal capsule includes a proximal capsule assembly having an inner proximal capsule and an outer proximal capsule, the outer proximal capsule being longitudinally movable in relation to the inner proximal capsule.

[0244] In an application, the inner proximal capsule fits snugly within the outer proximal capsule.

[0245] In an application:

[0246] the inner proximal capsule defines a proximal portion of the internal threading,

[0247] the outerproximal capsule defines a distal portion of the internal threading; and

[0248] rotation of the capsule catheter in a first direction facilitates:

[0249] rotation of the proximal disc in the first direction, for a first helical distance along the internal threading, facilitating longitudinal movement of the proximal capsule assembly with respect to the disc-assembly, and

[0250] rotation of the proximal disc in the first direction, for a second helical distance along the internal threading, facilitating longitudinal movement of the outer proximal capsule with respect to the inner proximal capsule and to the disc-assembly.

[0251] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in a delivery state for transluminally delivering the delivery tool to the heart:

[0252] the prosthetic valve is restrained in the compressed state by the delivery tool, and

[0253] the proximal capsule and the distal capsule are oriented with respect to each other such that:

[0254] an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule, and

[0255] a segment of the tubular portion is disposed at the inter-capsule gap.

[0256] In an application, the distal capsule and the proximal capsule are each coupled to the shaft such that the distal capsule is translatable toward the proximal capsule, such that the open end of the proximal capsule and the open end of the distal capsule meet, thereby closing the inter-capsule gap.

[0257] In an application, the distal capsule and the proximal capsule are each coupled to the shaft such that prior to disengagement of the prosthetic valve from the shaft, the distal capsule is not translatable toward the proximal capsule, such that the open end of the proximal capsule and the open end of the distal capsule meet, thereby closing the inter-capsule gap.

[0258] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the delivery state, the inter-capsule gap is greater than 5 mm and less than 25 mm in length.

[0259] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the delivery state, the inter-capsule gap is greater than 10 mm in length.

[0260] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the delivery state, the inter-capsule gap is less than 15 mm in length.

[0261] In an application, the proximal capsule and the distal capsule are each coupled to the shaft such that the distal portion of the delivery tool is transitionable from the delivery state to a deployment state such that (a) the inter-capsule gap while the distal portion is in the deployment state is longer than (b) the inter-capsule gap while the distal portion is in the delivery state.

[0262] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the deployment state, the inter-capsule gap is 50-200 percent greater than the gap while the distal portion is in the delivery state.

[0263] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the deployment state, the inter-capsule gap is 100-200 percent greater than the gap while the distal portion is in the delivery state.

[0264] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the deployment state, the inter-capsule gap is greater than 15 mm and less than 40 mm in length.

[0265] In an application, the delivery tool is configured such that while the distal portion of the delivery tool is in the deployment state, the inter-capsule gap is greater than 20 mm and less than 35 mm in length.

[0266] There is further provided, in accordance with an application of the present invention, a method for use at a heart of a subject, the method including:

[0267] transluminally advancing a delivery system to the heart, the delivery system including:

[0268] a delivery tool, the delivery tool having a distal portion that defines a distal portion axis, the distal portion including a proximal capsule and a distal capsule, each of the capsules having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule; and

[0269] an implant including:

[0270] a proximal-implant portion,

[0271] a distal-implant portion, and

[0272] a flange having a flange end-portion,

[0273] the implant being restrained by the delivery tool such that:

[0274] the proximal-implant portion and the flange end-portion are disposed within the proximal capsule, and

[0275] the distal-implant portion of the implant is disposed within the distal capsule; and

[0276] deploying the implant at the heart by:

[0277] proximally retracting the proximal capsule, with respect to the implant, such that the flange end-portion is released from the proximal capsule,

- [0278] subsequently, proximally retracting the distal portion of the delivery tool, such that the flange end-portion contacts tissue of the heart.
- [0279] subsequently, further proximally retracting the proximal capsule, with respect to the implant, such that the proximal-implant portion is released from the proximal capsule, and
- [0280] subsequently, distally advancing the distal capsule, with respect to the implant, such that the distal-implant portion is released from the distal capsule.
- [0281] In an application:
- [0282] the step of transluminally advancing includes transluminally advancing the distal portion of the delivery tool to the heart while the proximal capsule and the distal capsule are aligned along the distal portion axis such that an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule.
- [0283] In an application, the step of transluminally advancing includes transluminally advancing the distal portion of the delivery tool such that a segment of the implant is disposed at the inter-capsule gap.
- [0284] In an application:
- [0285] the delivery tool includes a flexible sheath, the sheath circumscribing the inter-capsule gap such that the sheath covers the segment of the implant, and
- [0286] the method includes, prior to proximally retracting the proximal capsule with respect to the implant, exposing the segment from the sheath by proximally retracting the sheath.
- [0287] In an application:
- [0288] the implant includes a frame, the frame being restrained by the delivery tool such that:
- [0289] the proximal-implant portion includes a proximal portion of the frame, and
- [0290] the distal-implant portion includes a distal portion of the frame; and
- [0291] the step of further proximally retracting the proximal capsule includes further proximally retracting the proximal capsule, with respect to the proximal portion of the frame, such that the proximal portion of the frame is released from the proximal capsule; and
- [0292] the step of distally advancing the distal capsule includes distally advancing the distal capsule, with respect to the distal portion of the frame, such that the distal portion of the frame is released from the distal capsule.
- [0293] In an application:
- [0294] the frame is an inner frame,
- [0295] the flange is a first flange of a plurality of flanges.
- [0296] the implant includes an outer frame, the outer frame defining the plurality of flanges,
- [0297] each of the flanges:
- [0298] is coupled to the inner frame at a respective coupling point that is longitudinally between the proximal portion of the inner frame and the distal portion of the inner frame, and
- [0299] extends from the coupling point to a respective flange end-portion; and
- [0300] the step of proximally retracting the proximal capsule includes proximally retracting the proximal capsule, with respect to the implant, such that the respective flange end-portions are released from the proximal capsule; and
- [0301] the step of proximally retracting the distal portion of the delivery tool includes proximally retracting the distal portion of the delivery tool, such that the respective flange end-portions contact tissue of the heart.
- [0302] In an application:
- [0303] the implant includes a prosthetic valve and:
- [0304] the inner frame has a tubular portion that defines a lumen.
- [0305] a plurality of prosthetic leaflets are disposed within the lumen,
- [0306] the proximal portion of the inner frame includes an upstream support portion that extends from the tubular portion, and
- [0307] the distal portion of the inner frame includes a downstream end of the tubular portion:
- [0308] the distal portion of the delivery tool includes a shaft, the proximal capsule and the distal capsule each being coupled to the shaft, and
- [0309] the step of transluminally advancing includes:
- [0310] transluminally delivering the delivery tool to the heart while the delivery tool is in a delivery state in which the prosthetic valve is restrained in a compressed state by the delivery tool, such that:
- [0311] the tubular portion is engaged with a portion of the shaft,
- [0312] the portion of the shaft and the downstream end of the tubular portion are restrained within the distal capsule,
- [0313] the upstream support portion and the flange end-portions are restrained within the proximal capsule,
- [0314] the proximal capsule and the distal capsule are aligned along the distal portion axis such that an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule, and
- [0315] a segment of the tubular portion is disposed at the inter-capsule gap; and transluminally advancing the delivery tool to a ventricle of the heart such that the distal capsule is disposed within the ventricle; and
- [0316] deploying the implant at the heart includes deploying the prosthetic valve at a native valve of the heart, such that the prosthetic valve automatically expands from the compressed state to an expanded state, by:
- [0317] proximally retracting the proximal capsule, with respect to the shaft, such that the respective flange end-portions expand radially outward.
- [0318] proximally retracting the distal portion of the delivery tool, such that the respective flange end-portions contact tissue of the native valve,
- [0319] further proximally retracting the proximal capsule, with respect to the shaft, such that:
- [0320] the upstream support portion is released from the proximal capsule, and expands radially outward, and
- [0321] tissue of the native valve is squeezed between the upstream support portion and the flange end-portions; and

- [0322] distally advancing the distal capsule, with respect to the shaft, such that:
- [0323] the downstream end of the tubular portion is released from the distal capsule, and
- [0324] the tubular portion expands radially outward.
- [0325] In an application:
 - [0326] subsequently to distally advancing the distal capsule with respect to the shaft, withdrawing the delivery tool from the heart, by:
 - [0327] distally advancing the proximal capsule, with respect to the shaft, such that the open end of the proximal capsule abuts the open end of the distal capsule, and
 - [0328] subsequently, proximally retracting the distal portion of the delivery tool, through the lumen of the tubular portion.
- [0329] In an application, the method includes, prior to distally advancing the proximal capsule with respect to the shaft:
 - [0330] proximally retracting the distal capsule, with respect to the shaft.
- [0331] In an application:
 - [0332] the delivery system includes a guidewire,
 - [0333] the delivery tool includes a nosecone having a flexible distal end-portion,
 - [0334] the method includes transluminally advancing the guidewire to the heart.
 - [0335] transluminally advancing the delivery system to the heart includes transluminally advancing the delivery tool along the guidewire, such that the guidewire enters the distal end-portion of the nosecone, and
 - [0336] deploying the implant at the heart includes, prior to retracting the proximal capsule with respect to the implant, proximally withdrawing the guidewire from within the distal end-portion of the nosecone.
- [0337] In an application, the method includes, subsequently to distally advancing the distal capsule, with respect to the implant, advancing the guidewire into the distal end-portion.
- [0338] In an application:
 - [0339] the distal portion of the delivery tool includes a shaft, the proximal capsule and the distal capsule each being coupled to the shaft,
 - [0340] the proximal capsule is shaped to define:
 - [0341] a longitudinal track, and
 - [0342] internal threading, the internal threading traversing the longitudinal track:
 - [0343] the delivery tool includes:
 - [0344] a capsule catheter extending proximally from the distal portion of the delivery tool, and
 - [0345] a disc-assembly, the disc-assembly including:
 - [0346] a proximal disc fixedly coupled to the capsule catheter, the proximal disc shaped to define external threading that is complementary to the internal threading, and
 - [0347] a distal disc, the distal disc:
 - [0348] dimensioned to engage the longitudinal track.
 - [0349] fixedly coupled to the shaft, and
 - [0350] rotatably coupled to the proximal disc;
 - [0351] the disc-assembly is disposed within the proximal capsule such that the external threading fits the internal threading; and
 - [0352] proximally retracting the proximal capsule with respect to the implant includes rotating the capsule catheter in a first direction such that:
 - [0353] the proximal disc rotates, in the first direction, along the internal threading, and
 - [0354] the proximal capsule moves along the distal portion axis, with respect to:
 - [0355] the disc-assembly, and
 - [0356] the implant.
 - [0357] In an application:
 - [0358] the proximal capsule includes a proximal capsule assembly having an inner proximal capsule and an outer proximal capsule, and
 - [0359] the step of proximally retracting the proximal capsule includes:
 - [0360] proximally retracting the proximal capsule assembly, with respect to the disc-assembly, and
 - [0361] proximally retracting the outer proximal capsule, with respect to the inner proximal capsule and to the disc-assembly.
 - [0362] In an application:
 - [0363] the inner proximal capsule defines a proximal portion of the internal threading;
 - [0364] the outer proximal capsule defines a distal portion of the internal threading; and
 - [0365] proximally retracting the proximal capsule assembly, with respect to the disc-assembly includes:
 - [0366] rotating the capsule catheter in the first direction for a first helical distance along the internal threading, facilitating longitudinal movement of the proximal capsule assembly with respect to the disc-assembly and the implant, and
 - [0367] rotating the capsule catheter in the first direction for a second helical distance along the internal threading, facilitating longitudinal movement of the outer proximal capsule with respect to:
 - [0368] the inner proximal capsule.
 - [0369] the disc-assembly, and
 - [0370] the implant.
 - [0371] In an application:
 - [0372] the distal disc includes a locking pin that is disposed within the longitudinal track, and
 - [0373] rotating the capsule catheter in the first direction includes advancing the locking pin along the longitudinal track.
 - [0374] There is further provided, in accordance with an application of the present invention, apparatus for use at a heart of a subject, the apparatus including:
 - [0375] a delivery tool, the delivery tool having a distal portion that defines a distal portion axis and includes:
 - [0376] a proximal capsule and a distal capsule, each of the capsules:
 - [0377] having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, such that, while the distal portion of the delivery tool is in a delivery state for transluminally delivering the delivery tool to the heart, an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule; and
 - [0378] an implant, the implant being restrainable in a compressed state by the delivery tool, and
 - [0379] a segment of the implant is disposed at the inter-capsule gap.

- [0380] In an application, the implant includes a prosthetic valve.
- [0381] In an application, the prosthetic valve includes:
- [0382] a tubular portion that defines a lumen;
 - [0383] a plurality of prosthetic leaflets disposed within the lumen;
 - [0384] an upstream support portion that extends from the tubular portion; and
 - [0385] a plurality of flanges, each of the flanges coupled to the tubular portion at a respective coupling point that is downstream of the upstream support portion, and extending from the coupling point to a respective flange end-portion of the flange.
- [0386] In an application, the distal portion is configured such that while the delivery tool is in the delivery state, the prosthetic valve is engaged with the delivery tool such that:
- [0387] a downstream end of the tubular portion is disposed within the distal capsule,
 - [0388] the upstream support portion and the flange end-portions are disposed within the proximal capsule, and
 - [0389] the tubular portion is the segment of the implant disposed at the inter-capsule gap.
- [0390] In an application, the apparatus includes a flexible sheath, the sheath circumscribing the inter-capsule gap such that the sheath covers the segment of the implant.
- [0391] In an application, the proximal capsule is covered by the sheath.
- [0392] In an application, a distal end of the sheath abuts the distal capsule.
- [0393] In an application, a distal end of the sheath is partially disposed within the distal capsule.
- [0394] In an application, the sheath includes a polymer.
- [0395] In an application, the sheath includes a fabric.
- [0396] There is further provided, in accordance with an application of the present invention, apparatus for percutaneous delivery of an implant to a subject, the apparatus including:
- [0397] a guidewire; and
 - [0398] a delivery tool having a proximal portion and a distal portion, the delivery tool including:
 - [0399] at the proximal portion of the delivery tool, an extracorporeal controller;
 - [0400] a delivery catheter, the delivery catheter connecting the extracorporeal controller to the distal portion of the delivery tool, the delivery catheter configured such that the guidewire is extendable through the delivery catheter, and
 - [0401] at the distal portion of the delivery tool, a capsule configured to house the implant, the capsule including a nosecone having a flexible distal end-portion, and:
 - [0402] in an absence of the guidewire from the distal end-portion, the distal end-portion has a curled resting shape, and
 - [0403] while the guidewire is positioned within the distal end-portion, the distal end-portion is straightened.
- [0404] There is further provided, in accordance with an application of the present invention, apparatus for percutaneous delivery of an implant to a subject, the apparatus including a delivery tool, the delivery tool including:
- [0405] at a proximal portion of the delivery tool, an extracorporeal controller;
 - [0406] at a distal portion of the delivery tool, a capsule that defines a chamber therein; and
 - [0407] a shaft, extending from the extracorporeal controller to the capsule, and including:
 - [0408] a rigid proximal shaft segment that extends distally from the extracorporeal controller,
 - [0409] a flexible shaft segment that extends distally from the rigid proximal shaft segment, and
 - [0410] a rigid distal shaft segment that extends distally from the flexible shaft segment, and extends through at least part of the chamber of the capsule, each rigid shaft segment being more rigid than the flexible shaft segment.
- [0411] In an application, the delivery tool includes at least one pull-wire, the pull-wire operatively connecting the distal portion of the delivery tool to the controller, such that operating the controller facilitates using the pull-wire to steer the distal portion.
- [0412] In an application, the rigid distal shaft segment extends distally out of the chamber of the capsule.
- [0413] In an application, the rigid distal shaft segment is a first rigid distal shaft segment, the delivery tool includes a second rigid distal shaft segment, and the first rigid distal shaft segment and the second rigid distal shaft segment are configured to slide telescopically with respect to each other.
- [0414] In an application, the apparatus includes the implant, and:
- [0415] the delivery tool includes a mount, the mount attached to the rigid distal shaft segment, and
 - [0416] the implant is:
 - [0417] engaged with the mount, and
 - [0418] compressed onto a portion of the rigid distal shaft segment.
- [0419] In an application, the implant is housed at least partially within the chamber of the capsule.
- [0420] In an application, a length of the rigid proximal shaft segment is greater than 50 cm, and less than 100 cm.
- [0421] In an application, the length of the rigid proximal shaft segment is greater than 70 cm, and less than 75 cm.
- [0422] In an application, a length of the flexible shaft segment is greater than 5 cm, and less than 10 cm.
- [0423] In an application, the length of the flexible shaft segment is greater than 6 cm, and less than 8 cm.
- [0424] In an application, a length of the rigid distal shaft segment is greater than 2 cm, and less than 10 cm.
- [0425] In an application, the length of the rigid distal shaft segment is greater than 4 cm, and less than 7 cm.
- [0426] There is further provided, in accordance with an application of the present invention, a method, including:
- [0427] advancing, into a subject, an implant disposed within a capsule, the capsule coupled to a flexible capsule catheter that extends through a flexible second catheter and out of a second-catheter distal portion of the second catheter, the second-catheter distal portion abutting the capsule;
 - [0428] subsequently, exposing a capsule-catheter distal portion of the capsule catheter, from the second catheter, by axially separating the second catheter from the capsule; and
 - [0429] subsequently, releasing the implant from the capsule by moving the capsule proximally away from the implant, by retracting the capsule-catheter distal portion into the second catheter.

[0430] In an application, the step of advancing includes advancing the implant into the subject while:

[0431] the second-catheter distal portion is disposed within a flexible first catheter,

[0432] the capsule is disposed distally from the second catheter, and

[0433] a first-catheter distal portion abuts the capsule.

[0434] In an application, the method includes, subsequently to the step of advancing and prior to the step of releasing the implant, exposing the second-catheter distal portion from the first catheter by axially separating the second catheter from the capsule.

[0435] In an application, the method includes, subsequently to exposing the second-catheter distal portion, and prior to releasing the implant from the capsule:

[0436] bending the second-catheter distal portion with respect to the first catheter by actuating a bend actuator of an extracorporeal control system.

[0437] In an application:

[0438] the bend actuator is a second-catheter bend-actuator, operably coupled to one or more second-catheter bend-control elements that extend from the second-catheter bend-actuator, along the second catheter to the second-catheter distal portion, and

[0439] bending the second-catheter distal portion with respect to the first catheter includes bending the second-catheter distal portion with respect to the first catheter by tensioning at least one of the second-catheter bend-control elements, by actuating the second-catheter bend actuator.

[0440] In an application, the method includes, prior to releasing the implant from the capsule, bending the first-catheter distal portion by actuating a first-catheter bend actuator of the extracorporeal control system, the first-catheter bend actuator being operably coupled to one or more first-catheter bend-control elements that extend from the first-catheter bend actuator, along the first catheter to the first-catheter distal portion, and bending the first-catheter distal portion includes bending the first-catheter distal portion by tensioning at least one of the first-catheter bend-control elements, by actuating the first-catheter bend actuator.

[0441] In an application, during the advancing, the implant is coupled to a mount, the mount being coupled to a shaft that extends through the capsule catheter and into the capsule, and retracting the capsule-catheter distal portion into the second catheter includes sliding the capsule-catheter distal portion proximally over the shaft.

[0442] In an application:

[0443] the capsule is a first capsule that has an open distal end.

[0444] a rod extends distally out of the shaft.

[0445] a second capsule is coupled to a distal portion of the rod, and includes a circumferential wall that extends proximally from the distal portion of the rod to define (i) a chamber, and (ii) an open proximal end that faces the open distal end of the first capsule,

[0446] during the advancing, a first part of the implant is disposed within the first capsule, and a second part of the implant is disposed within the second capsule, and

[0447] the method includes releasing the second part of the implant from the second capsule by moving the second capsule distally with respect to the mount, by moving the rod distally through the shaft.

[0448] In an application, the rod defines a screw thread, and moving the rod distally through the shaft includes rotating the rod such that the screw thread transforms the rotation of the rod into axial movement of the rod.

[0449] In an application, releasing the second part of the implant from the second capsule includes releasing the second part of the implant from the second capsule prior to releasing the first part of the implant from the first capsule.

[0450] In an application, releasing the second part of the implant from the second capsule includes releasing the second part of the implant from the second capsule prior to releasing the capsule-catheter distal portion from the second catheter.

[0451] In an application, the method includes, subsequently to exposing the capsule-catheter distal portion, and prior to releasing the implant from the capsule, bending the capsule-catheter distal portion with respect to the second catheter by actuating a bend actuator of an extracorporeal control system.

[0452] In an application:

[0453] during the advancing, the implant is coupled to a mount that is coupled to a shaft that extends through the capsule catheter and into the capsule.

[0454] the bend actuator is a shaft bend-actuator, operably coupled to one or more shaft bend-control elements that extend along the shaft to a shaft distal portion, and

[0455] bending the capsule-catheter distal portion with respect to the second catheter includes bending the capsule-catheter distal portion with respect to the second catheter by bending the shaft distal portion while the shaft distal portion is disposed within the capsule-catheter distal portion.

[0456] There is further provided, in accordance with an application of the present invention, apparatus, the apparatus including a delivery tool for use with an implant, the delivery tool including:

[0457] an extracorporeal control system at a proximal portion of the delivery tool, the control system including a second-catheter bend-actuator and a shaft bend-actuator;

[0458] a flexible second catheter, extending distally from the control system, and including one or more first-tube bend-control elements that are operably coupled to the first-tube bend-actuator, and that extend from the control system and along the second catheter to a first-tube distal portion of the second catheter;

[0459] a flexible capsule catheter, extending distally from the control system through the second catheter to a capsule-catheter distal portion of the capsule catheter; and

[0460] a flexible shaft, extending distally from the control system through the capsule catheter, and including one or more shaft bend-control elements that are operably coupled to the shaft bend-actuator, and that extend from the control system along the shaft to a shaft distal portion of the shaft; and:

[0461] via the control system:

[0462] the first-tube distal portion is axially slidable (i) distally over the capsule-catheter distal portion to ensheath the capsule-catheter distal portion within the first-tube distal portion, and (iii) proximally off of

the capsule-catheter distal portion to expose the capsule-catheter distal portion from the second catheter;

[0463] the capsule-catheter distal portion is axially slidable (i) distally over the shaft distal portion to ensheath the shaft distal portion within the capsule-catheter distal portion, and (ii) proximally off of the shaft distal portion to expose the shaft distal portion from the capsule catheter;

[0464] actuation of the first-tube bend-actuator actively bends the first-tube distal portion via the first-tube bend-control elements,

[0465] actuation of the shaft bend-actuator actively bends the shaft distal portion via the shaft bend-control elements.

[0466] the control system does not include a capsule-catheter bend actuator, and the capsule catheter does not include bend-control elements via which the capsule-catheter distal portion is actively bendable.

[0467] In an application:

[0468] each of the first-tube bend-control elements includes a respective pull-wire that extends from the first-tube bend-actuator and through a respective secondary lumen of the second catheter, and is fixed to the second catheter at the first-tube distal portion, and

[0469] each of the shaft bend-control elements includes a respective pull-wire that extends from the shaft bend-actuator and through a respective secondary lumen of the shaft, and is fixed to the shaft at the shaft distal portion.

[0470] In an application, each of the shaft bend-control elements includes a respective pull-wire that extends from the shaft bend-actuator and through a respective secondary lumen of the shaft, and is fixed to the shaft distally from the capsule catheter.

[0471] In an application, the capsule-catheter distal portion is sufficiently flexible such that, while the shaft distal portion is ensheathed in the capsule-catheter distal portion, bending of the shaft distal portion by actuation of the shaft bend-actuator causes the capsule-catheter distal portion to passively bend.

[0472] In an application, the capsule-catheter distal portion is sufficiently flexible such that, while the capsule-catheter distal portion is ensheathed in the first-tube distal portion, bending of the first-tube distal portion by actuation of the first-tube bend-actuator causes the capsule-catheter distal portion to passively bend.

[0473] In an application:

[0474] the delivery tool includes a rod and a capsule.

[0475] the rod extends distally out of the shaft,

[0476] the capsule is coupled to a distal portion of the rod, and includes a circumferential wall that extends proximally from the distal portion of the rod to define a chamber, and

[0477] the rod is axially movable with respect to the shaft, axial movement of the rod with respect to the shaft moving the capsule axially with respect to the capsule catheter.

[0478] In an application, the delivery tool includes a capsule coupled to the capsule catheter distally from the second catheter, and dimensioned to house at least part of the implant, the shaft extends distally through the capsule, and axial sliding of the capsule-catheter distal portion proxi-

mally off of the shaft distal portion causes axial sliding of the capsule proximally along the shaft distal portion.

[0479] In an application, each of the shaft bend-control elements includes a respective pull-wire that extends from the shaft bend-actuator and through a respective secondary lumen of the shaft, and is fixed to the shaft within the capsule.

[0480] In an application, the second-catheter distal portion is axially slidable distally such that it abuts the capsule.

[0481] In an application:

[0482] the control system includes a first-catheter bend-actuator,

[0483] the delivery tool includes a flexible first catheter, extending distally from the control system, and including one or more first-catheter bend-control elements that are operably coupled to the first-catheter bend-actuator, and that extend from the control system and along the first catheter to a first-catheter distal portion of the first catheter.

[0484] the shaft extends distally from the control system through the first catheter to the shaft distal portion, and is axially slidable (i) proximally through the first catheter such that the shaft distal portion becomes ensheathed in the first-catheter distal portion, and (ii) distally through the first catheter such that the shaft distal portion becomes exposed from the first catheter, and

[0485] actuation of the first-catheter bend-actuator actively bends the first-catheter distal portion via the first-catheter bend-control elements.

[0486] In an application, the control system includes an outer-first juxtaposition actuator, operatively coupled to the first catheter and to the second catheter, such that actuation of the outer-first juxtaposition actuator slides the second catheter axially with respect to the first catheter.

[0487] In an application:

[0488] the second catheter is rotationally locked to the first catheter by (i) a proximal lock defined by the control system, and (ii) a distal lock at which the first catheter includes a first-catheter coupling, and

[0489] the second catheter includes a second-catheter coupling that is rotationally-locked to the first-catheter coupling.

[0490] In an application, the second catheter is rotationally locked to the first catheter.

[0491] There is further provided, in accordance with an application of the present invention, apparatus, including:

[0492] a delivery tool for use with an implant, the delivery tool including:

[0493] a tubular shaft;

[0494] a rod:

[0495] extending from within the shaft out of a distal end of the shaft,

[0496] having a distal portion disposed outside of the distal end of the shaft, and

[0497] operatively coupled to the shaft such that rotational movement of the rod with respect to the shaft is converted into axial movement of the rod with respect to the shaft, and

[0498] a capsule, coupled to the distal portion of the rod, and including a circumferential wall that extends proximally from the distal portion of the rod to define a chamber and

- [0499] an accessory, including a detent, the accessory being couplable to the capsule such that the detent rotationally locks the capsule to the rod.
- [0500] In an application:
- [0501] the delivery tool has an extended state and a retracted state, axial movement of the rod distally with respect to the shaft extending the delivery tool from the retracted state toward the extended state,
- [0502] in the retracted state, a part of the shaft is disposed within the chamber, and
- [0503] in the extended state, the part of the shaft is disposed outside of the chamber.
- [0504] In an application:
- [0505] the apparatus includes a catch, coupled to the rod.
- [0506] the capsule defines a lateral detent-hole that extends from outside the capsule toward the catch, and
- [0507] the accessory is couplable to the capsule such that the detent extends through the detent-hole and engages the catch, rotationally locking the capsule to the rod.
- [0508] In an application, the accessory includes a clip, and is couplable to the capsule by the clip being clipped to the capsule such that the detent rotationally locks the capsule to the rod.
- [0509] In an application, the accessory includes a e-shaped clip, and is couplable to the capsule by the c-shaped clip being placed over the capsule such that the detent rotationally locks the capsule to the rod.
- [0510] In an application, the delivery tool includes a mount, coupled to the shaft, extending radially outward from the shaft, and shaped to define a plurality of implant-receiving slots arranged circumferentially, each implant-receiving slot of the plurality of implant-receiving slots being shaped to receive a respective portion of an implant.
- [0511] In an application:
- [0512] the delivery tool has an extended state and a retracted state, axial movement of the rod distally with respect to the shaft extending the delivery tool toward the extended state, and axial movement of the rod proximally with respect to the shaft retracting the delivery tool toward the retracted state,
- [0513] in the retracted state, the plurality of implant-receiving slots is disposed within the chamber, and
- [0514] in the extended state, the plurality of implant-receiving slots is disposed outside of the chamber.
- [0515] In an application, the rod is shaped to define an external thread, and the operative coupling of the rod to the shaft is provided by the external thread.
- [0516] In an application, the shaft is shaped to define an internal thread, and the operative coupling of the rod to the shaft is provided by mating between the internal thread and the external thread.
- [0517] In an application, the accessory includes:
- [0518] a first component that includes the detent and is couplable to the capsule such that the detent rotationally locks the capsule to the rod, and
- [0519] a knob, couplable to the first component after the first component is coupled to the capsule, and facilitating rotation, by hand, of the accessory, the capsule, and the rod, by gripping and rotating the knob by hand.
- [0520] In an application, the knob is shaped to define an opening dimensioned (i) to allow passage of a distal tip of the capsule through the opening, and (ii) to receive and engage the first component.
- [0521] There is further provided, in accordance with an application of the present invention, a method, including:
- [0522] placing an implant on a distal portion of a delivery tool, the delivery tool: (a) having a proximal portion, and (b) including a capsule at the distal portion and a controller at the proximal portion;
- [0523] extracorporeally ensheathing the implant in the capsule by moving the capsule helically over the implant;
- [0524] subsequently advancing the ensheathed implant and the distal portion of the delivery tool into a subject, while retaining the proximal portion of the delivery tool outside of the subject; and
- [0525] subsequently, intracorporeally deploying the implant from the capsule by moving the capsule linearly off of the implant.
- [0526] In an application, ensheathing the implant in the capsule includes ensheathing the implant in the capsule by moving the capsule helically over the implant by applying, at the distal portion of the delivery tool, a rotational force to the capsule.
- [0527] In an application, moving the capsule linearly off of the implant includes moving the capsule linearly off of the implant by applying, at the proximal portion of the delivery tool, an unsheathing force to the controller.
- [0528] In an application, the delivery tool includes a rod that extends between the distal portion and the proximal portion of the delivery tool, and moving the capsule helically over the implant includes rotating the rod in a first rotational direction.
- [0529] In an application, the delivery tool includes a shaft from which a distal portion of the rod extends distally, and rotating the rod in the first rotational direction includes screwing the distal portion of the rod into the shaft.
- [0530] In an application, moving the capsule linearly off of the implant includes rotating the rod in a second rotational direction, the second rotational direction being opposite to the first rotational direction.
- [0531] In an application, rotating the rod in the first rotational direction includes driving rotation of the rod from the distal portion of the delivery tool, and rotating the rod in the second rotational direction includes driving rotation of the rod from the proximal portion of the delivery tool.
- [0532] In an application:
- [0533] rotating the rod in the first rotational direction includes rotating the rod in the first rotational direction while the capsule is rotationally locked with respect to the rod, and
- [0534] the method includes, subsequently to extracorporeally ensheathing the implant, and prior to intracorporeally deploying the implant, rotationally unlocking the capsule with respect to the rod.
- [0535] In an application, rotationally unlocking the capsule with respect to the rod includes rotationally unlocking the capsule with respect to the rod prior to the step of advancing.
- [0536] In an application:
- [0537] rotating the rod in the first rotational direction while the capsule is rotationally locked with respect to the rod includes rotating the rod in the first rotational

direction while an accessory that includes a detent is coupled to the capsule such that the detent rotationally locks the capsule to the rod, and

[0538] rotationally unlocking the capsule with respect to the rod includes decoupling the accessory from the capsule.

[0539] In an application, rotating the rod in the first rotational direction includes driving rotation of the rod using the accessory.

[0540] There is further provided, in accordance with an application of the present invention, a method, including:

[0541] placing an implant on a distal portion of a delivery tool, the delivery tool having a proximal portion, and including a capsule at the distal portion, and a controller at the proximal portion;

[0542] extracorporeally ensheathing the implant in the capsule by extracorporeally applying an ensheathing force to the distal portion of the delivery tool;

[0543] subsequently, advancing the ensheathed implant and the distal portion of the delivery tool into a subject, while retaining the proximal portion of the delivery tool outside of the subject; and

[0544] subsequently, intracorporeally deploying the implant from the capsule by extracorporeally applying an unsheathing force to the controller.

[0545] In an application, extracorporeally applying the ensheathing force to the distal portion of the delivery tool includes extracorporeally applying the ensheathing force to the capsule.

[0546] In an application, extracorporeally applying the ensheathing force to the distal portion of the delivery tool includes rotating the capsule.

[0547] In an application, the delivery tool includes a rod that extends between the distal portion and the proximal portion of the delivery tool, and extracorporeally applying the ensheathing force to the distal portion of the delivery tool includes rotating the rod in a first rotational direction by extracorporeally applying the ensheathing force to the distal portion of the delivery tool.

[0548] In an application, extracorporeally applying the unsheathing force to the controller includes rotating the rod in a second rotational direction by extracorporeally applying the unsheathing force to the controller, the second rotational direction being opposite to the first rotational direction.

[0549] In an application:

[0550] rotating the rod in the first rotational direction includes rotating the rod in the first rotational direction while the capsule is rotationally locked with respect to the rod, and

[0551] the method includes, subsequently to extracorporeally ensheathing the implant, and prior to intracorporeally deploying the implant, rotationally unlocking the capsule with respect to the rod.

[0552] In an application, rotationally unlocking the capsule with respect to the rod includes rotationally unlocking the capsule with respect to the rod prior to the step of advancing.

[0553] In an application:

[0554] rotating the rod in the first rotational direction while the capsule is rotationally locked with respect to the rod includes rotating the rod in the first rotational direction while an accessory that includes a detent is coupled to the capsule such that the detent rotationally locks the capsule to the rod, and

[0555] rotationally unlocking the capsule with respect to the rod includes decoupling the accessory from the capsule.

[0556] In an application, the ensheathing force is a rotational force, and rotating the rod in the first rotational direction while the capsule is rotationally locked with respect to the rod includes applying the rotational force to the accessory such that the accessory imparts the rotational force to the rod.

[0557] There is further provided, in accordance with an application of the present invention, a method, including:

[0558] using a delivery tool having a proximal portion and a distal portion, and including:

[0559] a capsule at the distal portion.

[0560] a shaft, and

[0561] a rod that extends from the proximal portion, through the shaft, and out of a distal end of the shaft, and is coupled to the capsule;

[0562] placing an implant on the delivery tool such that the implant circumscribes the shaft;

[0563] extracorporeally ensheathing the implant in the capsule by rotating the capsule with respect to the shaft but not with respect to the rod; and

[0564] subsequently, intracorporeally deploying the implant from the capsule by rotating the rod with respect to the shaft and with respect to the capsule.

[0565] There is further provided, in accordance with an application of the present invention, a method, including:

[0566] using a delivery tool having a proximal portion and a distal portion, and including:

[0567] a capsule at the distal portion,

[0568] a shaft, and

[0569] a rod that extends between the proximal portion and the distal portion, and is coupled to the capsule;

[0570] placing an implant on the delivery tool, proximally from the capsule;

[0571] subsequently, extracorporeally ensheathing the implant in the capsule by rotating the rod such that the rod and the capsule move proximally with respect to the implant;

[0572] subsequently, at the proximal portion, engaging a rod-controller with a proximal region of the rod, and

[0573] subsequently, intracorporeally deploying the implant from the capsule by using the rod-controller to rotate the rod with respect to the shaft such that the rod and the capsule move distally with respect to the implant.

[0574] In an application:

[0575] the delivery tool includes a shaft, and the rod extends from the proximal region, through the shaft, and out of a distal end of the shaft, and is coupled to the capsule distally from the shaft, and

[0576] placing the implant on the delivery tool includes placing the implant on the delivery tool such that the implant circumscribes the shaft, proximally from the capsule.

[0577] In an application, the method includes, prior to placing the implant on the delivery tool, moving the capsule distally with respect to the shaft while the rod-controller is not engaged with the rod.

[0578] In an application, the method includes disengaging the rod-controller from the rod prior to moving the capsule distally with respect to the shaft.

[0579] In an application, moving the capsule distally with respect to the shaft includes twirling the proximal region of the rod between a finger and a thumb.

[0580] There is further provided, in accordance with an application of the present invention, apparatus, including:

[0581] a first-catheter controller;

[0582] a first catheter:

[0583] extending distally from the first-catheter controller, and

[0584] having a first-catheter distal portion that includes a first-catheter distal end that is operably coupled to the first-catheter controller so as to be bendable, in a first-catheter steering plane, by actuation of the first-catheter controller;

[0585] a second-catheter controller; and

[0586] a second catheter:

[0587] extending distally from the second-catheter controller, through the first catheter, and

[0588] having a second-catheter distal portion that includes a second-catheter distal end that extends distally out of the first-catheter distal end, and that is operably coupled to the second-catheter controller so as to be bendable, in a second-catheter steering plane, by actuation of the second-catheter controller, and:

[0589] the second catheter is rotationally oriented with respect to the first catheter such that, while the first-catheter distal end is bent in the first-catheter steering plane, bending of the second-catheter distal end by actuation of the second-catheter controller causes the second-catheter distal end to rotate with respect to the first-catheter distal end such that the second-catheter steering plane moves toward being perpendicular to the first-catheter steering plane.

[0590] In an application:

[0591] the second catheter is rotationally locked to the first catheter by (i) a proximal lock defined by the control system, and (ii) a distal lock at which the first catheter includes a first-catheter coupling, and

[0592] the second catheter includes a second-catheter coupling that is rotationally-locked to the first-catheter coupling.

[0593] In an application, the second catheter is rotationally locked to the first catheter.

[0594] There is further provided, in accordance with an application of the present invention, a method, including:

[0595] using a catheter system, the catheter system including:

[0596] a first-catheter controller;

[0597] a first catheter:

[0598] extending distally from the first-catheter controller, and

[0599] having a first-catheter distal portion that includes a first-catheter distal end that is operably coupled to the first-catheter-controller;

[0600] a second-catheter controller; and

[0601] a second catheter:

[0602] extending distally from the second-catheter controller, through the first catheter, and

[0603] having a second-catheter distal portion that includes a second-catheter distal end that extends

distally out of the first-catheter distal end, and that is operably coupled to the second-catheter controller;

[0604] bending the first-catheter distal end, in a first-catheter steering plane, by actuation of the first-catheter controller;

[0605] bending the second-catheter distal end, in a second-catheter steering plane, by actuation of the second-catheter controller, and

[0606] bending the second-catheter distal end in the second-catheter steering plane includes rotating the second-catheter distal end with respect to the first-catheter distal end such that the second-catheter steering plane moves toward being perpendicular to the first-catheter steering plane.

[0607] In an application:

[0608] the catheter system includes a lock configured to rotationally lock the first catheter with respect to the second catheter, and

[0609] the method includes:

[0610] prior to bending the first-catheter distal end, rotationally locking the first catheter with respect to the second catheter; and

[0611] prior to bending the second-catheter distal end, rotationally unlocking the first catheter with respect to the second catheter.

[0612] The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

[0613] FIGS. 1A-C are schematic illustrations showing a delivery tool, in accordance with some applications of the invention;

[0614] FIG. 2A is a schematic illustration showing, in frames A and B thereof, a two-catheter system, as is known in the prior art, and in frame C thereof, a hypothetical state of the two-catheter system;

[0615] FIG. 2B is a schematic illustration showing a catheter system, in accordance with some applications of the invention;

[0616] FIGS. 3A-D, 4A-H, 5A-B, 6, and 7 are schematic illustrations showing the delivery tool in various states thereof for use with an implant, in accordance with some applications of the invention;

[0617] FIGS. 8A-G and 9 are schematic illustrations showing at least some steps of loading the implant into a capsule assembly of the delivery tool, in accordance with some applications of the invention;

[0618] FIGS. 10A-E are schematic illustrations showing a delivery tool, in accordance with some applications of the invention;

[0619] FIG. 11 is a schematic illustration showing the delivery tool while a prosthetic valve has assumed an expanded state, in accordance with some applications of the invention;

[0620] FIGS. 12 and 13A-B are schematic illustrations showing the prosthetic valve being restrained in a compressed state by the delivery tool, in accordance with some applications of the invention;

[0621] FIG. 14A-J are schematic illustrations showing the delivery tool being used to deploy the prosthetic valve at a tricuspid valve of a head of a subject, in accordance with some applications of the invention;

[0622] FIGS. 15A-B are schematic illustrations showing a delivery tool, in accordance with some applications of the invention;

[0623] FIGS. 16A-I are schematic illustrations showing some steps of loading a prosthetic valve onto a distal portion of a delivery tool, in accordance with some applications of the invention;

[0624] FIGS. 16J-K are schematic illustrations showing advancement of an alignment mechanism over a catheter system of the delivery tool, in accordance with some applications of the invention;

[0625] FIGS. 17A-B are schematic illustrations showing use of the delivery tool to advance the prosthetic valve toward the heart, in accordance with some applications of the invention; and

[0626] FIGS. 18A-O are schematic illustrations showing use of the delivery tool to deploy the prosthetic valve at the tricuspid valve of the heart, and use of the alignment mechanism to facilitate withdrawal of the delivery tool from the subject in accordance with some applications of the invention;

[0627] FIGS. 19A-T are schematic illustrations showing use of a delivery tool to deploy the prosthetic valve at the tricuspid valve of the heart, and use of an alignment mechanism to facilitate withdrawal of the delivery tool from the subject, in accordance with some applications of the invention; and

[0628] FIGS. 20A-F are schematic illustrations showing a delivery tool being used to deploy a prosthetic valve at the tricuspid valve of the heart, in accordance with some applications of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS

[0629] Reference is made to FIGS. 1A-C, which are schematic illustrations of a delivery tool 100, in accordance with some applications of the invention.

[0630] As shown in FIG. 1A, delivery tool 100 is a multi-catheter transluminal (e.g., transfemoral) delivery tool, comprising two primary components: a catheter system 110, and an implantation instrument 160.

[0631] Catheter system 110 comprise % a first catheter unit 120 that comprises a first catheter (e.g., an outer catheter) 122 coupled at a proximal end thereof to a first-catheter handle 124; and a second catheter unit 130 that comprises a second catheter 132 coupled at a proximal end thereof to a second-catheter handle 134. A proximal opening of second catheter 132 is accessible proximally from first catheter 122, and the second catheter extends distally through the lumen of first catheter 122, and out of a distal end of the first catheter. Typically, second-catheter handle 134 is disposed proximally from first-catheter handle 124. Typically, handles 124 and 134 are mounted on a mount 108 to stabilize the handles during use. Further typically, the handles are mounted in a manner that facilitates selective adjustment of the axial and/or rotational position of the handles, and therefore their corresponding catheters.

[0632] Typically, each of catheters 122 and 132 is steerable, and this steerability is controlled by respective controller 126, 136 (which may be alternatively referred to as bend-actuators) of the respective catheter unit, each of the controllers being operably coupled to a steerable distal end-portion of its respective catheter via one or more bend-control elements, such as pull-wires, that extend along and within the respective catheter. This is described in more

detail hereinbelow. It is to be noted that the term steerable (including the specification and the claims) means actively steerable (e.g., by an extracorporeal control system), not merely sufficiently flexible to be bent responsively when pressed against a surface. Controllers 126 and 136 are typically mounted on the respective handle of their respective catheter unit. As shown, controllers 126 and 136 may be rotatable controllers such as wheels.

[0633] Implantation instrument 160 (FIGS. 1A and 1C) has a proximal portion 161 that is typically disposed proximally from handles 124 and 134, and that is typically also mounted on mount 108. Handle 124, handle 134, and proximal portion 161 are disposed at a proximal part 104 of delivery tool 100 that is configured to remain outside the subject during use. Proximal part 104 may be considered to be an extracorporeal control system. A distal part 102 of delivery tool 100 (e.g., a distal portion 163 of implantation instrument 160) is configured to be advanced into the subject, and comprises a capsule assembly 200 that houses implant 20 during this advancement.

[0634] Instrument 160 comprises a plurality of tubular members that extend distally from proximal portion 161, which are coaxial about a central longitudinal axis ax1 of delivery tool 100, and which are discussed in more detail hereinbelow. The outermost of these tubular members is typically a capsule catheter 162 that extends distally from proximal portion 161, through catheter 132, out of an open distal end of catheter 132, to distal part 102, where it abuts, and/or is coupled to a proximal capsule 202 of capsule assembly 200 (FIG. 1B). Proximal capsule 202 comprises a circumferential wall that extends distally from capsule catheter 162 to define a chamber of the proximal capsule. Capsule assembly 200 further comprises a distal capsule 204. Each of capsules 202 and 204 has a respective open end that faces the open end of the other capsule (see description hereinbelow of open ends 1065 and 1067 with reference to FIG. 14i).

[0635] As shown in the cross-sectional view shown in the upper inset of FIG. 1C, the tubular members of instrument 160 include a shaft 164 that extends distally from proximal portion 161, coaxially through capsule catheter 162. As shown in FIG. 3A, shaft 164 typically extends through proximal capsule 202, and out of the open end of the proximal capsule.

[0636] As described hereinbelow (e.g., with reference to FIGS. 4A-H), for delivery of implant 20, the implant is housed, compressed around shaft 164, within capsule assembly 200. For some applications, a mount 172 (FIGS. 1B, 3A-D) to which the implant may be engaged, is fixedly coupled to a distal end of shaft 164. Alternatively or in addition to the mount engaging the implant, the implant may be engaged with a portion of the shaft, mutatis mutandis.

[0637] Typically, a downstream portion (e.g., downstream end 1016 of prosthetic valve 1036 described hereinbelow with reference to frame A of FIG. 11) of an implant is disposed within distal capsule 204 and engaged with mount 172 (e.g., implant-engaging slots 175 thereof), and an upstream portion (e.g., upstream end 1014 of the prosthetic valve described with reference to frame A of FIG. 11) of the implant is disposed within proximal capsule 202.

[0638] A rod 168 (upper inset of FIG. 1C) is disposed coaxially through shaft 164. Rod 168 may define a guidewire lumen therethrough, and may therefore be another of the tubular members of instrument 160. Rod 168 extends

out of the distal end of shaft **164** (FIG. 1B), such that a distal portion of the rod is disposed outside of the distal end of the shaft. Rod **168** is operatively coupled to shaft **164** such that rotational movement of the rod with respect to the shaft is converted into axial movement (e.g., along longitudinal axis **ax1**) of the rod with respect to the shaft. This is typically achieved by shaft **164** defining an internal screw thread, and rod **168** defining a complementary external screw thread **167** (FIG. 1B).

[0639] For some applications, and as shown, shaft **164** has a rigid distal portion **164d** within which the internal screw thread is defined. For such applications, more proximal portions of shaft **164** (indicated by reference numeral **164p** in FIG. 1B) are flexible. Despite the difference in flexibility of portions **164p** and **164d**, these portions are typically axially and rotationally locked, and define a continuous lumen throughout the entirety of shaft **164**.

[0640] Distal capsule **204** is coupled to the distal portion of rod **168**, and comprises a circumferential wall that extends proximally from the distal portion of the rod to define a chamber of the distal capsule. Distal capsule **204** is typically axially locked with respect to rod **168**, meaning that axial movement of the rod distally or proximally moves the distal capsule axially distally or proximally. However, distal capsule **204** is rotationally coupled to and rotationally movable with respect to rod **168**, meaning that rotation of the rod does not necessarily rotate the distal capsule (e.g., if the distal capsule encounters rotational resistance). It is hypothesized by the inventors that this advantageously facilitates generally axial distal sliding of distal capsule **204** off of an implant that is disposed within distal capsule **204** (e.g., rather than the sliding off requiring helical rotation of the distal capsule with respect to the implant, which might increase an amount of abrasion between the distal capsule and the implant).

[0641] For some applications, and as shown in the lower inset of FIG. 1C, this axial locking and rotational coupling is provided by pins **170** that extend transversely through distal capsule **204**, laterally from rod **168**. Pins **170** are typically disposed distally from the chamber of the distal capsule. Pins **170** are typically axially aligned with a circumferential recess **169** defined in rod **168**, such that the pins are sufficiently close to the rod to inhibit axial movement of the rod with respect to the pins, while providing sufficient clearance between the pins and the rod (e.g., recess **169**) to allow the rod to rotate with respect to the pins.

[0642] Reference is made to FIGS. 3A-D, 4A-H, 5A-B, 6, and 7, which are schematic illustrations showing delivery tool **100** in various states thereof for use with an implant **20**, in accordance with some applications of the invention.

[0643] FIGS. 3A-D show capsule assembly **200** of delivery tool **100** in various states thereof. FIGS. 4A-H show delivery tool **100** being used to deliver implant **20**, and being transitioned between the various states in order to implant the implant. Implant **20** is described herein as being a prosthetic valve, but for some applications may be a different implant. For some applications, implant **20** is a prosthetic valve **1036** described hereinbelow, and/or may be identical to implant (prosthetic valve) **420** of WO 2019/026059 to Hariton et al., which is incorporated herein by reference. Implant **20** is typically self-expanding.

[0644] FIG. 3A shows distal portion **163** of instrument **160** (e.g., capsule assembly **200** thereof) in a closed state thereof. In this state, and with implant **20** disposed within capsule

assembly **200**, the capsule assembly is advanced transluminally (e.g., transfemorally) to a native valve **10** of the heart **4** of a subject (FIGS. 4A-B). Although FIGS. 4A-1-H show native valve **10** as the mitral valve, delivery tool **100** may alternatively be used to deliver an implant (e.g., a prosthetic valve) to another native valve of the heart, such as the tricuspid valve, the aortic valve, or the pulmonary valve, *mutatis mutandis*.

[0645] For some applications, and as shown in FIGS. 4A-B, during transluminal advancement of delivery tool **100**, capsule assembly **200** may be retracted proximally so as to abut the distal open end of second catheter **132**, and/or the distal open end of first catheter **122**. FIG. 4A shows delivery tool **100** being advanced in this manner, with catheter **132** and capsule catheter **162** hidden inside catheter **122**. Once within atrium **6**, upstream of native valve **10** (in this case the left atrium, upstream of the mitral valve), catheter **132** is extended from catheter **122** (FIG. 4B), and is steered to point capsule assembly **200** toward and through the native valve.

[0646] For some applications, extension of catheter **132** from catheter **122** (e.g., by sliding catheter **132** with respect to catheter **122**) is actuated using a juxtaposition actuator **176** (FIGS. 1A and 1C). For some such applications, catheter **132** is extended from catheter **122** while the catheters are rotationally locked with respect to each other. For example, catheters **122**, **132** may be rotationally locked via a proximal lock defined by actuator **176**. Alternatively or in addition, a distal lock defined by respective couplings of catheters **122**, **132** may rotationally lock catheters **122**, **132** with respect to each other. For example, the couplings defining the distal lock may be in certain ways similar to those described in U.S. Pat. No. 9,949,828 to Sheps et al. (e.g., with reference to FIG. 1 thereof), which is incorporated herein by reference.

[0647] For some applications, and as shown in FIGS. 4B-H, after the initial positioning of capsule assembly **200** (FIG. 4B), catheter **122** and **132** remain stationary throughout subsequent manipulation of the capsule assembly. For such applications, advancement and retraction of capsule catheter **162** from and into catheter **132** facilitates advancement and retraction of capsule assembly **200** as a whole, and of proximal capsule **202**, independently of distal capsule **204**.

[0648] Subsequently, and as shown in FIGS. 4C and 38, distal capsule **204** is advanced distally, in order to release flanges **54** of implant **20**, allowing the flanges to automatically expand radially outward. It is to be noted that the upstream end of implant **20** remains within proximal capsule **202**, and mount **172** and the downstream end of the implant remain within distal capsule **204** at this stage. Because mount **172** is fixedly coupled to shaft **164**, and distal capsule **204** is axially locked with respect to rod **168**, this distal advancement of distal capsule **204** off of implant **20** can be achieved by distally advancing the rod with respect to the shaft (e.g., by screwing the rod through the shaft).

[0649] For some applications, and as shown, this step of deploying flanges **54** is performed while the flanges (and the seam between capsules **202** and **204**) are disposed within atrium **6**. For such applications, while the deployment state of capsule assembly **200** typically remains as shown in FIG. 3B, the capsule assembly is subsequently advanced distally, downstream through native valve **10** (FIGS. 4D-E), until it is determined via imaging (e.g., fluoroscopy) that the leaflets **12** of the native valve are coapting upstream of flanges **54**

during ventricular systole (FIG. 4E). This is hypothesized by the inventors to facilitate reliable placement of the flanges downstream of the leaflets, while minimizing the distance downstream of the leaflets that the deployed flanges are advanced, thereby advantageously reducing a likelihood of inadvertently ensnaring ventricular tissue such as chordae tendineae.

[0650] Subsequently, and while the deployment state of capsule assembly 200 typically remains as shown in FIG. 38, the capsule assembly is retracted proximally, upstream, until it is determined (e.g., via imaging, such as fluoroscopy) that flanges 54 have engaged leaflets 12 (FIG. 4F).

[0651] Subsequently, as shown in FIG. 4G, an upstream support portion 40 of implant 20 is deployed by releasing it from proximal capsule 202, by retracting the proximal capsule proximally with respect to mount 172 (and therefore with respect to the implant) (FIG. 3C). This may be achieved by moving capsule catheter 162 proximally (as shown), or may be achieved by rotating the capsule catheter (e.g., as described hereinbelow for delivery tool 1020, mutatis mutandis). Upstream support portion 40 typically comprises a plurality of radial arms, and optionally a flexible sheet covering the arms, and becomes disposed over the upstream surface of the annulus of native valve 10 (FIG. 4G). Leaflets 12 therefore become at least lightly sandwiched between upstream support portion 40 and flanges 54.

[0652] Subsequently, as shown in FIG. 4H, implant 20 is fully deployed by releasing the distal end of the implant from distal capsule 204, by advancing the distal capsule distally with respect to mount 172 (FIG. 3D). That is, mount 172 is typically shaped to engage the distal end of the implant (e.g., by slots 175 defined by the mount receiving adaptors 22 defined by the distal end of the implant, as described hereinbelow with reference to FIGS. 8E-G), such that exposing the mount from the distal capsule fully releases the implant from the distal capsule.

[0653] For some such applications, distal advancement of distal capsule 204 is accomplished via axial movement of rod 168. Typically for such applications, the axial movement of the rod transitions delivery tool 100 from a retracted state (FIGS. 3A-C), in which a part of shaft 164 and/or implant-receiving slots 175 are within distal capsule 204, to an extended state (FIG. 3D), in which the part of the shaft and/or the slots are outside the distal capsule.

[0654] As shown in FIG. 4H, expansion of implant 20 opens a central channel of the implant to blood flow, and allows leaflets of the implant (not shown) to provide one-way valve functionality. The expansion also typically further squeezes leaflets 12 between upstream support portion 40 and flanges 54, thereby securing implant 20 in place, and inhibiting paravalvular leakage.

[0655] FIGS. 5A-B, 6, and 7 show respective views of distal part 102 of delivery tool 100, at the stage of deployment shown in FIG. 4B, but with certain differences as noted, in order to illustrate some flexibility that delivery tool 100 provides, and which the inventors hypothesize to be advantageous. In both FIG. 4B and FIGS. 5A-B, catheter 122 extends through fossa ovalis 14 into atrium 6, and is steered toward a position that is overhead of valve 10 (e.g., overhead of the center of valve 10). Also, in both FIG. 4B and FIGS. 5A-B, catheter 132 extends out of catheter 122, and is steered downward toward valve 10, such that capsule assembly 200 is disposed between leaflets 12 of the valve. However, in contrast to FIG. 4B, the cardiac anatomy in the

example shown in FIGS. 5A-B is such that, in order to position capsule assembly 200 between the leaflets, the capsule assembly has been advanced away from catheter 132 and no longer abuts the catheter. This may be advantageous, for example, if fossa ovalis 14 is particularly high above native valve 10. In contrast to FIG. 4B, the cardiac anatomy in the example shown in FIG. 6 is such that, in order to position capsule assembly 200 between the leaflets, additional steering is desirable. In the example shown in FIG. 7, the cardiac anatomy is such that the distance between fossa ovalis 14 and native valve 10 is shorter than that in FIG. 4B, and the cumulative length of the steerable distal portion of catheter 122 and the steerable distal portion of catheter 132 would be too great to obtain the desired angle of attack.

[0656] FIGS. 6 and 7 illustrate an additional steering feature that, for some applications, is included in delivery tool 100. This steering feature is the steerability of shaft 164 (or at least a steerable distal region of proximal portion 164p (FIG. 1B)). It is hypothesized that this steering feature advantageously increases the flexibility of delivery tool 100, and its suitability to a greater range of anatomies.

[0657] As shown in the upper inset of FIG. 1C, steerability of shaft 164 is provided by pull-wires 364a and 364b that extend from the steerable distal region of the shaft, proximally within the shaft to a controller 166 of proximal portion 161 of implantation instrument 160, similarly to the steerability of catheter 122 and 132, mutatis mutandis. This is described in more detail hereinbelow.

[0658] In the example shown in FIG. 6, steering of shaft 164 is used in addition to steering of catheter 132. In the example shown in FIG. 7, steering of shaft 164 is used instead of steering of catheter 132, with catheter 132 barely exposed from catheter 122. It is to be noted that, as shown, while shaft 164 is steerable (meaning actively steerable), it is disposed within catheter 162, which is flexible (but not itself steerable) and therefore passively bends in response to the steering of shaft 164.

[0659] Reference is again made to FIG. 1C, which includes a cross-section of delivery tool 100 that illustrates the arrangement of the various tubular members and pull-wires thereof. Two pull-wires 322a and 322b extend proximally from a steerable distal portion of catheter 122 to controller 126, actuation of which steers the steerable portion of catheter 122. Two pull-wires 332a and 332b extend proximally from a steerable distal portion of catheter 132 to controller 136, actuation of which steers the steerable portion of catheter 132. For applications in which shaft 164 is steerable, two pull-wires 364a and 364b extend proximally from a steerable portion of shaft 164 to controller 166 (e.g., a shaft bend-actuator thereof), actuation of which steers the steerable portion of shaft 164.

[0660] Reference is made to FIG. 2A, which is a schematic illustration showing, in frames A and B thereof, a two-catheter system 110', as is known in the prior art. Elements of two-catheter system 110' are labeled with the same reference numerals as corresponding elements of catheter system 110, with the addition of an apostrophe: '. As shown, pull-wire plane p3' passes through both pull-wires 332a' and 332b' of inner catheter 132', and is rotationally offset with respect to pull-wire plane p2' that passes through both pull-wires 322a' and 322b' of outer catheter 122'.

[0661] Typically, pull-wire planes p2' and p3' define respective steering planes along which catheters 122', 132'

can be bent, and therefore, along which catheter system 110' can be steered. Outer catheter 122' and inner catheter 132' are typically rotationally oriented with respect to each other such that pull-wire planes p2' and p3' are offset by a 90-degree angle_1' (frame A of FIG. 2A).

[0662] As shown in frame B of FIG. 2A, tensioning of pull-wire 322b' of outer catheter 122' may not significantly alter the rotational orientation of inner catheter 132' with respect to the outer catheter. That is, while catheter system 110' is steered along pull-wire plane p2' of outer catheter 122', both the outer catheter and inner catheter 132' may bend along pull-wire plane p2', such that angle alpha_1' is maintained at 90 degrees.

[0663] Reference is made to frame C of FIG. 2A, which shows a hypothetical state of two-catheter system 110'. As shown, when inner catheter 132' is steered along a steering plane that is different from plane p2', bending of the inner catheter may yield rotational slippage of the inner catheter with respect to the outer catheter (rotational arrows). Thus, steering of inner catheter 132' within bent outer catheter 122' may cause pull-wire planes p2' and p3' to become closer to being co-planar, such that alpha_1' deviates from 90 degrees (e.g., becomes acute in frame C). This rotational slippage may therefore reduce the range of steering of catheter system 110' and/or not yield the desired final steering angle of inner catheter 132'.

[0664] It is therefore hypothesized by the inventors that an improved placement of the pull-wires (when the catheter system is at rest) to address this issue is not at a 90-degree offset.

[0665] Reference is made to FIG. 2B, which is a schematic illustration showing catheter system 110, in accordance with some applications of the invention. In catheter system 110, the pull-wires are positioned such that, at rest (e.g., with no steering or bending of the catheters), pull-wire planes p2 and p3 are not offset by 90 degrees. For example, and as shown in frame A, an obtuse angle alpha_1 formed by the intersection of plane p3 and plane p2 may be greater than 95 degrees and/or less than 120 degrees (such as about 110 degrees). For some applications in which catheters 122, 132 are rotationally lockable via a proximal lock and/or a distal lock, as described hereinabove with reference to FIGS. 4A-B, the proximal and/or distal locks may keep pull-wire planes p2 and p3 offset such that angle alpha_1 remains obtuse along a length of the catheters.

[0666] Tensioning of pull-wire 322b of first catheter 122 may not significantly alter the rotational orientation of second catheter 132 with respect to the first catheter. That is, while catheter system 110 is steered along pull-wire plane p2 of first catheter 122, both the first catheter and second catheter 132 may bend along pull-wire plane p2, such that angle alpha_1 is maintained as an obtuse angle.

[0667] However, because angle alpha_1 is obtuse when catheter system 110 is at rest, rotational slippage of second catheter 132 with respect to first catheter 122 (rotational arrows in frame C of FIG. 2B) resulting from steering the second catheter 132 along plane p3 brings angle alpha_0.1 closer to 90 degrees, thereby improving the ease of steering of catheter system 110, relative to catheter system 110'. For some applications in which catheters 122, 132 are rotationally lockable via a proximal lock and/or a distal lock, the proximal and/or distal locks may be unlocked prior to steering the second catheter 132 along plane p3, thereby allowing at least a portion of the second catheter to rotate

with respect to the first catheter, such that plane p3 moves toward being perpendicular to plane p2.

[0668] It is therefore hypothesized by the inventors that positioning pull-wires 322a, 322b. 332a, 332b such that, while at rest, pull-wire planes p2 and p3 are not offset by 90 degrees, facilitates bi-planar steering of catheter system 110. Even if rotational slippage in catheter system 110 continues past the 90-degree position, such that, angle alpha_1 becomes less than 90 degrees, this resulting angle is advantageously greater than that of steeling system 110', in which the initial angle is 90 degrees.

[0669] Reference is now made to FIGS. 8A-G and 9, which are schematic illustrations showing at least some steps of loading implant 20 into capsule assembly 200 of delivery tool 100, in accordance with some applications of the invention. It is hypothesized by the inventors that it is advantageous to load implant 20 into distal capsule 204 in at least two steps. In a first step (FIGS. 8A-13), the implant is slid proximally over distal capsule 204. In a second step (FIGS. 8E-G), distal capsule 204 is slid over the implant by manipulating the distal capsule from distal part 102 of delivery tool 100 (as described hereinbelow), rather than from proximal part 104 of the delivery tool. That is, whereas an unsheathing force applied by the operator during deployment of implant 20 is applied to actuator 176 of proximal portion 161 as described hereinabove (e.g., FIGS. 4C-H), it is hypothesized by the inventors that it is advantageous to apply the ensheathing force directly to distal part 102, e.g., such that greater force can be applied to the implant, and/or such that the application of the ensheathing force is performed near to the site of manipulation of implant 20, thereby improving visibility of the implant to the person performing the ensheathing and/or improving control of the loading process.

[0670] For some applications, and as shown in FIG. 8A, actuator 176 (visible in FIGS. 1A and 1C) is entirely removed from instrument 160, prior to ensheathing implant 20 in distal capsule 204. (Alternatively, instrument 160 may be provided with actuator 176 initially separate from the rest of the instrument.)

[0671] As described hereinabove, distal capsule 204 is rotationally coupled to rod 168 (FIGS. 1B and 3). Therefore, rotating distal capsule 204 directly (e.g., by grasping the distal capsule by hand) would not necessarily rotate rod 168, and therefore would not result in movement of the distal capsule proximally over mount 172 and implant 20. Therefore, for some applications, an accessory 240 is provided (FIG. 8B), attachment of which to distal portion 163 of implantation instrument 160 (e.g., to distal capsule 204) rotationally locks distal capsule 204 to rod 168, thereby allowing ensheathing of implant 20 within the distal capsule by rotating the distal capsule directly.

[0672] FIG. 8A shows distal portion 163 of implantation instrument 160, with capsule assembly 200 open in the configuration shown in FIG. 3D, such that mount 172 is released proximally from the open end of distal capsule 204. Implant 20 is introduced over the distal end of catheter system 110, over and past distal capsule 204 (FIG. 8B). For applications in which implant 20 is a prosthetic valve, distal capsule 204 typically moves with respect to the leaflets of the valve in an upstream-to-downstream direction (e.g., from an upstream end of the prosthetic valve to a downstream end of the prosthetic valve, as described hereinbelow with reference to prosthetic valve 1036 shown in frame A of

FIG. 11). Subsequently, accessory **240** is attached to distal portion **163** of implantation instrument **160** (e.g., to distal capsule **204**) (FIGS. 8B-C).

[0673] Typically, accessory **240** comprises (or defines) a detent **242**, and is configured to be attached to distal capsule **204** such that the detent rotationally locks the distal capsule to rod **168**. For some applications, distal capsule **204** defines a detent-hole **180**, and the attachment of accessory **240** to the distal capsule is such that detent **242** extends through the detent-hole to rotationally lock the distal capsule to rod **168**. For some such applications, delivery tool **100** (e.g., catheter system **110** thereof) comprises a catch to which detent **242** may be engaged. For example, catheter system **110** may define a recess **178** (FIG. 1B) into which detent **242** becomes disposed. For some applications, catheter system **110** comprises a ring **174** that is rotationally and axially fixed with respect to rod **168**, and that defines recess **178** (see FIGS. 1B, and 3A-D). For some applications, rod **168** defines ring **174** and/or recess **178**.

[0674] For some applications, accessory **240** comprises a c-shaped clip **244**, and is attached to distal capsule **204** by being placed over the distal capsule (e.g., snapped into place). For such applications, detent **242** is attached to clip **244**, and typically extends radially inward from its point of attachment to the clip.

[0675] For some applications, a knob **250** is subsequently introduced over the distal end of catheter system **110** (FIGS. 8C-D), and is engaged with clip **244** (and optionally with distal capsule **204**), to facilitate rotation by hand of distal capsule **204** with respect to rod **168**.

[0676] As shown in FIG. 8E, implant **20** is then compressed (“crimped”) such that the implant engages mount **172**, e.g., with adaptors **22** being received by respective slots **175**.

[0677] Subsequently, and as shown in FIG. 8F, distal capsule **204** and rod **168** are hand-rotated in a first rotational direction (e.g., by grasping the distal capsule, accessory **240**, and/or knob **250**), such that rod **168** screws into shaft **164**, thereby screwing the distal capsule proximally over implant **20** and mount **172**. In this way, at least the downstream end of the implant is ensheathed by the distal capsule, while maintaining the engagement between the implant and the mount. Therefore, implant **20** is ensheathed in distal capsule **204** by moving the distal capsule helically over the implant, but is unsheathed by moving the distal capsule linearly off of the implant, as described hereinabove with reference to FIGS. 3A-D and 4C-H. For some applications, and as shown in FIG. 8G, proximal capsule **202** is advanced distally over the upstream end of implant **20**, further ensheathing the implant within capsule assembly **200**.

[0678] Subsequently to ensheathing the implant, and yet prior to capsule assembly **200** being used to deliver implant **20**, accessory **240** (and knob **250**, if present) is removed (FIG. 9). Removing accessory **240** from distal capsule **204** rotationally unlocks the distal capsule with respect to rod **168**, such that the unsheathing force (e.g., rotation of the rod in a second rotational direction) moves the distal capsule linearly off of the implant.

[0679] In any case, if actuator **176** was initially not engaged with rod **168** or was disengaged by the operator from rod **168** prior to ensheathing implant **20** in distal capsule **204**, then actuator **176** is engaged (or reengaged) with rod **168** prior to implantation of the implant. Typically

for such applications, the unsheathing force is then applied to distal portion **163** of implantation instrument **160** via actuator **176**.

[0680] Reference is made to FIGS. 10A-E, which are schematic illustrations showing a delivery tool **1020**, in accordance with some applications of the invention.

[0681] FIG. 10A shows delivery tool **1020** assembled, and FIG. 10B shows an exploded view of a distal portion **1024** of the delivery tool. As shown, delivery tool **1020** comprises an extracorporeal controller **1021** and distal portion **1024** that is dimensioned for transluminal (e.g., transfemoral) delivery to a subject.

[0682] Delivery tool **1020** bears certain similarities to delivery tool **100** described hereinabove. Particularly, distal portion **1024** of delivery tool **1020** is in certain ways similar to distal portion **163** of implantation instrument **160** of delivery tool **100**. Components that are identically named between delivery tools **100**, **1020** typically share similar features and serve similar functions as each other.

[0683] As shown, distal portion **1024** comprises a shaft **1034** (e.g., extending distally from within a capsule catheter **1072**) to which a proximal capsule **1064** and a distal capsule **1066** (collectively defining a capsule assembly **1063**) are coupled. As shown in FIG. 10B, each capsule **1064**, **1066** has a respective open end **1065**, **1067**, such that open end **1065** of proximal capsule **1064** faces the proximal end of distal capsule **1066**. For some applications, and as shown, open end **1067** is the proximal end of distal capsule **1066**, such that open end **1065** of proximal capsule **1064** faces the open end of the distal capsule.

[0684] Similarly to distal portion **163** of instrument **160** described hereinabove, distal portion **1024** further comprises a mount **1028** dimensioned (e.g., defining slots **1029**) to engage an implant. Typically, capsules **1064**, **1066** can be moved with respect to the mount (e.g., along a distal portion axis **ax1018**), via extracorporeal controller **1021**. For some applications, extracorporeal controller **1021** controllably moves proximal capsule **1064** and/or distal capsule **1066** both distally (“advanced”) and proximally (“retracted”), with respect to mount **1028**.

[0685] For some applications, and as shown in FIG. 10B, distal portion **1024** comprises a rod **1168** that extends out of a distal end of shaft **1034**. Similarly to as described hereinabove with reference to rod **168** of instrument **160**, delivery tool **1020** is configured to distally advance distal capsule **1066** with respect to mount **1028** by screwing the rod through shaft **1034**.

[0686] For some such applications, and further similarly to distal capsule **204** of instrument **160**, distal capsule **1066** is rotationally coupled to and rotationally movable with respect to rod **1168**, such that rotation of rod **1168** does not necessarily rotate distal capsule **1066**. Typically for such applications, pins **1170** are fitted within distal capsule **1066**, in relation to a recess **1169** defined by rod **1168**, so as to axially fix the distal capsule with relation to the rod, while allowing rotation of the rod with respect to the pins, as described hereinabove with reference to instrument **160** regarding FIG. 1C and FIGS. 3A-D.

[0687] For some such applications, and as shown, distal capsule **1066** defines a window **1110**, and a hole **1180**. Typically for such applications, hole **1180** is shaped to facilitate attachment of accessory **240** and/or knob **250** (not shown in FIG. 10B, yet described hereinabove with reference to FIGS. 8B-G) in order to rotationally lock the distal

capsule with respect to rod **1168**. For example, rotationally locking distal capsule **1066** to rod **1168** may be facilitated by extending a portion of accessory **240** through hole **1180** such that the portion occupies a recess **1178** defined by a ring **1174** that is fixedly coupled to the rod.

[**0688**] Notwithstanding similarities between delivery tools **100** and **1020**, the description below of delivery tool **1020** focuses upon features that are particular to delivery tool **1020**. A difference between delivery tools **100** and **1020** lies in distal portion **1024** being configured to be transluminally advanced to the heart while in a delivery state (FIGS. **10A** and **10C**) in which an inter-capsule gap **1071a** exists between proximal capsule **1064** and distal capsule **1066** (e.g., between respective open ends **1065**, **1067**).

[**0689**] Typically for applications in which capsules **1064**, **1066** can be both advanced and retracted, the capsules may be moved axially to a range of positions, relative to mount **1028**. For some such applications, and as shown in FIG. **10D**, segments **1034c** and **1034d** of shaft **1034** may be slidably coupled to each other, thereby facilitating axial movement of capsules **1064**, **1066**. Thus, and as shown in FIG. **10D**, segments **1034c** and **1034d** may slide telescopically with respect to each other, such that distal portion **1024** assumes a deployment state in which inter-capsule gap **1071b** is longer than gap **1071a** when the distal portion is in the delivery state (FIG. **10C**).

[**0690**] Inter-capsule gap **1071b** is typically at least 50% (e.g., at least 100%) and/or less than 200% (e.g., less than 150%) greater than inter-capsule gap **1071a**.

[**0691**] For some applications, while distal portion **1024** is in the delivery state, inter-capsule gap **1071a** is greater than 1 mm (e.g., greater than 5 mm, e.g., greater than 10 mm, e.g., greater than 15 mm, e.g., greater than 20 mm) and/or less than 25 mm in length (e.g., less than 20 mm, e.g., less than 15 mm, e.g., less than 10 mm, e.g., less than 5 mm). For example, inter-capsule gap **1071a** may be 10-20 mm.

[**0692**] For some such applications, while distal portion **1024** is in the deployment state, inter-capsule gap **1071b** is greater than 15 mm (e.g., greater than 20 mm, e.g., greater than 25 mm, e.g., greater than 30 mm, e.g., greater than 35 mm) and/or less than 40 mm in length (e.g., less than 35 mm, e.g., less than 30 mm, e.g., less than 25 mm, e.g., less than 20 mm). For example, inter-capsule gap **1071b** may be 20-35 mm.

[**0693**] For some applications, proximal capsule **1064** may be further advanced, and/or distal capsule **1066** may be further retracted, such that distal portion **1024** assumes a withdrawal state in which the inter-capsule gap is shorter than when the distal portion is in the delivery state (e.g., such that the gap is closed or nearly-closed, as shown in FIG. **10E**). Consequently, for some such applications, a capsule-assembly length **1074c** from a proximal end of proximal capsule **1064** to a distal end of distal capsule **1066** is lesser while distal portion **1024** is in the withdrawal state (FIG. **10E**) than capsule-assembly length **1074a** while the distal portion is in the delivery state (FIG. **10C**).

[**0694**] Although the states (e.g., delivery state, deployment state, withdrawal state) of delivery tool **1020** are described with respect to distal portion **1024**, they are typically implemented by controller **1021**. For example, controller **1021** may define these states as discrete states by enforcing only certain operations and/or degrees of movement of control elements (e.g., knobs, switches, levers, wheels etc.) of controller **1021**, the control elements being

operatively coupled to capsules **1064** and **1066**, e.g., by wires, rods, and/or cables. Furthermore, for some applications, the orders of operation described hereinbelow are facilitated and/or enforced by controller **1021**, e.g., by the controller selectively and/or sequentially locking and/or unlocking locks that selectively and/or sequentially enable and/or disable the control elements of the controller.

[**0695**] Reference is made to FIG. **11**, which is a schematic illustration showing delivery tool **1020** while prosthetic valve **1036** has assumed an expanded state, in accordance with some applications of the invention. Prosthetic valve **1036** comprises a frame assembly **1022** (shown in frame A), within which prosthetic leaflets **1058** are disposed (frame B).

[**0696**] Prosthetic valve **1036** is in certain ways similar to that described in WO 2019/026059 to Hariton et al. (e.g., with reference to FIGS. **1**, **2**, and **18** thereof), which is hereby incorporated by reference. For some applications, delivery tool **1020** may be used to deliver implant (prosthetic valve) **420** of WO 2019/026059 to Hariton et al. Prosthetic valve **1036** is typically self-expanding.

[**0697**] As shown in frame A of FIG. **11**, frame assembly **1022** comprises an inner valve frame **1030** nested within an outer frame **1060**. Valve frame **1030** is typically shaped to define: (i) a tubular portion **1032** that defines a lumen **1038** between an upstream end **1014** and a downstream end **1016**, and (ii) a plurality of arms **1046** that collectively form an upstream support portion **1040** that extends upstream from the tubular portion. For some applications, and as shown, valve frame **1030** further defines adaptors **1025**, and mount **1028** engages adaptors **1025** (e.g., by receiving adaptors **1025** into slots **1029** defined by the mount).

[**0698**] As shown, outer frame **1060** comprises flanges **1054**, which are each coupled to tubular portion **1032** at a respective coupling point **1052** that is disposed downstream of upstream support portion **1040**. In this way, each flange **1054** extends upstream from coupling point **1052**, to a respective flange end-portion **1068**.

[**0699**] Typically, and as shown in frame B, prosthetic valve **1036** comprises a plurality of prosthetic leaflets **1058**, which are disposed within lumen **1038** so as to facilitate unidirectional blood flow from upstream end **1014** to downstream end **1016**. For some applications, and as shown, prosthetic valve **1036** also comprises an upstream covering **1048**, disposed over arms **1046** to define an upstream skin, in order to reduce a risk of paravalvular leakage.

[**0700**] Reference is made to FIGS. **12** and **13A-B**, which are schematic illustrations showing prosthetic valve **1036** being restrained in a compressed state by delivery tool **1020**, in accordance with some applications of the invention. FIG. **12** therefore shows delivery system **1010** in a delivery state (described hereinabove in reference to FIGS. **10A** and **10C**) in which the system is configured to be transluminally advanced to the heart of a subject.

[**0701**] In the delivery state, a distal-implant portion **1100** comprising a distal portion of valve frame **1030** (e.g., downstream end **1016** of tubular portion **1032**) is engaged with mount **1028** (e.g., by slots **1029** receiving adaptors **1025**), such that the downstream end and the mount are both disposed within distal capsule **1066** (e.g., within a chamber defined by the distal capsule), with the distal capsule restraining the downstream end compressed against the mount, thereby maintaining engagement of the downstream end with the mount.

[0702] In the delivery state, a proximal-implant portion **1102** comprising a proximal portion of valve frame **1030** (e.g., upstream support portion **1040**) is disposed within (e.g., restrained by) proximal capsule **1064**. Additionally, in the delivery state, each flange end-portion **1068** is disposed within (e.g., restrained by) proximal capsule **1064**.

[0703] Typically, a segment **1056** of prosthetic valve **1036** is disposed at inter-capsule gap **1071a**. That is, segment **1056** is exposed by inter-capsule gap **1071a**. Typically, segment **1056** includes part of tubular portion **1032**, part of each flange **1054**, and/or coupling points **1052**.

[0704] For some applications, and as shown in FIG. 13A, delivery tool **1020** comprises a flexible sheath **1044** (e.g., comprising a polymer and/or a fabric) that covers segment **1056** by circumscribing inter-capsule gap **1071a**. For some such applications, a distal end of the sheath may abut, and/or be partially disposed within, distal capsule **1066**.

[0705] For some such applications, and as shown, sheath **1044** extends proximally from inter-capsule gap **1071a**, covering proximal capsule **1064**. Sheath **1044** may extend into and through a delivery catheter **1050** that connects distal portion **1024** to extracorporeal controller **1021** (FIGS. 14A-C), e.g., with a proximal end of the sheath remaining outside of the subject.

[0706] For some applications, distal portion **1024** of delivery tool **1020** comprises a nosecone **1026** having a flexible distal end-portion **1027**. For some such applications, and as shown in FIG. 13A, distal end-portion **1027** has a resting shape (e.g., in the absence of a straightening force that may be provided by a more rigid element such as a guidewire **1023**) that is curled. FIG. 13B shows distal end-portion **1027** having been straightened by guidewire **1023** having been extended through capsule catheter **1072** and shaft **1034**, and into the distal end-portion. Typically for such applications, an axial length d1025, d1025b of nosecone **1026** is greater when guidewire **1023** is disposed within the distal end-portion (FIG. 13B), than in the absence of the guidewire (e.g., length d1025, d1025a in FIG. 13A). For some such applications, shape-memory of distal end-portion **1027** tends to maintain the distal end-portion in the resting (e.g., curled) shape. That is, even after having been straightened by guidewire **1023**, distal end-portion **1027** automatically assumes the curled shape when the guidewire is removed from the distal end-portion. It is hypothesized by the inventors that this curling of nosecone **1026** advantageously allows the nosecone to be longer (and therefore have a shallower taper-angle) than a similar nosecone that does not curl, because it is possible to allow the nosecone to curl during steps in which a long axial length of a nosecone would otherwise be disadvantageous e.g., as described hereinbelow with respect to FIGS. 14A-B.

[0707] Reference is made to FIGS. 14A-J, which are schematic illustrations showing delivery tool **1020** being used to deploy prosthetic valve **1036** at a tricuspid valve **1096** of a heart **1090** of a subject, in accordance with some applications of the invention.

[0708] FIG. 14A shows distal portion **1024** of delivery tool **1020** having been transluminally advanced, through inferior vena cava **1092** and right atrium **1094** of heart **1090**, such that nosecone **1026** and distal capsule **1066** have passed through tricuspid valve **1096** to enter right ventricle **1098**.

[0709] For some applications, delivery tool **1020** is transluminally advanced along guidewire **1023** (e.g., after the

guidewire is advanced to heart **1090**). In this way, guidewire **1023** extends from extracorporeal controller **1021** to delivery catheter **1050**. For some applications, controller **1021** is used to manipulate guidewire **1023** (e.g., to steer the guidewire while the guidewire advances to the heart). Alternatively or in addition, steering of distal portion **1024** may be facilitated by delivery tool **1020** comprising at least one pull-wire operatively connecting distal portion **1024** to controller **1021**. For example, delivery catheter **1050** may be implemented using catheter system **110** described hereinabove with reference to FIGS. 1C and 2B, such that capsule catheter **1072** and shaft **1034** extend through the catheter system, mutatis mutandis.

[0710] For some applications, steering of distal portion **1024** may be further facilitated by shaft **1034** having segments distinguished by their relative rigidity. Typically for such applications, shaft **1034** extends distally from a proximal portion (e.g., from extracorporeal controller **1021**) of delivery tool **1020** (e.g., within delivery catheter **1050** and capsule catheter **1072**, as shown in the insets on the left side of FIG. 14A). For some such applications, a rigid proximal shaft segment **1034a** (lower left inset of FIG. 14A) extends distally from extracorporeal controller **1021**. Typically for such applications, rigid proximal shaft segment **1034a** is greater than 50 cm (e.g., e.g., greater than 60 cm. e.g., greater than 70 cm. e.g., greater than 80 cm. e.g., greater than 90 cm) and/or less than 100 cm (e.g., e.g., less than 9) cm. e.g., less than 80 cm. e.g., less than 70 cm. e.g., less than 60 cm) in length. It is hypothesized by the inventors that rigidity of rigid proximal shaft segment **1034a** facilitates transfer of force from the proximal portion of delivery tool **10210** (e.g., from extracorporeal controller **1021**).

[0711] For some applications, as described hereinabove, a distal shaft segment **1034b** is relatively less rigid than proximal shaft segment **1034a**, and is configured to be sufficiently flexible to turn from the vena cava toward tricuspid **1096** (as shown in FIG. 14A). Flexible shaft segment **1034b** of shaft **1034** extends distally from rigid proximal shaft segment **1034a** (upper left inset of FIG. 14A). For some such applications, flexible shaft segment **1034b** is greater than 5 cm (e.g., greater than 6 cm. e.g., greater than 8 cm.) and/or less than 10 cm (e.g., less than 8 cm. e.g., less than 6 cm) in length.

[0712] For some applications, a rigid distal shaft segment extends distally from flexible shaft segment **1034b**, such that the rigid distal shaft segment reaches distal portion **1024** of delivery tool **1020**. That is, as shown in FIGS. 10C-D, rigid distal shaft segments **1034c** and/or **1034d** extend through at least part of capsules **1064**, **1066**. For example, rigid distal shaft segments **1034c** and/or **1034d** may extend distally out of proximal capsule **1064**, it is hypothesized by the inventors that rigidity of rigid distal shaft segments **1034c** and/or **1034d** may ensure alignment of capsules **1064**, **1066** along distal portion axis ax1018, thereby facilitating axial movement of capsules **1064**, **1066** along the distal portion axis.

[0713] For some such applications, mount **1028** is attached to the rigid distal shaft segment (e.g., rigid distal shaft segment **1034d**, as shown). Typically for such applications, prosthetic valve **1036** is compressed upon rigid distal shaft segments **1034c** and/or **1034d**.

[0714] For some such applications, rigid distal shaft segments **1034c**, **1034d** may slide telescopically with respect to each other, as described hereinabove in reference to FIG. 10D. Therefore, while distal portion **1024** is in the delivery

state, rigid distal shaft segments **1034c**, **1034d** may together be greater than 2 cm (e.g., greater than 3 cm, e.g., greater than 5 cm, e.g., greater than 8 cm) and/or less than 10 cm (e.g., less than 6 cm, e.g., less than 4 cm) in length.

[0715] For some such applications, rigid proximal shaft segment **1034a** (bottom left inset of FIGS. **14A-C**), as well as rigid distal shaft segments **1034c** and **1034d**, may each be more rigid than flexible shaft segment **1034b** (upper left inset of FIGS. **14A-C**).

[0716] It is hypothesized by the inventors that the relative flexibility of flexible shaft segment **1034**, **1034b** facilitates steering of distal portion **1024**, particularly from inferior vena cava **1092** to right ventricle **1098**. It is further hypothesized by the inventors that the relative rigidity of rigid proximal shaft segment **1034**, **1034a** provides support (e.g., a resistance force) that facilitates steering of distal portion **1024**. Additionally, the relative rigidity of rigid distal shaft segments **1034**, **1034c**, **1034d** is further hypothesized by the inventors to facilitate maintenance of the alignment of the capsules along linear distal portion axis **ax1018**, e.g., while distal portion **1024** transitions between delivery state, deployment state and withdrawal state, as described hereinabove.

[0717] FIG. **14A** shows distal end-portion **1027** as it is straightened by guidewire **1023**, as described hereinabove. For some applications, it may be desirable to reduce axial length **d1025** of nosecone **1026** (FIGS. **13A-B**), prior to deploying prosthetic valve **1036** at the native valve. Typically for such applications, and as shown in FIG. **14B**, guidewire **1023** is withdrawn from at least distal end-portion **1027**, thereby reducing axial length **d1025** of nosecone **1026** (as described hereinabove with reference to FIGS. **13A-B**). It is hypothesized by the inventors that reducing length **d1025** by withdrawing guidewire **1023** may facilitate deployment of prosthetic valve **1036** by reducing an amount of space within right ventricle **1098** required to maneuver distal portion **1024** (e.g., distal capsule **1066** thereof).

[0718] FIG. **14B** shows distal portion **1024** of delivery tool **1020** after guidewire **1023** has been proximally withdrawn from distal end-portion **1027** of nosecone **1026**.

[0719] FIG. **14C** shows flexible sheath **1044** having been subsequently retracted, exposing segment **1056** and proximal capsule **1064**. For the sake of clarity, and similarly to as in FIG. **12**, distal portion **1024** is shown as if proximal capsule **1064** and distal capsule **1066** were transparent, in order to visualize the orientation of prosthetic valve **1036** within the respective capsules. As shown, mount **1028** and the downstream end of tubular portion **1032** are disposed within distal capsule **1066**, whereas upstream support portion **1040** and flange end-portions **1068** are disposed within proximal capsule **1064**, as described hereinabove in reference to FIG. **12**.

[0720] Subsequently, proximal capsule **1064** is partially retracted with respect to mount **1028**, such that flange end-portions **1068** are released from the proximal capsule (FIG. **14D**). Since outer frame **1060** typically comprises a shape-memory elastic material (e.g., Nitinol), flanges **1054** (e.g., end-portions **1068** thereof) automatically expand radially outward from coupling points **1052** (FIG. **11**), upon release from proximal capsule **1064**. However, since the distal end of tubular portion **1032** is still restrained by distal capsule **1066**, and upstream support portion **1040** is still restrained by proximal capsule **1064**, valve frame **1030** remains in the compressed state.

[0721] The inset of FIG. **14D** shows a mechanism by which, for some applications, proximal capsule **1064** is retracted. (The proximal capsule is shown in the inset without the prosthetic valve, for the sake of simplicity.)

[0722] For some applications, and as shown, delivery tool **1020** (e.g., controller **1021** thereof) is configured to retract and/or advance proximal capsule **1064** by translating rotational motion of capsule catheter **1072** into longitudinal motion of the proximal capsule along axis **ax1018**. For this purpose, a disc-assembly **1086**, comprising a proximal disc **1080** that is rotationally coupled to and rotationally movable with respect to a distal disc **1082**, is typically fitted within proximal capsule **1064**. Further typically, the exterior of proximal disc **1080** defines external screw threading that is complementary to internal screw threading **1089** defined by the interior of proximal capsule **1064**. Further typically for such applications, and as shown, the proximal capsule **1064** is shaped to define a longitudinal track **1088** that traverses internal threading **1089**.

[0723] Since proximal disc **1080** is fixedly coupled to capsule catheter **1072**, rotation of capsule catheter **1072** with respect to shaft **1034** (e.g., via controller **1021**) screws the proximal disc along internal threading **1089** of proximal capsule **1064**. At the same time, distal disc **1082** is inhibited from rotating because distal disc **1082**: (i) is fixedly coupled to shaft **1034**, and (2) engages track **1088** of proximal capsule **1064** (e.g., by locking pin **1084** having been fitted into the track). Thus, screwing of proximal disc **1080** pushes distal disc **1082** along track **1088**, thereby translating rotational movement of the proximal disc into axial movement **1078** (e.g., retraction) of proximal capsule **1064** with respect to disc-assembly **1086**, as well as to mount **1028**.

[0724] FIG. **14E** shows continued deployment of prosthetic valve **1036** at tricuspid valve **1096**. Distal portion **1024** has been retracted as a whole, relative to tissue of heart **1090**, and to delivery catheter **1050**. In this way, flanges **1054** (e.g., end-portions **1068** thereof) now engage tissue (e.g., leaflets) of tricuspid valve **1096**.

[0725] In the following deployment step shown in FIG. **14F**, proximal capsule **1064** has been further retracted, thereby releasing upstream support portion **1040** from the proximal capsule, such that the upstream support portion expands radially outward. As shown, proximal capsule **1064** is retracted proximally with respect to capsule catheter **1072** and delivery catheter **1050**. Proximal capsule **1064** therefore typically has an internal diameter that is large enough to allow for the proximal capsule to be retracted over capsule catheter **1072** and/or delivery catheter **1050**.

[0726] Similarly to outer frame **1060**, valve frame **1030** also typically comprises a shape-memory material, such that upstream support portion **1040** expands automatically upon release from proximal capsule **1064**. In this way, tissue of tricuspid valve **1096** is squeezed between upstream support portion **1040** and the flanges **1054**.

[0727] FIG. **14G** shows distal capsule **1066** having been advanced with respect to mount **1028**, such that distal portion **1024** assumes the deployment state described hereinabove in reference to FIG. **10D**. Release of mount **1028** and tubular portion **1032** from the distal capsule allows tubular portion **1032** to automatically expand radially outward, such that frame assembly **1022** (and therefore prosthetic valve **1036** as a whole) has assumed its expanded state.

[0728] Once prosthetic valve 1036 is fully expanded at tricuspid valve 1096, it is desirable to withdraw distal portion 1024 from heart 1090. In order to reduce a likelihood of distal capsule 1066 (e.g., open end 1067 thereof) undesirably engaging valve 1036 (e.g., leaflets thereof) during upstream retraction through lumen 1038, proximal capsule 1064 is first advanced downstream through lumen 1038, thereby closing inter-capsule gap 1071b (e.g., such that open end 1065 of the proximal capsule abuts open end 1067 of the distal capsule), as shown in FIGS. 10E and 14I. As described hereinabove in reference to FIG. 10D, slidable coupling of segments 1034c and 1034d facilitates closure of gap 1071b. For some applications, and as shown in FIG. 14H, distal capsule 1066 is partially retracted (e.g., such that the distal capsule again houses mount 1028), to facilitate closure of inter-capsule gap 1071b.

[0729] Alternatively or in addition to closing inter-capsule gap 1071b, withdrawal of distal portion 1024 from heart 1090 may be facilitated by guidewire 1073 being m-advanced into distal end-portion 1027 of nosecone 1026, either prior to or during retraction of distal portion 1024 through lumen 1038 of tubular portion 1032. Readvancing guidewire 1073 into distal end-portion 1027 typically straightens the distal end-portion, as described hereinabove in reference to FIGS. 13B and 14A. It is hypothesized by the inventors that the straightening of nosecone 1026 may facilitate withdrawal of distal portion 1024 in a retrograde direction through prosthetic valve 1036 (i.e., against the direction for which prosthetic leaflets 1058 are configured to allow blood flow through prosthetic valve 1036), e.g., by reducing a likelihood of the nosecone ensnaring the prosthetic valve compared to when the nosecone is curled.

[0730] FIG. 14J shows the subsequent retraction of distal portion 1024 through lumen 1038 of tubular portion 1032. It is again noted that capsule assembly-length 1074c (FIG. 10E), while distal portion 1024 assumes the withdrawal state, is less than capsule assembly-length 1074a of the distal portion in the delivery state (FIG. 10C). It is therefore hypothesized by the inventors that closing the inter-capsule gap 1071b facilitates transluminal, removal of delivery tool 1020 from the heart.

[0731] Reference is made to FIGS. 15A-B, which are schematic illustrations showing a delivery tool 2020, in accordance with some applications of the invention.

[0732] Except where noted, delivery tool 2020 is typically identical to delivery tool 1020 described hereinabove with reference to FIGS. 10A-B, and is used similarly to the use of delivery tool 1020, mutatis mutandis. Components that are identically named between the systems typically share similar features and serve similar functions as each other. As such, the description below of delivery tool 2020 focuses upon features that are particular to delivery tool 2020.

[0733] FIG. 15A shows delivery tool 2020 assembled, and FIG. 15B shows an exploded view of a distal portion 2024 of the delivery tool. As shown, delivery tool 2020 comprises an extracorporeal controller 2021 and a distal portion 2024 that is dimensioned for transluminal (e.g., transfemoral) delivery to a subject.

[0734] As shown, distal portion 2024 comprises a tubular shaft 2034 (e.g., extending distally from within a capsule catheter 2072) to which a proximal capsule 2064 and a distal capsule 2066 (collectively defining a capsule assembly 2063) are coupled. For some applications, and in contrast to

shaft 1034, shaft 2034 does not necessarily comprise segments that are distinguishable by their relative rigidity.

[0735] As shown, each capsule 2064, 2066 has a respective open end 2065, 2067, such that open end 2065 of proximal capsule 2064 faces open end 2067 of distal capsule 2066. Typically, capsules 2064, 2066 are axially moveable with respect to the shaft (e.g., along a central longitudinal axis ax2018), via extracorporeal controller 2021. For some applications, proximal capsule 2064 and/or distal capsule 2066 can be moved both distally (“advanced”) and proximally (“retracted”), with respect to shaft 2034.

[0736] For some applications, and as shown, distal portion 2024 further comprises a mount 2028 that surrounds shaft 2034 and that is dimensioned (e.g., defining slots 2029) to engage an implant. For some applications, distal capsule 2066 is shaped to define an opening (e.g., a window) 2110 that facilitates use of delivery capsule assembly 2063 with an implant (e.g., by allowing a user to visualize mount 2028 and/or a portion of the implant), as described hereinbelow with reference to FIGS. 16E-G.

[0737] Typically, and as shown in FIG. 15B, distal portion 2024 comprises a rod 2168 having a distal portion that extends out of a distal end of shaft 2034. Similarly to as described hereinabove with reference to delivery tool 1020, delivery tool 2020 is configured to distally advance distal capsule 2066 with respect to mount 2028 by screwing the rod through shaft 2034.

[0738] For some applications, and further similarly to delivery tool 1020, distal capsule 2066 is rotationally moveable with respect to rod 2168, such that rotation of rod 2168 does not necessarily rotate distal capsule 2066. Typically for such applications, and as shown, pins 2170 are fitted within distal capsule 2066, into a recess 2169 defined by rod 2168, so as to axially fix the distal capsule with relation to the rod, while allowing rotation of the rod with respect to the pins, as described hereinabove.

[0739] In contrast to delivery tool 1020, and as shown, delivery tool 2020 comprises a delivery stent 2200 that is fixedly coupled to shaft 2034. Typically, delivery stent 2200 comprises a shape-memory material, such that when an implant is crimped over the delivery stent and shaft 2034 (as described hereinbelow with reference to FIGS. 16E-I), the delivery stent assumes a compressed state within the implant. Delivery stent 2200 is shown in FIGS. 15A-B in an expanded state, without an implant.

[0740] Reference is made to FIGS. 16A-I, which are schematic illustrations showing some steps of loading a prosthetic valve 2036 onto a distal portion 2024 of a delivery tool 600, in accordance with some applications of the invention.

[0741] Except where noted, delivery tool 600 is in many ways similar to delivery tool 100 described hereinabove with reference to FIGS. 8A-G, and is used similarly as delivery tool 100, mutatis mutandis. Components that are identically named between the systems typically share similar features and serve similar functions as each other. As such, the description below of delivery tool 600 focuses upon features that are particular to delivery tool 600.

[0742] As shown in FIG. 16A, delivery tool 600 is a multi-catheter transluminal (e.g., transfemoral) delivery tool, comprising two primary components: a catheter system 610, and an implantation instrument 660 at a proximal portion 604 of the delivery tool. Similarly to proximal part 104 of delivery tool 100 described hereinabove, implanta-

tion instrument **660** may be considered to be an extracorporeal control system of delivery tool **600**, and distal portion **663** is configured to be advanced into the subject.

[0743] Catheter system **610** comprises an outer catheter **622** coupled at a proximal end thereof to implantation instrument **660**. Instrument **660** comprises a plurality of tubular members that extend distally from proximal portion **604**, which are coaxial about a central longitudinal axis **ax1** of delivery tool **60**, and which are discussed in more detail hereinbelow. The outermost of these tubular members is typically a delivery catheter **2050** that extends distally from proximal portion **604**, through outer catheter **622**, out of an open distal end of catheter **622**.

[0744] Typically, and as shown, capsule catheter **2072** extends distally through delivery catheter **2050**, to proximal capsule **2064** of capsule assembly **2063**. As described hereinbelow with reference to FIGS. 17A-B, capsule assembly **2063** is used to encase prosthetic valve **2036** during advancement toward the heart. Catheter system **610** further comprises an alignment mechanism **2300** that is used to align proximal and distal capsules **2064**, **2066** during advancement to and/or withdrawal from the heart, as described hereinbelow.

[0745] FIGS. 16B-D show attachment of an accessory, e.g., distal-capsule ensheathing tool **540**, to ensheath a downstream end of prosthetic valve **2036** in distal capsule **2066**. For example, distal-capsule ensheathing tool **540** comprises a clip **244** and a knob **250**. Clip **244** is shaped to define a detent **242**, a portion of which is extended within a detent-hole **2080**. By extending detent **242** through detent-hole **2080**, the detent occupies at least a portion of a recess **2178** defined by a ring **2174** (FIGS. 15A-B) that is fixedly coupled to rod **2168**. In this way, distal capsule **2066** is rotationally locked with respect to rod **2168**. Knob **250** is typically attached over clip **244**, to facilitate manual rotation of the clip, and therefore of distal capsule **2066** and rod **2168**, with respect to shaft **2034**. As described hereinabove with reference to delivery tool **100**, rotation of the rod with respect to the shaft screws the rod into the shaft, resulting in linear (e.g., proximal) movement of distal capsule **2066** with respect to the shaft and to prosthetic valve **2036**.

[0746] As shown in FIGS. 16E-G, prosthetic valve **2036** is then compressed (“crimped”) using a crimping tool, around a distal portion of shaft **2034**, such that the downstream end of the prosthetic valve engages mount **2028**, e.g., with adaptors **2022** being received by respective slots **2029** (FIG. 16E). As prosthetic valve **2036** is crimped (or subsequently thereto), an ensheathing force is applied to distal-capsule ensheathing tool **540**, e.g., by rotating knob **250** that is directly coupled to distal capsule **2066**.

[0747] Direct application of the ensheathing force to distal capsule **2066** may be desirable over applying the ensheathing force using implantation instrument **660** over the length of catheter system **610**, since direct application of the ensheathing force typically avoids resistance that may be encountered over a length of the catheter system.

[0748] For some applications, and as shown, distal capsule **2066** defines an opening (e.g., window **2110**). As shown in FIGS. 16E-G, a user may use the opening to monitor proximal advancement of distal capsule **2066** over mount **2028** and prosthetic valve **2036**. It is hypothesized by the inventors that use of window **2110** to visualize the portion of prosthetic valve **2036** (e.g., a downstream end of the prosthetic valve) that is ensheathed by distal capsule **2066**

increases the reliability of delivery tool **600**, e.g., by reducing a risk of premature release of the prosthetic valve from the distal capsule that may be caused by the user estimating which portion of the prosthetic valve is ensheathed within the distal capsule. Instead, window **2110** allows the user to monitor the ensheathed portion of prosthetic valve **2036**. For example, and as shown, distal capsule **2066** is advanced over prosthetic valve **2036** until a portion of mount **2028** and/or the prosthetic valve (e.g., adaptors **2022** thereof) are visible through window **2110**. In this way, ensheathing prosthetic valve **2036** by distal capsule **2066** serves to maintain coupling between the prosthetic valve and mount **2028** by ensuring that: (i) the mount surrounds adaptors **2022**, and (ii) each adaptor remains within a respective slot **2029** of the mount.

[0749] For some applications, and as shown, after ensheathing the downstream end of prosthetic valve **2036** in distal capsule **2066**, the upstream end of the prosthetic valve is ensheathed in proximal capsule **2064**. The crimping tool is typically used to compress the proximal portion of prosthetic valve **2036**, such that the proximal portion assumes the compressed state, as shown in FIG. 16H.

[0750] For some applications, a second ensheathing force is applied directly to distal portion **2024** of delivery tool **600**. Similarly to as described hereinabove with reference to ensheathing the distal end of prosthetic valve **2036** within distal capsule **2066**, application of the ensheathing force directly to distal portion **2024** may be desirable over applying the ensheathing force along the length of catheter system **610**, to avoid resistance that may be encountered over the length of the catheter system.

[0751] For some applications, a second accessory, e.g., a proximal-capsule ensheathing tool such as cuff **700**, is attached to distal portion **2024** (FIG. 16I), for applying the second ensheathing force directly to distal portion **2024**. For some applications, and as shown, cuff **700** comprises a user grip **708** that is shaped to facilitate rotation of the cuff with respect to distal portion **2024**. For example, and as shown, cuff **700** may be directly coupled to a proximal disc **2092** of a disc assembly **2086** of distal portion **2024**, in order to convert rotational movement of cuff **700** into axial motion of proximal capsule **2064** over a proximal portion of prosthetic valve **2036**.

[0752] Typically for such applications, and as shown, cuff **700** further comprises a distal coupling portion **706** that is configured to reversibly couple the cuff to proximal disc **2092** of disc assembly **2086**. For example, and as shown, distal coupling portion **706** comprises one or more pins **702** shaped so as to fit within respective holes **704** defined by the proximal disc **2092**. Alternatively or in addition, distal coupling portion **706** is sized to fit within an opening in proximal disc **2092**.

[0753] Similarly to disc assembly **1086** (FIG. 14I3), disc assembly **2086** (FIG. 15B) comprises (i) proximal disc **2092** that defines external screw threading that is complementary to internal screw threading defined by the interior of proximal capsule **2064**, and (ii) a distal disc **2090** that is rotationally coupled to and rotationally movable with respect to the proximal disc. Distal disc **2090** is inhibited from rotating because the distal disc: (i) is fixedly coupled to shaft **2034**, and (2) engages track **2088** of proximal capsule **2064** (e.g., by a locking pin **2084** having been fitted into the track). Thus, screwing of proximal disc **2092** using cuff **700** pushes distal disc **2090** and locking pin **2084** along track **2088**,

translating rotational movement of the proximal disc into advancement of proximal capsule 2064 with respect to shaft 2034 and over the proximal portion of prosthetic valve 2036 (FIGS. 16H-I).

[0754] Reference is made to FIGS. 16J-K, which are schematic illustrations showing advancement, of an alignment mechanism 2300 over catheter system 610 of delivery tool 600, in accordance with some applications of the invention.

[0755] Typically, and as shown, alignment mechanism 2300 comprises a supplemental tube 2310 that is coupled to a distal end of an elongate oversheath 2320 at connecting portion 2318. Oversheath 2320 is shaped so as to define an elongate-oversheath lumen, through which catheter system 610 (e.g., capsule catheter 2072 thereof) is slidably passed, and supplemental tube 2310 is shaped so as to define a supplemental-tube lumen that is sized for encasing at least a portion of a housing (e.g., capsule assembly 2063) during transluminal delivery of distal portion 663 of the delivery tool to the heart, and while retracting the housing out of a body of the subject, as described in greater detail hereinbelow.

[0756] Reference is made to FIGS. 17A-B, which are schematic illustrations showing use of delivery tool 600 to advance prosthetic valve 2036 toward heart 1090 of a subject, in accordance with some applications of the invention.

[0757] FIGS. 17A-13 show an operator using implantation instrument 660 to transfemorally advance distal portion 663 of delivery tool 600 to inferior vena cava 1092, toward heart 1090. As shown, supplementary tube 2310 of alignment mechanism 2300 encases a portion of capsule assembly 2063 (e.g., proximal capsule 2064). For example, and as shown, supplementary tube 2310 may abut distal capsule 2066 while distal portion 663 is advanced toward the heart. Alternatively, supplementary tube 2310 may encase a portion of distal capsule 2066 during the advancement.

[0758] Typically, while distal portion 2024 is in a delivery state (FIG. 18A), an upstream portion of prosthetic valve 2036 is ensheathed by proximal capsule 2064 and a downstream portion of the prosthetic valve is ensheathed by distal capsule 2066, such that an exposed segment 2056 of prosthetic valve 2036 is disposed at an inter-capsule gap that separates open end 2065 of the proximal capsule from open end 2067 of distal capsule 2066.

[0759] The upper inset of FIG. 17A shows distal portion 2024 of delivery tool 600 in a delivery state in which an alignment tube 2314 of alignment mechanism 2300 is disposed between oversheath 2320 and capsule catheter 2072 of catheter system 610. While distal portion 2024 is in the delivery state, an aligner 2312 of alignment mechanism 2300 is typically disposed between alignment tube 2314 and supplemental tube 2310. For some applications, and as shown, aligner 2312 is shaped to form a ring that fits around alignment tube 2314. Aligner 2312 typically comprises a material that is stiffer than capsule catheter 2072. For example, aligner 2312 may comprise a metal or a polycarbonate.

[0760] Typically, and as shown, by occupying a space between capsule catheter 2072 and supplemental tube 2310, aligner 2312 is positioned to align the capsule catheter with respect to the supplemental tube (e.g., the aligner keeps a distal portion of the capsule catheter generally parallel with

the supplemental tube). For some applications, and as shown, aligner 2312 is coupled to a distal end of alignment tube 2314.

[0761] For some applications, aligner 2312 and a distal portion of alignment tube 2314 are axially slidable along the catheters (e.g., along capsule catheter 2072) of catheter system 610 (e.g., independently of supplemental tube 2310). For some such applications, aligner 2312 and the distal portion of alignment tube 2314 are axially slidable within the elongate-oversheath lumen and within the supplemental-tube lumen. For example, aligner 2312 may be advanced distally to align capsule catheter 2072 with respect to supplemental tube 2310 before capsule assembly 2063 is encased within the supplemental tube.

[0762] Elements comprising catheter system 610 and alignment mechanism 2300 are typically dimensioned in order to facilitate sliding aligner 2312 between the capsule catheter 2072 and supplemental tube 2310. Therefore, for some applications, an inner diameter di2312 of aligner 2312 is 0.05-3.0 mm, e.g., 0.15 mm, larger than an outer diameter do2072 of capsule catheter 2072 (i.e., a largest catheter of catheter system 610 that passes through supplemental tube 2310 and through aligner 2312), and/or an alignment-tube outer diameter do2314 of alignment tube 2314 is 1.9-5.5 mm, e.g., 4.9 mm, smaller than a supplemental-tube inner diameter di2310 of supplemental tube 2310. An outer diameter of capsule catheter 2072 is about 6.7 mm and an inner diameter of aligner 2312 is about 6.8 mm by way of illustration and not limitation. A length of aligner 2312 is typically between 1-10 mm, e.g., 8 mm.

[0763] Typically, supplemental tube 2310 comprises material that is stiffer than elongate oversheath 2320 of capsule catheter 2072. Alignment-tube outer diameter do2314 of alignment tube 2314 is 1.9-2.4 mm smaller than an inner diameter of elongate oversheath 2320.

[0764] Typically for such applications, and as shown, a supplemental-tube outer diameter do2310 is larger than both (i) outer diameter do2072 of capsule catheter 2072, and (ii) an outer diameter do2320 of elongate oversheath 2320.

[0765] For some applications, during entry of delivery tool 600 within the body, supplemental tube 2310 surrounds proximal capsule 2064 and at least a proximal portion of distal capsule 2066, as well as exposed segment 2056 of prosthetic valve 2036. Typically for such applications, subsequently to entering the body, proximal capsule 2064 and the proximal portion of distal capsule 2066 are exposed from within supplemental tube 2310 (FIG. 17B) and are advanced toward the heart by distally advancing capsule catheter 2072 with respect to oversheath 2320 (e.g., by pushing capsule catheter 2072 distally while retaining oversheath 2320 in place and/or by retracting the oversheath proximally with respect to capsule catheter 2072).

[0766] Reference is made to FIGS. 19A-C, which are schematic illustrations showing a delivery tool 36(x) that has an alignment mechanism 3300, in accordance with some applications of the invention.

[0767] Delivery tool 3600 is in many ways identical to delivery tool 600, and is used in a similar way to delivery tool 600, mutatis mutandis. Components bearing identical reference numerals are typically interchangeable between delivery tools 600 and 3600. For example, delivery tool 3600 comprises implantation instrument 660, and capsule assembly 2063 at distal portion 2024 of the delivery tool that is used to ensheath prosthetic valve 2036. Components that

are identically named between the systems typically share similar features and serve similar functions as each other. As such, the description below of delivery tool **3600** focuses upon features that are particular to delivery tool **36(x)**.

[0768] Similarly to alignment mechanism **2300**, alignment mechanism **3300** is used to align proximal and distal capsules **2064**, **2066** during advancement and/or retraction of delivery tool **3600**, as described hereinbelow. Further similarly to alignment mechanism **23M**, alignment mechanism **3300** comprises a supplemental tube **3310** that is used to encase a portion of capsule assembly **2063** (FIG. 19B). As shown, and as described hereinabove regarding delivery tool **600**, while distal portion **2024** of delivery tool **3600** is in a delivery state, an aligner **3312** of alignment mechanism **3300** is typically disposed between an alignment tube **3314** and supplemental tube **3310**, to align capsule catheter **3072** with respect to the supplemental tube. Further similarly to alignment mechanism **2300** of tool **600**, proximal capsule **2064** and the proximal portion of distal capsule **2066** may be exposed from within supplemental tube **3310** (FIG. 19C) and advanced toward the heart (not shown).

[0769] Delivery tool **3600** comprises an aligner locking mechanism **3400** that is operatively connected to alignment mechanism **3300**. For some applications, and as shown, locking mechanism **3400** comprises a sliding lock **3440** that is: i) longitudinally fixed with respect to aligner **3312**, and ii) slidable along a housing **3420** of the locking mechanism **3400**, and with respect to supplemental tube **3310**. For example, and as shown, locking mechanism **3400** may be fixedly connected to a proximal end of alignment tube **3314**, and aligner **3312** may be fixedly connected to a distal end of the alignment tube. In this way, longitudinal movement (e.g., along a proximal-to-distal axis) of sliding lock **3440** results in similar longitudinal movement of aligner **3312**. Therefore, sliding lock **3440** from a proximal position (e.g., abutting a proximal endcap **3412**, FIG. 19B) to a distal position (e.g., abutting a distal endcap **3410**, FIG. 19C) causes aligner **3312** to slide distally within supplemental tube **3310** from a proximal position (FIG. 19B) to a distal position (FIG. 19C).

[0770] In some cases, it may be desirable for aligner **3312** to remain in the distal position while advancing and/or withdrawing distal portion **2024**. Therefore, alignment mechanism **3300** typically has a locked state (FIG. 19C) in which the aligner is fixedly disposed within the supplemental tube, as well as an unlocked state (FIG. 19B) in which aligner **3312** is longitudinally slidable within supplemental tube **3310**. For some applications, the distal portion of alignment tube **3314** transitions together with aligner **3312** between the unlocked state and the locked state.

[0771] For example, and as shown, aligner **3312** may be transitioned into the locked state (e.g., “locked”) by sliding lock **3440** past a catch **3430** that protrudes from housing **3320** of locking mechanism **3400**. In this way, lock **3440** is held snugly between distal endcap **3410** and catch **3430**.

[0772] Reference is made to FIGS. 18A-O and to FIGS. 19D-T, which are schematic illustrations showing use of delivery tools **600**, **3600** to deploy prosthetic valve **2036** at tricuspid valve **1096** of the heart, and use of alignment mechanism **2300**, **3300** to facilitate withdrawal of the delivery tool from the subject, in accordance with some applications of the invention.

[0773] Typically, and as shown, capsule catheter **2072**, **3072** and capsule assembly **2063** encasing prosthetic valve

2036 are advanced along a guidewire **2023** through inferior vena cava **1092** and into right atrium **1094** of the heart. Further typically, prosthetic valve **1036** remains ensheathed at least until distal capsule **2066** is advanced into right ventricle **1098** of the heart (FIG. 18A, 19D).

[0774] Similarly to prosthetic valve **1036** described hereinabove with reference to FIG. 11, prosthetic valve **2036** typically comprises a tubular portion **2032** in which a plurality of prosthetic leaflets are disposed, and that defines a lumen between an upstream end and a downstream end. For some applications, and as shown in FIG. 18A, tubular portion **2032** and an upstream support portion **2040** together define a valve frame **2030**.

[0775] Typically for such applications, and as shown, a plurality of flanges **2054** are coupled to tubular portion **2032** at coupling points that are downstream of the upstream support portion. As shown, prosthetic valve **2036** is engaged with delivery tool **600**, **3600** such that a downstream end of tubular portion **2032** is disposed within distal capsule **2066**, and upstream support portion **2040** and such that end-portions **2068** of flanges **2054** are disposed within proximal capsule **2064**.

[0776] FIGS. 18H and 19E show capsule assembly **2063** having been advanced further distally, such that exposed segment **2056** is partially disposed in right atrium **1094**, and partially disposed in right ventricle **1098**. Typically, and as shown, at least a portion of flanges **2054** (e.g., end-portions **2068** thereof) are still ensheathed within proximal capsule **2064** at this stage.

[0777] FIGS. 18C and 19F show capsule assembly **2063** after guidewire **2023** has been proximally withdrawn from distal end-portion **2027** of nosecone **2026**. As described hereinabove with reference to FIGS. 13A-B, withdrawal of guidewire **2023** from distal end-portion **2027** reduces an axial length of nosecone **2026**, facilitating deployment of prosthetic valve **2036** by reducing an amount of space within right ventricle **1098** required to maneuver capsule assembly **2063** (e.g., distal capsule **2066** thereof).

[0778] FIGS. 18D and 19G show proximal capsule **2064** having been partially retracted with respect to mount **2028**, such that end-portions **2068** of flanges **2054** are released from the proximal capsule. Typically, an unsheathing force is applied extracorporeally to a controller, e.g., a knob or a dial of implantation instrument **660** in order to retract proximal capsule **2064**. As shown, flanges **2054** typically automatically expand radially outward from respective coupling points upon release from proximal capsule **2064**. However, since the distal end of tubular portion **2032** is still restrained by distal capsule **2066**, and upstream support portion **2040** is still restrained by proximal capsule **2064**, valve frame **2030** remains in the compressed state.

[0779] FIGS. 18E and 19H show distal portion **2024** having been advanced distally, such that flanges **2054** enter right ventricle **1098** (e.g., such that the flanges reach a point distal of leaflets **12** of tricuspid valve **1096**). FIGS. 18F and 19I show the distal portion having been retracted as a whole, relative to tricuspid valve **1096**, such that flanges **2054** (e.g., end-portions **2068** thereof) engage tissue (e.g., the leaflets) of the tricuspid valve.

[0780] Subsequently, proximal capsule **2064** is further retracted with respect to mount **2028**, such that upstream support portion **2040** is unsheathed from the proximal capsule (FIGS. 18G, 19J), and distal capsule **2066** is advanced with respect to mount **2028**, such that tubular

portion **2032** is allowed to expand radially outward, such that prosthetic valve **2036** assumes its expanded state. Typically, the unsheathing force is applied extracorporeally via implantation instrument **660** of delivery tools **600** and **3600**, as described hereinabove with reference to FIG. **14D**, *mutatis mutandis*.

[0781] As shown in FIGS. **18H** and **19K**, distal movement of distal capsule **2066** with respect to adaptors **2022** and/or mount **2028** (e.g., to a point that is further distal from the mount **2028**, or at least to a point that is further distal from slots **2029** thereof) releases the downstream end of tubular portion **2032** from within distal capsule **2066**. As shown, tubular portion **2032** radially expands, allowing delivery stent **2200** to expand to the expanded state.

[0782] FIGS. **18I** and **19L** show proximal capsule **2064** having advanced distally such that open end **2067** of the proximal capsule meets delivery stent **2200**. FIGS. **18J** and **19M** show distal capsule **2066** having advanced proximally such that open end **2065** of the distal capsule meets delivery stent **2200**. Typically, delivery stent **2200** is configured to fit snugly between proximal and distal capsules **2064**, **2066**, in order to facilitate smooth retraction of capsule assembly **2063** (e.g., the distal capsule thereof) through prosthetic valve **2036** (FIG. **18K**, **19N**), with less risk of damaging the prosthetic leaflets that are disposed within tubular portion **2032**. For some applications, delivery stent **2200** comprises a fabric covering (not shown) that further facilitates smooth retraction of capsule assembly **2063** through prosthetic valve **2036**.

[0783] FIGS. **18L-M** and **19O-P** show further retraction of distal portion **2024** of delivery tool **600**, **3600** from within prosthetic valve **2036** and into inferior vena cava **1092**. For some applications, distal portion **2024** (e.g., capsule assembly **2063** thereof) is retracted proximally toward supplemental tube **2310**, **3310** by proximally retracting capsule catheter **2072**, **3072** and/or delivery catheter **2050**.

[0784] For some applications, and as shown in FIG. **18M**, a distal portion of capsule catheter **2072** is configured (e.g., is sufficiently flexible) to assume a curved orientation while the capsule catheter is retracted through the vasculature. As shown in the lower inset of FIG. **18M**, there is sufficient space within the supplemental-tube lumen and capsule catheter **2072** such that a distal portion of the capsule catheter **2072** may assume the curved orientation. That is, supplemental tube **2310** does not necessarily apply an aligning force to the distal portion of capsule catheter **2072**. It is hypothesized by the inventors that fully retracting capsule assembly **2063** into supplemental tube **2310** while the distal portion of capsule catheter **2072** is in the curved orientation may result in an imperfect fit of the capsule assembly into supplemental tube **2310**, which may complicate withdrawal of distal portion **2024** from the body.

[0785] For some applications, to promote better fit of capsule assembly **2063** into supplemental tube **2310**, and to facilitate withdrawal of distal portion **2024** from the body, alignment tube **2314** is positioned to orient aligner **2312** so as to straighten the distal portion of capsule catheter **2072**. FIG. **18N** shows aligner **2312** having moved distally e.g., by pushing alignment tube **2314** proximally and/or by distally retracting capsule catheter **2072**.

[0786] As shown in FIG. **18O**, aligner **2312** applies the aligning force upon capsule catheter **2072**, straightening the distal portion of the capsule catheter. Typically for such applications, straightening the distal portion of capsule

catheter **2072** causes capsule assembly **2063** and/or capsule catheter **2072** to be concentrically disposed with respect to supplemental tube **2310** while the capsule assembly is retracted into the supplemental tube, thereby avoiding entry of capsule assembly **2063** into the supplemental tube at an angle.

[0787] Typically, distal portion **2024** is then extracted from the body while the distal portion is housed within supplemental tube **2310** (e.g., while supplemental tube **2310** surrounds proximal capsule **2064**, and while a distal end of the supplemental tube abuts the distal capsule).

[0788] As described hereinabove, and shown in FIG. **19Q**, catch **3430** protrudes from housing **4320** of locking mechanism **3400**, holding lock **3440** against distal endcap **3410**, and thereby locking aligner **3312** in the distal position. It is hypothesized by the inventors that locking aligner **3312** in the distal position while withdrawing distal portion **2024** from the body may simplify operation of delivery tool **3600**, e.g., by obviating manual application of force in order to keep aligner **3312** in the distal position.

[0789] As described hereinabove with reference to FIG. **18O**, at least a portion of capsule assembly **2063** is typically encased within supplemental tube **3310** while distal portion **2024** is withdrawn from the body. In order to allow for capsule assembly **2063** to enter supplemental tube **3310**, aligner **3312** must typically be slid distally. Alignment mechanism **3300** must therefore transition e.g., be “unlocked”) from the locked state to the unlocked state.

[0790] The embodiment of alignment mechanism **3300** shown in FIG. **19R** is unlocked by rotating lock **3440** of locking mechanism **3400** from state “A” to state “B.” As shown, lock **3440** is rotated such that a groove **3444** defined by the wall of the lock is aligned with catch **3430**. In this way, lock **3440** is no longer held snugly between catch **3430** and distal endcap **3410**, such that the lock and aligner **3312** are allowed to move longitudinally in response to a force (e.g., in response to a pulling force being applied to the lock).

[0791] For some such applications, and as shown, rotation of lock **3440** to unlock alignment mechanism **3300** is facilitated by a pin **3450**, that is fixedly coupled to alignment tube **3314**, being disposed within a slot **3442** that is defined by the lock. As shown, slot **3442** is dimensioned such that rotation of lock **3440** in alternate directions, such that pin **3450** contacts opposite ends of the slot, reversibly locks or unlocks alignment mechanism **3300**.

[0792] Rotation of lock **3440** into state “B.” such that groove **3444** is aligned with catch **3430**, facilitates movement of pin **3450** along a longitudinal track **3422**, such that lock **3440** and aligner **3312** each occupy their respective proximal positions FIG. **19S**). In this way, aligner **3312** aligns capsule catheter **3072** as capsule assembly **2063** is retracted into supplemental tube **3310**, and distal portion **2024** may be extracted from the body.

[0793] Reference is made to FIGS. **20A-F**, which are schematic illustrations showing a delivery tool **4020** being used to deploy prosthetic valve **1036** at tricuspid valve **1096** of the heart, in accordance with some applications of the invention.

[0794] Except where noted, delivery tool **4020** is typically identical to delivery tool **1020** described hereinabove, and is used similarly to delivery tool **1020**, *mutatis mutandis*. Components that are identically named between the systems typically share similar features and serve similar functions

as each other, and components bearing identical reference numerals are typically interchangeable between delivery tools **1020** and **4040**. As such, the description below of delivery tool **4020** focuses upon features that are particular to delivery tool **4020**.

[0795] FIG. 20A shows delivery tool **4020** within the heart, at a stage of deployment of prosthetic valve **1036** comparable to that described hereinabove with reference to delivery tool **1020** in FIG. 14C. In contrast to tool **1020**, the proximal capsule of delivery tool **4020** comprises a proximal capsule assembly **4064** that has an inner proximal capsule **4064a** and an outer proximal capsule **4064b**. Typically, outer proximal capsule **4064b** is longitudinally movable in relation to inner proximal capsule **4064a**. For some applications, and as shown, inner proximal capsule **4064a** fits snugly within outer proximal capsule **4064b**.

[0796] For some applications, and as shown in FIG. 20B, proximal capsule assembly **4064** defines an interior threading **4089** similar to interior threading **1089** that is defined by proximal capsule **1064** of delivery tool **1020**. For some such applications, and as shown in the insets of FIG. 201B a proximal portion **4089p** of interior threading **4089** is defined by inner proximal capsule **4064a**, and a distal portion of the interior threading is defined by outer proximal capsule **4064b**. Typically for such applications, both proximal portion **4089p** and distal portion **4089a** of internal threading **4089** are dimensioned to engage the external threading that is defined by proximal disc **4080** of disc assembly **4086**, as the proximal disc is screwed along the internal threading. For example, and as shown, threaded insets **4070** of outer proximal capsule **4064b** fit within windows defined by inner proximal capsule **4064a** so as to engage proximal disc **4080** (FIG. 20C).

[0797] Similarly to as described hereinabove regarding delivery tool **1020** with reference to FIG. 14D, rotation of capsule catheter **4072** with respect to shaft **4034** screws proximal disc **4080** along internal threading **4089** of proximal capsule assembly **4064**. At the same time, distal disc **4082** is inhibited from rotating because distal disc **4082**: (i) is fixedly coupled to shaft **4034**, and (2) engages track **4088** of proximal capsule assembly **4064** (e.g., by locking pin **4084** having been fitted into the track). Thus, screwing of proximal disc **4080** pushes distal disc **4082** along track **4088**, thereby translating rotational movement of the proximal disc into axial movement (e.g., retraction) of proximal capsule assembly **4064** with respect to disc-assembly **4086**, as well as to prosthetic valve **1036**.

[0798] As shown in FIG. 20C, rotation of capsule catheter **4072** screws proximal disc **4080** along internal threading **4089**, causing proximal capsule assembly **4064** to retract proximally with respect to disc-assembly **4086**, which releases flange end-portions **1068** from the proximal capsule assembly, as described hereinabove with reference to FIG. 14D. As shown, proximal capsule assembly **4064** is retracted proximally with respect to capsule catheter **4072** and delivery catheter **4050**. Proximal capsule assembly **4064** therefore typically has an internal diameter that is large enough to allow for the proximal capsule assembly to be retracted over capsule catheter **4072** and/or delivery catheter **4050**.

[0799] FIG. 20D shows continued deployment of prosthetic valve **1036** at tricuspid valve **1096**, in which distal portion **4024** has been retracted as a whole, such that flanges **1054** (e.g., end-portions **1068** thereof) now engage tissue (e.g., leaflets) of tricuspid valve **1096**.

[0800] In FIG. 20E, further screwing of proximal disc **4080** along internal threading **4089** causes outer proximal capsule **4064b** to retract proximally with respect to inner proximal capsule **4064a**, disc-assembly **4086** and prosthetic valve **1036**. As shown in the inset of FIG. 20E, outer proximal capsule **4064b** is retracted over inner proximal capsule **4064a** as proximal disc **4080** advances from (a) proximal portion **4089p** of internal threading **4089** that is defined by the inner proximal capsule, to (b) distal portion **4089a** of the internal threading that is defined by the outer proximal capsule.

[0801] Retraction of outer proximal capsule **4064b** over inner proximal capsule **4064a** causes proximal capsule assembly **4064** to have a shorter length after releasing upstream support portion **1040** than prior to releasing the upstream support. It is hypothesized by the inventors that this shortening of proximal capsule assembly **4064** during deployment of prosthetic valve **1036** may facilitate deployment, inter alia since a shorter proximal capsule assembly may be more easily retracted over a bent portion of capsule catheter **4072** and/or delivery catheter **4050**.

[0802] As shown, retraction of outer proximal capsule **4064b** releases upstream support portion **1040**, such that the upstream support portion expands radially outward. FIG. 20F shows distal capsule **4066** having been advanced with respect to mount **4028**, such that distal portion **4024** assumes the deployment state described hereinabove in reference to FIGS. 10D and 14G. Release of mount **4028** and tubular portion **1032** from the distal capsule allows the tubular portion to automatically expanded radially outward, such that frame assembly **1022** (and therefore prosthetic valve **1036** as a whole) has assumed its expanded state.

[0803] It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

1-68. (canceled)

69. Apparatus for use at a heart of a subject, the apparatus comprising:

- a delivery tool dimensioned for percutaneous delivery to the heart, the delivery tool having a distal portion that defines a distal portion axis and comprises:

- a shaft; and

- a proximal capsule and a distal capsule, each of the capsules:

- having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule, and

- being coupled to the shaft in a manner that facilitates axial movement of the capsule with respect to the shaft, along the distal portion axis; and

- a prosthetic valve comprising:

- a tubular portion that defines a lumen;

- a plurality of prosthetic leaflets disposed within the lumen;

- an upstream support portion that extends from the tubular portion; and

- a plurality of flanges, each of the flanges:

coupled to the tubular portion at a respective coupling point that is downstream of the upstream support portion, and

extending from the coupling point to a respective flange end-portion of the flange;

wherein the prosthetic valve is restrainable in a compressed state by the delivery tool, such that:

the shaft and a downstream end of the tubular portion are disposed within the distal capsule, and

the upstream support portion and the flange end-portions are disposed within the proximal capsule.

70. The apparatus according to claim **69**, wherein the distal capsule is coupled to the shaft in a manner that facilitates proximal and distal movement of the distal capsule, with respect to the shaft.

71. The apparatus according to claim **69**, wherein the proximal and distal capsules are coupled to the shaft in a manner that facilitates proximal and distal movement of the proximal and distal capsules, with respect to the shaft.

72. The apparatus according to claim **69**, wherein the proximal capsule is coupled to the shaft in a manner that facilitates proximal and distal movement of the proximal capsule, with respect to the shaft.

73. The apparatus according to claim **69**, further comprising:

a capsule catheter extending proximally from the distal portion of the delivery tool; and

a disc-assembly coupled to the capsule catheter, the disc-assembly comprising:

a proximal disc fixedly coupled to the capsule catheter, the proximal disc shaped to define external threading, and

a distal disc, the distal disc rotatably coupled to the proximal disc.

74. The apparatus according to claim **73**, wherein the proximal capsule is shaped to define:

a longitudinal track configured to engage the distal disc, and

internal threading, the internal threading: complementary to the external threading, and traversing the longitudinal track;

wherein the disc-assembly is disposed within the proximal capsule such that:

the external threading fits the internal threading, and rotation of the capsule catheter in a first direction facilitates:

rotation of the proximal disc in the first direction, and longitudinal movement of the proximal capsule, with respect to the disc-assembly.

75. The apparatus according to claim **74**, wherein the distal disc comprises a locking pin that is disposed within the longitudinal track.

76-78. (canceled)

79. The apparatus according to claim **69**, wherein the delivery tool is configured such that while the distal portion of the delivery tool is in a delivery state for transluminally delivering the delivery tool to the heart:

the prosthetic valve is restrained in the compressed state by the delivery tool, and

the proximal capsule and the distal capsule are oriented with respect to each other such that:

an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule, and

a segment of the tubular portion is disposed at the inter-capsule gap.

80. The apparatus according to claim **196**, wherein the distal capsule and the proximal capsule are each coupled to the shaft such that the distal capsule is translatable toward the proximal capsule, such that the open end of the proximal capsule and the open end of the distal capsule meet, thereby closing the inter-capsule gap.

81. (canceled)

82. The apparatus according to claim **79**, wherein the delivery tool is configured such that while the distal portion of the delivery tool is in the delivery state, the inter-capsule gap is greater than 5 mm and less than 25 mm in length.

83-89. (canceled)

90. A method for use at a heart of a subject, the method comprising:

transluminally advancing a delivery system to the heart, the delivery system including:

a delivery tool, the delivery tool having a distal portion that defines a distal portion axis, the distal portion comprising a proximal capsule and a distal capsule, each of the capsules having a respective open end, the open end of the proximal capsule facing the open end of the distal capsule; and

an implant comprising:

a proximal-implant portion,

a distal-implant portion, and

a flange having a flange end-portion,

the implant being restrained by the delivery tool such that:

the proximal-implant portion and the flange end-portion are disposed within the proximal capsule, and

the distal-implant portion of the implant is disposed within the distal capsule; and

deploying the implant at the heart by:

proximally retracting the proximal capsule, with respect to the implant, such that the flange end-portion is released from the proximal capsule,

subsequently, proximally retracting the distal portion of the delivery tool, such that the flange end-portion contacts tissue of the heart,

subsequently, further proximally retracting the proximal capsule, with respect to the implant, such that the proximal-implant portion is released from the proximal capsule, and

subsequently, distally advancing the distal capsule, with respect to the implant, such that the distal-implant portion is released from the distal capsule.

91. The method according to claim **90**, wherein:

the step of transluminally advancing comprises transluminally advancing the distal portion of the delivery tool to the heart while the proximal capsule and the distal capsule are aligned along the distal portion axis such that an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule.

92. The method according to claim **91**, wherein the step of transluminally advancing comprises transluminally advancing the distal portion of the delivery tool such that a segment of the implant is disposed at the inter-capsule gap.

93. The method according to claim **92**, wherein:
the delivery tool further includes a flexible sheath, the sheath circumscribing the inter-capsule gap such that the sheath covers the segment of the implant, and
the method further comprises, prior to proximally retracting the proximal capsule with respect to the implant, exposing the segment from the sheath by proximally retracting the sheath.

94. The method according to claim **90**, wherein:
the implant includes a frame, the frame being restrained by the delivery tool such that:
the proximal-implant portion includes a proximal portion of the frame, and
the distal-implant portion includes at a distal portion of the frame; and
the step of further proximally retracting the proximal capsule comprises further proximally retracting the proximal capsule, with respect to the proximal portion of the frame, such that the proximal portion of the frame is released from the proximal capsule; and
the step of distally advancing the distal capsule comprises distally advancing the distal capsule, with respect to the distal portion of the frame, such that the distal portion of the frame is released from the distal capsule.

95. The method according to claim **94**, wherein:
the frame is an inner frame,
the flange is a first flange of a plurality of flanges,
the implant further includes an outer frame, the outer frame defining the plurality of flanges,
each of the flanges:

is coupled to the inner frame at a respective coupling point that is longitudinally between the proximal portion of the inner frame and the distal portion of the inner frame, and

extends from the coupling point to a respective flange end-portion; and

the step of proximally retracting the proximal capsule comprises proximally retracting the proximal capsule, with respect to the implant, such that the respective flange end-portions are released from the proximal capsule; and

the step of proximally retracting the distal portion of the delivery tool comprises proximally retracting the distal portion of the delivery tool, such that the respective flange end-portions contact tissue of the heart.

96. The method according to claim **95**, wherein:
the implant includes a prosthetic valve wherein:
the inner frame has a tubular portion that defines a lumen,
a plurality of prosthetic leaflets are disposed within the lumen,
the proximal portion of the inner frame includes an upstream support portion that extends from the tubular portion, and
the distal portion of the inner frame includes a downstream end of the tubular portion;

the distal portion of the delivery tool further includes a shaft, the proximal capsule and the distal capsule each being coupled to the shaft, and

the step of transluminally advancing comprises:
transluminally delivering the delivery tool to the heart while the delivery tool is in a delivery state in which the prosthetic valve is restrained in a compressed state by the delivery tool, such that:

the tubular portion is engaged with a portion of the shaft,

the portion of the shaft and the downstream end of the tubular portion are restrained within the distal capsule,

the upstream support portion and the flange end-portions are restrained within the proximal capsule,

the proximal capsule and the distal capsule are aligned along the distal portion axis such that an inter-capsule gap separates the open end of the proximal capsule from the open end of the distal capsule, and

a segment of the tubular portion is disposed at the inter-capsule gap; and

transluminally advancing the delivery tool to a ventricle of the heart such that the distal capsule is disposed within the ventricle; and

deploying the implant at the heart comprises deploying the prosthetic valve at a native valve of the heart, such that the prosthetic valve automatically expands from the compressed state to an expanded state, by:

proximally retracting the proximal capsule, with respect to the shaft, such that the respective flange end-portions expand radially outward,

proximally retracting the distal portion of the delivery tool, such that the respective flange end-portions contact tissue of the native valve,

further proximally retracting the proximal capsule, with respect to the shaft, such that:

the upstream support portion is released from the proximal capsule, and expands radially outward, and

tissue of the native valve is squeezed between the upstream support portion and the flange end-portions; and

distally advancing the distal capsule, with respect to the shaft, such that:

the downstream end of the tubular portion is released from the distal capsule, and

the tubular portion expands radially outward.

97. The method according to claim **96**, further comprising:

subsequently to distally advancing the distal capsule with respect to the shaft, withdrawing the delivery tool from the heart, by:

distally advancing the proximal capsule, with respect to the shaft, such that the open end of the proximal capsule abuts the open end of the distal capsule, and
subsequently, proximally retracting the distal portion of the delivery tool, through the lumen of the tubular portion.

98. The method according to claim **97**, further comprising, prior to distally advancing the proximal capsule with respect to the shaft:

proximally retracting the distal capsule, with respect to the shaft.

99. The method according to claim **90**, wherein:

the delivery system further includes a guidewire,

the delivery tool further includes a nosecone having a flexible distal end-portion,

the method further comprises transluminally advancing the guidewire to the heart,

transluminally advancing the delivery system to the heart comprises transluminally advancing the delivery tool along the guidewire, such that the guidewire enters the distal end-portion of the nosecone, and

deploying the implant at the heart comprises, prior to retracting the proximal capsule with respect to the implant, proximally withdrawing the guidewire from within the distal end-portion of the nosecone.

100. The method according to claim **99**, further comprising, subsequently to distally advancing the distal capsule, with respect to the implant, advancing the guidewire into the distal end-portion.

101. The method according to claim **90**, wherein: the distal portion of the delivery tool further includes a shaft, the proximal capsule and the distal capsule each being coupled to the shaft,

the proximal capsule is shaped to define:

a longitudinal track, and
internal threading, the internal threading traversing the longitudinal track;

the delivery tool further includes:

a capsule catheter extending proximally from the distal portion of the delivery tool, and

a disc-assembly, the disc-assembly including:

a proximal disc fixedly coupled to the capsule catheter, the proximal disc shaped to define external threading that is complementary to the internal threading, and

a distal disc, the distal disc:
dimensioned to engage the longitudinal track,
fixedly coupled to the shaft, and
rotatably coupled to the proximal disc;

wherein the disc-assembly is disposed within the proximal capsule such that the external threading fits the internal threading; and

proximally retracting the proximal capsule with respect to the implant comprises rotating the capsule catheter in a first direction such that:

the proximal disc rotates, in the first direction, along the internal threading, and

the proximal capsule moves along the distal portion axis, with respect to:

the disc-assembly, and

the implant.

102-103. (canceled)

104. The method according to claim **101**, wherein:

the distal disc includes a locking pin that is disposed within the longitudinal track, and

rotating the capsule catheter in the first direction comprises advancing the locking pin along the longitudinal track.

105-195. (canceled)

196. The apparatus according to claim **79**, wherein the distal capsule and the proximal capsule are each coupled to the shaft such that the distal capsule is translatable toward the proximal capsule, thereby minimizing the inter-capsule gap.

197. The apparatus according to claim **69**, wherein the prosthetic valve comprises:

an inner frame comprising the tubular portion; and

an outer frame comprising the plurality of flanges, wherein the outer frame and the inner frame are coupled to each other and wherein each of the flanges extend from a respective coupling point that is longitudinally between a proximal portion of the inner frame and a distal portion of the inner frame.

* * * * *